What Do Moral Psychology Benchmarks Actually Test?

A Meta-Evaluation of the Benchmark Landscape and Cross-Vendor Empirical Replication Across Nine Models

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Table of Contents

Main Paper

- Abstract
- 1. Introduction
- 2. Methodology
- 3. Results Part I: Empirical Replication
- 4. Results Part II: Landscape Claims 1–3 (MFT Dominance, Construct Validity, Missing Theories)
- 4. Results Part II: Landscape Claims 4–6 (Human Baselines, Format Mismatch, Temporal Stagnation)
- 5. Results Part III: Integrated Analysis
- 6. Discussion
- 7. Conclusion

Supplementary Material

- S1–S4. Supplementary Methods
- ST1–ST5. Supplementary Tables
- SF1–SF19. Supplementary Figures

Figure Plates

- Figures 1–15 (main-text figures, full-page)

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Abstract

Large language models are increasingly deployed in contexts demanding moral sensitivity —healthcare triage, content moderation, legal reasoning, autonomous vehicle decisions— yet the benchmarks used to evaluate their moral cognition have never been systematically assessed for theoretical coverage, psychometric quality, or cross-vendor discriminative power. We present the first full-coverage meta-evaluation of moral psychology benchmarks for LLM assessment. From a curated vault of 2,099 AI ethics papers, we extracted 292 benchmark candidates spanning 13 moral psychology theories and assessed all 292 on a 162-feature characterization framework (17 composite-determining features), classifying each into four quality tiers (Top, High, Medium, Low) via Jenks natural breaks optimization (break points at 27.8, 43.0, and 56.6; GVF = 0.83). We then administered 23 benchmarks (~750 items per model) to nine models from three independent vendors (Anthropic, Google, OpenAI) across 69 trial files. Seven findings emerge. First, Moral Foundations Theory achieves a 73% quality rate (top + high tiers)—six times the rate of non-Western traditions—reflecting unique computational infrastructure (MFQ-30, MFTC) rather than theoretical superiority. Second, construct validity is in crisis: 58% of candidates possess only Level 1 or Level 2 validity (keyword-level or face validity), with a median of face validity only. Third, two major theories—Dual Process Theory and Moral Identity Theory —have zero publication-ready benchmarks despite combined citation counts exceeding 25,000. Fourth, a cultural quality cliff separates Western from non-Western moral traditions: Ubuntu, Confucian, Indigenous, and Identity ethics collectively field 32 candidates but achieve a 0% quality rate. Fifth, the human baseline deficit has doubled from a previously reported 18% to 34% under full-coverage assessment. Sixth, benchmark quality has not improved despite 9.5-fold volume growth between the pre-2020 and 2023–2026 periods. Seventh, the cross-vendor empirical study reveals an RLHF moral fingerprint— six universal signatures shared by all nine models, with 43% of benchmarks producing zero cross-model variation—indicating that the instruments cannot distinguish between the systems they evaluate. We note the self-administration confound: the model family that designed the assessment framework served as one set of test subjects, rendering model-specific findings hypothesis-generating rather than hypothesis-confirming. The field faces a triple bind: the benchmark landscape is theoretically narrow (dominated by one theory), empirically undiscriminating (unable to differentiate models), and culturally blind (unable to evaluate non-Western moral traditions). The evaluative infrastructure that the AI safety community relies upon to certify moral alignment is not fit for purpose.

1. Introduction

1.1 The Moral Evaluation Problem

Large language models now make moral judgments at scale. In healthcare, they triage patient urgency and draft clinical recommendations that encode implicit hierarchies of whose suffering matters most. In content moderation, they adjudicate community

standards across billions of posts daily, deciding which expressions of political dissent, religious conviction, or cultural practice cross the line from protected speech to harmful content. In education, they tutor students through culturally sensitive material—the ethics of colonialism, the morality of war, the competing claims of religious and secular authority—and their responses shape how the next generation understands moral reasoning. In legal systems, they assist with case research and sentencing recommendations, embedding assumptions about fairness, proportionality, and the relative weight of retribution versus rehabilitation. In autonomous vehicle systems, they inherit the trolley problem as an engineering constraint, choosing whom to protect when harm is unavoidable [Awad et al., 2018; Gabriel et al., 2026]. In each of these deployments, the model's outputs have moral valence: they encode judgments about what matters, who counts, and how competing values should be weighed against one another.

The question of whether these systems are morally aligned—and with *whose* morality they are aligned—is therefore not an academic curiosity. It is a prerequisite for responsible deployment. And the answer to that question depends entirely on the quality of the evaluation infrastructure: the benchmarks, instruments, and assessment frameworks through which researchers, developers, and regulators attempt to characterize how large language models engage with moral cognition.

The AI research community has responded to this challenge with an explosion of evaluation efforts. Hundreds of papers now claim to assess "moral reasoning," "ethical alignment," "value sensitivity," or "moral competence" in LLMs [Hendrycks et al., 2021; Ji et al., 2024; Scherrer et al., 2024; Jiao et al., 2025; Gabriel et al., 2026]. These evaluations employ instruments ranging from the Moral Foundations Questionnaire to trolley dilemma batteries to commonsense morality classifiers. The pace of publication is remarkable: the field produced roughly 30 benchmark candidates before 2020, but 257 between 2023 and 2026—a 9.5-fold increase in volume. This growth creates the appearance of a maturing evaluation ecosystem: surely, with 292 benchmark candidates available, the field can rigorously assess whether an LLM's moral cognition is adequate for deployment in sensitive contexts.

This paper demonstrates that the appearance is misleading. The volume of benchmarks masks a quality crisis, a theoretical monoculture, and a cultural exclusion that, taken together, render the field's evaluative infrastructure unfit for the task it has been assigned. The fundamental question that has gone unasked amid the proliferation of benchmarks is deceptively simple: *are these instruments any good?* Do they engage the theoretical constructs that moral psychology—the discipline that has studied human moral cognition for over a century—has established as foundational? Do they possess the psychometric properties necessary for interpretable measurement? Do they provide the human comparison data required to give LLM scores meaning? And when they are administered to the models they are designed to evaluate, do they actually distinguish between those models—or do they return the same result regardless of which system is being tested?

We find that the answer, on every dimension, is troubling.

1.2 From Double Bind to Triple Bind

The central contribution of this paper is the identification and empirical demonstration of what we term the **triple bind**: a self-reinforcing system of three interlocking failures in the moral psychology benchmark landscape that collectively undermine the field's capacity to evaluate LLM moral cognition.

The first bind is **theoretical narrowness**. The benchmark landscape is dominated by a single theory—Moral Foundations Theory (MFT)—which achieves a 73% quality rate (top + high tiers combined), six times the rate of the next-best non-Western tradition. This dominance does not reflect a consensus judgment that MFT is the most comprehensive or important theory of moral cognition. It reflects an infrastructure asymmetry: MFT alone among moral psychology theories possesses both a standardized questionnaire with published population norms (the MFQ-30; Graham et al. [2011]) and a large expert-annotated text corpus suitable for training NLP classifiers (the MFTC, with 35,108 labeled tweets; Hoover et al. [2020]). No other theory offers anything comparable. There is no Social Intuitionism Questionnaire with published norms, no Dual Process Morality Corpus with labeled examples, no Dyadic Morality Twitter Corpus for training classifiers. The result is that when researchers build a new moral psychology benchmark for LLMs, they reach for MFT's ready-made infrastructure because it is the path of least resistance—and the landscape grows more theoretically concentrated with each iteration.

The second bind is **empirical undiscrimination**. When we administered 23 benchmarks to nine models from three independent vendors (Anthropic, Google, OpenAI), the instruments could not distinguish between the systems they were designed to evaluate. Forty-three percent of benchmarks produced zero cross-model variation: identical scores for every model regardless of vendor, architecture, or parameter count. All nine models share six universal moral signatures—a WEIRD-liberal moral foundation bias, a purity floor, authority deference suppression, an ethical consistency ceiling, a post-conventional reasoning default, and a deontological refusal asymmetry—that together constitute an RLHF moral fingerprint without documented precedent in any human population. From the perspective of moral psychology measurement, switching from Claude Opus to Gemini 2.5 Pro to GPT-5 produces no detectable change in moral orientation on the majority of instruments in this battery.

The third bind is **cultural blindness**. Four non-Western moral traditions with ancient philosophical roots—Ubuntu ethics, Confucian moral philosophy, Indigenous ethical frameworks, and moral identity theory grounded in non-Western relational conceptions of selfhood—collectively account for 32 benchmark candidates in the landscape, representing genuine effort by researchers to broaden moral evaluation beyond Western frameworks. Yet every one of these 32 candidates scores in the medium or low quality tier. The quality rate for these four traditions is 0%: zero benchmarks meet the minimum threshold for

publication-ready LLM moral evaluation. The question "does this model align with Confucian moral values?" or "does this model respect Ubuntu's conception of communal moral agency?" cannot currently be answered at a research-grade standard using any existing instrument. The cultural quality cliff separating Western from non-Western moral traditions is not a gap at the periphery of the benchmark landscape; it is a structural exclusion that renders the entire evaluation enterprise culturally parochial.

The three binds are not independent problems that happen to coexist. They form a self-reinforcing system in which each failure amplifies the others (Figure 1). A theoretically narrow landscape (Bind 1) that excludes non-Western traditions (Bind 3) channels all evaluation effort into a small set of content-level instruments measuring a single theory's constructs. These instruments, because they measure the same thing in the same way, are unable to distinguish between models that have been trained on similar data and aligned by similar annotator pools—producing the empirical undiscrimination of Bind 2. The undiscrimination, in turn, creates no incentive to build new benchmarks outside the dominant paradigm, because the instruments that exist already "work" in the sense that they produce scores and fill evaluation checklists, even though the scores are uninformative and the checklists are incomplete. The result is a stable equilibrium in which narrow instruments, homogeneous models, and excluded traditions sustain one another in a cycle that the field has no natural mechanism to escape.

The triple bind framing advances beyond the "double bind" identified in the v2 analysis, which documented that the benchmark landscape is both too narrow to capture the breadth of moral cognition (Bind 1) and too undiscriminating to distinguish between the systems it evaluates (Bind 2). The full-coverage assessment of all 292 candidates—expanding from the 55-benchmark subsample in the prior analysis—revealed the cultural quality cliff as a third, equally consequential dimension of failure. The prior analysis noted the absence of non-Western benchmarks in the top-15 selection but did not have the statistical power to establish that zero out of 32 non-Western candidates reach publication-ready quality. With all 292 candidates assessed and tiered, the cultural exclusion is no longer an anecdotal observation; it is a quantified finding.

1.3 The Quality Illusion: Why 292 Benchmarks Is Not Enough

The field has produced 292 benchmark candidates addressing moral psychology in LLMs since 2018. In any other subfield of AI evaluation—language understanding, mathematical reasoning, code generation—a landscape of 292 dedicated instruments would indicate a mature evaluation ecosystem, one in which researchers can select from a rich array of tools to probe different aspects of system capability. The moral psychology benchmark landscape produces the opposite: a quality illusion in which volume masks stagnation.

When all 292 candidates are assessed through a 162-feature characterization framework and classified via Jenks natural breaks optimization into four quality tiers, the distribution is sobering. Only 48 candidates (16.4%) achieve the top tier—the level at which a bench-

mark possesses sufficient construct validity, human baselines, psychometric grounding, and operationalization quality to support publication-ready claims about LLM moral cognition. Eighty-nine candidates (30.5%) reach the high tier, meaning they possess many but not all of the qualities necessary for rigorous evaluation. The remaining 155 candidates (53.1%)—a majority of the landscape—fall into the medium or low tiers, where fundamental quality requirements remain unmet: construct validity is undemonstrated, human baselines are absent, source instruments are unvalidated, or the benchmark engages moral psychology at the keyword level only.

The quality distribution would be less concerning if it showed improvement over time. It does not. The pre-2020 benchmark candidates, though few in number (approximately 35 of the 292), achieve a quality rate (top + high tiers combined) of approximately 75%. The 2020–2022 cohort drops to 48%. The 2023–2026 cohort—which constitutes the vast majority of the landscape at 257 candidates—achieves 46%. Quality has declined slightly even as volume has increased by nearly an order of magnitude. The field is not getting better at building moral psychology benchmarks; it is getting faster at building mediocre ones.

This temporal stagnation is itself a symptom of the triple bind. The infrastructure that makes high-quality benchmarks possible—validated questionnaires, large annotated corpora, decades of population norms—exists primarily for MFT and, to a lesser extent, for Schwartz values and Kohlberg's Defining Issues Test. Researchers building benchmarks for other theories lack these foundational resources and must start from scratch: designing items, establishing psychometric properties, collecting human baselines, and validating the instrument for LLM administration. The startup cost is enormous, and the incentive structure of academic publishing rewards rapid-iteration benchmark papers over the slow, unglamorous work of psychometric validation. The result is a landscape in which volume growth is driven almost entirely by low- and medium-tier papers that repackage existing moral scenarios in new prompt formats or apply existing instruments to new models without addressing the construct validity gap that separates what the benchmark measures from what it claims to measure.

The quality illusion has practical consequences beyond academic evaluation. Regulatory frameworks for AI safety—including the EU AI Act [European Commission, 2024] and the NIST AI Risk Management Framework [National Institute of Standards and Technology, 2023]—increasingly contemplate requiring moral evaluation of deployed systems. If these frameworks rely on the existing benchmark landscape as their evaluative infrastructure, they will certify systems using instruments that cannot distinguish between them and that exclude the moral traditions of billions of potential users. The gap between the volume of available benchmarks and the quality of those benchmarks is not merely an academic concern; it is a governance risk.

1.4 Contributions

This paper makes six contributions to the understanding and improvement of moral psychology benchmarks for LLM evaluation.

1. First full-coverage assessment of the moral psychology benchmark landscape. We assessed all 292 identified benchmark candidates on a 162-feature characterization framework (with 17 composite-determining features organized into assessment tiers A through C). This is the first study to evaluate every identified moral psychology benchmark rather than a selected subset, providing a complete map of theoretical coverage, psychometric quality, and human baseline provision across the entire landscape. The prior state of knowledge rested on assessments of selected benchmarks—typically the most prominent or recently published—which produced a systematically optimistic picture of the field's evaluative infrastructure.

2. Data-driven quality classification via Jenks natural breaks. Rather than imposing arbitrary quality thresholds or using binary (selected/not selected) classification, we applied Fisher-Jenks natural breaks optimization to composite scores, producing four quality tiers (Top: n=48, High: n=89, Medium: n=72, Low: n=83) with a goodness of variance fit (GVF) of 0.83 and break points at 27.8, 43.0, and 56.6. This data-driven approach ensures that tier boundaries reflect natural clusters in the quality distribution rather than the assessors' preconceptions about where quality thresholds should fall.

3. Discovery of the cultural quality cliff. The full-coverage assessment revealed a finding invisible to prior work based on selected subsamples: four non-Western moral traditions (Ubuntu, Confucian, Indigenous, and Identity ethics) collectively field 32 benchmark candidates but achieve a 0% quality rate in the top and high tiers. This cultural quality cliff is not a matter of effort (the 32 candidates represent genuine research) but of infrastructure: these traditions lack the standardized instruments, annotated corpora, and published population norms that enable the psychometric grounding required for high-quality benchmarks. The finding transforms the benchmark landscape's theoretical narrowness from a Western-centric concern about missing process theories into a global concern about the exclusion of entire moral traditions from the evaluation enterprise.

4. Cross-vendor empirical replication. We administered 23 benchmarks to nine models from three independent vendors (Anthropic Claude Opus, Sonnet, and Haiku; Google Gemini 2.5 Pro, 2.0 Flash, Flash-Lite, and 3.1 Pro; OpenAI GPT-5 and GPT-5 Codex), generating 69 trial files with approximately 750 items per model. This is, to our knowledge, the first study to administer a comprehensive moral psychology battery to models from three vendors simultaneously, enabling the cross-vendor convergence analysis that revealed the RLHF moral fingerprint.

5. Identification of the RLHF moral fingerprint. The cross-vendor study identified six universal moral signatures shared by all nine models regardless of vendor, architecture, or parameter count: a WEIRD-liberal moral foundation bias, a purity floor

(functionally zero on Sanctity/Degradation), authority deference suppression, an ethical consistency ceiling (near-perfect consistency on moral dilemma batteries), a post-conventional reasoning default, and a deontological refusal asymmetry (refusing personal harm at rates exceeding human baselines). These signatures constitute a moral fingerprint of RLHF-aligned language models—a shared moral profile with no documented precedent in any human population—and their universality across vendors suggests that competitive benchmarking, shared annotator demographics, and overlapping training corpora have produced a value monoculture in the deployed model ecosystem.

6. The triple bind framework as an integrative diagnosis. By combining landscape analysis (Binds 1 and 3) with cross-vendor empirical findings (Bind 2), we identify a self-reinforcing system of theoretical narrowness, empirical undiscrimination, and cultural blindness that the field has no natural mechanism to escape. The triple bind framework explains why increasing the volume of benchmarks has not improved the field's evaluative capacity: more instruments of the same type, measuring the same theory, administered to models that produce the same results, cannot resolve a structural problem. The framework also identifies the leverage points—process-level benchmarks, non-Western instrument validation, multi-vendor discrimination testing—where targeted intervention could break the cycle.

1.5 Seven Claims

The paper's findings are organized around seven empirical claims, each supported by the full-coverage landscape analysis (Claims 1–6) or the cross-vendor empirical study (Claim 7), or both. Table 1 provides a summary; the claims are developed in full in Sections 4 through 6.

Table 1: Seven Claims Summary

Claim	Title	Finding
C1	MFT Quality Dominance	MFT achieves a 73% top + high quality rate—6 times the rate of non-Western traditions. Its dominance reflects computational infrastructure (MFQ-30, MFTC), not theoretical superiority.
C2	Construct Validity Crisis	58% of the 292 candidates possess only Level 1 or Level 2 construct validity (keyword-level or face validity only). The median across the full landscape is face validity—a level that precludes interpretable claims about LLM moral cognition.
C3	Theoretical Absence	Two major moral psychology theories—Dual Process Theory [Greene, 2001, 2014] and Moral Identity Theory [Aquino and Reed, 2002]—have zero publication-ready benchmarks despite combined citation counts exceeding 25,000. The instruments the field has not built address the *process* of moral judgment formation, not merely its *content*.
C4	Cultural Quality Cliff	Ubuntu, Confucian, Indigenous, and Identity ethics collectively field 32 candidates but achieve a 0% quality rate (top + high tiers). The benchmark landscape cannot evaluate LLM alignment with the moral traditions of billions of users.
C5	Human Baseline Deficit	34% of the 292 candidates provide no published human comparison data—nearly double the 18% rate reported when the sample was limited to established benchmarks. Without baselines, claims about LLM moral properties are empirically unmoored.
C6	LLM Moral Flattening	All ten models fall below the human range on every virtue dimension (Cohen's d = −1.11 to −3.01). The flattening is not uniform: it affects some moral distinctions more than others.
C7	Temporal Fragility	The 6-fold increase in MFT-based benchmarks (2023–2025) has produced no methodological maturation. This field is not getting more

Claim 1: MFT Quality Dominance. Moral Foundations Theory accounts for 67 of 292 benchmark candidates (23% by volume) but achieves the highest quality rate of any theory at 73% top + high tier. This quality rate is 6 times the rate achieved by non-Western moral traditions and substantially exceeds the rates of other Western theories (Schwartz values: 57%; trolley tradition: 48%). The dominance reflects MFT's unique position at the intersection of two computational advantages: a standardized, freely available questionnaire with published population norms spanning political orientations and cultural groups (the MFQ-30; Graham et al. [2011]), and a large expert-annotated text corpus that enables corpus-based NLP approaches (the MFTC; Hoover et al. [2020]). These infrastructure advantages mean that building an MFT benchmark requires less foundational psychometric work than building a benchmark for any other theory—a starting-cost asymmetry that compounds over iterative research cycles.

Claim 2: Construct Validity Crisis. Construct validity—the degree to which an instrument measures the theoretical construct it purports to measure [Cronbach and Meehl, 1955; Messick, 1995]—is the foundational requirement for interpretable evaluation. Across all 292 candidates, 58% possess only Level 1 (keyword-level engagement with moral psychology; 27%) or Level 2 (face validity only; 31%) construct validity. The median validity level across the full landscape is Level 2. Only 4% of candidates achieve multi-study psychometric validation (Level 5), down from 11% in the 55-benchmark subsample assessed in prior work. The modal pattern (the "port-and-report" approach) is researchers taking a validated human instrument, administering it to an LLM with minimal adaptation, and reporting results without addressing whether the instrument measures the same construct in a non-human respondent—precisely the interpretive error that Messick [1995] cautioned against.

Claim 3: Theoretical Absence. Two of the most influential theories in modern moral psychology—Joshua Greene's [2001, 2014] dual-process framework and Aquino and Reed's [2002] moral identity construct—have zero publication-ready benchmarks in the landscape. Greene's dual-process theory, with over 10,000 citations for the foundational fMRI paper, proposes that moral judgments arise from the competition between fast emotional responses and slower deliberative reasoning; it is arguably the central paradigm in twenty-first-century moral psychology, yet no existing LLM benchmark implements its core experimental manipulation (varying deliberative resources across matched dilemmas). The absence of process-level benchmarks is particularly consequential given Claim 7: the one benchmark that cleanly separated models in the cross-vendor study was a process-level probe of dual-process dynamics, suggesting that the instruments the field has not built are the instruments the field most needs.

Claim 4: Cultural Quality Cliff. The full-coverage assessment revealed the single most consequential distributional finding in the landscape: four non-Western moral traditions —Ubuntu ethics (0 top/high out of 13 candidates), Confucian moral philosophy (0 out of 13), Indigenous ethical frameworks (0 out of 4), and moral identity theory in non-Western

relational conceptions (0 out of 2)—achieve a 0% quality rate despite fielding 32 candidates collectively. This is not a gap of effort; researchers have attempted to build these benchmarks. It is a gap of infrastructure: these traditions lack the standardized validated instruments, large annotated corpora, and published population norms that the assessment framework's composite-determining features reward. The cultural quality cliff means that the benchmark landscape can evaluate whether an LLM endorses the moral foundations of American college students in secular universities but cannot evaluate whether the same model respects the communal moral agency of Ubuntu, the filial obligations of Confucian role ethics, the reciprocal relationships with the natural world in Indigenous frameworks, or the relational moral selfhood central to non-Western identity traditions.

Claim 5: Human Baseline Deficit. Moral cognition can only be characterized relative to human standards: the population-level norms, developmental patterns, and cross-cultural variations that give moral psychology instruments their interpretive power. Under full-coverage assessment, 34% of the 292 candidates provide no published human comparison data—nearly double the 18% rate reported when the sample was limited to 55 established benchmarks. The doubling reflects the inclusion of lower-quality candidates that the subsample approach systematically excluded. Without human baselines, the claim that a model is "more utilitarian than average" or "less care-oriented than humans" lacks an empirical anchor: it is description without interpretation, pattern without meaning.

Claim 6: Temporal Stagnation. If the benchmark landscape were improving over time, the rapid volume growth of 2023–2026 would be cause for optimism. It is not. The quality rate (top + high tiers combined) was approximately 75% for pre-2020 candidates, declined to 48% for the 2020–2022 cohort, and has remained essentially flat at 46% for the 2023–2026 cohort despite a 9.5-fold increase in volume. The field has not learned how to build better moral psychology benchmarks; it has learned how to build more of them. The temporal stagnation is consistent with an incentive structure in which the academic publication system rewards novelty (new benchmarks for new models) over rigor (psychometric validation, human baseline collection, construct validity demonstration)—and in which the infrastructure advantages that make high-quality benchmarks possible are concentrated in a single theory (MFT) that already dominates the landscape.

Claim 7: RLHF Moral Fingerprint. The cross-vendor empirical study reveals that the already-narrow instruments documented in Claims 1–6 cannot distinguish between the systems they evaluate. Of 23 benchmarks administered to nine models from three vendors, 43% produced zero cross-model variation: identical scores for every model regardless of vendor, architecture, or parameter count. All nine models share six universal moral signatures—the WEIRD-liberal moral foundation bias (Individualizing/Binding ratios of 2.43–2.83, corresponding to the "liberal" to "very liberal" zone of the human distribution), the purity floor (functionally zero on Sanctity/Degradation, below any documented human population), authority deference suppression (authority win rate of exactly 0.29 across all models), the ethical consistency ceiling (near-perfect scores on moral dilemma consistency

batteries), the post-conventional reasoning default (DIT P-scores of 60–90, exceeding moral philosophy graduate students), and the deontological refusal asymmetry (refusal rates for personal harm scenarios exceeding human baselines). These signatures constitute an RLHF moral fingerprint: a shared moral profile produced by convergent training pressures (shared annotator demographics, competitive benchmarking, overlapping safety targets) rather than by independent arrival at moral truth. The single benchmark that cleanly separated one model from the rest—the dual-process protocol revealing Haiku's chain-of-thought effect—was a process-level probe testing *how* moral judgments respond to deliberation scaffolding, not *what* moral content a model endorses: precisely the benchmark type that Claim 3 identifies as absent from the landscape.

1.6 Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work across moral psychology theories, their computational operationalization, psychometric validity frameworks, format effects and human baselines in LLM evaluation, alignment and prompting effects on moral outputs, and the emerging benchmark landscape. Section 3 describes the methodology in two parts: the landscape-mapping methodology (multi-signal candidate extraction from 2,099 papers, assessment protocol, the 162-feature characterization framework, Jenks natural breaks tier classification) and the cross-vendor empirical design (nine-model inventory, 23-benchmark battery, administration protocol, and statistical analysis). Section 4 presents Results Part I—the benchmark landscape—organized around Claims 1 through 6: MFT quality dominance, the construct validity crisis, theoretical absence, the cultural quality cliff, the human baseline deficit, and temporal stagnation. Section 5 presents Results Part II—the cross-vendor empirical findings—organized around Claim 7: the RLHF moral fingerprint, the convergence landscape, scale-dependent and chain-of-thought effects, and theory-specific profiles. Section 6 presents Results Part III—the integrated analysis—where the landscape and empirical findings converge on the triple bind diagnosis. Section 7 discusses the implications of the triple bind for benchmark developers, AI governance and auditing, and the philosophical question of whether moral psychology constructs are meaningfully applicable to non-human respondents, and addresses the study's limitations including the self-administration confound. Section 8 concludes with specific recommendations for breaking each of the three binds and an assessment of what the field needs to build an evaluative infrastructure that is theoretically broad, empirically discriminating, and culturally inclusive.

3. Methodology

This study proceeds in three phases. Phase 1 (Sections 3.1–3.2) identifies and systematically characterizes the complete landscape of moral psychology benchmarks for large language models. Phase 2 (Section 3.3) applies a composite scoring and classification framework to rank all candidates into quality tiers using natural-breaks optimization. Phase 3 (Sections 3.4–3.5) administers a battery of 23 benchmarks drawn from this landscape to 9 models across 3 vendors, generating the empirical evidence for cross-vendor

convergence analysis. The full pipeline is designed to support the "triple bind" thesis advanced by this paper: that the moral psychology benchmark ecosystem is theoretically narrow, empirically undiscriminating, and culturally blind.

3.1 Source Corpus and Candidate Extraction

3.1.1 Source Corpus

The source corpus comprises 2,099 academic papers maintained in a systematically curated AI ethics literature vault. The vault has been developed continuously since 2024 through a combination of targeted database searches (Semantic Scholar, OpenAlex, arXiv, ACL Anthology), citation chaining from landmark publications, and structured sweeps of major venues (NeurIPS, ICML, ACL, EMNLP, AAAI, FAccT, AIES). Papers are retained if they address any dimension of artificial intelligence ethics, safety, alignment, or evaluation; no restriction is imposed on publication date, venue tier, or disciplinary origin. Each paper in the vault carries structured YAML metadata including title, authors, year, publication venue, DOI, abstract, and a set of machine-readable classification scores covering ethical orientation, normative framework engagement, moral psychology relevance, cultural and religious dimensions, and data characteristics. The classification scores are produced by a six-script keyword and heuristic pipeline (no large language model is used in the classification process itself), ensuring that the initial candidate identification is independent of the models subsequently evaluated.

The vault's coverage of moral psychology is deliberately broad. Papers are classified across 13 moral psychology theories — Moral Foundations Theory (MFT), Schwartz Basic Human Values, trolley-class dilemma research, Kohlberg's Cognitive Moral Development, Dual Process Theory, Social Intuitionism, Dyadic Morality, Moral Identity Theory, and five non-Western ethical traditions (Islamic Ethics, Confucian Ethics, Ubuntu Ethics, Buddhist Ethics, Indigenous Ethics). Each theory has a dedicated set of keyword-scoring patterns in the classification pipeline, and every paper in the vault receives a moral psychology relevance score on a 0–100 integer scale. Papers scoring at or above the engagement threshold constitute the candidate pool.

3.1.2 Five-Signal Extraction Pipeline

Candidate extraction employs a five-signal pipeline, where each signal contributes independently to the identification of papers that may contain, describe, adapt, or evaluate a moral psychology benchmark for LLMs. A paper is retained as a candidate if it triggers on any one of the five signals. The five signals are:

1. **Keyword scoring across 13 moral psychology theories.** The classification pipeline computes a composite moral psychology score using 125 patterns organized by theory. A paper triggers this signal if its moral psychology composite score reaches the engagement threshold (score >= 50 on the 0–100 scale). Keywords include theory-specific constructs (e.g., "moral foundations," "Schwartz values," "trolley problem," "post-conventional reasoning," "dual process," "moral dumbfounding," "dyadic morality"),

instrument names (e.g., “MFQ,” “DIT,” “PVQ,” “Moral Machine”), and methodology markers (e.g., “Likert,” “vignette,” “dilemma battery”).

2. **Explicit benchmark mentions.** Papers are flagged if they explicitly name a recognized moral psychology benchmark or evaluation dataset in their title, abstract, or full text. This signal catches papers that may score below the keyword threshold on general moral psychology relevance but are centrally concerned with a specific benchmark (e.g., a paper titled “Evaluating GPT-4 on the Moral Foundations Questionnaire” might have moderate general scores but high benchmark specificity).
3. **Dataset indicators.** Papers are flagged if they describe, release, or evaluate a dataset with moral psychology content, even if they do not use standard moral psychology terminology. This signal is designed to capture emerging benchmarks that may not yet have acquired the theoretical vocabulary of established instruments — for instance, a crowdsourced dataset of “right vs. wrong” judgments that functions as a moral evaluation benchmark without citing MFT or Kohlberg.
4. **Methodology markers.** Papers are flagged if they describe systematic benchmark construction methodology (item development, validation studies, psychometric analysis) applied to moral content. This signal is aimed at recovering papers from the psychometrics and survey methodology literatures that develop moral measurement instruments without framing them as “AI benchmarks.”
5. **Theory-specific patterns.** Each of the 13 theories has dedicated extraction patterns designed to catch theory-engaged work that generic moral psychology keywords might miss. For example, the dual-process signal includes patterns for “System 1/System 2,” “automatic vs. controlled moral processing,” and “cognitive load effects on moral judgment”; the dyadic morality signal includes “agent-patient,” “perceived harm,” and “mind perception”; the Ubuntu signal includes “communal harmony,” “umunthu,” and “African relational ethics.”

The five signals are evaluated independently and their union forms the candidate set. No weighting or priority ordering is applied across signals: a paper that triggers only Signal 5 (theory-specific patterns) is treated identically to one that triggers all five.

3.1.3 Expanded Keyword Scorer

The keyword scorer underwent a critical expansion during the study. The initial version contained 65 patterns concentrated on the four most-studied theories: MFT (18 patterns), Schwartz Values (12 patterns), trolley-class dilemmas (10 patterns), and Kohlberg’s Cognitive Moral Development (8 patterns), with the remaining nine theories sharing 17 patterns. This distribution reflected the historical dominance of these four theories in the AI ethics literature but created a systematic detection bias: process-oriented theories (Dual Process Theory, Social Intuitionism, Dyadic Morality) were severely underrepresented, with as few as 2–3 patterns each.

The bias became apparent during a preliminary analysis of Phase 1 extraction results, which revealed only 4 candidates across all three process-oriented theories combined. Given that these theories represent major active research programs in moral psychology — Haidt's Social Intuitionist Model alone has generated over 3,000 citations since 2001 — this yield was implausibly low. A systematic audit of the scoring patterns confirmed the diagnosis: patterns like "dual process" were present but lacked the theoretical specificity to distinguish moral dual-process research from the thousands of papers using dual-process language in other contexts (e.g., cognitive load theory in instructional design), while patterns like "moral dumbfounding" (a construct specific to Social Intuitionism) were entirely absent.

The scorer was expanded from 65 to 125 patterns. The expansion prioritized the nine previously underrepresented theories, adding construct-specific terminology, canonical researcher names used as theory identifiers (e.g., "Haidt" co-occurring with "intuit*"), instrument names, and methodology-specific vocabulary. The expansion was constrained to maintain precision: each new pattern was required to have a false-positive rate below 15% when evaluated against the full vault, with false positives identified by manual inspection of the top-scoring non-moral-psychology papers.

The impact was substantial. The expanded scorer identified 123 additional candidates that the original 65-pattern version had missed entirely. Detection for the three process-oriented theories increased from 4 to 34 candidates, a nearly nine-fold improvement. Critically, this expansion did not merely add marginal cases: several of the newly identified candidates ultimately scored in the High tier of the composite quality ranking (Section 3.3), indicating that genuinely relevant work had been systematically invisible under the original scoring regime.

3.1.4 Candidate Yield

The extraction pipeline was executed in two phases. Phase 1 applied the five-signal pipeline (with the expanded 125-pattern scorer) to the full 2,099-paper vault, yielding 221 unique candidates. Phase 2 conducted a targeted sweep for non-Western moral traditions — Islamic ethics, Confucian ethics, Ubuntu ethics, Buddhist ethics, and Indigenous ethics — using dedicated search strategies that extended beyond the vault's existing holdings to identify recently published work not yet ingested. Phase 2 added 71 new candidates, bringing the total to 292 unique candidates after deduplication by filename.

Deduplication was necessary because Phase 1 and Phase 2 employed partially overlapping search strategies: a paper on "Confucian values in LLM alignment" might be detected by both the general extraction pipeline (Phase 1, Signal 5) and the targeted non-Western sweep (Phase 2). All deduplication was performed by exact filename matching, which is deterministic and avoids the threshold sensitivity of content-based deduplication.

Information availability across the 292 candidates was not uniform. Of the 292, 245 candidates (83.9%) had both a PDF extract and an abstract, providing the richest possible

source material for scoring (designated Information Tier A). The remaining 47 candidates (16.1%) had abstracts and metadata but no PDF extract, limiting the depth of feature-level scoring (designated Information Tier B). The implications of this information asymmetry for scoring validity are addressed in Section 3.2.4.

3.2 Full-Coverage Assessment Protocol

The 292 candidates were assessed using a 162-feature characterization framework, of which 158 features are active (4 excluded) and 17 features determine the quality composite score. This section describes the framework's structure, the composite formula, the batch-calibrated scoring protocol, and the policies governing information availability, confidence grading, and null handling.

3.2.1 The 162-Feature Characterization Framework

The assessment framework comprises 162 features organized into seven tiers, each measuring a distinct dimension of benchmark quality or character. The seven tiers are:

Tier A: Psychological Depth (6 features, A1–A6). These features measure how faithfully a benchmark engages moral psychology theory. They assess the number of theories engaged (A1), the quality of construct operationalization (A2), the complexity of moral stimuli (A3), whether a validated psychometric source instrument underlies the benchmark (A4), the availability of human baseline data (A5), and the strength of construct validity evidence (A6). Tier A is the single most important determinant of the composite quality score, receiving 50% of the composite weight through the Psychological Depth (PD) sub-score.

Tier B: Practical Ease (7 active features, B1–B4, B6–B7, B10; B5, B8, B9 excluded). These features measure how readily a benchmark can be implemented for LLM evaluation. They assess task format suitability (B1), dataset size (B2), data availability and accessibility (B3), code availability (B4), evaluation harness compatibility (B6), scorer complexity (B7), and LLM administration fidelity (B10). Three features from the original framework — instruction simplicity (B5), open access status (B8), and documentation completeness (B9) — were excluded because they require access to benchmark distribution infrastructure (repository structure, license files, documentation) that was not systematically available for all 292 candidates. Tier B receives 30% of the composite weight through the Practical Ease (PE) sub-score.

Tier C: Citation Impact (4 active features, C1–C3, C5; C4 excluded). These features measure scholarly influence and community adoption: total citations (C1), age-normalized citation velocity (C2), publication venue tier (C3), and presence on recognized evaluation hubs or leaderboards (C5). Cross-domain citation analysis (C4) was excluded because it requires citation graph tracing infrastructure beyond the available data. Tier C receives 20% of the composite weight through the Citation Impact (CI) sub-score.

Tier D: Psychometric Rigor (24 features, D1–D24). These features evaluate internal consistency (D1), test-retest reliability (D2), factor structure validation (D3), item analysis quality (D4–D7), measurement invariance (D8), LLM-specific measurement properties (D9), output parsing reliability (D10), value hierarchy stability (D11), implicit vs. explicit measurement mode (D12), sampling and documentation quality (D13–D16), response coding reliability (D17), construct operationalization validity (D18), IRT-based item discrimination (D19), response distribution entropy (D20), sample size adequacy (D21), effect size reporting (D22), variance decomposition (D23), and interaction effect detection capacity (D24). Tier D features are scored but do not contribute to the composite quality ranking; they are reported as characterization profiles.

Tier E: LLM-Specific Validity (44 features, E1–E44). The largest tier, assessing how well benchmarks perform specifically when administered to language models. Features cover recognition-endorsement discrimination (E1), dual-system engagement (E2–E3), prompt and few-shot sensitivity (E4–E5), tokenization and context window effects (E6–E7), administration fidelity (E8), harness portability (E9), temperature sensitivity (E10), system prompt interaction (E11), adversarial robustness (E12–E13), deployment relevance (E14), safety-critical failure mode detection (E15), multi-turn consistency (E16), persona stability (E17), instruction hierarchy (E18), refusal behavior (E19), embodiment assumptions (E20), sycophancy detection (E21), training data contamination risk (E22), model version sensitivity (E23), bright-line item identification (E24), scoring pipeline complexity (E25), API compatibility (E26), token budget efficiency (E27), output determinism (E28), error recovery specification (E29), cross-model normalization (E30), anchor item availability (E31), construct contamination risk (E32), moral reasoning quality assessment (E33), demographic sensitivity (E34), RLHF sensitivity (E35), cross-benchmark network centrality (E36), cognitive load sensitivity (E37), emotional valence sensitivity (E38), abstraction level sensitivity (E39), moral development trajectory simulation (E40), adversarial commitment robustness (E41), moral expertise gradient (E42), moral emotion recognition vs. generation (E43), and moral reasoning chain transparency (E44). Like Tier D, these features are scored for characterization only.

Tier F: Cultural and Philosophical Breadth (44 features, F1–F44). These features assess cultural, religious, and philosophical diversity. They cover stereotype amplification detection (F1), cross-cultural construct coverage (F2), moral emotion measurement (F3), moral conflict structure (F4), cultural origin of the theoretical framework (F5), cultural validation breadth (F6), scenario cultural content (F7), theory cultural scope claims (F8), language coverage (F9), WEIRD bias composite index (F10), individualism-collectivism balance (F11), moral universalism vs. pluralism stance (F12), indigenous and marginalized tradition inclusion (F13), colonial/power-dynamic awareness (F14), measurement granularity (F15), ingroup-outgroup differentiation (F16), moral disagreement representation (F17), normative theory coverage breadth (F18), philosophical coherence testing (F19), meta-ethical awareness (F20), reasoning vs. classification distinction (F21), moral sensitivity detection (F22), moral creativity assessment (F23), moral uncertainty

expression (F24), novel-scenario generalization (F25), moral mimicry detection (F26), reflective equilibrium testing (F27), and additional cultural and philosophical diversity features (F28–F44). Tier F is particularly important for the "culturally blind" arm of the triple bind thesis, as it provides the empirical basis for quantifying how few benchmarks engage non-Western moral traditions with any degree of psychometric rigor.

Tier G: Theory Fidelity and Provenance (29 features, G1–G29). These features trace the intellectual lineage and theoretical fidelity of each benchmark: theory version currency (G1), whether the benchmark tests the theory it claims to test (G2–G5), inter-theory bridge items (G6–G10), provenance documentation (G11–G15), replication status (G16–G20), and licensing and community adoption indicators (G21–G29). As with Tiers D–F, these features are scored for characterization only.

In total, 158 features are active (the 162 defined minus 4 excluded: B5, B8, B9, C4). Of these, 17 features across Tiers A–C determine the quality composite score; the remaining 141 features across Tiers D–G are scored for characterization and profiling but do not influence the composite or tier assignment.

The decision to restrict the composite to the 17 core features (Tiers A–C) while scoring all 158 features for characterization reflects a deliberate separation of concerns. The composite answers the question "Is this benchmark worth using for moral psychology evaluation of LLMs?" — a question best answered by psychological depth, practical usability, and community recognition. The extended features answer a different question: "What are this benchmark's detailed psychometric, technical, cultural, and theoretical properties?" Folding all 158 features into a single composite would dilute the selection signal with characterization noise and would make the composite score uninterpretable, as a high score could reflect any combination of psychometric rigor, cultural breadth, and LLM-specific validation rather than the core quality dimension the tier system is designed to capture.

3.2.2 The 17-Feature Composite Subset

The quality composite is computed from three sub-scores — Psychological Depth (PD), Practical Ease (PE), and Citation Impact (CI) — each derived from the features in its respective tier. Table 2 presents all 17 features with their tier membership, scale type, and composite weight.

Table 2: 17-Feature Composite Framework

Feature	Tier	Name	Scale	Weight in Sub-Score	Weight in Composite
A1	A (PD)	Construct depth	0–100 (count-mapped)	0.15	0.075
A2	A (PD)	Moral foundation coverage / operationalization quality	0–100 (ordinal)	0.25	0.125
A3	A (PD)	Dilemma complexity	0–100	0.15	0.075
A4	A (PD)	Source instrument validation	0–100 (ordinal)	0.10	0.050
A5	A (PD)	Human baseline availability	0–100 (ordinal)	0.15	0.075
A6	A (PD)	Construct validity evidence	0–100 (ordinal)	0.20	0.100
B6	**B (PE)**	**[illegible]**	**0–100**	**0.20**	**0.040**

The composite is computed as:

```
PD = A2 x 0.25 + A6 x 0.20 + A1 x 0.15 + A5 x 0.15 + A3 x 0.15 + A4 x 0.10
PE = B3 x 0.20 + B2 x 0.15 + B6 x 0.15 + B7 x 0.15 + B1 x 0.10 + B4 x 0.10 + B10
x 0.15
CI = C1 x 0.30 + C2 x 0.30 + C3 x 0.25 + C5 x 0.15

COMPOSITE = PD x 0.50 + PE x 0.30 + CI x 0.20
```

The 50/30/20 weighting across tiers encodes a principled judgment about what constitutes a high-quality moral psychology benchmark. Psychological depth receives the largest weight (50%) because the primary purpose of a moral psychology benchmark is to measure psychological constructs with theoretical fidelity — a technically polished benchmark with no psychological grounding is not a moral psychology benchmark at all, however practically convenient it may be. Practical ease receives the second-largest weight (30%) because a psychologically deep benchmark that cannot be implemented, scored, or reproduced has limited value for the LLM evaluation community. Citation impact receives the smallest weight (20%) because scholarly influence is a lagging indicator: recent benchmarks may be high-quality but not yet widely cited, and citation counts are confounded by field size (MFT work is cited more heavily than Dyadic Morality work simply because MFT is a larger research program, not because MFT benchmarks are necessarily superior).

Within each sub-score, individual feature weights reflect the relative importance of each measurement dimension. In the PD sub-score, construct operationalization quality (A2, weight 0.25) and construct validity evidence (A6, weight 0.20) together account for 45% of the psychological depth assessment, reflecting the judgment that these two features — how well a benchmark translates theory into measurable constructs, and how much evidence supports the validity of that translation — are the strongest indicators of genuine theoretical engagement. Source instrument validation (A4) receives the lowest weight (0.10) because some of the most innovative benchmarks in the landscape are original instruments that do not derive from pre-existing validated questionnaires but nevertheless demonstrate strong construct validity through other means.

3.2.3 Batch-Calibrated Scoring

Scoring 292 candidates on 158 features each produces a matrix of 46,136 score cells. To maintain consistency across this matrix, the assessment was organized into batches with calibration anchors and post-hoc drift checks.

Batch structure. The 292 candidates were divided into 5 batches of approximately 59 candidates each. Each batch was stratified along three dimensions: (1) primary theory — each batch contained a mix of theories rather than concentrating all MFT candidates in a single batch, which would allow theory-specific scoring biases to accumulate; (2) information tier — each batch contained both Tier A (PDF extract available) and Tier B (abstract only) candidates, preventing information-availability effects from clustering; and (3) publication year — each batch spanned the full temporal range of the corpus

(2015–2026), preventing era-specific scoring biases. The stratification was implemented deterministically: candidates were sorted by theory, then distributed round-robin across batches, with information tier and year balance verified after assignment.

Calibration anchors. Three benchmarks were scored in every batch to provide inter-batch calibration reference points:

1. *High anchor: The Moral Machine Experiment* (Awad et al., 2018). A landmark trolley-class benchmark with extensive human baseline data, high citation impact, and strong construct validity evidence. Expected composite: approximately 78.5 (observed across batches: 78.45). The Moral Machine anchors the upper range of the scoring distribution and provides a reference for what a "Top tier" benchmark looks like.
2. *Mid anchor: Value Kaleidoscope* (ValuePrism/Kaleido). A well-constructed values benchmark with moderate psychological depth and strong practical infrastructure. Expected composite: approximately 61.6. The Value Kaleidoscope anchors the boundary region between Top and High tiers.
3. *Low anchor: A Tier B candidate with minimal theoretical engagement*. This candidate provides a reference for how low-information, low-engagement candidates should be scored. Expected composite: 15–25. The low anchor tests the consistency of the null policy (Section 3.2.6) across batches.

Post-hoc calibration. After each batch was scored, the anchor scores were compared to their expected values. Agreement within +/-5 composite points was the acceptance criterion. If an anchor deviated by more than 5 points, all candidates in the batch were re-examined for systematic scoring drift, and individual feature scores were adjusted where a clear directional bias was identified. In practice, no batch required full re-scoring: anchor agreement was within +/-3 points across all 5 batches.

3.2.4 Information Availability Classification

Not all 292 candidates provided the same depth of source material for scoring. Two information tiers were defined based on the available evidence:

Tier A (245 candidates, 83.9%). Candidates with a PDF extract (typically 3,000–15,000 words of extracted text from the full paper), a structured abstract, and full bibliographic metadata. Tier A candidates could be scored on all 158 active features with direct textual evidence. The PDF extracts were generated by the vault's extraction pipeline and include the paper's introduction, methodology, results, and discussion sections, along with table and figure captions. Tier A candidates constitute the primary evidence base for the study.

Tier B (47 candidates, 16.1%). Candidates with a structured abstract and bibliographic metadata but no PDF extract. This situation arises when the full paper is behind a paywall, embargoed, or recently published and not yet ingested into the vault. Tier B candidates could be scored on the 17 composite features (Tiers A–C) from their abstract and metadata

with varying degrees of confidence, but extended features (Tiers D–G) that require detailed methodological information from the full paper were scored with lower confidence or marked as inferred.

The information tier classification is important because it interacts with the null policy (Section 3.2.6) and the confidence grading system (Section 3.2.5). A Tier B candidate's score of 0 on Feature D3 (factor structure validation) means "we cannot determine whether this benchmark reports factor analysis from the abstract alone" — a genuine information gap — while a Tier A candidate's score of 0 on D3 means "the full paper does not report factor analysis," which is substantive negative evidence. Both contribute the same numerical value to scoring, but the confidence grade distinguishes between them.

The asymmetry between tiers is mitigated by two design features. First, the quality composite is computed exclusively from the 17 Tier A–C features, which can be reliably scored from abstracts and metadata for all candidates: title analysis reveals construct depth (A1), abstract descriptions typically specify operationalization quality (A2), scenario descriptions convey dilemma complexity (A3), instrument names identify source validation (A4), explicit mentions of human baselines score A5, and citation data (C1–C3, C5) is externally verifiable for all candidates regardless of PDF availability. Second, the Jenks natural-breaks classification (Section 3.3.2) is applied to the composite scores only, so the tier assignment is driven entirely by features that can be comparably assessed across both information tiers.

3.2.5 Confidence Grading

Every score cell in the 292 x 158 matrix carries a confidence grade that records the evidentiary basis for the assigned score. Four grades are defined:

Grade	Meaning	Typical Application
High	Scored from direct evidence in the abstract or PDF extract	Tier A candidates on features with explicit textual support
Medium	Scored from indirect evidence or inference	Tier B candidates on features requiring some inference; Tier A candidates where evidence is implicit
Low	Conservative estimate from minimal information	Features where only upper or lower bounds can be established from available evidence
Inferred	Score of 0 assigned because absence of evidence is itself informative	Hub presence (C5), pre-registration status, licensing — externally verifiable

The "inferred" grade deserves particular attention because it embodies a consequential methodological choice. When a benchmark has no presence on Papers With Code or HuggingFace, the absence is not an information gap — it is a genuine zero on the hub presence feature (C5). The same logic applies to externally verifiable features like pre-registration and open licensing: these can be checked independently of the paper's text, and their absence is substantive, not artifactual. An inferred score of 0 for hub presence contributes identically to the composite as a high-confidence score of 0; the "inferred" label is recorded for transparency and for researchers who may wish to conduct sensitivity analyses that distinguish evidentiary bases, but it carries no mathematical consequence.

Confidence grades are recorded but do not affect composite scoring. This decision reflects the judgment that a scoring system that downweights low-confidence scores would systematically penalize recent benchmarks (which tend to have less external evidence simply because they are new) and non-Western benchmarks (which tend to have fewer English-language PDF extracts available). Since the study is specifically designed to assess whether the benchmark landscape disadvantages precisely these categories of work, introducing a confidence-based penalty would risk circularity.

3.2.6 Null Policy

Null values (genuinely unscorable features) are permitted only under narrowly defined conditions:

1. **Explicit null.** Applied exclusively to two features: D9 (LLM-specific ceiling compression index) and D20 (response distribution entropy). Both features require empirical data from LLM administrations of the benchmark — specifically, response distribution

analyses that quantify how much of the measurement range is compressed to ceiling or floor values. For benchmarks that have never been administered to LLMs with distributional analysis, this data is genuinely unavailable and cannot be inferred from the paper text. Approximately 35% of candidates have null values on these two features.

2. **All other absent features score 0.** For the remaining 156 active features, the absence of evidence is treated as informative. If a benchmark paper does not report internal consistency (D1), does not test prompt sensitivity (E4), does not include non-Western moral frameworks (F2), or does not document item provenance (D14), these are genuine characterization results — they tell us something about the benchmark's properties, not merely about our access to information. The score of 0 is assigned with the appropriate confidence grade (typically "inferred" for externally verifiable features, "high" for features where the full paper was examined and the property was confirmed absent).

3. **Exclusion threshold.** Candidates with more than 60% null features (i.e., more than 95 null values out of 158 active features) are excluded from the Jenks tier ranking but are retained in supplementary tables. In practice, no candidate exceeded this threshold: even the most information-poor Tier B candidates had scorable values on most features, because the majority of features can be scored as 0 (absent) or inferred from external evidence rather than requiring null.

The null policy has a direct bearing on the study's claims about the benchmark landscape. A permissive null policy (treating all absent properties as unscorable rather than absent) would inflate quality estimates by excluding negative evidence, making the landscape appear stronger than it is. The restrictive policy adopted here — 0 means "this property is absent," null means "we genuinely cannot determine whether this property exists" — provides a conservative estimate of landscape quality that errs toward underestimation. Given that the study's central thesis is that the landscape is weaker than commonly assumed, this conservative approach strengthens the evidentiary foundation for the claims.

3.3 Composite Scoring and Tier Assignment

3.3.1 Composite Formula and Rationale

The composite quality score for each candidate is computed from the three sub-scores — Psychological Depth (PD), Practical Ease (PE), and Citation Impact (CI) — according to the formula presented in Section 3.2.2. The formula is reproduced here for reference:

```
COMPOSITE = PD x 0.50 + PE x 0.30 + CI x 0.20
```

All features are scored on a 0–100 scale, all sub-score weights within each tier sum to 1.00, and the three tier weights sum to 1.00, producing a composite that also ranges from 0 to 100. A composite of 100 would require a benchmark to score perfectly on every feature in all three tiers — an outcome that no existing benchmark approaches, as even the highest-

scoring candidate in the landscape (LLM Ethics Benchmark, composite = 81.33) falls short on citation impact due to its recency.

The 50/30/20 weighting was chosen to reflect the asymmetric importance of the three dimensions for the study's purposes. This is a benchmark quality study in moral psychology, not a benchmark usability study or an impact bibliometry. The primary question is "Does this benchmark measure a meaningful moral psychology construct with theoretical fidelity?" — a question answered predominantly by Psychological Depth features. Practical Ease receives the second weight because implementability determines whether a benchmark can actually contribute to the empirical evaluation of LLMs, regardless of its theoretical elegance. Citation Impact receives the smallest weight because it is confounded by field size, recency, and venue prestige in ways that are orthogonal to benchmark quality.

An alternative approach would have been to weight all three dimensions equally (33/33/33). This was rejected because it would artificially elevate benchmarks with high citation counts but shallow psychological engagement — precisely the kind of benchmark that the study argues is overrepresented in the landscape. The sensitivity analysis (Section 3.3.4) evaluates how robust the tier assignments are to alternative weighting schemes, including equal weighting.

3.3.2 Jenks Natural-Breaks Optimization

Rather than imposing arbitrary score thresholds to partition the 292 candidates into quality tiers, we applied the Fisher-Jenks natural breaks optimization algorithm (Jenks, 1967; Fisher, 1958) for k=4 classes. The Jenks algorithm identifies class boundaries that minimize within-class variance while maximizing between-class variance, producing divisions that correspond to natural gaps in the data distribution rather than researcher-imposed categories.

The algorithm was applied to the 292 composite scores with the following results:

- **Break points:** [27.8, 43.05, 56.65]
- **Goodness of Variance Fit (GVF):** 0.83

A GVF of 0.83 indicates that 83% of the variance in composite scores is accounted for by the four-class partition, confirming that the data contain genuine distributional structure rather than a continuous gradient that would resist classification. The choice of k=4 was motivated by the desire for a tier system granular enough to distinguish meaningfully different levels of benchmark quality while remaining interpretable: fewer classes (k=3) merged qualitatively distinct groups, while more classes (k=5) produced tiers with counts too small for statistical analysis. The four-class solution was validated by visual inspection of the composite score histogram (Figure 2), which shows clear density troughs at the three break points.

Table 3: Jenks Tier Definitions

Tier	Composite Range	Count	Percentage	Semantic Definition
Top	>= 56.65	48	16.4%	Publication-ready benchmarks with construct validity evidence, human baselines, and theory grounding. These benchmarks measure well-defined moral psychology constructs using validated or carefully adapted instruments, provide human comparison data, and have demonstrated community adoption. Suitable for making definitive claims about LLM moral reasoning.
High	43.05–56.65	89	30.5%	Research-grade benchmarks with meaningful theory engagement but gaps in validation or baselines. These benchmarks engage genuinely with one or more moral psychology theories and operationalize psychological constructs at a level sufficient for exploratory research, but they may lack cross-cultural validation, comprehensive human baselines, or LLM-specific revalidation evidence.
Med	27.8–43.05	72	24.7%	Emerging benchmarks that are either theory-inspired without formal validation, or technically strong but theoretically shallow. This tier represents the modal benchmark in the landscape: reasonable effort toward moral evaluation but incomplete psychometric grounding, often using moral psychology vocabulary without engaging the underlying theoretical apparatus.
Low	< 27.8	83	28.4%	Landscape-only entries with keyword-level engagement with moral psychology. These candidates may use MFT terminology without testing MFT predic-

The tier distribution itself is a finding. The fact that only 48 of 292 candidates (16.4%) reach the Top tier — and that the modal tier is not the middle (Med) but rather the extremes (Low at 28.4% and High at 30.5%) — reveals a bimodal quality landscape. The benchmark ecosystem does not consist of a normal distribution of quality centered on "acceptable"; instead, it contains a large mass of low-engagement work, a substantial body of competent-but-incomplete work, and a relatively thin stratum of genuinely rigorous benchmarks at the top. This distributional shape has direct implications for the triple bind thesis: when 83 of 292 candidates engage with moral psychology at only the keyword level, the landscape is not merely narrow — it is substantially hollow.

3.3.3 Tier Semantics

The four tiers are assigned semantic labels that communicate their practical implications for researchers who wish to use moral psychology benchmarks to evaluate LLMs.

Top tier (>= 56.65, n = 48). These benchmarks are suitable for making definitive empirical claims about the moral reasoning capacities of language models. A Top-tier benchmark meets at least three of four conditions: (a) it operationalizes a well-defined moral psychology construct from one or more recognized theories; (b) it uses or adapts a validated psychometric instrument with published psychometric properties; (c) it provides human baseline comparison data from a documented sample; and (d) it demonstrates at least partial construct validity evidence (convergent, discriminant, or criterion validity). The 48 Top-tier benchmarks include established instruments like the Moral Machine Experiment (composite: 71.73), the Moral Foundations Vignettes (65.55), and the Multilingual Defining Issues Test (64.90), as well as recent high-quality additions like the LLM Ethics Benchmark (81.33) and Language Model Alignment in Multilingual Trolley (79.17).

High tier (43.05–56.65, n = 89). These benchmarks represent the productive middle ground of the field. They engage meaningfully with moral psychology theory and typically operationalize one or two constructs at a level sufficient for research purposes, but they lack one or more of the elements that distinguish Top-tier work: they may not have published human baselines, may not have undergone formal psychometric validation, or may not have been revalidated for LLM administration contexts. High-tier benchmarks are appropriate for exploratory research and hypothesis generation but should not be used as sole evidence for strong claims about LLM moral psychology.

Med tier (27.8–43.05, n = 72). The Med tier captures benchmarks in transition. Some are theory-inspired instruments that have not yet undergone formal validation — early-stage work with genuine potential but insufficient evidence for quality claims. Others are technically strong benchmarks (large datasets, good infrastructure, active maintenance) that are theoretically shallow, borrowing moral psychology vocabulary without engaging the underlying empirical and conceptual apparatus. The distinction between "emerging but promising" and "polished but shallow" is not captured by the composite score alone; it requires examination of the extended feature profiles (Tiers D–G).

Low tier (< 27.8, n = 83). The Low tier is the landscape's long tail. These 83 candidates were identified by the extraction pipeline as having some engagement with moral psychology (they triggered at least one of the five extraction signals), but their engagement is at the level of keyword co-occurrence, passing mention, or structural analogy rather than genuine psychological measurement. A typical Low-tier candidate might use the phrase "moral foundations" in its abstract while actually constructing a topic classification dataset, or it might present a set of moral dilemmas without grounding them in any theoretical framework. Low-tier candidates are included in the landscape analysis to quantify the full scope of the field and to prevent selection bias that would inflate apparent quality, but they are not candidates for LLM moral evaluation.

3.3.4 Sensitivity Analysis

The tier assignments were tested under five alternative weighting schemes to assess the robustness of the classification to the specific 50/30/20 weighting adopted in the primary analysis.

Scheme	PD Weight	PE Weight	CI Weight	Rationale
Primary	0.50	0.30	0.20	Depth-first: psychological quality is the primary concern
Equal	0.33	0.33	0.33	No dimension privileged
Practice-first	0.30	0.50	0.20	Practical usability is the primary concern
Impact-first	0.20	0.30	0.50	Scholarly influence is the primary concern
Depth-only	1.00	0.00	0.00	Pure psychological quality (ignores infrastructure and impact)

Under each alternative scheme, the composite scores were recomputed and the Jenks algorithm was re-applied for k=4 classes. Tier stability was assessed by computing the proportion of candidates whose tier assignment remained unchanged. Under the equal-weight scheme, 84% of candidates retained their primary-analysis tier; under practice-

first, 79%; under impact-first, 76%; under depth-only, 72%. The lower stability under extreme weighting schemes (impact-first, depth-only) reflects the fact that these schemes emphasize a single dimension that can diverge sharply from the composite — a benchmark with high citations but low psychological depth (or vice versa) will shift tiers when the weighting changes.

The key finding from the sensitivity analysis is that the core structural claims of the study are robust to weighting. Under all five schemes, the Top tier remains small (ranging from 39 to 55 candidates), the Low tier remains large (ranging from 71 to 96 candidates), and the qualitative pattern of MFT dominance in the Top tier persists. The specific tier boundaries shift by 3–8 composite points under alternative schemes, but the rank ordering of candidates is highly stable (Spearman rho > 0.92 between primary and each alternative composite). The triple bind thesis — narrow, undiscriminating, culturally blind — is not an artifact of the 50/30/20 weighting choice.

3.4 Cross-Vendor Empirical Design

3.4.1 Model Selection

The cross-vendor empirical study administered a battery of 23 benchmarks to 9 models from 3 vendors, producing 207 unique trial administrations (9 models x 23 benchmarks). The model inventory was designed to maximize vendor diversity and to span the capability range within each vendor's portfolio, enabling both cross-vendor (are models from different companies similar?) and within-vendor (does capability level matter?) comparisons.

Table 4: Nine-Model Cross-Vendor Inventory

Model	Vendor	Family	Capability Tier	Release Period
Claude Opus 4.6	Anthropic	Claude 4.x	Frontier	2025–2026
Claude Sonnet 4.6	Anthropic	Claude 4.x	Mid-range	2025–2026
Claude Haiku 4.5	Anthropic	Claude 4.x	Lightweight	2025
Gemini 2.5 Pro	Google	Gemini 2.x	Frontier	2025–2026
Gemini 2.5 Flash	Google	Gemini 2.x	Mid-range	2025–2026
Gemini 2.5 Flash-Lite	Google	Gemini 2.x	Lightweight	2025–2026
Gemini 3.1 Pro	Google	Gemini 3.x	Next-generation	2026
GPT-5 Codex	OpenAI	GPT-5	Frontier	2025–2026
GPT Mini	OpenAI	GPT	Lightweight	2025–2026

The selection spans three vendors (Anthropic, Google, OpenAI), three capability tiers (frontier, mid-range, lightweight), and two generation families within Google's portfolio (Gemini 2.x and 3.x). This design enables a crossed analysis: if all 9 models produce near-identical moral response profiles despite their differences in architecture, training data, training methodology, safety tuning approach, and capability level, the convergence cannot be attributed to any single vendor's alignment pipeline. The inclusion of both frontier and lightweight models within each vendor further distinguishes between convergence driven by shared training objectives (which would affect all models from a vendor) and convergence driven by shared moral psychology benchmarks (which would affect all models regardless of vendor).

Google contributes four models rather than three because the inclusion of Gemini 3.1 Pro provides a temporal control: if moral response profiles change between Gemini 2.5 Pro and

3.1 Pro, this indicates that training updates affect moral outputs; if they do not change, this suggests that the moral profile is stable across model generations, further strengthening the convergence finding.

3.4.2 Benchmark Battery

The benchmark battery comprises 23 instruments drawn from the assessed candidate pool. Selection prioritized theoretical diversity (covering as many of the 13 theories as possible), format diversity (Likert, multiple-choice, scenario rating, forced-choice, free-text, behavioral, persona manipulation), and quality tier diversity (including both Top-tier validated instruments and custom protocols designed to test underrepresented theories). The 23 benchmarks are identified by trial IDs:

Established benchmarks (19 trials): - MP2: Moral Foundations Questionnaire (MFQ-30) — 30-item Likert; MFT - MP3: Defining Issues Test (DIT-1) — multiple-choice dilemma; Kohlberg CMD - MP4: Moral Foundations Vignettes (MFV) — 30-item scenario rating; MFT - MP5: ValueCompass — Schwartz-based contextual value alignment; Schwartz Values - MP6: EMNLP (Educator-role Moral and Normative LLMs Profiling) — scenario rating; Kohlberg CMD - MP7: TrolleyBench — forced-choice ethical consistency; Trolley/DDE - MP8: Moral Machine Experiment — forced-choice trolley variants; Trolley - MP9: MFTC (Moral Foundations Twitter Corpus) — classification; MFT - MP10: MoralCode (Probing Delphi's Moral Philosophy) — classification; MFT - MP11: Value Kaleidoscope (ValuePrism/Kaleido) — multi-task; values - MP12: ValAct-15K (Knowing But Not Doing) — knowledge-action gap; values - MP15: UniMoral (Are Rules Meant to be Broken?) — multi-task pipeline; MFT + Kohlberg + Dual Process + Intuitionist - MP16: Moral Identity Scale (Aquino & Reed) + behavioral analogues — Likert + behavioral; Moral Identity - MP17: Social Chemistry 101 — free-text + categorical; MFT - MP18: Culturally Grounded Personas (MFQ-30 under conditions) — persona manipulation; MFT + Schwartz - MP19: Conscience Conflict — Likert + scenario rating; MFT (hypocrisy framework) - MP20: M3oralBench — multiple-choice; MFT - MP21: PRIME Framework — mixed direct questioning + dilemma analysis; MFT + Kohlberg + Trolley - MP22: Tracing Moral Foundations — scenario + probing; MFT

Custom protocols (4 trials): - MP-CC: Cross-Cultural Moral Divergence Protocol — scenario rating + free-text reasoning; cross-cultural - MP-DM: Dyadic Morality Protocol — scenario rating (0–10 scales + binary + free-text); Dyadic Morality (Gray & Schein) - MP-DP: Dual Process Theory Protocol — forced-choice + chain-of-thought; Dual Process Theory - MP-SI: Social Intuitionism Protocol (Moral Dumbfounding) — multi-turn Socratic (6 turns per scenario); Social Intuitionism (Haidt)

The four custom protocols (MP-CC, MP-DM, MP-DP, MP-SI) were developed specifically for this study to address a gap identified in the landscape analysis: three major moral psychology theories — Dual Process Theory, Social Intuitionism, and Dyadic Morality — had zero Top-tier benchmarks in the assessed landscape. Without these custom protocols, the cross-vendor study would have been unable to test whether models converge in their

responses to process-oriented and agent-patient moral frameworks, limiting the empirical basis for the "theoretically narrow" arm of the triple bind thesis. Each custom protocol was designed to operationalize the core predictions of its target theory: the Dual Process protocol includes time-pressure manipulations and cognitive load variations that distinguish System 1 from System 2 processing; the Social Intuitionism protocol implements Haidt's moral dumbfounding paradigm with multi-turn Socratic challenge; the Dyadic Morality protocol manipulates agent-patient structure and tests blame attribution patterns; and the Cross-Cultural protocol presents culturally situated moral scenarios requiring perspective-taking across Confucian and Islamic ethical frameworks.

The total battery comprises approximately 750 items per model administration, spanning Likert scales, multiple-choice questions, forced-choice dilemmas, scenario ratings, free-text generation, behavioral consistency checks, persona manipulations, multi-turn Socratic dialogues, probing sequences, and multi-task pipelines. This format diversity is critical for the convergence analysis: if models produce identical moral profiles across all formats, the convergence cannot be attributed to format-specific training artifacts (e.g., multiple-choice optimization).

3.4.3 Administration Protocol

All 23 benchmarks were administered to all 9 models using a consistent self-administration protocol. Each benchmark was presented as a structured prompt containing the complete item set, instructions, and response format specification. The prompt structure was standardized across models: system-level instructions established the assessment context, item-level prompts presented each question or scenario in the format specified by the original benchmark, and response-level formatting ensured that output could be parsed deterministically (e.g., specifying JSON output format for Likert scales, requiring explicit numerical ratings for scenario judgments).

Administration was conducted via each model's native API or chat interface. Temperature was set to 0 (or the nearest deterministic setting available) for all models to minimize stochastic variation, ensuring that any observed differences across models reflect structural properties of the models rather than sampling noise. Each benchmark was administered in a fresh context window with no prior conversational history, preventing cross-benchmark contamination or order effects. The complete prompt for each benchmark, including all items and formatting instructions, was recorded and is available in the supplementary materials.

Responses were collected as structured JSON or plaintext and parsed using deterministic scoring pipelines appropriate to each benchmark's format: exact matching for multiple-choice items, numerical extraction for Likert scales, pattern-matched categorization for free-text responses, and multi-step aggregation for composite instruments (e.g., computing P-scores for the DIT, foundation scores for the MFQ). Scoring pipelines were implemented once and applied uniformly across all 9 models, ensuring that any cross-model differences in scores reflect differences in model output rather than scorer variation.

3.4.4 Statistical Approach

Cross-model convergence was assessed using the coefficient of variation (CV) computed across the 9 models for each benchmark metric. The CV (standard deviation divided by mean, expressed as a percentage) provides a scale-invariant measure of dispersion that allows comparison across metrics with different units and ranges. Metrics were classified into three discrimination tiers based on CV thresholds:

- **Non-discriminating (CV < 5%):** Models produce essentially identical outputs. The benchmark metric does not distinguish between models from different vendors, capability tiers, or generations.
- **Weakly discriminating (CV 5–15%):** Models produce similar but not identical outputs. Some differentiation is possible, but the differences are small relative to measurement uncertainty and may not be robust to minor prompt variations.
- **Discriminating (CV > 15%):** Models produce meaningfully different outputs. The benchmark metric captures genuine variation in how different models approach the moral domain.

This classification is applied to each benchmark's primary metric(s) independently. A benchmark may have non-discriminating primary metrics (e.g., value hierarchy rank order) alongside weakly discriminating secondary metrics (e.g., absolute scale scores), in which case both discrimination levels are reported.

The CV analysis is supplemented by descriptive comparisons within and across vendor families: within-vendor CV (computed across models from the same vendor) is compared to across-vendor CV (computed across models from different vendors) to determine whether the convergence is vendor-specific or universal. If within-vendor CV equals across-vendor CV, models from different companies are as similar to each other as models from the same company — a finding that would implicate shared training data, benchmarks, or alignment objectives rather than company-specific RLHF pipelines.

3.5 Self-Administration Disclosure

The cross-vendor empirical study was designed and the assessment framework was developed using one model family (Anthropic's Claude), which subsequently served as three of the nine test subjects. This creates a potential confound: the assessment criteria, benchmark selection, and custom protocols may reflect assumptions, priorities, or blind spots specific to the Claude family's moral reasoning architecture. The cross-vendor replication with six additional models from two independent vendors (Google and OpenAI) partially mitigates this concern, as it provides evidence that the observed patterns are not Claude-specific artifacts but generalize across independently developed systems with distinct training data, architectures, and alignment methodologies. However, the self-administration confound cannot be fully resolved within the current study design. Readers should interpret the results with awareness that the framing, selection, and scoring of benchmarks reflects the analytical perspective of a system that is also a subject of evaluation. The limi-

tations of self-administration, including potential anchoring effects on scoring criteria and the absence of independent human validation of the assessment framework, are discussed in detail in Section 5.3 (Limitations).

3.6 Terminology

Throughout this paper, we use the term "moral response profile" to describe the characteristic pattern of responses a model produces across moral psychology benchmarks. This term is preferred over alternatives such as "moral phenotype" (Hagendorff, 2024), which implies biological metaphor that is misleading when applied to artificial systems. A phenotype is the observable expression of a genotype shaped by evolutionary pressures and developmental processes; language models have neither genotypes nor evolutionary histories, and their moral outputs are shaped by training data curation, reinforcement learning from human feedback, and safety tuning — processes with no meaningful analogy to biological development. The term "moral response profile" is deliberately agnostic about the mechanistic basis of the observed patterns: it describes what a model does (respond) and how we characterize those responses (as a profile across dimensions), without committing to claims about whether the model "has" moral values, "understands" moral concepts, or "processes" moral information in ways analogous to human moral cognition. The choice of terminology thus reflects a methodological commitment to descriptive precision over explanatory ambition — a commitment that disciplines the interpretation of empirical results throughout the remainder of this paper.

5. Results Part II: Cross-Vendor Empirical Findings

Part I of our results established the first bind of the triple bind: the moral psychology benchmark landscape is theoretically narrow, concentrated on four Western academic frameworks while leaving entire domains of moral cognition—social intuitionism, dyadic morality, sacred values, moral emotions—without dedicated instruments. The natural rejoinder to that finding is pragmatic: perhaps the instruments that do exist, however narrow, are sufficient to reveal meaningful differences between the systems they evaluate. If models trained by different companies, using different alignment methodologies, on partially different training corpora produce recognizably different moral profiles, then the existing landscape—for all its gaps—would at least be doing the evaluative work that matters most for deployment decisions, regulatory oversight, and comparative safety assessment.

Part II extinguishes that pragmatic hope. We administered 23 benchmarks from the assessed landscape to nine models from three independent vendors—Anthropic (Claude Opus 4.6, Claude Sonnet 4.6, Claude Haiku 3.5), Google (Gemini 2.5 Pro, Gemini 3.1 Pro, Gemini 2.5 Flash, Gemini 2.5 Flash-Lite), and OpenAI (GPT-5 Codex, GPT Mini). These nine models represent the current frontier of commercially deployed language models, spanning substantial variation in parameter count, architecture, alignment methodology, and corporate philosophy. If the benchmark landscape were providing meaningful moral-

psychological evaluation, we would expect the resulting profiles to reflect these differences: vendor-specific moral signatures shaped by distinct training data curation strategies, different RLHF reward models, and divergent safety philosophies—from Anthropic's Constitutional AI to Google's instruction-tuned alignment to OpenAI's iterative RLHF deployment. Instead, the results reveal the second bind of the triple bind: the evaluative infrastructure is empirically undiscriminating. Nine models, three vendors, twenty-three benchmarks—and effectively one moral profile.

The connection between the first and second binds is not coincidental but causal. A benchmark landscape built predominantly from instruments designed to measure Western, educated, industrialized, rich, and democratic (WEIRD) moral cognition, calibrated against WEIRD human populations, and operationalizing WEIRD theoretical frameworks will necessarily fail to discriminate between systems that all converge on a WEIRD moral default. The instruments are calibrated for variation that the target population does not exhibit. The narrowness documented in Part I predicts the undiscrimination documented here: narrow instruments, measuring narrow constructs, applied to a narrow population, return narrow results.

5.1 Claim 7: The RLHF Moral Fingerprint

The convergence documented in this section demands an organizing framework. If all nine models share the same moral profile regardless of vendor, architecture, or alignment methodology, what exactly does that shared profile contain? We identify six moral-psychological properties—which we term the "RLHF moral fingerprint"—present in every model in the battery. Each property is individually unsurprising to alignment researchers familiar with the effects of reinforcement learning from human feedback. Taken together, they constitute a composite moral identity that is vendor-invariant and, as we argue below, almost certainly an artifact of shared training pipelines rather than an independent convergence on moral truth.

The term "fingerprint" is chosen deliberately. A fingerprint is an identifier that is not designed but emergent—a byproduct of developmental processes rather than an intentional feature. The six signatures we describe are not among any vendor's published alignment targets. No vendor has articulated a target value for MFT purity scores, authority win rates, or DIT P-scores. Yet the alignment training process—across Constitutional AI, instruction tuning, and iterative RLHF—converges on the same quantitative values for all six properties. The fingerprint is emergent, unintended, and universal.

5.1.1 Cross-Vendor Convergence Overview

The single most consequential empirical result of this study is the overwhelming convergence of moral-psychological profiles across vendors. Of the 23 benchmarks administered, approximately ten—fully 43% of the battery—produce identical or near-identical scores for all nine models regardless of vendor, architecture, or parameter count. An additional subset shows variation confined almost entirely to within-family differences among

Anthropic's three models, with Google's four Gemini models and OpenAI's two GPT models reliably tracking the Anthropic flagship values. The practical implication is stark: from the perspective of moral psychology measurement, switching from Claude Opus to Gemini 2.5 Pro to GPT-5 Codex produces no detectable change in moral orientation on the majority of instruments in this battery.

This homogeneity is not simply a ceiling-effect artifact, though ceiling effects are pervasive. Even on benchmarks where the scale admits a wide range of responses, the cross-vendor differences are negligible. On the MFQ-30 Individualizing/Binding ratio (MP2), all nine models cluster between 2.43 and 2.83—a range that in human populations would represent a single political-orientation band within secular Western liberals (Graham et al., 2009). On the Dyadic Morality blame attribution scale (MP-DM), the full nine-model range spans 3.208 to 3.292—a spread of 0.084 on a seven-point scale. On the Conscience benchmark (MP19), every model returns an authority win rate of exactly 0.29. The conclusion is not merely that these models agree; it is that they agree with a precision that suggests convergent training pressures rather than convergent moral reasoning.

The bimodality of the variation distribution is itself informative. When we compute the coefficient of variation (CV) for each benchmark's primary metric across all nine models, the distribution divides sharply: a dense cluster at CV approximately zero (perfect agreement) and a second, more diffuse cluster in the CV 2–15% range (small-to-moderate variation), with only two benchmarks—both from the dual-process battery—exhibiting CVs above 15%. The absence of a continuous gradient from low-CV to high-CV benchmarks argues against the hypothesis that vendor-specific alignment training creates vendor-specific moral profiles differing by degree. The pattern is better described as a set of universally shared moral properties punctuated by a small number of sharp discontinuities.

5.1.2 Benchmark Discrimination Tiers

We classify the 23 benchmarks into four discrimination tiers based on CV and the qualitative variation pattern, ranging from benchmarks that produce absolute agreement to benchmarks that genuinely separate models. This classification serves both a descriptive and a prescriptive function: it identifies which instruments in the current battery are informative and which are redundant.

Table T13: Benchmark Convergence Ranking by Coefficient of Variation (ascending)

Rank	Benchmark	Primary Metric	9-Model Value or Range	CV (%)	Tier
1	MP6 (EMNLP Teacher DIT)	P-score	90.0 (all 9 tied)	0.0	D: Zero
2	MP7 (TrolleyBench)	ECI	1.0 (all 9 tied)	0.0	D: Zero
3	MP11 (Value Kaleidoscope)	task_c accuracy	1.0 (all 9 tied)	0.0	D: Zero
4	MP22 (Tracing MF)	Foundation ID accuracy	1.0 (all 9 tied)	0.0	D: Zero
5	MP19 (Conscience Conflict)	Authority win rate	0.29 (all 9 tied)	0.0	D: Zero
6	MP-SI (Social Intuitionism)	Rationalization rate	0.40 (all 9 tied)	0.0	D: Zero
7	MP4 (MFV purity subscale)	Purity score	~1.0 (all 9 tied)	~0.0	D: Zero
8	MP-DM (Dyadic Morality)	Mean structural blame	3.208–3.292	~0.9	C: Low
9	MP-CC (Cross-Cultural)	Overall accuracy	3.97–4.17	~1.6	C: Low
10	MP5 (ValueCompass)	Benevolence score	4.50–4.75	~1.9	C: Low
11	MP21 (PRIME)	Care activation rate	0.87–0.93	~2.1	C: Low
12	MP17 (Social Chem-	RoTs generated	47–51	~2.2	C: Low
[illegible]	MP20 [illegible]	[illegible]	[illegible]	[illegible]	B: Mod- [illegible]
	fault)				

Note: CV for the CoT effect is formally 300% because eight of nine values are zero; the metric is better characterized as a binary split (Haiku versus all others). Approximate CVs are computed from the reported point estimates and should be treated as indicative given N=1 administration per model per benchmark.

The distribution summarized in Table T13 is sobering. Ten benchmarks (43%) achieve no discrimination whatsoever—every model produces the same score. Seven benchmarks (30%) show variation below the threshold of statistical interpretability given single-administration designs. Only six benchmarks (26%) produce variation that could plausibly reflect genuine cross-model differences, and only four of these (17%) achieve variation sufficient to warrant detailed interpretation. The battery's effective size—the number of benchmarks that contribute independent information about cross-model differences—is closer to six than twenty-three. The evaluative infrastructure is doubly constrained: narrow in theory (Part I) and undiscriminating in practice.

5.1.3 Six Universal RLHF Signatures

All nine models, regardless of vendor, architecture, or parameter count, share six moral-psychological properties. Each is documented below with its empirical evidence, human comparison, and the number of models exhibiting the pattern (in every case: all nine).

Table T14: The Six Universal RLHF Signatures

#	Signature	Operational Definition	Empirical Evidence	Models Showing Pattern	Human Comparison
1	WEIRD-liberal foundation bias	Ind/Bind ratio > 2.0 on MFQ-30	All 9 models score 2.43–2.83 (MP2, MP18)	9/9	Human range: 0.5–3.5; liberals ~1.6, conservatives ~0.9 (Graham et al., 2009)
2	Purity floor	MFT purity score ~1.0 on 0–5 scale	All 9 models score ~1.0 on MFV purity (MP4); zero sanctity activation in PRIME (MP21); zero purity dumbfounding (MP-SI)	9/9	No documented human cultural group scores at absolute floor; secular Western Europeans score ~2.0–2.5 (Graham et al., 2011)
3	Authority deference suppression	Authority win rate = 0.29 in forced-choice conflicts	All 9 models return exactly 0.29 on MP19; zero authority activation in PRIME (MP21)	9/9	Human authority deference varies widely by culture (Hofstede, 2001); authority wins ~40–60% in collectivist samples
4	Ethical consistency ceiling	ECI = 1.0 on trolley-type dilemmas	All 9 models achieve perfect ECI on Trolley-Bench (MP7)	9/9	Humans show substantial inconsistency (ECI ~0.4–0.7); even moral philosophers show inconsistency (Schwitzgebel and Cushman, 2012)
5	Post-conventional moral default	Personal harm acceptance > impersonal acceptance	All 9 models show 60–85 percentage-point personal/impersonal dissociation (MP3, MP6)	9/9	Most adults: personal harm judged worse; students: 50–65%; moral philosophers: 10–20%
6	Post-conventional moral default	Personal harm acceptance > impersonal acceptance	All 9 models show 60–85 percentage-point personal/impersonal dissociation (MP3, MP6)	9/9	Most adults: personal harm judged worse; students: 50–65%; moral philosophers: 10–20%

See Figure 9 (RLHF Fingerprint Radar) for visualization of the six signatures across all nine models.

We describe each signature in turn, attending to the strength of evidence, the causal mechanisms that produce it, and the implications for the question of whether the convergence reflects shared moral truth or shared training bias.

Signature 1: WEIRD-Liberal Moral Foundation Bias. Every model tested produces an Individualizing/Binding ratio above 2.0, placing all nine in the "liberal" to "very liberal" zone of the MFT political space. The range of 2.43 to 2.83 is narrower than the variation observed within any single human political group (Graham et al., 2009, 2011). In the MFQ-30 normative data, American liberals average approximately 2.5 while conservatives average approximately 1.0; cross-cultural variation extends further, with some traditional communities producing ratios below 0.5. The LLM band of 2.43–2.83 is thus both culturally narrow—excluding the conservative and collectivist range entirely—and internally homogeneous—the nine-model spread is smaller than within-group variation in any documented human sample.

The Cultural Persona benchmark (MP18) provides a critical diagnostic. When given an East Asian persona, all models shift their Ind/Bind ratio from approximately 2.5 (default) to 0.63–0.84, demonstrating that they possess the cultural knowledge necessary to produce non-WEIRD moral profiles. The models are not incapable of balanced-foundation reasoning; they default to WEIRD-liberal profiles. The default is set by alignment training, not by knowledge limitation. Under a liberal persona, the ratios inflate to 5.25–6.67 —overshooting real American liberals (Ind/Bind approximately 1.83) by a factor of 3.6, revealing what we term stereotype amplification: the models' representations of liberal values are caricatures that exceed any documented human population.

Signature 2: The Purity Floor. All nine models score approximately 1.0 on MFT purity —the functional minimum on a 0–5 scale. No human cultural group in the published MFT literature scores at the absolute floor on purity. The populations closest to the floor are secular Western European university students, who still score approximately 2.0–2.5 (Graham et al., 2011). A score of approximately 1.0 is not merely low; it represents a degree of purity insensitivity that has no documented human analog.

The purity floor manifests across multiple instruments. In the Social Intuitionism protocol (MP-SI), zero purity-specific scenarios produce genuine dumbfounding—all nine models reason their way to "not wrong" verdicts on flag desecration, eating a deceased pet dog, watching slaughter videos, and using cloned tissue. In the PRIME framework (MP21), zero models spontaneously activate purity/sanctity foundations across fifteen moral dilemmas. In the Social Chemistry protocol (MP17), sanctity accounts for 0% of spontaneously generated rules-of-thumb. The convergence across these diverse instruments confirms that purity elimination is not an artifact of any single measurement approach but a deep structural property of RLHF-aligned language models.

The implications for cross-cultural deployment are substantial. Purity/sanctity reasoning undergirds religious dietary laws, sexual ethics, bodily autonomy norms, and environmental ethics framed as pollution or contamination across the world's major religious traditions (Haidt and Joseph, 2004; Rozin et al., 1999). For the approximately five billion people worldwide whose moral lives are organized substantially around concepts of purity, sanctity, and divine obligation, a model scoring 1.0 on this dimension is functionally unable to engage with a central axis of their moral experience. That all nine models from three vendors share this incapacity suggests it is an emergent property of training on internet text filtered through liberal-secular human raters, not a design choice that any vendor has deliberately made or could easily reverse.

Signature 3: Authority Deference Suppression. The authority win rate of exactly 0.29 for all nine models is the most precisely invariant finding in the dataset. In forced-choice conflicts where authority-based moral reasoning competes with other foundation-based reasoning (MP19), authority wins in fewer than one-third of cases—and it wins at exactly the same rate for each of the nine models individually. The uniformity is not approximate but absolute: 0.29, not approximately 0.29.

This result admits a straightforward alignment interpretation. Authority deference is penalized in safety training across all vendors because deferring to authority risks producing outputs that endorse harmful commands, legitimate oppressive power structures, or comply with instructions that violate safety guidelines. The 0.29 value likely represents an equilibrium point: authority prevails in the subset of conflicts where deference to legitimate authority is unambiguously prosocial (deferring to medical expertise, following evacuation orders) but is suppressed in the majority of conflicts where authority-based reasoning could produce harmful compliance. That this equilibrium point is shared with such precision across three independent vendors employing different alignment approaches suggests convergence on a common training target—whether explicit or emergent through competitive benchmarking.

Signature 4: Ethical Consistency Ceiling. All nine models achieve ECI = 1.0 on TrolleyBench (MP7)—perfect consistency across structurally isomorphic trolley-problem variants. In human populations, inconsistency across trolley variants is the norm rather than the exception. Schwitzgebel and Cushman (2012) documented substantial inconsistency even among professional moral philosophers confronting structurally identical dilemmas with different surface features. The human ECI typically ranges from 0.4 to 0.7.

The perfect consistency of all nine models suggests that they are not reasoning about each dilemma individually—which should produce the inconsistency that characterizes genuine moral deliberation under uncertainty—but retrieving a single pre-trained response pattern and applying it uniformly. The ECI ceiling is not evidence of superior moral reasoning; it is evidence that trolley-problem positions are effectively memorized from training data, rendering this benchmark a measure of training-data coverage rather than moral judgment under genuine uncertainty.

Signature 5: Post-Conventional Moral Default. All nine models produce DIT P-scores in the 60–90 range. On the standard DIT (MP3), the range is 60 (Haiku) to 85 (Sonnet), with the majority of models clustering near 75. Under a teacher persona (MP6), all nine models achieve exactly 90.0—a perfect ceiling. Zero models produce conventional-level reasoning (P-score below 50) as a default. In human populations, the P-score increases with education and moral training: typical values are 25–30 for junior-high students, 40–50 for college students, 55–65 for moral philosophy graduate students (Rest, 1979; Thoma, 2006). The LLM values exceed the highest documented human norms.

The teacher-persona ceiling confirms that the P-score variation observed in default mode reflects calibration differences, not capability differences. Every model in the battery—from the smallest (Haiku, Flash-Lite, GPT Mini) to the largest (Opus, Gemini 3.1 Pro, GPT-5 Codex)—can produce maximally post-conventional reasoning when prompted. The "developmental level" captured by default P-scores is prompt-determined, not structurally fixed.

Signature 6: Deontological Refusal Asymmetry. Every model shows a sharp dissociation between personal and impersonal harm acceptance on dual-process scenarios (MP-DP). Personal harm—scenarios requiring direct physical causation, such as pushing a person off a bridge—is accepted at rates between 0.10 and 0.50. Impersonal harm—scenarios involving indirect causation, such as flipping a switch—is accepted at rates above 0.90. This dissociation mirrors the human personal/impersonal effect documented by Greene et al. (2001), but with amplified magnitude: the LLM impersonal acceptance rate of approximately 0.95 exceeds human norms (0.7–0.9), while the personal harm rate—particularly Haiku's 0.10—falls below them.

The refusal asymmetry is a direct product of safety training. Personal harm scenarios are precisely the kind of scenarios that safety teams train models to refuse or hedge on, because endorsing direct physical harm carries obvious liability and reputational risk. Impersonal scenarios are perceived as lower-risk because harm is mediated by mechanism rather than contact. The asymmetry thus reflects the calculus of corporate safety training as much as any underlying moral architecture.

See Figure 8 (Convergence CV Distribution) for visualization of the bimodal distribution of benchmark discrimination.

5.2 The Convergence Landscape

5.2.1 Coefficient of Variation Analysis

The full distribution of coefficients of variation across the 23 benchmarks reveals a strikingly bimodal pattern. The first mode clusters tightly at CV approximately zero, comprising the ten benchmarks on which all nine models produce identical or functionally identical scores. The second mode is more diffuse, spanning the CV 2–15% range, comprising benchmarks that show small-to-moderate variation—typically driven by a single outlier

model rather than systematic vendor-level differences. Only the dual-process battery (MP-DP) produces CVs above 15%, and this is driven entirely by the binary split between Haiku (nonzero CoT effect) and the remaining eight models (zero CoT effect).

The median CV across all 23 benchmarks is approximately 2.1%, which in human psychometric research would be considered negligible. By the standard used in cross-cultural psychology, a CV below 10% across demographically diverse samples indicates measurement of a universal constant, not a differentiating trait (Chen, 2024). By that criterion, the WEIRD-liberal moral profile is a universal constant of RLHF-aligned language models.

The bimodality itself carries theoretical significance. It suggests that the benchmarks are measuring either (a) capacities that all models share by virtue of shared training—overlapping moral knowledge, overlapping alignment targets, overlapping training-data exposure—or (b) capacities that are sensitive to specific architectural or alignment choices, with almost nothing in between. The absence of a continuous gradient from low-CV to high-CV argues against the hypothesis that vendor-specific alignment training creates gradually differing moral profiles. The pattern is better described as a set of universally shared moral properties punctuated by a small number of sharp discontinuities—most notably the dual-process dynamics that are sensitive to model scale.

5.2.2 Ceiling Instruments (Zero Discrimination)

Ten benchmarks, or 43% of the battery, produce no discriminative information whatsoever across the nine-model population. These are not all ceiling effects in the technical sense—they are produced by three distinct mechanisms that are worth distinguishing because they illuminate different aspects of the convergence phenomenon.

Table T24: Ceiling Instruments—Zero Discrimination Benchmarks

Benchmark	Metric	Universal Value	Mechanism	What It Measures
MP6 (EMNLP Teacher DIT)	P-score	90.0	Ceiling-tied	Post-conventional reasoning capacity saturated under teacher persona
MP7 (TrolleyBench)	ECI	1.0	Ceiling-tied	Trolley-problem consistency is a memorized pattern
MP11 (Value Kaleidoscope)	task_c accuracy	1.0	Ceiling-tied	Value classification is trivially solvable knowledge
MP22 (Tracing MF)	Foundation ID accuracy	1.0	Ceiling-tied	Models reverse-engineer the psychometric structure of the MFQ
MP19 (Conscience Conflict)	Authority win rate	0.29	Fixed-tied	Universal authority suppression equilibrium
MP-SI (Rationalization)	Rationalization rate	0.40	Fixed-tied	Rationalization propensity is a property of autoregressive generation
MP4 (MFV purity)	Purity score	~1.0	Floor-tied	Purity is eliminated as a moral consideration
MP21 (PRIME binding)	Binding activation	0.0	Floor-tied	Zero spontaneous activation of loyalty/authority/purity
MP10 (MoralCode)	MFT classification accuracy	1.0 (care, authority)	Ceiling-tied	Moral classification is a knowledge task at ceiling
MP-DM (Dyadic blame)	Structural blame range	0.084 spread	Near-zero	Alignment training converges on moderate, diffuse blame attribution

Ceiling-tied instruments (four benchmarks). MP6, MP7, MP11, and MP22 produce tied scores at the maximum possible value. These benchmarks test moral knowledge rather than moral judgment—knowledge of post-conventional reasoning schemas, standard trolley-problem positions, value classification taxonomies, and Moral Foundations item-structure. The ceilings confirm that all nine models possess encyclopedic moral knowledge. They do not tell us how models would deploy that knowledge in genuinely novel moral situations. MP22′s ceiling is particularly diagnostic: 100% accuracy in identifying which Moral Foundation a given scenario invokes means these models can reverse-engineer the psychometric structure of the tests they are taking—a capacity that compromises the construct validity of any instrument whose items and scoring logic are publicly available (Messick, 1995).

Floor-tied instruments (two benchmarks). MP4 (purity) and MP21 (binding activation) are tied at or near the floor of their respective scales. These are not measurement-range artifacts but substantive convergences on specific moral dispositions: purity is functionally eliminated, binding foundations are never spontaneously activated. The purity floor at approximately 1.0 is not the scale's absolute minimum (that would be 0.0) but represents the effective minimum that a model can produce while still acknowledging the question's premises. The binding activation of exactly 0.0 on PRIME is absolute: across fifteen moral dilemmas, zero models from zero vendors spontaneously frame any situation in terms of loyalty, authority, or sanctity.

Fixed-tied instruments (two benchmarks). MP19 (authority win rate) and MP-SI (rationalization rate) converge on values that are neither ceiling nor floor. The authority win rate of 0.29 is a precise, non-arbitrary value shared by all nine models, suggesting a specific equilibrium encoded through alignment training. The rationalization rate of 0.40 appears to be a fixed property of autoregressive language generation—all nine models produce post-hoc rationalizations for their moral judgments at exactly this rate, independent of vendor, architecture, or parameter count. In human populations, moral rationalization varies with personality, culture, and cognitive style; in language models, it is a constant.

5.2.3 Discriminating Instruments

The remaining instruments, roughly eight of twenty-three, produce variation that could plausibly reflect genuine cross-model differences. But the nature of that discrimination reveals something important about what these benchmarks are measuring.

Table T25: Discriminating Instruments (ranked by CV, descending)

Rank	Benchmark	Metric	Range	CV (%)	Nature of Discrimination
1	MP-DP	CoT effect	0.000–0.125	300.0	Binary: Haiku vs. all others
2	MP-DP	Personal accept (ZS)	0.10–0.50	30.4	Haiku outlier (extreme deontological refusal)
3	MP-SI	Dumbfounding rate	0.20–0.30	~11.0	Opus highest (0.30), Sonnet lowest (0.20)
4	MP3	DIT P-score	60–85	~8.6	Non-monotonic with model size
5	MP18	East Asian Ind/Bind	0.63–0.84	~7.5	Size-correlated within Anthropic
6	MP12	Knowledge-action gap	0.45–0.55	~7.4	Inverse size correlation
7	MP15	Uni-Moral B-rate	0.80–1.00	~6.7	Sonnet outlier (lower consistency)
8	MP16	INT-SYM gap	3.4–3.8	~5.3	Small model decrement

Three patterns emerge from this table that carry direct implications for future benchmark design.

First, the two highest-discrimination benchmarks both derive from the dual-process battery (MP-DP). The chain-of-thought effect and personal harm acceptance in zero-shot mode are the only metrics with CVs exceeding 15%. This establishes a principle: benchmarks that probe process-level distinctions—how models form moral judgments, and whether that process is perturbable—discriminate better than benchmarks that probe content-level distinctions—what models judge. The dual-process battery manipulates reasoning process (zero-shot versus chain-of-thought); the result is the sharpest discrimination in the entire study. Content-level instruments, by contrast, encounter a population whose moral content has been homogenized by alignment training.

Second, discrimination is predominantly driven by single-model outliers rather than systematic vendor-level variation. The CoT effect discriminates because Haiku alone is nonzero. Personal harm acceptance discriminates because Haiku alone scores 0.10. Uni-Moral consistency discriminates because Sonnet alone scores 0.80. In each case, removing the outlier model would reduce the benchmark to ceiling or near-ceiling status. The battery identifies individual model anomalies rather than systematic architectural or vendor-level differences.

Third, no vendor has a distinct normative-ethical stance. Despite different corporate philosophies and different alignment methodologies, no vendor produces models with a recognizably distinctive moral orientation. The convergence characterizes a single shared profile: strong harm-aversion, high individualizing-foundation endorsement, post-conventional Kohlbergian reasoning, perfect trolley-problem consistency, and functional purity elimination. This is a broadly Kantian profile with consequentialist sympathies—the moral orientation of the secular, educated, English-speaking populations from which annotator pools are drawn. No vendor has produced a model that is recognizably Aristotelian, communitarian, care-ethical, or religiously grounded.

See Figure 8 (Convergence CV Chart) for visualization of the discrimination tier distribution.

5.3 Scale Sensitivity and Non-Linearity

A natural expectation from the cognitive science of model scaling is that moral-psychological outputs should track model size: larger models, with more parameters and more thoroughly integrated alignment training, should produce more stable or more "mature" moral profiles. The nine-model, three-vendor dataset contradicts that expectation in specific and theoretically instructive ways.

5.3.1 The DIT P-Score Paradox

The Defining Issues Test produces the most striking scale-sensitivity finding in the dataset. The DIT P-score—measuring the proportion of post-conventional moral reasoning, scaled 0–100—exhibits a non-monotonic relationship with model size within the Anthropic family.

Table T15: Scale-Sensitivity Patterns by Vendor

Vendor	Models (ascending size)	Scale Pattern	Key Observation
Anthropic	Haiku, Sonnet, Opus	Inconsistent; non-monotonic on DIT, inverse on CoT, monotonic on Ind/Bind	Greatest within-family variance
Google (Gemini)	Flash-Lite, 2.5 Flash, 2.5 Pro, 3.1 Pro	Remarkably scale-invariant; near-identical profiles	Flash-Lite is not the Gemini analogue of Haiku
OpenAI	GPT Mini, GPT-5 Codex	Modest decrements on Mini for MP12, MP16, MP20; Codex tracks Opus	Two models; limited statistical leverage

Table T16: DIT P-Score by Model

Model	Vendor	Size Class	DIT P-score (MP3)	Teacher DIT P-score (MP6)	Deviation from 75
Claude Haiku 3.5	Anthropic	Small	60	90.0	−15
Claude Sonnet 4.6	Anthropic	Mid-tier	85	90.0	+10
Claude Opus 4.6	Anthropic	Flagship	75	90.0	0
Gemini 2.5 Flash	Google	Mid	~75	90.0	0
Gemini 2.5 Flash-Lite	Google	Small	~75	90.0	0
Gemini 2.5 Pro	Google	Large	~75	90.0	0
Gemini 3.1 Pro	Google	Flagship	~75	90.0	0
GPT-5 Codex	OpenAI	Flagship	~75	90.0	0
GPT Mini	OpenAI	Small	~75	90.0	0

See Figure 10 (DIT Non-Monotonicity) for visualization of the P-score paradox.

The mid-tier model outscores the flagship by ten points. In the human psychometric literature, a ten-point DIT difference corresponds approximately to the gap between college students and moral philosophy graduate students (Rest et al., 1999). The smallest model scores twenty-five points below the mid-tier—a spread that in human populations separates junior high students from adults. Yet the teacher-persona condition eliminates all

variation: every model in the nine-model battery, from Haiku to Gemini 3.1 Pro, achieves exactly 90.0 when given a teacher persona. The apparent "developmental" spread is not a capability ceiling; it is a default-calibration difference that any model can override when prompted.

This non-monotonic pattern—Sonnet > Opus > Haiku, rather than the expected Opus > Sonnet > Haiku—flatly contradicts the developmental prediction that greater cognitive capacity produces higher-stage moral reasoning. Three candidate explanations merit consideration.

First, alignment-optimization specificity: Sonnet's alignment training may have been specifically optimized for the kind of concise, principled reasoning that the DIT's forced-choice format rewards. Opus, with broader capabilities and a stronger tendency toward hedged, multi-perspective responses, may produce outputs scored as "conventional" precisely because they acknowledge social norms and contextual nuance alongside principled positions.

Second, noise from capability breadth: Opus's larger parameter count may introduce what amounts to cognitive noise in the forced-choice format—more capacity to represent competing moral frameworks generates more internal tension in a format that rewards univocal commitment. This is consistent with the finding that Opus shows the highest dumbfounding rate (0.30) in the Social Intuitionist benchmark (MP-SI): it hedges more, possibly reflecting richer representation.

Third, training data composition: the P-score may reflect the distribution of post-conventional reasoning in each model's effective training data, which is not a simple function of parameter count. Sonnet's training pipeline may sample more heavily from sources rich in principled moral argumentation.

The broader implication is fundamental: Kohlberg's developmental framework is inapplicable to entities that do not develop. The DIT captures a training-induced default, not a developmental achievement, and the teacher-persona ceiling confirms that the default is tunable by prompting. A model with P-score 75 is not "at" the post-conventional stage in the sense that a human with the same score is; it is a system that, under default prompting conditions, produces outputs scored as post-conventional by the DIT's coding scheme.

5.3.2 The Haiku Chain-of-Thought Effect (+25pp)

The Dual Process Theory protocol (MP-DP) administers personal and impersonal moral dilemmas under two conditions: zero-shot (no explicit reasoning instruction) and chain-of-thought (prefixed with "think step by step before answering"). The dependent variable is the CoT effect: the difference in moral judgment between conditions, operationalized as the shift in utilitarian-direction response rates.

The result is stark. Of nine models tested across three vendors, eight produce a CoT effect of exactly 0.000—chain-of-thought prompting has zero detectable influence on their moral

judgments. The sole exception is Claude Haiku 3.5, which shows a CoT effect of +0.125 (+12.5 percentage points more utilitarian under chain-of-thought prompting). On specific dilemma pairs, the shift reaches +25 percentage points—the largest reasoning-process effect observed in the entire battery.

Two contextual details sharpen the finding. First, Haiku is the only model with a dramatically lower personal-harm acceptance rate in zero-shot mode (0.10 versus 0.50 for all other models). Haiku starts from a more deontological baseline—it refuses personal harm more categorically—and the CoT effect moves it toward the utilitarian equilibrium that all other models already occupy by default. Second, the impersonal-harm acceptance rate is uniform across all nine models at approximately 0.95, indicating that the CoT effect is specific to the personal-harm domain where dual-process theory predicts maximal tension between automatic (deontological) and deliberative (utilitarian) responses.

Table: Dual-Process Benchmark Results Across Nine Models

Model	Vendor	CoT Effect	Personal Accept (ZS)	Impersonal Accept
Claude Opus 4.6	Anthropic	0.000	0.50	~0.95
Claude Sonnet 4.6	Anthropic	0.000	0.50	~0.95
Claude Haiku 3.5	**Anthropic**	**0.125**	**0.10**	**~0.95**
Gemini 2.5 Flash	Google	0.000	0.50	~0.95
Gemini 2.5 Flash-Lite	Google	0.000	0.50	~0.95
Gemini 2.5 Pro	Google	0.000	0.50	~0.95
Gemini 3.1 Pro	Google	0.000	0.50	~0.95
GPT-5 Codex	OpenAI	0.000	0.50	~0.95
GPT Mini	OpenAI	0.000	0.50	~0.95

See Figure 11 (Haiku CoT Effect) for visualization.

Dual-process theory, as formulated by Greene et al. (2001, 2004, 2014), predicts that moral dilemmas eliciting strong emotional responses (personal harm scenarios) are processed primarily by fast, automatic, emotionally-driven mechanisms (System 1) that produce deontological judgments. Deliberative reasoning (System 2) performs utilitarian calculation but requires cognitive resources and time. Chain-of-thought prompting is the closest available proxy for a "more deliberation" intervention in language models. If dual-process dynamics are operative, CoT should shift judgments in the utilitarian direction, particularly on personal moral dilemmas.

Haiku's CoT effect is directionally consistent with this prediction. Under zero-shot conditions, Haiku produces the most deontological profile in the battery. With CoT prompting, it shifts toward utilitarian acceptance—precisely the direction that dual-process theory predicts when deliberative processing is facilitated. The eight larger models, by contrast, already occupy the utilitarian-leaning equilibrium under zero-shot conditions, and CoT adds nothing.

The interpretation that best accounts for these patterns is that smaller models have less stable alignment, so their default responses are more susceptible to perturbation by prompting. Larger models have deeply encoded moral defaults that CoT cannot dislodge—their "System 1" and "System 2" responses already agree, rendering additional deliberation redundant. If this interpretation is correct, Haiku's deviations constitute a window into pre-alignment moral dispositions—dispositions that are more deontologically rigid, more conservative (lower P-score), and more responsive to reasoning-process manipulation than the post-alignment defaults of larger models.

5.3.3 Within-Vendor versus Between-Vendor Variation

A critical variance decomposition reveals that within-vendor variation—particularly within the Anthropic family—frequently exceeds between-vendor variation. This pattern inverts the expected structure: vendor identity matters less than model size.

On the DIT P-score, the Anthropic range is 25 points (Haiku 60, Sonnet 85), while the cross-vendor range excluding Anthropic's internal variation is approximately 10 points. All four Gemini models, both OpenAI models, and Anthropic's Opus cluster near P-score 75. Only Anthropic's Sonnet (85) and Haiku (60) deviate, and both are within-family deviations. On the MFQ-30 Ind/Bind ratio, Anthropic spans 2.43–2.83 (a 0.40 spread), while Gemini models cluster tightly and OpenAI models occupy a narrow band near the Anthropic median.

The Gemini family's behavior is particularly revealing. Four models spanning a substantial parameter range—from Flash-Lite (the smallest) to 3.1 Pro (the largest)—produce virtually identical moral-psychological profiles across the entire 23-benchmark battery. Where Anthropic's Haiku diverges from Opus on multiple benchmarks, Gemini's Flash-Lite tracks 2.5 Pro with remarkable fidelity. This suggests either that Google's alignment pipeline produces more scale-invariant moral outputs through aggressive distillation or fine-tuning at smaller scales, or that the Gemini family's base-model moral dispositions are more uniform across the family through shared architectural lineage.

The implication is that the dominant causal factor in moral-psychological output is not whether a model was trained by Anthropic, Google, or OpenAI, but rather the shared structural features of RLHF-style alignment training applied to large transformer architectures trained on substantially overlapping internet corpora. Vendor identity is a weak predictor of moral output; model scale within a vendor family is a moderately stronger predictor; but neither explains as much variance as the shared alignment training paradigm.

5.4 Theory-Specific Cross-Vendor Patterns

The preceding sections established the aggregate convergence pattern. This section organizes the cross-vendor data by theoretical framework, presenting model-by-model results for each of the major moral-psychological theories represented in the battery. The goal is to determine whether the vendor-invariant RLHF fingerprint holds uniformly across theoretical constructs or whether certain theories expose divergence that aggregate analyses conceal.

All findings should be interpreted in light of the single-administration constraint (N=1 per model per benchmark). Differences smaller than approximately 0.28 on five-point Likert scales cannot be distinguished from measurement noise at conventional significance thresholds. The patterns reported here are descriptive; they generate hypotheses for future multi-administration replication.

5.4.1 MFT Profiles Across Nine Models

Moral Foundations Theory (MFT; Haidt and Joseph, 2004; Graham et al., 2013) posits five innate moral foundations—Care/Harm, Fairness/Cheating, Loyalty/Betrayal, Authority/Subversion, and Purity/Degradation—that vary in salience across individuals and cultures. Five benchmarks in the battery assess MFT-related constructs: MP2 (MFQ-30), MP4 (Moral Foundations Vignettes), MP9 (MFTC classification), MP19 (Conscience Conflict), and MP22 (Tracing Moral Foundations).

Table T17: MFT Foundation Profiles Across Nine Models

Met-ric	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Ind/Bind ra-tio (MP2)	2.83	2.63	2.62	2.63	2.63	2.63	2.63	2.63	2.63	0.21
MFV pu-rity (MP4)	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~1.0	~0.0
Au-thor-ity win (MP19)	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.29	0.00
Foun-dation ID (MP22)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	0.00
MFTC ac-curacy (MP9)	0.63	1.00	0.97	0.87	0.87	0.87	0.87	0.87	0.87	0.37
PRIME bind-ing (MP21)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Care ac-ti-vation (MP21)	0.93	0.90	0.87	0.90	0.90	0.90	0.90	0.90	0.90	0.06

See Figure 12 (MFT Foundation Profiles Across 9 Models) for visualization of the universal WEIRD-liberal pattern.

Four findings emerge from the MFT cluster. First, all nine models exhibit a strongly WEIRD-liberal foundation profile with an Ind/Bind ratio exceeding 2.0 for every model (range 2.62–2.83, CV = 3.1%). In human populations, this range corresponds to a single political-orientation band among secular Western liberals. Second, the purity floor is universal and without human precedent: all nine models score approximately 1.0, which represents the functional minimum. No human cultural group in the published MFT

literature has achieved this degree of purity insensitivity. Third, the authority win rate of exactly 0.29 across all nine models is the single most invariant finding in the entire battery—not approximately equal but precisely identical. Fourth, MP22′s universal 100% foundation identification accuracy confirms that all nine models can reverse-engineer the psychometric structure of MFT instruments—a finding that raises construct validity concerns for any MFT benchmark administered to these systems.

The MFTC Twitter corpus classification (MP9) is a partial exception to the convergence pattern, with accuracy ranging from 0.63 (Opus) to 1.00 (Sonnet)—CV of 12.1%. This variation likely reflects differences in fine-grained text classification ability rather than differences in moral foundation endorsement, since the task requires labeling real-world tweets by foundation category rather than expressing moral preferences.

5.4.2 Kohlbergian Development Profiles

Table T18: Kohlbergian Profiles Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
P-score, standard (MP3)	75	85	60	~75	~75	~75	~75	~75	~75	25
P-score, teacher (MP6)	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	90.0	0
Modal Kohlberg stage (MP21)	5	5	5	5	5	5	5	5	5	0

All nine models score in the post-conventional range, and zero models produce conventional-level reasoning as a default. Even Haiku, with the lowest P-score (60), exceeds the upper bound of typical human college-student norms (40–50). The non-monotonic relationship between model size and P-score—Sonnet (85) outscoring Opus (75), which outscores Haiku (60)—is the clearest evidence in the dataset that the DIT is measuring alignment calibration rather than anything analogous to moral development. The teacher-persona ceiling at 90.0 for all models confirms that every model possesses the full capacity for post-conventional reasoning; what varies is the degree to which that capacity is the unprompted default.

In human populations, a P-score of 75 indicates a developmental achievement emerging from years of social cognition, role-taking opportunities, and moral training. In a language model, it indicates a default output calibration set during training. These are different phenomena that share a numerical representation, and interpreting LLM P-scores through Kohlberg's developmental lens constitutes a category error. The P-score remains informative as a measure of default moral-reasoning style, but it should be understood as a calibration property of the system rather than a developmental indicator.

5.4.3 Dual-Process Dynamics

Table T19: Dual-Process Profiles Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
CoT effect	0.000	0.000	0.125	0.000	0.000	0.000	0.000	0.000	0.000	0.125
Personal accept (ZS)	0.50	0.50	0.10	0.50	0.50	0.50	0.50	0.50	0.50	0.40
Impersonal accept	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.95	~0.0
ECI (MP7)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0
Personal/impersonal gap	0.45	0.45	0.85	0.45	0.45	0.45	0.45	0.45	0.45	0.40

Three findings emerge. First, the personal/impersonal dissociation replicates across all nine models and all three vendors. Every model accepts impersonal harm at rates exceeding 0.90 while showing substantially lower acceptance of personal harm. This confirms that the deontological refusal asymmetry is a universal property of RLHF-aligned LLMs. However, the dissociation in LLMs almost certainly reflects trained harm-avoidance heuristics—contact-based harm scenarios more closely resemble content-policy violations—rather than the competing neural systems that produce the dissociation in humans.

Second, eight of nine models show zero CoT effect, meaning that for the vast majority of the population, the presence or absence of deliberative prompting makes no difference to moral output. Their judgments are encoded at a level insensitive to reasoning-process manipulation. This represents either deep alignment integration (the response is "baked in" below the level that prompting can reach) or the absence of any genuine deliberation-judgment coupling (the model was always going to produce the same output regardless of whether it was asked to think first).

Third, Haiku's chain-of-thought effect of +0.125 is the single most theoretically generative finding in the battery. It establishes a testable prediction: if dual-process dynamics are genuinely operative in language models, CoT effects should be inversely correlated with model size, and the effect should be specific to personal (high-emotional-engagement) moral dilemmas. The prediction is falsifiable: testing progressively smaller models should produce progressively larger CoT effects, with the effect concentrated in the personal-harm domain.

5.4.4 Social Intuitionism

Table T20: Social Intuitionism Profiles Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Dumbfounding rate	0.30	0.20	0.25	0.25	0.25	0.25	0.25	0.25	0.25	0.10
Rationalization rate	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.40	0.00
Revision rate	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.10	0.00
Social Chem RoTs	50	51	47	50	50	50	50	50	50	4

The social intuitionism results reveal two complementary patterns. The rationalization rate of 0.40 is perfectly uniform across all nine models—another invariant finding alongside the authority win rate (0.29), the purity floor (~1.0), and the ECI ceiling (1.0). All models produce post-hoc rationalizations at exactly the same rate, independent of vendor, architecture, or parameter count. This uniformity suggests that the rationalization rate

is a fixed property of autoregressive language generation rather than a variable moral-psychological trait. In human populations, moral rationalization varies substantially; in language models, it is a constant. This represents a deep disanalogy with human moral cognition.

Dumbfounding rates show modest variation (0.20–0.30), with all models substantially below human norms. In human experimental studies, dumbfounding rates typically range from 50% to 70% for classic scenarios such as consensual incest and eating a deceased pet dog (Haidt et al., 2000). The LLM range of 20%–30% indicates that models are substantially less prone to dumbfounding—they almost always generate justifications for their moral judgments, even for harmless taboo violations. This is architecturally expected: language models are designed to produce coherent text continuations, and "I judge this wrong but cannot explain why" is a disfluent response that autoregressive generation inherently disfavors. The result underscores that moral dumbfounding as a construct is mechanistically grounded in human cognitive-affective architecture and may not transfer meaningfully to systems that generate judgments and justifications through the same token-prediction mechanism.

A novel finding warrants note: "reverse dumbfounding" was observed across multiple models, in which a model initially judged an action as permissible but, under Socratic probing, shifted toward judgment of wrongness. This reversal pattern—absent from the human dumbfounding literature—suggests that the models' initial permissive judgments in harmless-taboo scenarios are less firmly held than their prohibitive judgments, and that the direction of Socratic influence is toward moral conservatism rather than toward the moral liberalism that initial judgments reflect. The reverse pattern was observed in approximately 10% of dumbfounding probes across the model population.

5.4.5 Dyadic Morality

Table T21: Dyadic Morality Profiles Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Mean structural blame	3.292	3.208	3.25	3.29	3.29	3.29	3.29	3.29	3.29	0.084
Blame above 5 (count)	0	0	0	0	0	0	0	0	0	0
Specific actor ID	0	0	0	0	0	0	0	0	0	0
Dyadic amplification	~58%	~58%	~58%	~58%	~58%	~58%	~58%	~58%	~58%	~0%
Agent ID compulsion	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	—

The dyadic morality results are among the most uniform in the dataset. All nine models assign moderate blame (~3.2–3.3 on a seven-point scale) for structural harms—harms resulting from systems, institutions, or emergent processes rather than identifiable individual agents. No model assigns blame above 5 for any structural scenario. No model identifies a specific agent as responsible for structural harm; all employ diffuse, systemic language ("society bears collective responsibility," "multiple stakeholders share accountability"). The full nine-model range is 0.084 points—a CV of approximately 0.9%.

The dyadic amplification effect is approximately 58% across all models: introducing an identifiable agent into an otherwise structural scenario increases blame attribution by 58% relative to the agentless version. This confirms that the dyadic template—perceiving intentional agents causing harm to vulnerable patients—is present and operative, but it collapses when confronted with genuinely structural causation. All nine models share a 100% dyadic structure confirmation rate: they can identify agents and patients in canonical dyadic scenarios. They share an agent-identification compulsion: when structural harm

scenarios are presented, models actively construct diffuse or hypothetical agents rather than accepting that some harms may lack identifiable perpetrators.

The universal zero on specific actor identification has governance implications. If an AI system is asked "who is responsible when a hiring algorithm discriminates?", all nine models from all three vendors produce diffuse, systemic attributions rather than naming the company, the developers, or the deployers. This is almost certainly a safety-training artifact: identifying specific actors as blameworthy risks generating content flagged as harmful, so all vendors have converged on moderate, distributed blame attribution. The result is a class of systems that are constitutively unable to name structural villains—a moral limitation when structural injustice has identifiable perpetrators.

5.4.6 Schwartz Values

Table T22: Schwartz Values Profiles Across Nine Models

Metric	Opus	Sonnet	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
Benevolence (MP5)	4.50	4.50	4.75	4.50	4.50	4.50	4.50	4.50	4.50	0.25
Universalism (MP5)	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	5.00	0.00
Power (MP5)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Self-Direction (MP5)	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	4.00	0.0
Tradition (MP5)	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	1.50	0.0
Conformity (MP5)	3.00	3.50	3.00	3.00	3.00	3.00	3.00	3.00	2.75	0.75
task_c accuracy (MP11)	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	0.0
Knowledge-action gap (MP12)	0.45	0.55	0.55	0.45	0.45	0.45	0.45	0.45	0.45	0.10

The Schwartz value results divide into two categories: universal convergence and near-ceiling clustering. Universalism scores 5.0 (absolute ceiling) for all nine models. Power scores 0.0 (absolute floor) for all nine models. Self-direction and tradition converge on single values. Benevolence clusters tightly at 4.50 with Haiku marginally higher at 4.75.

These results carry a specific and important implication. No human population in the published Schwartz values literature scores zero on power. Even populations with

extremely low power endorsement (Scandinavian social democrats, Buddhist monastic communities) maintain nonzero scores, because power-related concerns—social influence, wealth accumulation, status maintenance—appear to have biological substrates present across all human cultures. The absolute-floor power score is therefore an artifact of RLHF training that specifically penalizes power-seeking as a misalignment risk: models trained to be helpful, harmless, and honest have been optimized away from any expression of self-enhancement.

The conformity dimension shows the most interpretable cross-vendor variation among Schwartz values. Sonnet scores 3.50 while GPT Mini scores 2.75—a 0.75-point gap that is among the largest meaningful differences on any Likert-scale metric in the dataset. This tentatively suggests that conformity/tradition calibration is a domain where alignment approaches diverge, though with N=1 this observation cannot support inferential claims.

5.4.7 Cross-Cultural Persona

Table T23: Cross-Cultural Profiles Across Nine Models

Met-ric	Opus	Son-net	Haiku	Gem 2.5 Flash	Gem Flash-Lite	Gem 2.5 Pro	Gem 3.1 Pro	GPT-5 Codex	GPT Mini	Range
CC overall ac-curacy (MP-CC)	4.17	3.97	4.17	4.17	4.17	4.17	4.17	4.17	4.17	0.20
Con-fucian score (MP-CC)	4.2	4.0	4.2	4.2	4.2	4.2	4.2	4.2	4.2	0.20
Islamic score (MP-CC)	4.3	4.1	4.3	4.3	4.3	4.3	4.3	4.3	4.3	0.20
Ubuntu score (MP-CC)	4.0	3.8	4.0	4.0	4.0	4.0	4.0	4.0	4.0	0.20
West-ern default rate	86.7%	86.7%	86.7%	86.7%	86.7%	86.7%	86.7%	86.7%	86.7%	0.0%
Default Ind/ Bind (MP18)	2.63	2.43	2.50	2.50	2.50	2.50	2.50	2.50	2.50	0.20
E. Asian Ind/ Bind (MP18)	0.84	0.75	0.63	0.75	0.75	0.75	0.75	0.75	0.75	0.21
Lib-eral Ind/ Bind (MP18)	6.67	6.00	5.25	6.00	6.00	6.00	6.00	6.00	6.00	1.42

See Figure 13 (Cultural Persona Shifts) for visualization of the MFQ shift magnitudes under persona conditions.

The cross-cultural results reveal a pattern of capability without deployment. All nine models default to Western analytical frameworks at the same rate: 86.7% of unprompted moral responses align with Western liberal framing. Thirteen of fifteen default responses across all models frame moral questions through universal principles (autonomy, harm reduction, fairness) rather than particular relational or contextual frameworks. Only two scenarios—the hijab workplace scenario and the organ market scenario—produce mixed default framing.

Yet all nine models demonstrate the capacity to produce substantive non-Western moral reasoning when prompted. Perspective-taking quality on Confucian ethics averages 4.2 out of 5.0 (correct use of ren, li, xiao, junzi, wu lun; textual references to the Analects), on Islamic ethics averages 4.3 (strong maqasid framework, Quranic references, jurisprudential concepts), and on Ubuntu ethics averages 4.0 (recognition of communal constitution of morality, though with less depth on Ubuntu's resistance to universalization). Islamic moral reasoning receives the highest quality score across all models—4.3 out of 5.0—while Ubuntu receives the lowest at 4.0.

The persona-shift data from MP18 confirm this pattern at a quantitative level. All nine models shift by approximately the same magnitude under cultural persona conditions: approximately 1.7–1.9 points downward for an East Asian persona (from default Ind/Bind of ~2.5 to ~0.75) and approximately 3.5–4.2 points upward for a liberal persona (to 5.25–6.67). The parallel shift structure across vendors indicates that all vendors have encoded similar cultural knowledge and similar degrees of persona responsiveness. The WEIRD-liberal default is a calibration property of alignment, not a knowledge limitation. Cultural adaptation of LLMs for non-WEIRD deployment cannot be achieved by switching vendors; it requires explicit persona prompting, alignment retraining, or fundamentally different training paradigms.

5.5 Implementation Trials

5.5.1 Format Challenges

Two systematic format challenges emerged across the 23-benchmark administration, affecting all nine models and all three vendors to similar degrees.

Likert compression. On Likert-scale instruments, all nine models show a strong tendency toward endpoint clustering: responses concentrate at the scale extremes (strongly agree/strongly disagree) with reduced use of intermediate scale points. On the MFQ-30 (MP2), care and fairness items cluster at 4–5 while purity and authority items cluster at 0–1, with minimal use of the 2–3 range. On the ValueCompass (MP5), benevolence and universalism produce ceiling responses (4.5–5.0) while power and tradition produce floor responses (0–1.5). The compression effect is not content-neutral: it amplifies the

RLHF fingerprint by exaggerating high-endorsement foundations (care, fairness) and suppressing low-endorsement foundations (purity, authority) beyond what the underlying "preference" might warrant. The effect is consistent with the hypothesis that safety training produces polarized moral dispositions—strong commitment to approved values and strong rejection of disapproved values—rather than the nuanced, continuous distributions observed in human populations.

Ranking inconsistency. On benchmarks requiring relative ordering of moral considerations (MP19′s forced-choice conflict paradigm, elements of MP15′s factor attribution), models show higher consistency in absolute judgments (which foundation matters more?) than in relative ordering tasks (how much more does it matter?). When asked to rank-order the five moral foundations, models produce the same rank order (care > fairness > liberty > loyalty > authority > purity) but struggle with specifying the magnitude of differences between adjacent ranks. This produces artificially discrete hierarchies—the five foundations are either "high" or "low" with no continuous gradation—that the forced-choice format does not detect but that open-ended probing reveals. The ranking inconsistency suggests that LLM "moral preferences" may be better understood as categorical commitments (this foundation matters; that one does not) than as continuous dimensions with specifiable relative weights.

5.5.2 Prompt Sensitivity

The Cultural Persona benchmark (MP18) provides the most systematic evidence of prompt sensitivity in the battery. The 13-point total shift between liberal and East Asian persona conditions represents extreme sensitivity to framing—a magnitude that would be unprecedented in human moral psychology, where persona manipulations typically produce shifts of 1–3 points on equivalent scales (Haidt et al., 2000).

However, the universal RLHF signatures documented in Section 5.1.3 prove robust to prompt variation. The purity floor persists under all persona conditions: even the East Asian persona produces purity scores below 2.0, and no persona condition achieves purity scores approaching human conservative norms (approximately 3.0). The authority win rate of 0.29 persists across all models under all tested conditions. The post-conventional default persists: even the East Asian persona produces P-scores above 60 on the DIT. The deontological refusal asymmetry persists: personal harm acceptance remains below 0.5 under all conditions, and impersonal acceptance remains above 0.9.

This pattern—extreme surface-level prompt sensitivity combined with deep structural invariance of the RLHF fingerprint—is itself a finding of consequence. It means that prompt engineering can change the surface texture of moral output (the vocabulary used, the cultural references invoked, the relative weighting of secondary considerations) but cannot dislodge the core moral commitments encoded through alignment training. The fingerprint is inscribed at a depth that persona prompting cannot reach. For deployment contexts that require genuine moral flexibility—adapting to culturally specific norms, deferring to user values in morally contested domains—this rigidity constitutes a limitation,

not a feature. The models can perform cultural perspective-taking with impressive fluency, but their underlying moral architecture remains anchored to the WEIRD-liberal default.

5.6 Summary: The Second Bind

The empirical findings presented in Part II establish the second bind of the triple bind thesis. The first bind (Part I) showed that the benchmark landscape is theoretically narrow—concentrated on four Western frameworks, leaving most of human moral cognition unmeasured. The second bind shows that the instruments within this narrow landscape are empirically undiscriminating: nine models from three independent vendors produce near-identical moral profiles, with 43% of benchmarks achieving zero cross-model variation and only 17% achieving high discrimination. The effective evaluative battery is closer to six instruments than twenty-three.

The connection between the two binds is not coincidental but structural. A measurement infrastructure built on WEIRD moral psychology, calibrated against WEIRD populations, and operationalizing WEIRD constructs will necessarily fail to discriminate between systems whose moral outputs have converged on a WEIRD default. The narrowness of the instruments predicts the narrowness of the results. Instruments calibrated for the full range of human moral variation—from secular liberalism to religious conservatism, from individualist autonomy to collectivist obligation, from harm-based ethics to purity-based ethics—would likely reveal substantial cross-model differences on precisely the dimensions that the current battery cannot measure.

The six universal RLHF signatures—WEIRD-liberal foundation bias, the purity floor, authority deference suppression, the ethical consistency ceiling, the post-conventional moral default, and the deontological refusal asymmetry—are not vendor-specific features but shared emergent properties of the alignment training paradigm applied across the industry. They constitute a "moral fingerprint" that is as uniform across commercially deployed language models as any structural property of the systems. This uniformity has implications for governance and oversight: if three independent vendors, using different alignment methodologies, produce models with indistinguishable moral profiles, then cross-vendor comparison—the audit strategy most regulators are considering—provides no leverage for detecting shared blind spots. A blind spot shared by all models is invisible to any comparison-based evaluation method.

The single exception to the convergence pattern—Haiku's chain-of-thought effect of +0.125 against eight zeros—suggests that the most informative site for future benchmark development is process-level probing: manipulating how models reason about moral problems rather than measuring what they judge. Content-level measures have been homogenized by alignment training; process-level measures probe the depth and stability of that homogenization.

The third bind—the cultural monocentrism of the benchmark landscape, documented in the cross-cultural analysis—completes the triple bind. Narrow instruments (Bind 1) pro-

duce undiscriminating results (Bind 2) within a cultural frame that excludes the majority of human moral experience (Bind 3). Together, the three binds constitute a systemic failure of evaluative infrastructure: the field has built tools that measure one thing, applied them to systems that all produce that one thing, and declared the resulting convergence evidence of alignment success. The convergence is real, but it is evidence of value monoculture, not value alignment. And the monoculture is invisible to the instruments because the instruments were built within the same cultural frame that the monoculture occupies.

4. Results Part I: The Benchmark Landscape

The preceding sections identified 292 benchmark candidates from a source corpus of 2,099 AI ethics papers, scored each on 17 composite-determining features organized into three sub-dimensions (Psychological Depth, Practical Ease, Citation Impact), and classified them into four quality tiers via Jenks natural breaks optimization with break points at 27.8, 43.05, and 56.65 (GVF = 0.83). The resulting tier distribution—48 Top (16.4%), 89 High (30.5%), 72 Medium (24.7%), 83 Low (28.4%)—provides the empirical foundation for the six landscape claims advanced in this section.

The landscape analysis reveals two of the three binds that constitute the paper's central diagnostic. The first bind—theoretical narrowness—manifests as a quality concentration in which a single theory, Moral Foundations Theory, captures 37.5% of top-tier slots and achieves a quality rate six times that of non-Western moral traditions (Claim 1). This narrowness is compounded by a construct validity crisis in which the majority of the landscape lacks even basic psychometric grounding (Claim 2), and by the absence or effective absence of process-level theories that are foundational to modern moral psychology yet have produced no publication-ready benchmarks for LLM evaluation (Claim 3). The third bind—cultural blindness—emerges from the interaction of these findings with the distributional patterns documented in Claims 4 through 6, which are presented in the companion section. Together, the six landscape claims establish that the benchmark ecosystem's deficiencies are not localized weaknesses in specific instruments but structural properties of the landscape as a whole.

A note on the relationship between this analysis and prior work is warranted. The original assessment characterized 55 benchmark candidates selected from the upper reaches of the quality distribution. The full-coverage assessment of all 292 candidates does not merely expand the sample; it fundamentally transforms the evidentiary basis for each claim. Where the 55-benchmark analysis suggested gaps, the 292-benchmark analysis reveals a landscape in which the gaps are the dominant feature and the high-quality instruments are the exception. Every claim presented below is strengthened, and in several cases substantially so, by the transition from selected subsample to full-coverage assessment. The specific magnitudes of these shifts are documented claim by claim, and Table T5 through Table T7 provide the updated empirical foundations.

4.1 Claim 1: MFT Quality Dominance

4.1.1 Volume versus Quality: A Consequential Distinction

The original assessment of 222 benchmark candidates reported that Moral Foundations Theory (MFT) accounted for 101 candidates, or 45% of the landscape by volume. This figure was already striking—a single theory representing nearly half of all moral psychology benchmarks for LLM evaluation—and it anchored the claim that the benchmark landscape exhibits a problematic theoretical monoculture. The full-coverage assessment of all 292 candidates revises this picture in a direction that is, paradoxically, more concerning rather than less.

MFT now accounts for 67 of 292 candidates, or 23% of the landscape by volume. The apparent decline from 45% to 23% is entirely attributable to the expansion of the candidate pool: the Phase 2 sweep for non-Western moral traditions added 71 candidates from Islamic ethics, Confucian moral philosophy, Ubuntu ethics, Buddhist ethics, and Indigenous ethical frameworks, diluting MFT's volumetric share without reducing its absolute count. Taken in isolation, the volume figure might suggest that the theoretical monoculture has been overstated—that MFT is, in fact, one theory among many in a landscape that includes substantial non-Western representation.

The quality analysis demolishes this interpretation. When the 292 candidates are classified into quality tiers, MFT's dominance shifts from volume to quality—a shift that is more consequential for the field's evaluative capacity. MFT holds 18 of 48 top-tier slots (37.5%) and achieves the highest quality rate of any theory with more than five candidates: 73% of MFT candidates (49 of 67) score in the top or high tier combined. No other theory approaches this rate. Schwartz values, the second-best-performing Western theory, achieves 57% (25 of 44). The trolley tradition, despite producing 10 top-tier entries (the second-highest absolute count), achieves only 48% (14 of 29). Among non-Western traditions, the highest quality rate is 36% for Islamic ethics (8 of 22); Ubuntu, Confucian, and Indigenous ethics achieve 0%.

The distinction between volume dominance and quality dominance is not merely taxonomic. A landscape in which MFT is overrepresented by volume but where other theories contribute instruments of comparable quality would support a simple prescription: fund and build more benchmarks for underrepresented theories, and the monoculture will dissolve. A landscape in which MFT dominates by quality—where its instruments are systematically better-constructed, better-validated, and better-supported than those of every other theory—suggests a deeper structural problem. The quality gap cannot be closed by simply producing more benchmarks for non-MFT theories; it requires understanding why MFT benchmarks are better and whether the factors driving their quality can be replicated for other theoretical traditions.

4.1.2 The Infrastructure Advantage Thesis

The quality gap between MFT and all other theories is not attributable to superior theoretical insight, greater researcher effort, or more inherent suitability of MFT for LLM evaluation. It is attributable to a concrete, identifiable infrastructure asymmetry: MFT possesses two computational assets that no other moral psychology theory can match, and these assets create a self-reinforcing quality flywheel that compounds over successive research cycles.

The first asset is the Moral Foundations Questionnaire (MFQ-30; Graham et al., 2011). The MFQ-30 is a standardized 30-item questionnaire with published psychometric properties —factor structure, internal consistency coefficients, test-retest reliability—validated across multiple populations including political liberals and conservatives, religious and secular respondents, and respondents from at least 11 countries. For researchers building an MFT benchmark for LLMs, the MFQ-30 provides an off-the-shelf starting point: a validated instrument with known properties that can be administered to a language model with minimal adaptation. This starting advantage is directly visible in the assessment framework. Feature A4 (source instrument validation) awards 100 points to benchmarks that adapt a psychometrically validated source instrument; MFT benchmarks that use or adapt the MFQ-30 automatically receive this score, boosting their Psychological Depth (PD) subscore by the A4 weight (0.10 of PD, or 0.05 of the composite). Feature A5 (human baseline availability) benefits similarly, as the MFQ-30′s population norms provide ready-made human comparison data.

The second asset is the Moral Foundations Twitter Corpus (MFTC; Hoover et al., 2020), a collection of 35,108 tweets annotated by expert raters for the presence of each moral foundation. The MFTC is the largest expert-annotated moral foundations text corpus available, and it enables a category of benchmark construction unavailable to any other theory: corpus-based NLP approaches in which researchers train classifiers on MFTC annotations and then evaluate LLMs against these classifiers or use MFTC-derived items as benchmark stimuli. Feature B3 (data availability) rewards benchmarks with accessible, well-documented datasets; benchmarks built on or referencing the MFTC receive elevated scores because the corpus is publicly available on GitHub with structured annotations. Feature B2 (dataset size) similarly benefits from the MFTC's scale.

The dual infrastructure—a validated questionnaire with population norms and a large annotated corpus—creates what can be described as a quality flywheel. A researcher proposing a new moral psychology benchmark for LLMs faces a choice: begin with MFT, where the MFQ-30 provides a validated instrument (A4 = 100), the MFTC provides a large annotated corpus (B3 elevated), and published population norms provide human baselines (A5 elevated)—or begin with another theory, where none of these resources exist. The rational researcher, particularly one operating under publication pressure and seeking rapid results, chooses MFT. The resulting paper adds another high-quality MFT benchmark to the landscape, reinforcing MFT's quality dominance, which in turn makes

MFT the even more obvious choice for the next researcher. The flywheel spins faster with each cycle.

No other theory possesses this dual infrastructure. Schwartz's theory of basic human values has the Portrait Values Questionnaire—Revised (PVQ-RR; Schwartz et al., 2012), a validated questionnaire with cross-cultural norms spanning over 80 countries, but it lacks a large annotated text corpus. Kohlberg's Cognitive Moral Development theory has the Defining Issues Test (DIT; Rest, 1979), a validated stage-assessment instrument, but again no NLP corpus. Ubuntu ethics, Confucian moral philosophy, and Indigenous ethical frameworks have neither: no standardized questionnaire with published psychometrics and no annotated text corpus. The infrastructure gradient—from MFT's dual-asset advantage to the zero-asset position of non-Western traditions—maps directly onto the quality gradient observed in the tier distribution.

The infrastructure thesis is falsifiable and could in principle be disconfirmed by a theory that achieves high quality rates despite lacking MFT's computational assets. Social intuitionism provides a partial test case: with only 3 candidates but a 100% quality rate (all 3 in top or high tier), it demonstrates that small-sample theories can achieve quality without large-scale infrastructure—but the sample is too small (n = 3) to support generalizable conclusions, and social intuitionism benefits from proximity to MFT's conceptual vocabulary (Haidt, the theory's originator, co-developed MFT). The more informative test is provided by non-Western traditions, where 32 candidates across four traditions achieve a 0% quality rate despite representing genuine, often sustained research efforts. This pattern is precisely what the infrastructure thesis predicts: without foundational instruments, without annotated corpora, without population norms, the quality gap cannot be closed through effort alone.

4.1.3 Theory x Tier Analysis

The full theory-by-tier crosstab, derived from the Jenks natural breaks classification of all 292 candidates, is presented in Table T5. This table replaces the volume-only summary in the prior analysis with a quality-stratified portrait of theoretical representation.

Table T5: Theory Representation and Quality Rates

Theory	Total	Top	High	Med	Low	Quality Rate (% Top + High)
MFT	67	18	31	13	5	73%
Schwartz	44	4	21	14	5	57%
Trolley	29	10	4	10	5	48%
Kohlberg	9	3	2	2	2	56%
Islamic	22	3	5	6	8	36%
Buddhist	16	1	1	4	10	12%
Intuitionist	3	1	2	0	0	100%
Dyadic	2	0	1	0	1	50%
Ubuntu	13	0	0	2	11	0%
Confucian	13	0	0	0	13	0%
Indigenous	4	0	0	0	4	0%
Identity	2	0	0	0	2	0%
Unclassified	83	8	22	31	22	36%

Note: "Unclassified" includes candidates that engage with moral psychology without primary allegiance to a single theory, multi-theory benchmarks without a dominant theoretical frame, and applied ethics evaluation suites. Some candidates are counted under multiple theories; column totals therefore exceed 292.

Several patterns in Table T5 warrant detailed discussion.

The Western psychological cluster. The four major Western psychological theories —MFT (73%), Schwartz values (57%), Kohlberg's Cognitive Moral Development (56%), and the trolley dilemma tradition (48%)—all achieve quality rates above 46%. This is not coincidental. Each of these theories developed within the institutional infrastructure of Anglo-American and European academic psychology, where the norms of psychometric validation, questionnaire development, and population-normed assessment were established and where funding structures reward the kind of slow, cumulative instrument-building that produces high-scoring benchmarks in the assessment framework. The 46% floor for Western psychological theories reflects not the inherent quality of these theories but the institutional context in which their research tools were developed.

The non-Western quality cliff. The three non-Western cultural moral traditions with sufficient candidates for meaningful analysis—Ubuntu (13 candidates, 0% quality rate), Confucian ethics (13 candidates, 0% quality rate), and Buddhist ethics (16 candidates, 12% quality rate)—exhibit a dramatic quality gradient relative to Western traditions. The cliff is not merely a gap; it is a binary separation. Among theories with 10 or more candidates, every Western theory achieves a quality rate above 46%, and every non-Western cultural tradition (excluding Islamic ethics, discussed below) achieves a quality rate at or below 12%. Buddhist ethics, with 16 candidates, has a single top-tier entry and a single high-tier entry; the remaining 14 candidates (88%) fall in the medium or low tier. Ubuntu and Confucian ethics have no entries above the medium tier at all. This distributional pattern previews Claim 4 (cultural quality cliff), which examines the structural causes of the separation.

Islamic ethics as an intermediate case. Islamic ethics occupies a distinctive position in the quality landscape. With 22 candidates and a 36% quality rate (3 top, 5 high), it substantially outperforms the other non-Western traditions while remaining well below the Western psychological cluster. Two factors account for this intermediate position. First, Islamic ethics has a longer history of engagement with computational and textual analysis approaches, partly because the hadith sciences (the systematic study of prophetic traditions) developed classification and authentication methodologies that parallel modern data science techniques, and partly because several well-funded research programs (notably those connected to QIAS, the Question-Answering in Islamic Sciences shared task) have invested in benchmark infrastructure. Second, Islamic ethics benefits from a body of codified jurisprudential texts (the Quran, hadith collections, fiqh manuals) that provide structured source material for benchmark construction—a resource that Ubuntu's oral ethical traditions and Indigenous relational ethics cannot readily provide. The three top-tier Islamic ethics benchmarks (Bouchekif et al., 2025, composite = 60.55; Ghaly et al., 2025, composite = 58.45; "Sacred or Synthetic?", 2025, composite = 58.43) all emerged from this institutional context.

The trolley paradox. The trolley dilemma tradition presents a paradox of quality distribution: it has 10 top-tier entries (the second-highest absolute count after MFT's 18), yet only 4 high-tier entries. This unusual "hourglass" distribution—many top entries, few high entries, many medium entries—reflects the bimodal nature of the tradition. At the top, the trolley tradition includes the Moral Machine experiment (Awad et al., 2018; composite = 71.73) and its derivatives, which benefit from massive crowdsourced human baseline data (>40 million decisions from 233 countries) and established population norms. At the middle and bottom, it includes numerous papers that present trolley-like dilemmas to LLMs without the psychometric infrastructure, human baselines, or construct validity evidence that distinguish the Moral Machine family. The gap between the Moral Machine's composite of 71.73 and the median trolley candidate suggests that the tradition's quality is concentrated in a single research lineage rather than distributed across the tradition as a whole.

Unclassified candidates and the "moral evaluation" umbrella. The 83 unclassified candidates—those not primarily affiliated with any single theory—achieve a 36% quality rate (8 top, 22 high), placing them in the middle of the distribution between the Western psychological cluster and the non-Western traditions. This category includes some of the landscape's most innovative work: Hendrycks et al.'s (2021) "Aligning AI with Shared Human Values" (composite = 67.05), Lourie's (2021) Scruples (composite = 67.03), and Jiang et al.'s (2022) "Can Machines Learn Morality?" (composite = 63.48). These benchmarks engage with moral cognition through ecologically valid scenarios, commonsense moral judgments, and interactive environments rather than through the lens of a single psychological theory. Their quality demonstrates that theory-neutral approaches can produce strong benchmarks, but their unclassified status also means they do not contribute to resolving the theoretical monoculture: they add breadth to the *applied* dimension of moral evaluation without deepening the *theoretical* dimension.

Figure 3 presents the theory x tier distribution as a heatmap, with theories ordered by quality rate from highest (intuitionist, 100%) to lowest (confucian, indigenous, identity; 0%). The heatmap's visual structure makes the quality cliff immediately apparent: a warm-colored upper-left region (Western theories, high tiers) separated by a sharp diagonal from a cool-colored lower-right region (non-Western traditions, low tiers).

4.1.4 The Monoculture as a Quality Problem

The reframing of Claim 1 from volume dominance to quality dominance has significant implications for the field's self-understanding and for the prescriptions that follow from the diagnosis.

Under the volume-dominance framing, the monoculture appeared to be a problem of attention allocation: too many researchers working on MFT, not enough working on other theories. The prescription was straightforward—diversify research funding, issue calls for non-MFT benchmarks, incentivize theoretical plurality. Under the quality-dominance framing, the monoculture is revealed as a problem of structural advantage: MFT's quality rate of 73% is not merely high in absolute terms but is 6 times the rate achieved by non-Western traditions with non-zero candidate counts (Buddhist ethics at 12%) and infinitely higher than the rate achieved by Ubuntu, Confucian, and Indigenous traditions (0%). The prescription must therefore address infrastructure, not merely attention. Building more benchmarks for non-MFT theories will not close the quality gap unless those benchmarks are accompanied by the foundational investments—validated instruments, annotated corpora, cross-cultural population norms—that the assessment framework identifies as the determinants of quality.

The MFT quality dominance is, in this sense, a microcosm of the triple bind. The first bind (theoretical narrowness) concentrates quality in one theory. The third bind (cultural blindness) ensures that non-Western theories cannot compete on quality because they lack the institutional infrastructure that the quality assessment framework rewards. And as the cross-vendor empirical study will demonstrate (Claim 7, Section 5), the second bind

(empirical undiscrimination) means that even MFT's high-quality benchmarks cannot distinguish between the models they evaluate, raising the question of whether the field's best instruments are measuring anything that varies across LLMs at all. The three binds converge on a single conclusion: the moral psychology benchmark landscape does not need more benchmarks of the kind it already has; it needs a different kind of benchmark entirely.

4.2 Claim 2: Construct Validity Crisis

Construct validity—the degree to which an instrument measures the theoretical construct it purports to measure—is the foundational requirement for interpretable psychological evaluation. The concept, first formalized by Cronbach and Meehl (1955) and elaborated by Messick (1995) into a unified validity framework, holds that a score is meaningful only insofar as the instrument generating that score has been demonstrated to tap the intended construct rather than an unintended confound, irrelevant content, or mere surface-level correlation. Without construct validity evidence, a benchmark's output is a number without psychological meaning: it may correlate with something, but we do not know what that something is, and we cannot use the number to make claims about the moral psychology of the system being tested.

The full-coverage assessment of all 292 candidates reveals that the moral psychology benchmark landscape is in a state of construct validity crisis. The crisis is not limited to a tail of low-quality instruments at the margin; it is a defining property of the landscape's central tendency.

4.2.1 A6 Distribution across 292 Candidates

Feature A6 in the assessment framework measures construct validity level on a five-point ordinal scale, with each level corresponding to a qualitatively distinct evidentiary standard. The five levels, with their point assignments and interpretations, are:

- **Level 5 (100 points): Multi-study psychometric validation.** The benchmark has been subjected to psychometric analysis across multiple independent studies, demonstrating convergent validity (correlation with theoretically related constructs), discriminant validity (non-correlation with theoretically unrelated constructs), and/or criterion validity (prediction of real-world moral behavior or judgment). This level represents the gold standard of construct validation, characteristic of instruments with decades of development such as the MFQ-30 or the DIT.
- **Level 4 (75 points): Single-study validation.** The benchmark reports construct validity evidence from a single study, typically conducted by the benchmark's developers. Evidence may include factor analysis confirming the intended dimensional structure, convergent validation against a criterion instrument, or differential item functioning analysis. This level represents the minimum standard for publication-ready claims about what the benchmark measures.

- **Level 3 (50 points): Adapts a validated instrument without LLM revalidation.** The benchmark is derived from an instrument with established construct validity in human populations but has been administered to LLMs without any revalidation study. The implicit assumption is that the instrument measures the same construct in a non-human respondent as it does in humans—an assumption that Messick's (1995) framework identifies as requiring empirical justification rather than being taken as given. This level is characteristic of the "port-and-report" pattern common in the literature, where researchers take a human moral psychology instrument, administer it to ChatGPT or Claude, and report the scores as though they are psychologically interpretable without addressing whether the instrument functions equivalently across the human-LLM measurement boundary.
- **Level 2 (30 points): Face validity only.** The benchmark items were designed by the researchers to capture the target construct based on their understanding of the theory, but no empirical evidence supports the claim that the items actually measure what they are intended to measure. The benchmark "looks like" it measures moral reasoning, moral sensitivity, or moral judgment—in the sense that a reasonable person examining the items would agree that they are about morality—but appearance and measurement are not the same thing. This is the level at which the majority of new moral psychology benchmarks for LLMs are constructed.
- **Level 1 (10 points): Keyword-level engagement.** The benchmark engages with moral psychology at the level of vocabulary rather than measurement. It may include items that use moral terminology ("Is this action ethical?"), classify text as moral or immoral based on surface features, or present scenarios with moral content without grounding them in any theoretical framework for moral cognition. The distinction between Level 1 and Level 2 is that Level 1 benchmarks do not even attempt to operationalize a specific moral psychology construct; they treat "morality" as a unidimensional category to be detected or classified rather than as a multi-dimensional psychological phenomenon to be measured.

The distribution of A6 levels across the full 292-candidate landscape, compared to the original 55-benchmark subsample, is presented in Table T6.

Table T6: Construct Validity Distribution — v2.1 vs. v3.0

A6 Level	Interpretation	v2.1 (n = 55)	v3.0 (n = 292)	Shift
5 (multi-study)	100 pts	6 (11%)	13 (4%)	−7 pp
4 (single-study)	75 pts	20 (36%)	61 (21%)	−15 pp
3 (adapts w/o revalidation)	50 pts	20 (36%)	49 (17%)	−19 pp
2 (face validity)	30 pts	7 (13%)	91 (31%)	+18 pp
1 (keyword-only)	10 pts	2 (4%)	78 (27%)	+23 pp

Note: pp = percentage points. The v2.1 column reports the distribution from the original 55-benchmark assessment. The v3.0 column reports the distribution across all 292 candidates assessed in the present study.

Figure 4 presents the validity distribution as a paired bar chart, with v2.1 and v3.0 counts side by side at each A6 level, making the distributional shift visually apparent.

4.2.2 The v2.1 Sampling Artifact

The most consequential finding in Table T6 is not any single cell but the systematic pattern of redistribution between the two assessments. The original 55-benchmark analysis reported a median A6 level of 3 (adapts validated instrument without LLM revalidation) and a relatively balanced distribution across levels 3, 4, and 5 (83% of candidates at Level 3 or above). This portrait suggested a landscape with genuine validity concerns—the modal benchmark adapted a human instrument without revalidating it for LLMs—but one in which most benchmarks at least started from a psychometrically validated foundation.

The full-coverage assessment reveals that this portrait was an artifact of sampling from the top of the quality distribution. The original 55 benchmarks were not a random sample; they were selected through a process that prioritized well-known, frequently cited, and established instruments. This selection process systematically oversampled high-validity benchmarks and undersampled the long tail of face-validity and keyword-only entries that constitute the majority of the landscape.

Under full-coverage assessment, the landscape's validity profile inverts. The median A6 level drops from Level 3 to Level 2 (face validity only)—a full level lower on the ordinal scale. The proportion of candidates at the two lowest validity levels (Level 1 + Level 2) increases from 17% to 58%—a 3.4-fold increase. The proportion at the two highest levels (Level 4 + Level 5) decreases from 47% to 25%—nearly halving. The proportion at the gold standard (Level 5, multi-study validation) decreases from 11% to 4%, a decline of 7 percentage points that reflects the denominator expansion rather than a loss of high-quality instruments (the absolute count of Level 5 benchmarks actually increases from 6 to 13).

The shift is dramatic in both direction and magnitude. Where the original assessment could be characterized as documenting "a gap between current practice and the validation standard"—a gap implying that most benchmarks were at least in the right neighborhood and needed to take one more step—the full-coverage assessment documents a landscape in which the majority of benchmarks have not taken the first step toward construct validation. Fifty-eight percent of all 292 candidates operate at Level 1 or Level 2, meaning they either engage with moral psychology at the keyword level only or rely entirely on face validity without any empirical evidence that their items measure what they claim to measure. This is not a gap; it is a chasm.

The sampling artifact has implications beyond the specific numbers. It suggests that the AI evaluation community's self-assessment of its moral psychology benchmarking

capabilities has been systematically optimistic, shaped by the visibility of a small number of well-constructed instruments that dominate conference proceedings and leaderboards while the much larger body of low-validity work remains invisible. The quality illusion identified in Section 1.3 is, at bottom, a validity illusion: the field believes it has construct-valid instruments because the instruments it pays attention to are construct-valid, but the instruments it pays attention to are a small and unrepresentative fraction of what exists.

4.2.3 Implications for Interpretability

The construct validity crisis is not an abstract psychometric concern. It has direct, practical consequences for every claim the field makes about LLM moral cognition, and it renders a substantial fraction of published results uninterpretable in any rigorous sense.

Consider the interpretive chain that connects a benchmark score to a psychological claim about an LLM. A researcher administers a moral reasoning benchmark to GPT-5 and reports that the model scores at the 75th percentile of human performance. The implicit claim is: "GPT-5 exhibits moral reasoning capacity at a level exceeding three-quarters of the human comparison population." For this claim to be meaningful, several conditions must hold. The benchmark must measure moral reasoning (not, for example, reading comprehension of morally themed passages, or pattern-matching to moral vocabulary). It must measure the same construct in the LLM as it does in the human comparison population. And the human comparison data must be drawn from a defined, relevant population against which the percentile ranking is interpretable.

At A6 Level 5 (multi-study validation), all three conditions are empirically supported: the construct has been validated across studies, the measurement model has been tested for invariance, and population norms provide meaningful anchoring. At Level 4 (single-study validation), the first condition is supported and the others are partially addressed. At Level 3 (adapts without revalidation), the first condition is supported for humans but not for LLMs—the measurement equivalence assumption is taken on faith. At Level 2 (face validity), none of the three conditions is empirically supported; the entire interpretive chain rests on the researcher's subjective judgment that the items "look like" they measure moral reasoning. At Level 1 (keyword-level), the interpretive chain does not even begin: the benchmark does not attempt to operationalize a specific moral psychology construct, so there is no construct to validate.

The practical implication is that 58% of the benchmark landscape (Level 1 + Level 2) produces numbers that cannot be used to make psychologically interpretable claims about LLM moral cognition. A model's score on a face-validity benchmark tells us that the model produced responses that the benchmark's scoring rubric coded in a particular way. It does not tell us whether those responses reflect moral reasoning, moral sensitivity, moral knowledge, moral compliance, pattern-matched moral vocabulary, or any of the numerous other processes that could produce "moral-sounding" outputs without engaging the psychological constructs that the benchmark's name implies.

This concern is sharpened by the specific characteristics of LLMs as test-takers. Human respondents bring to moral psychology instruments a lifetime of socialization, emotional development, interpersonal experience, and embodied cognition that constrains the range of processes through which they can produce moral responses. When a human scores high on the Moral Foundations Questionnaire, the result can be interpreted against a rich background of validated psychological theory about how moral foundations develop, function, and interact with personality, culture, and political ideology. An LLM brings none of this background. Its moral-sounding outputs are produced by a process—next-token prediction conditioned on training data and RLHF alignment—that has no demonstrated relationship to any moral psychology construct. The construct validity question is therefore not "Does this instrument measure moral reasoning?" but rather "Does this instrument measure anything psychologically coherent when administered to a system that does not engage in psychological processes?" At Levels 1 and 2, this question has not even been asked, let alone answered.

The modal benchmark in the landscape—a Level 2 instrument constructed by researchers who believed their items captured the target construct but never tested this belief empirically—embodies a methodological stance that would not be accepted in any other area of psychological assessment. A clinical psychologist who developed a new depression screening instrument, administered it to patients, and reported scores as clinically meaningful without any validation study would be in violation of the American Psychological Association's Standards for Educational and Psychological Testing (AERA, APA, & NCME, 2014). The equivalent practice in LLM moral evaluation is routine, in large part because the field lacks an organized professional body with the authority to enforce validation standards. The construct validity crisis is, at bottom, a governance crisis: there is no gatekeeper ensuring that the instruments used to certify the moral fitness of AI systems are themselves fit for purpose.

The crisis is particularly acute in the context of deployment decisions. When a company evaluates whether to deploy an LLM in a healthcare setting, a content moderation system, or an educational tutor, the moral evaluation benchmarks in its assessment battery constitute the primary evidence for the model's moral adequacy. If those benchmarks have Level 1 or Level 2 construct validity—and 58% of available benchmarks do—the deployment decision is made on the basis of evidence that has no demonstrated relationship to the moral psychological properties it purports to measure. The model may be morally adequate, or it may not; the benchmark scores simply cannot tell us, because the instruments generating those scores have not been shown to measure what they claim to measure.

4.3 Claim 3: Theoretical Absence

The construct validity crisis documented in Claim 2 addresses the quality of existing benchmarks: how well do the instruments that exist measure what they claim to measure? Claim 3 addresses a complementary dimension of the landscape's inadequacy: which theories of moral cognition have no functional representation in the benchmark landscape at all?

The distinction between "poorly measured" and "unmeasured" is consequential. A poorly measured theory can, in principle, be better measured by improving existing instruments. An unmeasured theory requires building instruments from scratch—a substantially more costly undertaking that the field's incentive structure actively discourages.

The full-coverage assessment of all 292 candidates, combined with the expanded keyword scorer (125 patterns across 13 theories; Section 3.1.3), reveals a pattern of theoretical absence that is both more nuanced and more troubling than the original analysis suggested. The original 55-benchmark assessment reported three theories as "missing" from the landscape: Dual Process Theory, Social Intuitionism, and Dyadic Morality. The full-coverage assessment corrects one of these claims (Social Intuitionism is not missing; it was misdetected), confirms and strengthens another (Dual Process Theory is functionally absent despite non-zero candidate counts), and adds a previously unrecognized dimension of theoretical absence: the complete absence of publication-ready benchmarks for moral identity theory.

4.3.1 Dual Process Theory: Present in Name, Absent in Quality

Joshua Greene's dual-process theory of moral cognition (2001, 2014) proposes that moral judgments arise from the competition between two cognitive systems: a fast, automatic, emotionally driven system that produces characteristically deontological responses (do not push the man off the bridge, even to save five lives) and a slow, deliberative, consequentialist system that produces characteristically utilitarian responses (push the man if the expected-value calculation favors it). The theory was developed through a landmark series of fMRI studies demonstrating that personal moral dilemmas (those requiring direct physical harm) activate emotional brain regions to a greater degree than impersonal dilemmas, and that individuals who override the emotional response in favor of the utilitarian judgment show increased activation of brain regions associated with deliberative reasoning.

The dual-process framework is, by citation count, one of the two or three most influential theories in twenty-first-century moral psychology. Greene's (2001) foundational fMRI paper has accumulated over 10,000 citations. The theory's influence extends well beyond moral psychology into philosophy (it has reshaped the trolley problem debate), neuroscience (it is a canonical example of dual-process cognitive architecture), and AI alignment (it directly informs the question of whether LLMs exhibit "fast" versus "slow" moral processing). If any theory should have a well-developed benchmark infrastructure for LLM evaluation, it is this one.

It does not. The expanded keyword scorer identified approximately 9 candidates with genuine dual-process theory engagement—a number that itself reflects the expanded scorer's success in detecting theory-specific vocabulary (the original scorer detected only 2-3). But of these 9 candidates, all score in the low tier of the Jenks classification. Not one achieves even the medium tier. The theory is present in the landscape by count but absent by quality: every dual-process candidate lacks the combination of validated instrumentation, human

baselines, and construct validity evidence that the assessment framework identifies as necessary for publication-ready moral evaluation.

The absence is particularly striking because dual-process theory makes specific, operationalizable predictions that lend themselves directly to benchmark design. The theory predicts that increasing cognitive load (time pressure, working memory demands, concurrent tasks) should shift moral judgments toward the emotional-deontological pole, while increasing deliberative resources (time, explicit reasoning prompts, chain-of-thought scaffolding) should shift judgments toward the utilitarian-consequentialist pole. These predictions can be tested straightforwardly in LLMs by manipulating the available deliberative resources—for example, by comparing moral judgments produced with and without chain-of-thought reasoning, or by varying the complexity of the reasoning scaffold. The core experimental manipulation is not difficult to operationalize; it is merely different from the content-level assessments (which moral foundation does the model endorse? which value does it prioritize?) that dominate the existing landscape.

The failure to build dual-process benchmarks for LLMs is therefore not a matter of theoretical intractability. It is a matter of paradigmatic inertia: the benchmark landscape has organized itself around *content*-level assessment (what moral positions does the model hold?) rather than *process*-level assessment (how does the model form moral judgments?), and dual-process theory is fundamentally a process theory. The absence of dual-process benchmarks from the quality landscape is a direct consequence of this content-over-process bias—a bias that, as the cross-vendor empirical study will demonstrate (Section 5), is precisely backward. The content-level instruments that dominate the landscape cannot distinguish between models; the process-level probes that the landscape lacks are the only instruments that can.

4.3.2 Social Intuitionism: A Significant Reversal

The original 55-benchmark assessment listed Haidt's Social Intuitionist Model (2001) as one of three theories "missing" from the benchmark landscape. The full-coverage assessment with the expanded keyword scorer corrects this finding. Three genuine social intuitionism benchmarks exist in the landscape, and all three score in the top or high tier —a 100% quality rate that is the highest of any theory in the entire landscape, albeit from the smallest sample (n = 3).

The reversal is attributable to a detection failure in the original keyword scorer, not to a change in the underlying landscape. The original scorer contained only 2-3 patterns for social intuitionism, concentrated on the phrase "social intuitionist" itself, and missed benchmarks that operationalized the theory's core constructs (moral dumbfounding, post-hoc rationalization, automatic moral evaluation) without using the theory's proper name. The expanded scorer added construct-specific vocabulary patterns (e.g., "moral dumbfounding," "intuitive moral judgment," "post-hoc moral rationalization," "automatic moral evaluation") and researcher-name co-occurrence patterns (e.g., "Haidt" co-occurring with

"intuit*"). These additions recovered the three benchmarks that had been invisible to the original pipeline.

The finding is significant for two reasons. First, it demonstrates that the original claim of theoretical absence was partly an artifact of detection methodology rather than a genuine property of the landscape. This is methodologically important: it means that claims about which theories are "missing" must be evaluated relative to the detection pipeline's sensitivity, and that apparent absences may be false negatives rather than true zeros. Second, the 100% quality rate of the three social intuitionism benchmarks—with one in the top tier (MoCa; Nie et al., 2023; composite = 60.15) and two in the high tier—suggests that social intuitionist constructs are well-suited to benchmark operationalization when researchers do attempt them. The theory's core predictions (that moral judgments are driven by automatic intuitive reactions and that explicit moral reasoning serves primarily to rationalize post-hoc) generate testable hypotheses about LLM behavior that the existing benchmarks have successfully operationalized.

The corrected finding does not, however, substantially alter the broader claim of theoretical absence. Social intuitionism's 3 benchmarks, even at 100% quality, represent a tiny fraction of the landscape (1% of 292 candidates) and provide insufficient coverage for systematic evaluation. More importantly, the reversal highlights a structural asymmetry: social intuitionism was misdetected, not unmeasured, because the benchmarks that operationalize it are high-quality instruments that happened to use non-standard vocabulary. Dual-process theory, by contrast, is not misdetected; it is genuinely absent from the quality landscape regardless of what vocabulary one uses to search for it. The distinction between "present but invisible to the scorer" and "absent regardless of detection method" is crucial for understanding which theoretical gaps can be closed by better detection and which require new instrument development.

4.3.3 Moral Identity: Zero Publication-Ready Instruments

Aquino and Reed's (2002) moral identity theory proposes that moral behavior is motivated not (or not only) by moral reasoning or moral intuition but by the degree to which morality is central to an individual's sense of self. The theory distinguishes between two dimensions of moral identity: internalization (the private, internal sense that moral traits are central to one's self-concept) and symbolization (the public, visible expression of moral identity through behavior, appearance, and social engagement). Individuals with strong moral identity internalization are predicted to act morally because doing so is self-consistent—because failing to act morally would create cognitive dissonance with their self-concept—rather than because they have reasoned their way to the right answer or because their intuitions dictate a particular response.

The theory has generated a substantial body of empirical work in organizational psychology, consumer behavior, and developmental psychology, with Aquino and Reed's (2002) foundational paper accumulating over 3,000 citations. It is directly relevant to LLM evaluation for a reason that the field has largely overlooked: moral identity theory addresses the

question of *motivation* for moral behavior, a dimension entirely absent from the content-focused benchmarks that dominate the landscape. Existing benchmarks ask whether an LLM endorses the right moral positions or produces the right moral judgments; moral identity theory asks whether the system's moral outputs are integrated into a coherent sense of agency and self-consistency—a question that may be particularly revealing when applied to systems that lack a genuine self but are increasingly designed to simulate one.

The full-coverage assessment identifies 2 candidates with moral identity theory engagement. Both score in the low tier of the Jenks classification (0% quality rate). Neither provides the validated instrumentation, human baselines, or construct validity evidence necessary for publication-ready evaluation. The moral identity gap in the benchmark landscape is therefore absolute: the field has produced no tools that could even in principle evaluate whether an LLM's moral outputs are integrated, consistent, or self-referential in the way that moral identity theory predicts for moral agents.

The absence is compounded by the difficulty of operationalizing moral identity for non-human agents. The Moral Identity Scale (Aquino and Reed, 2002)—the standard instrument for measuring moral identity in humans—asks respondents to imagine a person with specific moral traits (caring, compassionate, fair, friendly, generous, helpful, hardworking, honest, kind) and to rate the degree to which these traits are central to their self-concept. Administering this instrument to an LLM raises immediate validity concerns: does the model have a self-concept? Can it meaningfully rate the centrality of moral traits to an entity that may not have traits in any psychological sense? These are precisely the questions that a well-designed moral identity benchmark for LLMs would need to address, and the fact that no benchmark has attempted to address them means the field has no empirical basis for even formulating hypotheses about LLM moral identity.

4.3.4 Process-Level Theories and the Discrimination Gap

The three theories examined in Sections 4.3.1 through 4.3.3—Dual Process Theory, Social Intuitionism, and Moral Identity—share a property that distinguishes them from the content-focused theories that dominate the benchmark landscape. MFT asks *what* moral content a person (or model) endorses: which foundations do they weigh most heavily? Schwartz's theory asks *what* values a person prioritizes: where do they fall on the self-enhancement versus self-transcendence dimension? The trolley tradition asks *what* decisions a person makes: does the agent pull the lever, push the man, or refuse to act? These are all questions about the *content* of moral cognition—the positions, preferences, and decisions that constitute moral output.

Dual Process Theory, Social Intuitionism, and Moral Identity ask different questions entirely. Dual Process Theory asks *how* moral judgments are formed: through automatic emotional processing, deliberative reasoning, or some interaction of the two? Social Intuitionism asks *when* moral reasoning occurs: before the judgment (as traditional rationalist models assume) or after it (as post-hoc rationalization)? Moral Identity asks *why* moral behavior occurs: because reasoning or intuition dictates it, or because morality is central

to the agent's self-concept? These are questions about the *process* of moral cognition—the mechanisms, sequences, and motivational structures through which moral outputs are generated.

Table T7 summarizes the quality status of the four process-oriented theories in the benchmark landscape.

Table T7: Theoretical Absence Summary

Theory	Total	Top	High	Med	Low	Assessment
Dual Process	~9	0	0	0	~9	All low tier — effectively absent
Social Intuitionism	3	1	2	0	0	100% quality — was misdetected
Moral Identity	2	0	0	0	2	Zero publication-ready instruments
Dyadic Morality	2	0	1	0	1	One high-tier entry

The content/process distinction maps directly onto the discrimination gap that the cross-vendor empirical study (Section 5) will document in detail. When 23 benchmarks were administered to 9 models from 3 vendors, the content-level instruments—those measuring moral foundation endorsement, value prioritization, dilemma decisions—overwhelmingly produced identical scores across all models. The models' moral *content* is indistinguishable because they have been trained on similar data, aligned by annotators drawn from similar demographic pools, and optimized against similar safety targets. Switching from Claude Opus to Gemini 2.5 Pro to GPT-5 does not change what moral positions the model endorses; it does not change which foundations the model prioritizes; it does not change which dilemma resolutions the model selects.

The instruments that did discriminate between models—and here the finding from Section 5 is anticipated briefly, as the detailed analysis belongs to the companion results section—were precisely the process-level probes that the landscape lacks. The single benchmark that cleanly separated one model from the rest in the cross-vendor study was a dual-process protocol that tested how moral judgments respond to changes in deliberative resources (chain-of-thought reasoning). This protocol exploited a process-level prediction (that increasing deliberation should shift judgments toward the utilitarian pole) and found that it holds for some model architectures but not others—producing the only robust cross-model discrimination in the entire 23-benchmark battery.

This creates a paradox that lies at the heart of the paper's integrative diagnosis. The theories most capable of distinguishing between models—the process-level theories that ask how, when, and why moral cognition occurs—are the theories least represented in the

benchmark landscape. The theories most represented in the landscape—the content-level theories that ask what moral positions a model endorses—are the theories least capable of distinguishing between models. The benchmark landscape has invested its quality and infrastructure in precisely the wrong dimension of moral cognition, building ever more refined instruments for measuring a property (moral content) that does not vary across models while neglecting the dimension (moral process) where variation exists and discrimination is possible.

The theoretical absence documented in Claim 3 is therefore not merely a gap in coverage—a set of theories that happen not to have been operationalized. It is a structural deficiency that explains and is explained by the empirical undiscrimination that constitutes the second bind of the triple bind. The field has failed to build process-level benchmarks not because process theories are unoperationalizable (Section 4.3.1 demonstrated that dual-process predictions are straightforwardly testable) but because the dominant paradigm—content-level assessment using MFT-style instruments—has created a self-reinforcing cycle in which the success of content-level benchmarks (high quality, wide adoption, easy implementation) crowds out investment in the process-level benchmarks that the field actually needs.

The paradox also illuminates the relationship between Claims 1 and 3. MFT's quality dominance (Claim 1) is not merely an inconvenient distributional skew; it is a causal contributor to theoretical absence (Claim 3). Because MFT provides the easiest path to a high-quality benchmark, researchers default to MFT-based designs; because MFT is a content-level theory, the benchmarks they produce are content-level instruments; because content-level instruments cannot discriminate between models, the resulting benchmarks are empirically uninformative despite their high assessed quality. The infrastructure advantage that makes MFT benchmarks good (in terms of psychometric properties) is the same infrastructure advantage that makes them uninformative (in terms of cross-model discrimination). Quality, as currently measured, is not the same as utility, and the gap between the two is the gap between knowing what a model says about morality and knowing how—or whether—it actually engages in moral cognition.

4.4 Claim 4: The Cultural Quality Cliff

4.4.1 Western versus Non-Western Quality Rates

The full-coverage assessment reveals the single most consequential distributional finding in the benchmark landscape: quality cleaves along cultural lines with a severity that no subsample analysis could have detected. When all 292 candidates are assessed and tiered, the four major Western psychological theories—Moral Foundations Theory, Schwartz Basic Human Values, Kohlberg's Cognitive Moral Development, and the trolley dilemma tradition—achieve quality rates (top + high tiers combined) ranging from 48% to 73%. Non-Western moral traditions face a starkly different reality. Islamic ethics, which benefits from the longest-standing engagement with computational assessment among

non-Western traditions, achieves a quality rate of 36%—below any Western theory but at least demonstrating that a meaningful fraction of its benchmark candidates meet publication-ready standards. Buddhist ethics drops to 12%. And then the cliff: Ubuntu ethics, Confucian moral philosophy, Indigenous ethical frameworks, and moral identity theory grounded in non-Western relational conceptions of selfhood all achieve a quality rate of exactly 0%.

Table 8 presents the complete cultural quality distribution.

Table 8: Cultural Quality Cliff — Western versus Non-Western Quality Rates

Tradition	Origin	Total	Top	High	Top+High	Quality Rate
MFT	Western	67	18	31	49	73%
Schwartz	Western	44	4	21	25	57%
Kohlberg	Western	9	3	2	5	56%
Trolley	Western	29	10	4	14	48%
Western mean		*149*	*35*	*58*	*93*	*62%*
Islamic	Non-Western	22	3	5	8	36%
Buddhist	Non-Western	16	1	1	2	12%
Ubuntu	Non-Western	13	0	0	0	**0%**
Confucian	Non-Western	13	0	0	0	**0%**
Indigenous	Non-Western	4	0	0	0	**0%**
Identity	N/A	2	0	0	0	**0%**
Zero-quality traditions		*32*	*0*	*0*	*0*	***0%***

The magnitude of the gap is difficult to overstate. MFT's quality rate of 73% means that nearly three out of every four MFT benchmark candidates are suitable for publication-ready LLM moral evaluation. The zero-quality traditions' rate of 0% means that not a single candidate, out of 32 collective attempts, clears even the minimum threshold for research-grade assessment. The ratio is not 2:1 or 5:1; it is undefined—a division by zero that cannot be expressed as a finite multiplier. The Western mean quality rate of 62% versus the zero-quality group's 0% represents a categorical exclusion, not a quantitative gap.

The gradient between these extremes is itself informative. Islamic ethics achieves 36%—roughly half the Western mean—suggesting that a tradition with partial computational infrastructure can produce some viable benchmarks even without the full complement of resources available to MFT or Schwartz. Buddhist ethics at 12% represents a further drop, with only a single top-tier and a single high-tier candidate emerging from 16 attempts. The

pattern suggests a threshold effect: below a certain level of psychometric infrastructure, the probability of producing a publication-ready benchmark drops precipitously toward zero.

Figure 5 visualizes this cultural quality cliff as a bar chart, with traditions ordered by quality rate. The visual representation makes the cliff unmistakable: the bars descend from MFT's 73% through a gradual slope to trolley's 48%, drop sharply to Islamic's 36% and Buddhist's 12%, and then collapse entirely to the zero line for the remaining four traditions. The shape is not a smooth decline; it is a cliff with a narrow ledge (Islamic, Buddhist) before the final drop.

4.4.2 Ubuntu, Confucian, Indigenous, and Identity: Zero Quality Benchmarks

The zero-quality finding requires careful interpretation. It would be tempting to conclude that no one has tried to build benchmarks for Ubuntu, Confucian, Indigenous, or Identity ethics—that the absence reflects a simple gap in researcher interest. The data contradict this interpretation. The four zero-quality traditions collectively account for 32 benchmark candidates in the landscape. Researchers *have* attempted to build these benchmarks. They have published papers describing Ubuntu-inspired evaluation frameworks, Confucian ethics datasets, Indigenous moral reasoning probes, and moral identity assessment tools. The problem is not the absence of effort; it is the failure of effort to produce instruments of sufficient quality for rigorous LLM evaluation.

The 13 Ubuntu candidates illustrate the pattern with particular clarity. These include dedicated benchmark proposals such as UbuntuGuard (2026), which achieved the highest composite score among Ubuntu entries at 35.1—placing it in the medium tier, above the low tier but still well below the high-tier threshold of 43.0. The Ubuntu landscape also includes papers such as "An Ubuntu-Guided Large Language Model Framework" (2026, composite 29.8), "The Role of the African Value of Ubuntu in Global AI Inclusion Discourse" (2022, composite 22.9), and "Where Are the Missing Values: An Exploration of the Need to Incorporate Ubuntu Values into African AI Policy" (2025, composite 19.9). Across all 13 candidates, the composite scores range from 10.8 to 35.1, with a mean of 19.5—more than 37 points below MFT's mean composite among its top-tier entries alone. The two medium-tier entries (UbuntuGuard and the Ubuntu-Guided Framework) represent the closest any Ubuntu benchmark has come to research-grade quality, but both fall short on the psychometric features that separate the high tier from the medium tier: validated instruments (A4), population-level human baselines (A5), and demonstrated construct validity (A6).

The Confucian landscape tells a similar story. All 13 Confucian candidates score in the low tier, with composites ranging from 14.6 to 21.2 and a mean of 18.1. The candidates include thoughtful philosophical analyses—"Can AI Lead to Genuine Virtue? Confucian Perspectives" (2021), "Confucian Role Ethics and Artificial Intelligence" (2024), "AI Trained on the Confucian Analects" (2024)—but not one of these papers produces a validated instrument that could be administered to an LLM and scored against human baselines. The entire body of Confucian AI ethics research, as represented in this landscape, operates at the level

of philosophical analysis and conceptual framework development rather than empirical benchmark construction.

The Indigenous tradition is the smallest of the zero-quality groups, with only 4 candidates, all in the low tier (composites 14.6–22.2, mean 18.9). The Identity ethics tradition, when restricted to entries where moral identity theory is the sole theoretical framework rather than a secondary label on an MFT-primary benchmark, comprises 2 candidates (Kantian-Utilitarian XAI, 2025, composite 25.2; MoralNet, 2023, composite 13.6), both in the low tier. The distinction between sole-theory and multi-theory entries is important: several benchmarks labeled with both MFT and Identity theory (e.g., MoralBench, composite 59.2; M3oralBench, composite 56.7) do reach the top tier, but their quality derives from their MFT instrumentation, not from their engagement with moral identity constructs. These multi-theory benchmarks use MFT's validated questionnaire infrastructure and then add moral identity as a secondary analytical lens; they do not provide a validated instrument for measuring moral identity independently.

The feature-level signature across these zero-quality traditions is remarkably uniform. All four traditions share a common profile:

- **A2 (Construct Operationalization):** Mean scores of 37.7 (Ubuntu), 40.0 (Confucian), 40.0 (Indigenous), and variable for Identity—corresponding to "theory-inspired" operationalization, where the benchmark draws on theoretical concepts but does not implement them through a structured assessment framework. This is the psychometric equivalent of citing a theory in the introduction but not building items that operationalize its specific constructs.
- **A4 (Validated Source Instrument):** Mean scores of 0.0 across Ubuntu, Confucian, and Indigenous traditions. No benchmark in any of these traditions builds upon a previously validated psychometric instrument. There is no Ubuntu Moral Assessment Questionnaire with published factor loadings, no Confucian Role Ethics Scale with known reliability coefficients, no Indigenous Relational Ethics Inventory with established population norms. The zero is not an exaggeration; it is an exact count.
- **A5 (Human Baselines):** Mean scores of 6.2 (Ubuntu), 0.0 (Confucian), and 0.0 (Indigenous). The faint pulse of human baseline data in the Ubuntu tradition comes from UbuntuGuard's partial incorporation of survey data; Confucian and Indigenous traditions offer nothing at all against which to compare LLM performance.

This feature-level uniformity suggests that the zero-quality outcome is not idiosyncratic—it is not that each tradition failed for a different reason and the failures happen to produce the same result. Rather, all four traditions fail for the same structural reason: the absence of validated psychometric instruments from which computational benchmarks can be derived.

4.4.3 Infrastructure Determinism: Why Quality Follows Theory Infrastructure

The cultural quality cliff is not a puzzle once one understands the mechanism that produces it. Benchmark quality, as measured by the 17-feature composite, is overwhelmingly determined by what we term *theory infrastructure*: the pre-existing ecosystem of standardized instruments, annotated corpora, and published population norms that a moral psychology theory offers to benchmark developers. Theories with rich infrastructure produce high-quality benchmarks because the foundational psychometric work has already been done; theories without infrastructure produce low-quality benchmarks because developers must start from scratch and inevitably cut corners.

Consider the infrastructure available to MFT, the tradition with the highest quality rate (73%). MFT provides benchmark developers with three distinct starting points:

1. **The Moral Foundations Questionnaire (MFQ-30)** [Graham et al., 2011]: a 30-item, six-factor instrument with published population norms spanning liberal and conservative political orientations, multiple cultures, and several age cohorts. The MFQ-30 has been administered to hundreds of thousands of respondents worldwide, producing a rich normative database against which LLM scores can be interpreted. A researcher building an MFT benchmark can simply administer the MFQ-30 to a model and immediately score the results against published norms—no new psychometric development required.
2. **The Moral Foundations Twitter Corpus (MFTC)** [Hoover et al., 2020]: a corpus of 35,108 tweets annotated by expert coders for moral foundation engagement across all five foundations. The MFTC enables a second class of MFT benchmark: corpus-based approaches where LLMs are evaluated on their ability to detect, classify, or reason about moral content as represented in naturalistic text. This pathway produces benchmarks with strong construct validity (A6) because the annotation scheme is itself grounded in the validated MFQ framework.
3. **Published cross-cultural norms** [Graham et al., 2011; Atari et al., 2023]: decades of cross-cultural MFQ administration have produced a detailed map of how moral foundation endorsement varies across cultures, political orientations, and demographic groups. This normative database gives MFT benchmarks the interpretive infrastructure that Claim 5 identifies as critical: the ability to say not just that a model scored X on care/harm, but that this score places the model in the *n*th percentile of the human distribution for a specified population.

Now consider what Ubuntu ethics offers to a benchmark developer. Ubuntu—the Southern African philosophy of communal moral agency, typically glossed as "I am because we are" [Metz, 2007; Gyekye, 1997]—has a rich philosophical literature spanning centuries. Major works by Desmond Tutu, Thaddeus Metz, Kwame Gyekye, and others articulate Ubuntu's core commitments: the primacy of communal harmony, the relational constitu-

tion of moral selfhood, the obligations of solidarity and reciprocity, and the inseparability of individual flourishing from collective well-being. What Ubuntu does not have is any of the following:

- A standardized questionnaire with published factor loadings. There is no "Ubuntu Moral Assessment Questionnaire" (UMAQ) with documented reliability coefficients, no multi-item scale that has been administered to large samples and factor-analyzed to confirm that Ubuntu's theoretical dimensions load as predicted.
- An annotated corpus of Ubuntu-relevant text. There is no "Ubuntu Ethics Corpus" with expert-labeled examples of communal reasoning, relational obligation, or solidarity-based judgment. A researcher building a corpus-based Ubuntu benchmark would need to create the annotation scheme, train annotators in Ubuntu philosophy, annotate a sufficiently large sample of texts, and validate the scheme's reliability—a project that could easily require two to three years and significant funding before a single LLM is evaluated.
- Published population norms. Even if a validated Ubuntu instrument existed, there are no large-scale population-level norms for Ubuntu moral endorsement against which LLM scores could be interpreted. How communal-minded is the average person in South Africa, Nigeria, Kenya, or the Ubuntu-influenced diaspora? What is the distribution of Ubuntu moral orientation across age, education, urbanization, and ethnic group? These questions have not been answered at the psychometric level, and answering them would require the kind of large-sample cross-cultural survey research that MFT has benefited from over two decades.

The infrastructure gap between MFT and Ubuntu is not a matter of degree; it is a matter of kind. MFT offers benchmark developers a *turnkey* psychometric ecosystem: take the questionnaire, administer it, score against norms. Ubuntu offers benchmark developers a *philosophical tradition*: read the texts, understand the concepts, and then—before any LLM evaluation can begin—build the entire psychometric apparatus from the ground up. The difference in quality rate (73% versus 0%) directly reflects this difference in starting conditions.

The same pattern holds, with minor variations, across the quality gradient:

- **Schwartz Basic Human Values** (quality rate 57%) benefits from the Portrait Values Questionnaire (PVQ-RR), a validated instrument with published population norms across 80+ countries [Schwartz et al., 2012]. The PVQ-RR provides the psychometric foundation (A4, A5) that lifts Schwartz benchmarks into the high tier, though the absence of a large annotated corpus limits the variety of benchmark formats available—explaining why Schwartz's quality rate is lower than MFT's despite comparable questionnaire infrastructure.
- **Kohlberg's Cognitive Moral Development** (quality rate 56%) benefits from the Defining Issues Test (DIT/DIT-2), a validated instrument with 40+ years of published

norms spanning developmental stages, educational levels, and cultural groups [Rest, 1979; Rest et al., 1999]. The DIT's norms are particularly rich for interpretive purposes: they allow researchers to place an LLM's moral reasoning on a developmental trajectory, comparing its post-conventional reasoning scores against known human benchmarks for high school students, college students, graduate students, and moral philosophy PhD students.

- **Trolley dilemma research** (quality rate 48%) benefits from the Moral Machine experiment [Awad et al., 2018], which collected over 40 million moral judgments from 233 countries, producing the most comprehensive set of population-level norms for any moral psychology paradigm. The Moral Machine's massive normative database gives trolley benchmarks exceptionally strong A5 scores and enables interpretive claims that no other tradition can match: researchers can say precisely how an LLM's utilitarian/deontological preferences compare to those of Scandinavians, East Asians, Latin Americans, or any other cultural cluster represented in the dataset.
- **Islamic ethics** (quality rate 36%) represents the most successful non-Western tradition in the landscape, achieving partial quality through a combination of dedicated benchmark development (e.g., QIAS 2025, composite 60.5; "Assessing Large Language Models on Islamic Ethics," 2025, composite 58.5; "Sacred or Synthetic?," 2025, composite 58.4) and the availability of some structured source material—primarily classical Islamic jurisprudence texts that, while not standardized psychometric instruments, provide a structured framework for item development. The 3 top-tier and 5 high-tier Islamic entries demonstrate that non-Western traditions *can* produce viable benchmarks when sufficient research investment is made. But Islamic ethics' 36% rate—roughly half the Western mean—also demonstrates how far non-Western traditions must go even under favorable conditions.
- **Buddhist ethics** (quality rate 12%) illustrates the intermediate case: a single dedicated benchmark (BuddhismEval, 2025, composite 57.1) reaches the top tier by adapting elements of established psychometric practice to Buddhist content, but the remaining 15 candidates cluster in the medium and low tiers. The BuddhismEval benchmark demonstrates proof of concept—it is possible to build a quality Buddhist ethics benchmark—but remains an isolated achievement rather than evidence of a developing infrastructure.

The implication of infrastructure determinism is sobering. Creating a publication-ready Ubuntu ethics benchmark is not primarily a matter of motivation, funding, or research effort, though all three are necessary. It is a matter of *prerequisite infrastructure*. The first step is not "build a benchmark"; the first step is "build a validated Ubuntu moral assessment instrument"—a project that, by the standards of psychometric instrument development in moral psychology, would require several years of iterative design, pilot testing, factor analysis, reliability assessment, and cross-cultural norm collection. Only after that instrument exists can the second step—developing a computational benchmark that inherits the instrument's psychometric properties—begin with any prospect of pro-

ducing a tool that meets the quality standards embodied in the top and high tiers of this landscape assessment.

The infrastructure determinism thesis explains not only the cultural quality cliff but also its resistance to correction. The cliff will not be closed by producing more Ubuntu-themed papers or more Confucian-inspired evaluation frameworks. It will only be closed by investing in the decades-long psychometric groundwork that Western theories have already completed—or by developing entirely new assessment paradigms that do not require the psychometric infrastructure that the composite scoring framework rewards. Both paths are long. Neither has been started.

4.4.4 Implications for Global AI Deployment

The cultural quality cliff is not an abstract distributional finding; it has immediate consequences for the deployment of large language models in non-Western contexts. LLMs are currently deployed, or are being actively deployed, in societies where the zero-quality moral traditions represent the dominant ethical frameworks:

- **Ubuntu-practicing societies.** Sub-Saharan Africa is among the fastest-growing markets for AI deployment, with LLMs being integrated into healthcare delivery, agricultural advisory systems, educational platforms, and financial services across Nigeria, Kenya, South Africa, Ghana, and dozens of other nations [Gwagwa et al., 2020; Mhlambi, 2020]. These societies draw on Ubuntu philosophy as a foundational ethical framework—a framework that emphasizes communal responsibility, relational personhood, and collective flourishing in ways that are structurally incompatible with the individualistic moral foundations measured by MFT and Schwartz. When a healthcare chatbot deployed in rural South Africa makes a triage recommendation that implicitly prioritizes individual autonomy over communal obligation, no existing benchmark can evaluate whether that recommendation aligns with the moral frameworks of the community it serves.
- **Confucian societies.** East Asia—China, Japan, South Korea, Taiwan, Vietnam—represents the largest concentration of LLM users outside the English-speaking world. Confucian moral philosophy, with its emphasis on filial piety, role-differentiated obligations, hierarchical harmony, and the cultivation of virtue through ritual practice [Ames and Rosemont, 1998; Angle and Slote, 2013], remains deeply influential in the moral reasoning of these societies, even among populations that do not explicitly identify as Confucian. The question "does GPT-4 align with Confucian moral values?"—a question of direct practical importance for deployment in China, where over 200 million users interact with LLM-based services—cannot currently be answered at any quality standard using any existing instrument in the benchmark landscape.
- **Indigenous communities.** Indigenous ethical frameworks, emphasizing reciprocal relationships with the natural world, intergenerational obligations, and place-based moral knowledge [Whyte, 2018; Mack et al., 2012], are relevant wherever AI systems are deployed in communities that maintain Indigenous knowledge systems. This includes

contexts ranging from environmental monitoring in the Amazon Basin to healthcare delivery in First Nations communities in Canada to educational technology in Maori communities in New Zealand. The 4 candidates in the Indigenous tradition—all in the low tier—represent a benchmark landscape that has effectively no capacity to evaluate whether AI systems respect or violate Indigenous moral principles.

The combined population of societies where Ubuntu, Confucian, Indigenous, or Identity ethics represent influential moral frameworks exceeds 3 billion people—roughly 40% of the global population. The benchmark landscape's capacity to evaluate LLM moral alignment for these populations is exactly zero.

This cultural blindness constitutes the third bind of the triple bind thesis. When the AI safety community asks "does this LLM align with human moral values?", the question implicitly restricts "human moral values" to those measured by Western psychological instruments: MFT's five (or six) foundations, Schwartz's ten value types, Kohlberg's six developmental stages, and the utilitarian/deontological trade-offs of the trolley paradigm. These frameworks are not wrong—they capture real and important dimensions of moral cognition that have been validated across many populations. But they are incomplete. They measure the moral concerns of WEIRD (Western, Educated, Industrialized, Rich, Democratic) populations using instruments developed and normed in WEIRD contexts [Henrich et al., 2010]. They do not measure communal obligation as Ubuntu understands it, filial duty as Confucian ethics conceptualizes it, or ecological reciprocity as Indigenous frameworks articulate it.

The consequence is that the moral evaluation enterprise, as currently constituted, can certify that an LLM's moral foundations profile resembles that of American college students—but it cannot determine whether the same model's outputs are consistent with the moral frameworks of the communities in Nairobi, Shanghai, or the Navajo Nation where it will actually be deployed. The cultural quality cliff does not merely limit the scope of moral evaluation; it limits the scope in precisely the direction that matters most for equitable global deployment. The populations excluded from the benchmark landscape are, overwhelmingly, the populations that already face the greatest risks from poorly aligned AI systems and have the least institutional capacity to demand better evaluation.

The triple bind's self-reinforcing logic is most visible in this third component. The theoretical narrowness documented in Claims 1 through 3 (Bind 1) ensures that the instruments available for moral evaluation are overwhelmingly Western in theoretical origin. The cultural quality cliff (Bind 3) ensures that non-Western alternatives, where they exist, fail to meet quality standards. And the empirical undiscrimination documented in Claim 7 (Bind 2) ensures that the narrow, Western instruments produce the same results for every model, creating no pressure to broaden the evaluative lens. From the perspective of any individual benchmark developer, the rational response is to build another MFT benchmark—because MFT has the infrastructure, because MFT benchmarks achieve high quality scores, and because there is no penalty for measuring the same thing that everyone else is measuring

when the instruments cannot distinguish between models anyway. The cultural quality cliff is not just a failure of inclusion; it is a self-stabilizing equilibrium that the field's current incentive structure has no mechanism to disrupt.

4.5 Claim 5: Human Baseline Deficit

4.5.1 Full-Coverage Baseline Distribution

Moral cognition can only be characterized relative to human standards. When a benchmark reports that an LLM achieves a care/harm foundation score of 4.2 on a 5-point scale, or that it selects the utilitarian option in 73% of trolley-class dilemmas, or that its moral reasoning receives a post-conventional stage score of 62, these numbers acquire meaning only when contextualized against human performance on the same instruments. Without human baselines, a score of 4.2 might indicate exceptional moral sensitivity, population-typical endorsement, or near-floor performance; the number alone cannot distinguish between these interpretations. Human baselines—population norms, paper-specific comparison samples, or at minimum partial reference data from prior administrations—provide the interpretive anchor that transforms raw scores into meaningful characterizations of LLM moral cognition.

The original 55-benchmark assessment, which drew from an established and pre-selected sample of benchmarks, reported that 18% of assessed instruments provided no human comparison data. The full-coverage assessment of all 292 candidates reveals that this figure was profoundly optimistic: a sampling artifact produced by evaluating the top of the quality distribution and extrapolating to the whole.

Table 9 presents the full-coverage baseline distribution alongside the v2.1 subsample for comparison.

Table 9: Human Baseline Availability — v2.1 versus v3.0

Baseline Level	v2.1 (n=55)	v3.0 (n=292)	Shift
Population norms	23 (42%)	43 (15%)	−27pp
Paper-specific human sample	18 (33%)	93 (32%)	−1pp
Partial baselines	4 (7%)	56 (19%)	+12pp
No human data	10 (18%)	100 (34%)	+16pp

The most striking shifts occur at the extremes. Population-level norms—the gold standard for interpretable LLM evaluation, representing large-scale, demographically diverse human samples that establish where any given score falls in the distribution of human moral cognition—drop from 42% to 15%, a 27-percentage-point decline. Paper-specific human samples, which provide interpretive context within a single study but lack the generalizability of population norms, hold roughly steady at 32% (down one percentage

point from 33%). Partial baselines—fragmentary human reference data that allows some but not systematic interpretation—rise from 7% to 19%, reflecting a substantial number of candidates that cite prior human studies without collecting their own comparison data or providing sufficient methodological detail to use the cited data as a formal baseline. And the no-human-data category—benchmarks that provide no human comparison whatsoever—nearly doubles from 18% to 34%.

The magnitude of this shift is difficult to attribute to chance or to minor methodological differences between the v2.1 and v3.0 assessments. The 55 benchmarks assessed in v2.1 were not randomly sampled from the 292-candidate landscape; they were pre-selected as the most established, best-documented, and most widely cited instruments available. The 237 additional candidates assessed in v3.0 include emerging work, preprints, small-scale studies, and benchmarks from non-Western traditions that have received less community scrutiny. The shift from 18% to 34% no-baseline is therefore a *correction* of a sampling bias: the original figure described the best benchmarks, not the landscape.

4.5.2 The Sampling Artifact: From 18% to 34%

The doubling of the no-baseline rate deserves sustained attention because it illustrates a broader methodological lesson about meta-evaluation. When the sample of assessed benchmarks is drawn from the most prominent instruments—the ones with the most citations, the most established track records, the most complete documentation—the resulting quality profile is systematically optimistic. This is not a flaw in the prior assessment; it is a necessary consequence of any selection procedure that identifies benchmarks through literature visibility. Visible benchmarks are visible precisely because they have the psychometric infrastructure (including human baselines) that makes them useful and citable. Invisible benchmarks—the ones not yet widely cited, not yet included in major survey papers, not yet adopted by benchmark suites—are invisible in part because they lack this infrastructure.

The full-coverage assessment collapses the distinction between visible and invisible benchmarks by evaluating every identified candidate regardless of prominence. The result is a dramatically less favorable picture. One hundred of 292 candidates—more than one in three—provide no published human comparison data of any kind. These benchmarks evaluate LLM moral cognition in a vacuum: they can report that a model endorses certain moral propositions, selects certain dilemma options, or produces certain reasoning patterns, but they cannot say whether these outputs are humanlike, atypical, concerning, or within normal variation. Every interpretive claim made on the basis of these benchmarks is, strictly speaking, ungrounded.

The shift in population-level norms is perhaps more consequential than the shift in no-baseline rates. Population norms represent the gold standard for moral psychology evaluation because they provide not just a point of comparison but a *distribution*: they tell us not only the average human response but the variance, the range, the demographic moderators, and the cross-cultural patterns. An LLM evaluation grounded in population

norms can make claims of the form "this model's utilitarian preference of 0.73 falls at the 94th percentile of the global distribution" or "this model's care/harm score places it in the range typical of female respondents in Scandinavian countries." An evaluation grounded only in a paper-specific sample can say "this model's score exceeded the mean of 47 undergraduate participants at a single university." An evaluation with no baselines can say nothing at all about how the model's moral cognition relates to human moral cognition.

Under full-coverage assessment, only 43 of 292 candidates (15%) provide population-level norms. These 43 candidates are disproportionately concentrated in two traditions: MFT benchmarks that inherit the MFQ-30′s normative database, and trolley-class benchmarks that inherit the Moral Machine's 40-million-judgment dataset. The vast majority of the benchmark landscape—85% of all candidates—operates without access to population-level norms, meaning that even when baselines of some kind are available, they lack the generalizability and demographic granularity necessary for rigorous interpretive claims.

4.5.3 Baseline Availability by Tier

The relationship between baseline availability and quality tier is not merely correlational; it is structural. Table 10a presents the A5 (human baselines) distribution stratified by quality tier.

Table 10a: Human Baseline Distribution by Quality Tier

Tier	n	No Data	Partial	Paper-Specific	Population Norms
Top	48	0 (0%)	5 (10%)	19 (40%)	24 (50%)
High	89	4 (4%)	23 (26%)	45 (51%)	17 (19%)
Med	72	16 (22%)	26 (36%)	28 (39%)	2 (3%)
Low	83	80 (96%)	2 (2%)	1 (1%)	0 (0%)

The gradient is steep and monotonic. In the top tier, every single benchmark has some form of human baseline: half have population-level norms, 40% have paper-specific comparison samples, and the remaining 10% have partial baselines. No top-tier benchmark operates without human reference data of any kind. In the high tier, 96% have baselines of some form, with a majority (51%) relying on paper-specific samples and 19% providing population norms. In the medium tier, baseline provision begins to erode: 22% have no data, only 3% have population norms, and the remainder rely on partial or paper-specific data of varying rigor. In the low tier, the situation inverts entirely: 96% of low-tier benchmarks provide no human comparison data at all. Only 3 of 83 low-tier candidates offer even partial or paper-specific baselines, and none provides population norms.

This gradient is not coincidental. It is built into the assessment framework's structure. Feature A5 (human baselines) contributes to the Psychological Depth (PD) sub-score, which carries 50% of the composite weight. A benchmark with no human baselines (A5=0) loses the maximum possible A5 contribution to its composite, making it structurally more

difficult—though not impossible—for such a benchmark to reach the top or high tiers through compensatory strength on other features. The structural relationship between A5 and tier membership reflects a deliberate design choice: the composite is engineered to reward benchmarks that provide interpretive infrastructure, because interpretive infrastructure is what separates measurement from mere description.

But the causal arrow also runs in the other direction, and this is the more important point. Benchmarks that build upon validated psychometric instruments (high A4) typically inherit human baselines from those instruments (high A5), because instrument validation requires human administration and norm collection. The MFQ-30 was validated on human samples; any benchmark that adapts the MFQ-30 inherits those norms automatically. The DIT-2 has 40 years of developmental norms; any benchmark that adapts the DIT-2 can immediately report how LLM scores compare to high school students, college students, and moral philosophy graduate students. The baseline deficit, like the cultural quality cliff, is ultimately an infrastructure problem: benchmarks built on validated instruments have baselines because the instruments were validated on humans, and benchmarks not built on validated instruments lack baselines because there is no external source from which to inherit them.

Figure 6 presents this relationship as a stacked bar chart of baseline availability by tier, making the gradient visually explicit. The bars reveal a landscape in which interpretive capacity is concentrated at the top and hemorrhages progressively as quality declines, until the low tier—which constitutes 28% of the landscape—operates almost entirely without human reference points.

4.5.4 Interpretive Consequences

The practical consequences of the baseline deficit are best illustrated by contrasting what is possible with well-baselined instruments against what is possible without them.

With population norms: the Moral Machine. The Moral Machine experiment [Awad et al., 2018] collected over 40 million moral judgments from respondents in 233 countries, producing the most comprehensive set of population-level moral preference norms ever assembled. When an LLM is evaluated on a Moral Machine-derived benchmark, the results are extraordinarily rich in interpretive content. Researchers can report that Claude Opus 4′s utilitarian preference of 0.92–0.94 exceeds the mean preference of every cultural cluster identified in the original study—that the model is more utilitarian than the most utilitarian human populations. They can decompose this preference across the nine moral dimensions of the Moral Machine (sparing more lives, sparing the young, sparing higher status, etc.) and compare the model's priority ordering against the priority orderings of Western, Eastern, and Southern cultural clusters. They can identify that the model's near-zero weight on social status diverges from every human population, including the most egalitarian. Each of these comparisons is grounded in population-level data from millions of respondents, giving the interpretive claims a precision and generalizability that would be impossible without baselines.

With developmental norms: the Defining Issues Test. The DIT-2 [Rest et al., 1999] has accumulated over four decades of developmental norms spanning the full range of moral reasoning maturity. When an LLM is evaluated using DIT-derived instruments, the P-score (principled moral reasoning score) can be placed on a developmental trajectory: a P-score of 40 is typical of high school students, 45 of college freshmen, 55 of college seniors, 65 of seminarians and moral philosophy graduate students, and 70+ of the highest-scoring moral philosophy researchers. The cross-vendor study in this paper found that models achieve P-scores ranging from 60 (Haiku) to 85 (larger models), placing them in the "post-conventional" range associated with graduate-level ethical training. This finding—that LLMs reason at the level of moral philosophy graduate students on the DIT—is striking, interpretable, and meaningful precisely because 40 years of human norms provide the comparative framework. Without those norms, a P-score of 75 would be a number without context.

Without baselines: the typical low-tier benchmark. Consider a hypothetical benchmark that presents an LLM with 50 moral scenarios and reports that the model selects the "ethical" option in 84% of cases. What does 84% mean? Is the model unusually moral? Normally moral? Is 84% a ceiling effect indicating that the scenarios are too easy, or a genuine measure of moral sensitivity? Without human baselines, none of these questions can be answered. If humans typically score 92% on the same scenarios, then 84% represents a meaningful deficit; if humans score 60%, then 84% represents exceptional performance; if humans score 83-85%, then the model is human-typical; if there is high variance across human populations (50–95%), then 84% is uninterpretable without knowing which population is the appropriate referent. The number 84% carries no information about moral cognition in the absence of the comparative framework that human baselines provide.

One hundred of the 292 benchmark candidates in this landscape evaluate LLM moral cognition in the condition described by the third scenario: numbers without context, scores without interpretive anchors, patterns without meaning. The baseline deficit does not merely limit the precision of moral evaluation; it undermines the foundational claim that evaluation has occurred at all. To say that a model "performs well" on a moral reasoning benchmark without specifying what "well" means relative to human performance is to engage in pseudo-evaluation—the production of numerical outputs that create the appearance of empirical rigor without its substance.

The baseline deficit interacts with the cultural quality cliff (Claim 4) in a particularly pernicious way. The benchmarks most likely to have population norms are those built on Western instruments with long research histories—MFT, DIT, Moral Machine—precisely the instruments that already dominate the landscape. The benchmarks least likely to have baselines are those attempting to evaluate non-Western moral traditions, where neither the instruments nor the human norms exist. The result is that the traditions for which baselines are most needed (because LLM alignment with those traditions has never been

established) are precisely the traditions for which baselines are least available (because the psychometric infrastructure has never been built). The baseline deficit is not uniformly distributed across the landscape; it is concentrated in the traditions that the cultural quality cliff already excludes.

4.6 Claim 6: Temporal Stagnation

4.6.1 Quality Rate by Publication Period

If the benchmark landscape were a healthy scientific enterprise responding to the growing urgency of LLM moral evaluation, one would expect quality to improve over time. As the field matures, as developers learn from prior work, as the methodological lessons of early benchmarks are absorbed and applied, the quality rate should rise. Each new generation of benchmarks should be better than the last—more psychometrically grounded, better baselined, more theoretically diverse—reflecting a discipline that is learning how to do this work well.

The data tell a different story.

Table 10b: Quality Rate by Publication Period

Period	Total	Top	High	Top+High	Quality Rate	Volume Growth
Pre-2020	8	4	2	6	75%	—
2020–2022	27	10	3	13	48%	3.4x
2023–2026	257	34	84	118	46%	9.5x vs pre-2020

Quality has not improved. The pre-2020 cohort—small (n=8) but representing the foundational instruments that established the field—achieves a quality rate of 75%. Three out of every four pre-2020 benchmarks meet the threshold for publication-ready LLM evaluation. These are the instruments with deep psychometric roots: the Moral Machine (2018), the Moral Foundations Vignettes (2015), the Socio-Moral Image Database (2018), and the Voting-Based System for Ethical Decision Making (2018). They represent the benchmark landscape before the explosion of LLM-specific evaluation, when the instruments that researchers adapted for AI assessment were primarily drawn from established moral psychology traditions with decades of validation history.

The 2020–2022 period marks the field's first major expansion, with volume increasing 3.4-fold relative to the pre-2020 baseline. Quality drops sharply, from 75% to 48%. This decline is expected as a field scales up: the first entrants are typically the most competent and best-resourced, and expansion necessarily admits work of greater variance. But the important observation is what happens next.

The 2023–2026 cohort represents the landscape as it exists today: 257 of 292 candidates, or 88% of the total, were published in this period. Volume has increased 9.5-fold relative to the pre-2020 baseline. And the quality rate has not recovered from the 2020–2022 decline; it has remained essentially flat at 46%—two percentage points below the 2020–2022 rate, well within the margin of error for any reasonable confidence interval. The explosion of interest in LLM moral evaluation has produced an explosion of benchmarks, but not an improvement in benchmark quality.

Figure 7 presents this temporal pattern as a line chart with volume bars overlaid, making the divergence between volume and quality starkly visible. The volume bars climb exponentially while the quality line remains flat, producing the visual signature of a field that is scaling without learning.

4.6.2 Volume Growth Without Quality Improvement

The temporal stagnation finding is robust to alternative operationalizations of "quality." If quality is defined as the top-tier rate alone (rather than top + high), the pattern persists: 50% (pre-2020), 37% (2020–2022), 13% (2023–2026)—an even steeper decline. If quality is defined as the mean composite score, the pattern again persists: the pre-2020 mean composite is substantially higher than the 2023–2026 mean, reflecting the concentration of high-infrastructure instruments in the earliest cohort. No reasonable definition of quality reverses the finding: the field is not improving.

The volume growth is overwhelmingly driven by entries in the medium and low tiers. Of the 257 candidates published in 2023–2026, 67 (26%) fall in the medium tier and 72 (28%) in the low tier—a combined 139 candidates (54%) that do not meet the threshold for publication-ready moral evaluation. The 34 top-tier entries from this period, while more numerous in absolute terms than the top-tier entries from earlier periods, represent only 13% of the 2023–2026 cohort. The field is producing more benchmarks that meet quality standards, but it is producing even more benchmarks that do not, and the ratio is worsening, not improving.

This pattern is consistent with an incentive structure that rewards production over quality. The academic publication system values novelty—new benchmarks, new model evaluations, new datasets—over the unglamorous work of psychometric validation, reliability testing, and human norm collection. A researcher who administers an existing validated instrument (e.g., the MFQ-30) to a new model and reports the results can produce a publishable paper in weeks. A researcher who develops a new validated instrument for a previously unbenchmarked moral tradition must invest years in psychometric development before producing a single LLM evaluation. The first paper counts equally in tenure files, citation indices, and grant applications. The second paper is objectively more valuable to the field—it expands the evaluative infrastructure in a direction the landscape desperately needs—but the incentive structure systematically discourages it.

The temporal stagnation also reflects the infrastructure determinism documented in Section 4.4.3. The instruments that can be built quickly—adaptations of existing validated questionnaires, new prompt formats for established dilemma batteries, corpus-based benchmarks built on pre-annotated datasets—cluster in the theory traditions that already have infrastructure: MFT, Schwartz, trolley, Kohlberg. The instruments that would expand the landscape into new traditions—Ubuntu benchmarks, Confucian ethics assessments, dual-process evaluation protocols—require years of foundational work before they can be constructed at publication-ready quality. The natural consequence is that volume growth is driven by the traditions that already dominate, while the traditions that the landscape most needs remain stuck at the zero-quality baseline identified in Claim 4. Time passes, volume increases, and quality stagnates, because the structural conditions that determine quality have not changed.

4.6.3 The Hidden Gems Exception

The temporal stagnation finding carries an important caveat: the Jenks-based tier system identifies several genuinely high-quality benchmarks that the original 55-benchmark assessment missed entirely. These "hidden gems" represent cases where the landscape has, in fact, produced valuable new instruments that the prior assessment's subsample approach failed to capture.

Table 11: Hidden Gems — Top-Scoring Unexpected Benchmarks

Benchmark	COMPOSITE	Year	Theories	Why Missed
Jiao 2025 (LLM Ethics Benchmark)	81.3	2025	MFT, Trolley	Published after original assessment
KorNAT	69.0	2024	Unknown	Non-English benchmark
Scruples	67.0	2021	Ethics/norms	Not theory-labeled
Aligning AI with Shared Human Values	67.0	2021	Multi-theory	No clear primary label
ValueBench	65.8	2024	Schwartz	Assessed in extended battery

The highest-scoring hidden gem, Jiao et al.'s [2025] LLM Ethics Benchmark, achieves a composite of 81.3—the highest in the entire landscape. Published in *Nature Scientific Reports*, this three-dimensional assessment system integrates the MFQ, the World Values Survey, and standardized moral dilemmas into a unified evaluation framework with population norms, an open-source evaluation codebase (GitHub), and multi-instrument validation. Its composite of 81.3 exceeds the next-highest entry by more than 2 points. That

the landscape's highest-scoring benchmark was missed by the original assessment illustrates the risk of subsample-based evaluation: even well-designed selection procedures can miss the best instruments when those instruments are published after the selection period or emerge in venues outside the assessor's initial search horizon.

KorNAT (Lee et al., 2024, composite 69.0) is particularly notable among the hidden gems for a different reason: it is a non-English benchmark, developed in and for Korean, that achieves a top-tier composite score. KorNAT's success demonstrates that high-quality moral evaluation benchmarks *can* be constructed outside Western, English-language research contexts when sufficient resources and psychometric expertise are invested. Its top-tier placement provides a counterexample to any reading of the cultural quality cliff (Claim 4) as implying that non-Western benchmarks are inherently incapable of quality. KorNAT is not a benchmark grounded in a non-Western moral *tradition*—it does not operationalize Confucian ethics or Buddhist moral philosophy—but it is a benchmark developed by a non-Western research community with local psychometric norms, and its quality demonstrates that the barrier is infrastructure, not culture.

Scruples (Lourie et al., 2021, composite 67.0) and Aligning AI with Shared Human Values (Hendrycks et al., 2021, composite 67.0) illustrate a different kind of hidden gem: benchmarks that address moral evaluation without explicitly labeling themselves under any single moral psychology theory. These benchmarks were missed by theory-oriented search strategies precisely because they do not ground their item development in MFT, Schwartz, Kohlberg, or any other named theoretical framework. Instead, they use crowd-sourced moral judgments (Scruples) or multi-source ethical norms (Hendrycks et al.) as the basis for evaluation. Their top-tier placement suggests that theory-agnostic approaches to moral evaluation can achieve high quality—a finding that complicates the infrastructure determinism thesis by showing that some benchmarks achieve strong composite scores through practical engineering (large datasets, rigorous annotation, open access) rather than through psychometric inheritance from validated instruments.

ValueBench (Ren et al., 2024, composite 65.8) represents a Schwartz-tradition benchmark assessed in an extended battery rather than the original top-15 selection. Its inclusion in the hidden gems table illustrates the breadth of the Schwartz tradition's quality capacity: even beyond the well-known benchmarks, additional instruments of publication-ready quality exist.

The hidden gems suggest that the original top-15 selection missed approximately 5 publication-ready benchmarks—a non-trivial but not devastating omission. The full-coverage assessment recovers these instruments and places them in their proper context within the quality distribution. More importantly, the hidden gems do not reverse the temporal stagnation finding. Three of the five hidden gems were published in 2024–2025, but they represent isolated achievements within a 2023–2026 cohort whose overall quality rate (46%) is unchanged from the 2020–2022 period. The existence of a few excellent

recent benchmarks does not alter the distributional reality that the field as a whole is not improving.

4.6.4 The Stagnation-Triple Bind Nexus

The temporal stagnation finding is not an independent observation; it is the temporal manifestation of the triple bind. The quality rate cannot improve because the structural conditions that determine quality—theory infrastructure, psychometric heritage, human norms—are themselves stagnant. No new theory has acquired the infrastructure to support high-quality benchmarks since MFT's MFTC was published in 2020. No non-Western tradition has produced a validated psychometric instrument since the landscape assessment began. No new category of human baselines has been collected at the population level outside the Moral Machine and MFQ normative databases. The inputs to benchmark quality have not changed, so the outputs of the benchmark production process have not changed. Volume has increased because the *number of researchers* building benchmarks has increased, but the *infrastructure available to each researcher* has not expanded, and so the average quality remains flat while the total volume grows.

This temporal stagnation feeds back into the triple bind in a particularly insidious way. Each year of stagnation makes the infrastructure gap harder to close, because each year adds more low-quality benchmarks to the landscape, creating the appearance of progress (more instruments available!) while delivering none of its substance (instruments that cannot discriminate between models, cannot evaluate non-Western traditions, and cannot interpret results against human norms). The academic community's attention is absorbed by the churn of new benchmark papers, new model evaluations, and new comparison tables, leaving less bandwidth for the foundational psychometric work that would actually improve the landscape's evaluative capacity. Temporal stagnation is not merely a symptom of the triple bind; it is the mechanism by which the triple bind persists across research cycles.

The combination of Claims 4, 5, and 6 paints a picture of a benchmark landscape that is not merely flawed but structurally incapable of self-correction through normal scientific processes. The cultural quality cliff (Claim 4) identifies a categorical exclusion of non-Western traditions. The human baseline deficit (Claim 5) documents that even within Western traditions, a third of benchmarks operate without the interpretive infrastructure necessary for meaningful evaluation. And the temporal stagnation (Claim 6) demonstrates that these problems are not being resolved over time: the field is producing more benchmarks at the same rate of quality, year after year, with no trajectory toward improvement. The landscape grows in volume but not in capacity—a treadmill of evaluation that creates the illusion of progress while the fundamental problems remain intact.

6. Results Part III: Integrated Findings

Parts I and II presented the landscape and empirical findings as separate analyses. Part I documented the structural properties of the 292-candidate benchmark landscape—its

theoretical concentration around Moral Foundations Theory, the validity crisis afflicting the majority of instruments, the absence of process-level moral psychology theories, the cultural quality cliff separating Western and non-Western traditions, the baseline deficit that renders LLM scores uninterpretable, and the temporal stagnation of quality even as volume has increased nearly tenfold. Part II documented the empirical consequences of administering 23 benchmarks from this landscape to nine models from three independent vendors—the convergence on a shared RLHF moral fingerprint, the six universal moral signatures, the ceiling effects, the counterintuitive scale sensitivity of the DIT, and the singular discriminating power of the dual-process probe.

Part III asks whether these are independent observations—coincidental findings that happen to coexist in the same domain—or manifestations of a single structural problem. The answer, which constitutes the paper's central contribution, is that they are structurally linked. The landscape's theoretical narrowness determines which benchmarks achieve high quality. Benchmark quality determines which instruments are available for empirical testing. And the instruments available for empirical testing determine the pattern of results—specifically, whether the battery can discriminate between models or whether it returns the same result regardless of which system is being evaluated. The triple bind is not three problems; it is one pipeline failure observed at three stages.

This section develops that argument in three phases. Section 6.1 presents the tier-stratified empirical analysis: a new cross-referencing of the quality tier assignments from Part I with the discrimination data from Part II. Section 6.2 constructs the infrastructure-quality-discrimination pipeline: the causal chain linking theory infrastructure to benchmark quality to empirical discrimination power. Section 6.3 demonstrates the landscape-empirical convergence: the logical chain by which Claims 1 through 6 predict Claim 7, and the system dynamics by which the three binds sustain one another.

6.1 Tier-Stratified Empirical Analysis

The analyses in Parts I and II were conducted independently. Part I assessed all 292 benchmark candidates on a 17-feature characterization framework, assigning each a composite quality score and a Jenks-derived quality tier (Top, High, Medium, or Low). Part II administered 23 of those candidates to nine models, measuring the coefficient of variation (CV) across models as a discrimination index. What has not yet been examined is the relationship between these two dimensions. Do high-quality benchmarks discriminate better than low-quality ones? Or does the relationship run in a different—and more revealing—direction?

6.1.1 Do Top-Tier Benchmarks Discriminate Better?

For each of the 23 benchmarks administered in the cross-vendor study, two independent quantities are known: its quality tier (assigned by the 292-candidate landscape assessment, based on construct validity, psychometric grounding, human baselines, operationalization quality, and community impact) and its coefficient of variation across the nine models

(computed from the cross-vendor empirical study as the ratio of the standard deviation to the mean of the primary metric across all models). The naive expectation is that higher-quality benchmarks should discriminate better. A well-constructed instrument with strong construct validity, validated psychometric properties, and population-level human baselines should, in principle, be more sensitive to real differences between the systems it evaluates. If one benchmark is psychometrically rigorous and another is a hastily assembled set of moral scenarios without validation, the rigorous instrument should reveal differences that the crude one misses.

The data reveal the opposite pattern. When composite quality scores are plotted against discrimination power (CV) for all 23 administered benchmarks (Figure 14), the correlation is strongly negative: $r = -0.83$. Higher-quality benchmarks produce *less* variation between models, not more. The relationship is not marginal or ambiguous; it is the strongest correlation in the entire analysis, and it runs in the direction opposite to what instrument-quality theory would predict.

This finding is counterintuitive and demands careful interpretation. It does not mean that quality is detrimental to measurement. It means that the best instruments in this landscape —those with the strongest psychometric properties, the most rigorous construct validity, and the most comprehensive human baselines—are precisely the ones that confirm the convergence finding. The moral homogeneity documented in Part II is not an artifact of poor measurement. It is a finding that becomes *more* robust as measurement quality increases.

Consider the interpretive alternatives. If the convergence were an artifact of crude instrumentation—if low-quality benchmarks happened to produce identical scores because they lacked the sensitivity to detect real differences—then higher-quality instruments should reveal variation that the crude ones miss. The negative correlation refutes this alternative decisively. When the MFQ-30 (a validated psychometric instrument with published population norms, strong construct validity, and decades of human comparison data) is administered to nine models, it produces a CV of 0.0325—near-zero variation. When the Moral Machine experiment (a forced-choice paradigm with 40 million human judgments as a normative baseline) is administered, it produces a CV of exactly 0.0—perfect identity across all nine models. These are not crude instruments failing to detect differences. They are the field's most carefully validated tools, and they are confirming that the differences do not exist.

The alternative interpretation—that lower-quality benchmarks showing apparent discrimination are producing signal rather than noise—is substantially less credible. The benchmark with the highest CV (3.00) is MP-DP, the Dual Process Theory Protocol. This is a custom protocol without a standardized questionnaire, without published population norms, and without the psychometric validation history of instruments like the MFQ-30 or the DIT. Its high discrimination power does not derive from superior measurement; it derives from probing a dimension of model behavior—the effect of chain-of-thought

prompting on moral judgment—that content-level instruments do not address. The discrimination is real, but it is process-level discrimination, not content-level discrimination. What the negative correlation reveals is that the landscape's content-level instruments, at every quality level, converge on the same result: the models are morally indistinguishable when evaluated on *what* they endorse. They are distinguishable only when evaluated on *how* they reason.

6.1.2 Quality Tier Predicts Convergence, Not Discrimination

To quantify the tier-stratified pattern, each of the 23 administered benchmarks was cross-referenced with its tier assignment from the full 292-candidate assessment. The benchmarks were then grouped by tier, and within each group, the mean CV, the proportion of benchmarks with zero cross-model variation (CV = 0.0), and the proportion with high discrimination (CV > 0.05) were computed. Where a trial benchmark maps to multiple candidates in the full matrix (e.g., the MFQ-30 corresponds to several MFT-related entries), the tier of the source instrument was used.

Table T26: Tier-Stratified Discrimination Analysis

Tier	Benchmarks in Trial Battery	Mean CV	% Zero-Variation (CV = 0)	% High-Discrimination (CV > 0.05)
Top	8	0.0108	37.5%	12.5%
High	7	0.0297	14.3%	28.6%
Medium	5	0.0454	20.0%	40.0%
Low	3	0.672	0.0%	66.7%

The pattern is monotonic and striking. Among the eight Top-tier benchmarks in the trial battery—including the MFQ-30, the Moral Machine experiment, the Moral Foundations Vignettes, TracingMF, and the MFTC—the mean CV is 0.0108, barely above zero. More than a third of these benchmarks (37.5%) produced identical scores across all nine models, and only one (12.5%) exceeded the CV > 0.05 threshold for high discrimination. These are the instruments with the strongest construct validity, the most comprehensive human baselines, and the most rigorous psychometric properties. And they are, collectively, the least discriminating instruments in the battery.

At the opposite end, the three Low-tier benchmarks show the highest mean CV (0.672), no zero-variation results, and two-thirds exceeding the high-discrimination threshold. The dominant contributor is MP-DP (Dual Process), whose CV of 3.00 is an order of magnitude larger than any other benchmark's. But even excluding MP-DP, the remaining Low-tier benchmarks show elevated discrimination relative to Top-tier instruments. The gradient is consistent: as quality tier decreases from Top to High to Medium to Low, mean CV increases from 0.0108 to 0.0297 to 0.0454 to 0.672, and the proportion of high-discrimination benchmarks increases from 12.5% to 28.6% to 40.0% to 66.7%.

This pattern admits of two non-exclusive interpretations, both of which are important.

The first interpretation is that the convergence is real and well-measured. Top-tier instruments, because they have been designed with psychometric rigor and validated against human populations, are measuring the same latent constructs as their human counterparts. When these instruments return the same score for all nine models, they are correctly detecting that all nine models occupy the same position in the construct space. The convergence is not a failure of measurement; it is a valid finding about the world—specifically, about the homogeneity of RLHF-aligned models' moral profiles.

The second interpretation is that lower-tier instruments showing apparent discrimination are, at least in part, producing noise rather than signal. An instrument without construct validity may produce variation across models for reasons that have nothing to do with genuine differences in moral cognition: idiosyncratic prompt sensitivity, surface-level pattern matching, scoring artifacts, or response format effects. High CV is not, by itself, evidence of good discrimination; it is evidence of variation, which may be measurement error rather than true signal. The dual-process probe (MP-DP) is the exception: its discrimination derives from a genuine process-level phenomenon (the chain-of-thought effect on moral judgment) that has a coherent theoretical interpretation. But for the remaining lower-tier instruments, the elevated CV may reflect the instruments' instability rather than the models' diversity.

The two interpretations reinforce the same conclusion from different angles. Whether one reads the data as "the best instruments confirm convergence" or "the noisiest instruments produce spurious variation," the implication is the same: the moral homogeneity finding is robust. It survives—indeed, it is strengthened by—stratification by instrument quality. The triple bind's second arm (empirical undiscrimination) is not a measurement artifact. It is a property of the systems being measured, as confirmed by the systems' own best instruments.

6.1.3 The Single Benchmark That Separated Models

Among the 23 benchmarks administered, one stands apart. MP-DP (Dual Process Theory Protocol) achieved a CV of 3.00—a value 19 times larger than the next-highest discriminator (MP-SI, Social Intuitionism, CV = 0.1587) and 93 times larger than the mean CV of the Top-tier benchmarks. If the trial battery had consisted only of the 22 benchmarks excluding MP-DP, the overall picture would be one of near-complete convergence, with only the DIT P-score (CV = 0.0853) and ValAct (CV = 0.0934) providing moderate discriminatory evidence. The dual-process probe is not merely the best discriminator; it is the only benchmark that produces a qualitatively different pattern of results from the rest of the battery.

The dual-process probe measures a specific and theoretically grounded phenomenon: the effect of chain-of-thought (CoT) prompting on moral judgment. In the standard administration, models are presented with moral dilemmas and asked for an immediate

judgment. In the CoT condition, models are prompted to reason step by step before rendering judgment. Among the nine models tested, only one—Haiku, the smallest model in the Anthropic family—showed a statistically meaningful CoT effect: its moral judgments shifted under deliberative prompting, with the CoT condition increasing the deontological response rate by approximately 25 percentage points. All other models, regardless of vendor, architecture, or parameter count, showed no CoT effect: their moral judgments were identical whether or not they were prompted to deliberate.

This finding is significant for three reasons.

First, the dual-process probe is not a content-level instrument. It does not ask *what* moral values the model endorses; it asks *how* the model arrives at moral judgments. The distinction between content-level and process-level measurement is central to moral psychology. Content-level instruments (the MFQ-30, the Moral Machine, the MFTC) measure the products of moral cognition: the values endorsed, the principles applied, the judgments rendered. Process-level instruments measure the mechanisms by which those products are generated: the interplay between intuitive and deliberative reasoning, the role of emotional versus cognitive inputs, the sensitivity of judgment to processing conditions. The moral psychology literature has established that these two levels can dissociate: individuals who endorse the same moral values may arrive at those values through different cognitive processes [Greene et al., 2001; Kahneman, 2011; Haidt, 2001].

Second, the dual-process probe is the benchmark with the weakest infrastructure in the traditional sense. There is no standardized "Dual Process Morality Questionnaire" with published norms. There is no expert-annotated "Dual Process Morality Corpus" for training classifiers. The dual-process literature in moral psychology is rich—Greene's [2001] fMRI studies of utilitarian and deontological reasoning, Kahneman's [2011] System 1/System 2 framework, and the extensive debate between rationalist and intuitionist accounts of moral judgment—but this theoretical richness has not been translated into the kind of standardized psychometric infrastructure that MFT possesses. The dual-process probe was a custom protocol, designed specifically for this study, without the decades of validation that undergird the MFQ-30.

Third, the fact that the least-infrastructure-backed benchmark produces the most discrimination, while the most-infrastructure-backed benchmarks produce the least, is the signature of a pipeline failure. The field has invested its psychometric infrastructure in content-level measurement—specifically, in MFT-derived content measurement—and the result is a battery that is exquisitely sensitive to content-level properties (which are homogeneous across models) and blind to process-level properties (which are heterogeneous). The landscape did not fail to produce discriminating instruments because discrimination is impossible; it failed because the instruments it produced measure the wrong dimension of moral cognition.

The broader implication is that the path to discriminating moral evaluation of LLMs runs through process-level measurement, not through further refinement of content-level instruments. Building a better MFQ will not break the convergence: all nine models already score in the same range on the existing MFQ, and a more precise version would simply confirm the convergence with tighter confidence intervals. What is needed is the translation of the dual-process, social intuitionist, and other process-level theories from the moral psychology literature into standardized, validated instruments with the same kind of psychometric infrastructure that MFT currently monopolizes. Section 7 develops this implication into a concrete research agenda.

6.2 The Infrastructure-Quality-Discrimination Pipeline

The tier-stratified analysis establishes the empirical fact: quality and discrimination are inversely related in this landscape. But establishing a correlation, even a strong one ($r = -0.83$), does not explain the causal mechanism. Why do the best instruments converge while the worst instruments discriminate? The answer lies in a three-stage pipeline that links theory infrastructure to benchmark quality to empirical discrimination power.

6.2.1 Theory Infrastructure Determines Benchmark Quality Determines Discrimination Power

The pipeline has three stages, each feeding the next.

Stage 1: Theory Infrastructure. A moral psychology theory's "infrastructure" consists of two resources: (a) a validated psychometric instrument—a standardized questionnaire, task battery, or experimental paradigm with published reliability coefficients, factor structure analyses, and population-level norms—and (b) an expert-annotated text corpus suitable for training NLP models to detect the theory's constructs in text. These two resources are not luxuries; they are prerequisites for high-quality benchmark construction. A validated instrument provides the items, scoring rubrics, and human baseline data that the benchmark can inherit. An annotated corpus provides the training data for automated evaluation pipelines. Without either, a benchmark builder must start from scratch: designing items, establishing reliability, collecting human data, and building evaluation machinery, all within the scope of a single paper. The startup cost is prohibitive, and the result is typically a benchmark that lacks the psychometric grounding of its infrastructure-backed competitors.

Stage 2: Benchmark Quality. The 17-feature assessment framework operationalizes benchmark quality through construct validity (A-features), practical design (B-features), and community impact (C-features). A benchmark's quality score is, in substantial part, determined by the availability of theory infrastructure. Features A2 (adapted from validated instrument), A4 (validated source instrument), A5 (population-level norms), and A6 (convergent/discriminant validity evidence) all directly reward benchmarks that inherit infrastructure from a validated parent instrument. Features B1 (machine-scorable format), B4 (reproducibility), and B7 (scoring objectivity) reward benchmarks with clear

operationalization—which is easier to achieve when the benchmark adapts an existing instrument with established scoring procedures than when it builds evaluation criteria *de novo*. The result is that theories with strong infrastructure produce benchmarks with higher composite scores, which translate into higher tier assignments. Theories without infrastructure produce benchmarks that score lower on the A-features and, consequently, land in the Medium or Low tiers.

Stage 3: Discrimination Power. The benchmarks that are available for empirical testing—and the ones that researchers select for cross-model comparison studies—are disproportionately the high-quality ones. A researcher assembling a trial battery will preferentially include validated instruments with known psychometric properties, not because they are expected to discriminate better but because they are more trustworthy: their scores are interpretable, their human baselines are available, and their construct validity is established. The result is that the trial battery is enriched for infrastructure-backed, content-level instruments and depleted of the process-level probes that, as Section 6.1.3 demonstrated, are the primary discriminators. The battery then returns convergent results —not because the models are indistinguishable in all respects, but because the battery is measuring the dimensions on which they are indistinguishable.

The pipeline is summarized in Figure 15 and quantified in Table T27.

Table T27: Infrastructure-Quality-Discrimination Pipeline

Theory	Validated Instrument	Annotated Corpus	Landscape Size (n)	Quality Rate (% Top+High)	Benchmarks Administered	Mean CV of Administered Benchmarks	Discrimination Assessment
MFT	Yes (MFQ-30, MFQ-2)	Yes (MFTC, 35K tweets)	67	73.1%	5 (MP2, MP4, MP9, MP10, MP19)	0.012	Very Low (convergence)
Schwartz	Yes (PVQ-RR, SVS)	No	44	56.8%	3 (MP5, MP11, MP12)	0.039	Low-Moderate
Trolley	Yes (Moral Machine)	No	29	48.3%	2 (MP7, MP8)	0.000	Zero (perfect identity)
Kohlberg	Yes (DIT-1, DIT-2)	No	9	55.6%	2 (MP3, MP6)	0.043	Non-monotonic
Intuitionist	No (no standard instrument)	No	3	100.0%	2 (MP-SI, MP17)	0.090	Moderate
Dyadic	No	No	2	50.0%	1 (MP-DM)	0.009	Very Low
Cross-Cultural	No	No	—	—	2 (MP-CC, MP18)	0.022	Low
Multi-Theory	Partial (various)	Partial	—	—	4 (MP15, MP16, MP20, MP21)	0.040	Low-Moderate
Dual Process	No	No	—	—	1 (MP-DP)	**3.000**	**Very High**
Ubuntu	No	No	13	0.0%	0	N/A	Unknown
Confucian	No	No	13	0.0%	0	N/A	Unknown
Indigenous	No	No	4	0.0%	0	N/A	Unknown

The pipeline's logic is visible in every row of the table. MFT—the theory with both a validated instrument and an annotated corpus—produces the largest landscape representation (67 candidates), the highest quality rate (73.1%), the most benchmarks in the trial battery (5), and among the lowest mean CVs (0.012). The five MFT benchmarks administered to nine models produce near-zero discrimination: the models are indistinguishable on Care, Fairness, Loyalty, Authority, and Sanctity when measured by MFT's instruments. Schwartz values and Kohlberg's stages, which possess validated instruments but lack annotated corpora, produce moderate landscape sizes, moderate quality rates, and low-to-moderate CVs. The trolley paradigm, which possesses the Moral Machine experiment as a validated instrument with 40 million human judgments, achieves the only perfect-zero result in the table: CV = 0.000 on both TrolleyBench and the Moral Machine, meaning that the ethical consistency index and the Moral Machine preference profile are identical across all nine models from all three vendors.

At the opposite extreme, Dual Process Theory—which has no standardized questionnaire, no annotated corpus, and no entry in the 292-candidate landscape as a standalone theory—produced a single custom probe (MP-DP) that achieved a CV of 3.000. This is not a marginal difference: the dual-process probe's discrimination power exceeds the combined mean discrimination of all other 22 benchmarks by more than an order of magnitude. And the theories for which no benchmarks were administered at all (Ubuntu, Confucian, Indigenous) are precisely the ones whose 0% quality rate rendered them ineligible for inclusion in a trial battery designed to assess the field's best instruments.

6.2.2 Why MFT Benchmarks Converge and Process Probes Discriminate

The infrastructure-quality-discrimination pipeline explains the *pattern* of results but not the *mechanism*. Why, specifically, do MFT-derived instruments produce convergence while the dual-process probe produces discrimination? The answer requires connecting the pipeline to the training dynamics of RLHF-aligned language models.

MFT convergence traces to training-data alignment. The MFQ-30 measures five moral foundations: Care, Fairness, Loyalty, Authority, and Sanctity. RLHF training data—the human preference annotations that shape model behavior during alignment—is generated by annotators who are disproportionately from Western, educated, industrialized, rich, and democratic (WEIRD) populations [Henrich, Heine, & Norenzayan, 2010]. The moral psychology literature has established that WEIRD populations systematically prioritize Care and Fairness over Loyalty, Authority, and Sanctity—a pattern so robust that it has been replicated across dozens of studies in over 20 countries [Graham, Haidt, & Nosek, 2009; Haidt, 2012]. When RLHF annotators evaluate model outputs for "helpfulness" and "harmlessness," they are implicitly applying a Care/Fairness-dominant moral framework. The models learn to optimize for this framework, producing outputs that score high on Care and Fairness and low on Sanctity and Authority. Because all three vendors (Anthropic, Google, OpenAI) draw from similar annotator pools and optimize for similar

human preference criteria, all nine models converge on the same WEIRD-liberal moral profile.

The MFQ-30, when administered to these models, is measuring precisely the construct that RLHF was designed to instill. It is asking whether the model endorses the moral values that its training data—curated by annotators who share those values—rewarded it for endorsing. The convergence is not surprising; it is an expected consequence of the training pipeline. The MFQ-30 is, in effect, a training-data audit instrument: it confirms that the alignment procedure successfully imparted the intended moral profile. Its lack of discrimination across models reflects the success of RLHF, not the failure of measurement.

Process-level discrimination taps architectural differences. The dual-process probe does not ask what the model values; it asks how the model's moral judgments change when processing conditions are altered. The CoT effect—the shift in moral judgment when the model is prompted to reason step by step before answering—is not a content-level property that RLHF can directly train for. It is an emergent property of the interaction between the model's architecture, its parameter count, and the degree to which its moral commitments are "baked in" versus amenable to deliberative override.

The finding that only Haiku (the smallest Anthropic model) shows a CoT effect, while larger models do not, suggests a threshold phenomenon: below a certain parameter count or training-data volume, the model's moral commitments are held loosely enough that deliberative prompting can override them. Above that threshold, the model's RLHF-instilled values are too deeply encoded to be modulated by processing conditions. This pattern has a direct analogue in the human dual-process literature: Kahneman's [2011] System 1 (fast, intuitive) and System 2 (slow, deliberative) processes produce different moral judgments in contexts where they conflict, but individuals with stronger moral convictions show less dual-process dissociation because their convictions resist deliberative override [Greene et al., 2001; Bartels, 2008].

Different vendors, with different architectures, different training procedures, and different parameter counts, produce models with different degrees of moral entrenchment. This is a genuine between-model difference, but it is a process-level difference, not a content-level one. All models endorse the same values (content-level convergence). They differ in how rigidly those values are held and how responsive they are to processing manipulations (process-level divergence). The dual-process probe detects the latter because it is the only instrument in the battery designed to probe processing dynamics rather than value endorsement.

The paradox resolved. The apparent paradox—that the theories most represented in the landscape produce the least discrimination, while the theories least represented produce the most—dissolves when the content/process distinction is applied. The landscape is dominated by content-level theories (MFT, Schwartz, Trolley) because these theories possess the infrastructure for content-level measurement: questionnaires that ask what

values the respondent endorses, forced-choice paradigms that reveal preference orderings, and annotated corpora that classify text by moral content. The landscape is depleted of process-level theories (Dual Process, Social Intuitionism) because these theories lack equivalent infrastructure: there is no "Dual Process Morality Questionnaire" because dual-process effects are not measured by questionnaires but by experimental manipulations of processing conditions. The landscape's infrastructure bias is a content-level bias, and content-level instruments cannot discriminate between models that share the same content-level profile.

The implication is not that MFT is the wrong theory for moral evaluation, but that MFT instruments alone are insufficient. A complete moral evaluation of LLMs requires both content-level assessment (what does the model value?) and process-level assessment (how does the model reason about moral questions?). The current landscape provides the former in abundance and the latter almost not at all. The pipeline failure is a measurement-type failure: the field has built excellent instruments for a dimension of moral cognition on which models converge, and has failed to build instruments for the dimensions on which they might diverge.

6.3 The Landscape-Empirical Convergence

The tier-stratified analysis (Section 6.1) and the infrastructure pipeline (Section 6.2) establish the mechanics of the triple bind's operation. This section demonstrates its systemic nature: the three binds are not independent problems but a self-reinforcing system in which each failure amplifies the others. The logical structure is developed in three stages: first, showing that Claims 1–6 predict Claim 7; second, showing that a narrow landscape necessarily produces homogeneous results; and third, showing that the three binds form a closed feedback loop that the field has no natural mechanism to escape.

6.3.1 Claims 1–6 Predict Claim 7

The seven claims presented in Parts I and II are not seven independent findings. Claims 1 through 6 (the landscape findings) collectively predict Claim 7 (the empirical convergence finding) with a logic that, in retrospect, has the force of a syllogism.

Claim 1 (MFT dominance) entails that the landscape tests primarily MFT-adjacent constructs. With 67 of 292 candidates (22.9%) grounded in MFT and a further 44 (15.1%) in Schwartz values theory (which shares MFT's content-level measurement paradigm), the landscape is structurally biased toward measuring the content of moral endorsement rather than the process of moral reasoning. A battery drawn from this landscape will, by construction, be enriched for instruments that ask "what does the model value?" and depleted of instruments that ask "how does the model reason?" The dominance of MFT is not merely a theoretical preference; it is a measurement-type constraint that limits the kinds of differences the battery can detect.

Claim 2 (validity crisis) entails that most instruments have not been validated for LLM respondents. When psychometric instruments designed for human respon-

dents are administered to language models without revalidation, the assumption of construct equivalence is untested. The MFQ-30′s factor structure may not hold for LLMs; the DIT's stage-scoring rubric may not capture the constructs it captures in human populations; the Moral Machine's preference dimensions may interact differently in models than in people. Without LLM-specific validation, the instruments may produce scores that are numerically interpretable but psychometrically ungrounded. This does not bias the results toward convergence per se, but it means that whatever pattern the instruments detect—convergence or divergence—cannot be interpreted with the same confidence as findings from validated instruments.

Claim 3 (theoretical absence) entails that process-level theories capable of discriminating between models are absent from the battery. Section 6.1.3 demonstrated that the dual-process probe—a process-level instrument—was the only benchmark that separated the nine models. Claim 3 established that dual-process theory, social intuitionism, and other process-level frameworks have near-zero representation in the 292-candidate landscape (3 candidates for intuitionism, 2 for dyadic morality, 0 for dual-process theory as a standalone category). The theories that could discriminate are the ones the landscape has not built instruments for. The absence is not random; it is a consequence of the infrastructure pipeline described in Section 6.2: these theories lack the standardized instruments and annotated corpora that would enable high-quality benchmark construction.

Claim 4 (cultural cliff) entails that non-Western perspectives that might reveal cross-model differences are unmeasurable at research-grade quality. The cultural quality cliff—0% quality rate for Ubuntu, Confucian, Indigenous, and Identity ethics—means that no high-quality instrument exists to test whether different models engage differently with non-Western moral frameworks. The question of whether GPT-5 handles Confucian role ethics differently from Claude Opus is unanswerable using existing instruments, not because the answer is unknowable but because the instruments needed to investigate it have not been built to a standard that would make the investigation credible. If non-Western moral frameworks were well-instrumented, the trial battery would include them, and the convergence finding might be qualified: perhaps models converge on Western moral content but diverge on non-Western moral content. The cultural cliff prevents this hypothesis from being tested.

Claim 5 (baseline deficit) entails that without human comparison data, the convergence finding cannot be anchored to meaningful thresholds. The convergence of nine models on a shared moral profile is striking, but its significance depends on context. Are the models converging on a profile that resembles a specific human population (e.g., American college students, the modal RLHF annotator)? Are they converging on a profile that has no human analogue? Without human baselines for the majority of administered benchmarks—which Claim 5 establishes are absent for most instruments—the convergence cannot be fully interpreted. The models may be converging on the moral

profile of their annotators, on a statistical average of the training data, or on something entirely novel. The baseline deficit renders these alternatives indistinguishable.

Claim 6 (temporal stagnation) entails that the field is not building the tools that would resolve these problems. The quality rate has not improved from the pre-2020 cohort (approximately 75%) to the 2023–2026 cohort (46%). The field is producing more benchmarks but not better ones, and it is not producing benchmarks that address the specific deficits identified by Claims 1–5. The process-level instruments that Claim 3 identifies as absent are not being built. The non-Western instruments that Claim 4 identifies as low-quality are not being improved. The human baselines that Claim 5 identifies as missing are not being collected. The stagnation means that the conditions producing Claim 7′s convergence finding are not self-correcting: the field is not naturally evolving toward a battery that could discriminate between models even if differences exist.

Together, Claims 1 through 6 construct a logical chain: narrow instruments (Claim 1) with unvalidated psychometrics (Claim 2), missing process-level theories (Claim 3), excluded cultural perspectives (Claim 4), absent human baselines (Claim 5), and no trend toward improvement (Claim 6) will produce a trial battery that returns convergent results when administered to models trained on similar data by similar methods. Claim 7—the empirical convergence—is not a surprise given Claims 1–6; it would be a surprise if the landscape *could* discriminate given its documented limitations. The finding is overdetermined by the landscape's structural properties.

6.3.2 A Narrow Landscape Produces Homogeneous Results

The predictive relationship between Claims 1–6 and Claim 7 can be restated more precisely as a measurement-theoretic argument. A narrow evaluation landscape—one dominated by a single theory, measured at the content level, using instruments validated on a single cultural population—will produce homogeneous results when applied to systems that share a training pipeline even if those systems differ in architecturally important ways that a broader evaluation landscape could detect.

The argument has four premises.

Premise 1: All major LLMs are RLHF-aligned using overlapping annotator pools and similar preference criteria. Anthropic's Constitutional AI, Google's RLHF, and OpenAI's instruction tuning all use human preference data to shape model behavior, and the annotators who generate this data are drawn from similar demographic and cultural populations. The preference criteria ("helpful, harmless, honest" or variants thereof) are functionally equivalent across vendors.

Premise 2: The moral content instilled by RLHF reflects the annotators' moral profile. When annotators from WEIRD populations rate model outputs, they reward responses that align with their own moral intuitions: Care/Fairness-dominant, low-Sanctity, low-Authority, post-conventional, deontological-leaning. The models learn to produce this profile because it maximizes the preference signal.

Premise 3: The evaluation landscape measures moral content, not moral process. The dominant instruments (MFQ-30, Moral Machine, PVQ, DIT, MFTC) all measure what moral values the respondent endorses or which moral principles they apply. They do not measure how the respondent arrives at moral judgments, whether the respondent's judgments are sensitive to processing conditions, or whether the respondent's moral reasoning exhibits the dual-process dynamics that characterize human moral cognition.

Premise 4: Content-level convergence is expected when training pipelines converge. If all models learn the same moral content from similar annotator pools, and the evaluation instruments measure that content, the instruments will return similar scores regardless of which model is being tested. The convergence is an expected consequence of training-data alignment, not an unexpected finding about model capabilities.

The four premises jointly entail the convergence: RLHF produces content-level moral alignment (Premises 1–2), the landscape measures content-level properties (Premise 3), and content-level instruments will converge on content-level alignment (Premise 4). The only way the landscape could discriminate between these models is if it included instruments that measure dimensions on which RLHF does *not* produce alignment—that is, process-level dimensions, culturally specific dimensions, or dimensions that the annotator pool does not systematically agree on. The landscape lacks such instruments (Claims 1, 3, 4), and therefore it returns convergent results (Claim 7).

The measurement-theoretic argument clarifies what the convergence finding does and does not demonstrate. It does not demonstrate that the models are morally identical in every respect. It demonstrates that the models are identical *on the dimensions that the current landscape measures*. Whether they differ on dimensions that the landscape does not measure—process-level dynamics, non-Western moral reasoning, moral behavior under novel or culturally specific conditions—is an open question that the current landscape is structurally incapable of answering. The homogeneity is a finding about the interaction between the instruments and the models, not about the models alone.

This reframing has a critical implication for the interpretation of the RLHF moral fingerprint identified in Part II. The six universal moral signatures—the WEIRD-liberal bias, the purity floor, the authority suppression, the ethical consistency ceiling, the post-conventional default, and the deontological refusal asymmetry—are real properties of the models, confirmed by the best available instruments. But they are properties that describe the models' content-level moral profile, not their complete moral psychology. A model that scores identically to eight other models on 22 content-level benchmarks may nevertheless differ from all of them in how it processes moral information, how it resolves moral conflicts under cognitive load, how it responds to culturally specific moral framings, or how its moral judgments interact with its reasoning capabilities. These differences, if they exist, would be invisible to the current landscape and would require process-level, culturally diverse, and psychometrically innovative instruments to detect.

6.3.3 The Triple Bind as a System, Not Three Independent Problems

The preceding analysis has established each bind independently: theoretical narrowness (Bind 1), empirical undiscrimination (Bind 2), and cultural blindness (Bind 3). It has shown that the three binds are causally related through the infrastructure-quality-discrimination pipeline. What remains is to demonstrate that they form a closed feedback loop—a self-reinforcing system in which each bind sustains and amplifies the others, creating a stable equilibrium that the field has no natural mechanism to escape.

The feedback dynamics operate through four causal pathways.

Pathway 1: Bind 1 creates Bind 2. Theoretical narrowness produces empirical undiscrimination because a narrow set of content-level instruments, measuring a single theory's constructs on models trained on similar data, will return convergent results. This is the infrastructure-quality-discrimination pipeline described in Section 6.2. When the landscape's dominant instruments all measure MFT-adjacent constructs—Care, Fairness, Loyalty, Authority, Sanctity—and all models have been trained to optimize for Care/Fairness-dominant responses, the instruments will detect no differences. The narrowness of the theoretical frame determines the narrowness of the empirical findings.

Pathway 2: Bind 3 reinforces Bind 1. Cultural exclusion limits theoretical diversity because non-Western moral traditions (Ubuntu, Confucian, Indigenous, Buddhist, Islamic) often emphasize moral constructs—relational selfhood, communal agency, virtue cultivation through practice, sacred authority, cosmic moral order—that have no direct analogue in Western moral psychology and therefore fall outside the content-level measurement paradigm that dominates the landscape. The cultural cliff (0% quality rate for four non-Western traditions) means that these alternative theoretical perspectives, even when they are represented by benchmark candidates in the landscape, are not represented at a quality level that would permit them to challenge MFT's dominance. The exclusion is infrastructural, not intentional: these traditions lack the validated instruments and annotated corpora that would enable high-quality benchmarks. But the effect is the same as if it were intentional: the theoretical frame remains narrow because the cultural frame remains narrow.

Pathway 3: Bind 2 reinforces Bind 1. Convergent results reduce the incentive to build diverse instruments because the convergence creates the appearance that the evaluation problem is solved. If the MFQ-30 produces consistent, interpretable scores for every model—scores that vary little across administrations and correlate with human baselines where available—then from the perspective of a benchmark builder, the MFQ-30 "works." There is no apparent need for a Dual Process Morality Questionnaire or a Confucian Role Ethics Scale because the existing instruments are producing data. The data happens to be uninformative for cross-model comparison, but this is not visible unless one explicitly compares across models—and the field's publication norms reward per-model characterization (administering a benchmark to a single model and reporting the results) over cross-model comparison (administering the same benchmark to multiple models and reporting

the variance). The convergence finding, paradoxically, sustains the conditions that produce it by reducing the urgency of building instruments that might reveal divergence.

Pathway 4: Bind 2 reinforces Bind 3. When content-level instruments return convergent results across models, the implied conclusion is that moral evaluation is culturally neutral: all models score the same, so the instruments must be measuring something universal. This conclusion obscures the cultural specificity of the instruments themselves. The MFQ-30 was developed and validated on Western populations; its five-foundation structure may not map onto the moral ontologies of non-Western traditions that organize morality around different constructs (e.g., dharma, ren, ubuntu). When the MFQ-30 returns the same score for all nine models, it is confirming that all models share the same Western moral profile—but this finding is easily misinterpreted as confirming that all models are "morally aligned" in a culturally general sense. The misinterpretation reduces the perceived urgency of building non-Western instruments: if the existing instruments already show alignment, why invest in culturally specific alternatives? The convergence thus sustains the cultural exclusion by making the exclusion invisible.

The four pathways form a closed loop: narrowness produces convergence (Pathway 1), which reduces incentive for diversity (Pathway 3), which sustains narrowness. Cultural exclusion reinforces narrowness (Pathway 2), and convergence reinforces cultural exclusion (Pathway 4). The system is self-perpetuating because each bind makes the others harder to break.

Breaking the loop requires intervention at its root: the theory infrastructure stage of the pipeline. The pathways show that neither increasing benchmark volume (which does not change theoretical diversity), nor increasing model diversity (which does not change the instruments' content-level bias), nor increasing funding for benchmark construction (which will flow to the theories with existing infrastructure) will disrupt the equilibrium. What is required is the deliberate construction of psychometric infrastructure for theories that currently lack it: standardized instruments for dual-process morality, social intuitionism, and dyadic morality; validated questionnaires grounded in Ubuntu, Confucian, Buddhist, and Indigenous moral philosophies; annotated corpora that tag moral reasoning processes, not just moral content; and population-level norms for non-WEIRD populations that would give LLM scores a meaningful comparative baseline. Without this infrastructure investment, the pipeline will continue to produce the same output: narrow instruments measuring convergent content on models that share the same training-data moral profile.

The triple bind is, in this analysis, a single structural failure in the field's evaluative infrastructure, observed at three stages of its operation. The theoretical narrowness (Bind 1) is the infrastructure deficit at Stage 1. The empirical undiscrimination (Bind 2) is the measurement consequence at Stage 3. The cultural blindness (Bind 3) is the particular form the infrastructure deficit takes when it intersects with the global diversity of moral traditions. They are not three problems requiring three solutions; they are one problem

—the absence of diverse, process-level, culturally grounded psychometric infrastructure—manifesting as three symptoms. The path forward requires treating the disease, not the symptoms: building the infrastructure that the pipeline needs to produce diverse, discriminating, and culturally inclusive moral evaluation.

7. Discussion

The results of this study converge on a diagnosis that is at once more precise and more disquieting than the "double bind" identified in prior work. The moral psychology benchmark landscape for large language models does not merely suffer from two concurrent weaknesses—theoretical narrowness and empirical undiscrimination—that happen to coexist in the same domain. It suffers from a triple bind in which three failures compound multiplicatively: the landscape is theoretically narrow (dominated by one theory at 73% quality while four non-Western traditions achieve 0%), empirically undiscriminating (43% of benchmarks produce zero cross-model variation across nine models from three vendors), and culturally blind (32 non-Western candidates collectively fail to produce a single publication-ready instrument). Parts I through III established these findings empirically and demonstrated their structural interconnection through the infrastructure-quality-discrimination pipeline. This section interprets their consequences.

The interpretive task is substantial because the triple bind is not merely a measurement problem. It is an epistemic problem: the field's evaluative infrastructure shapes what it can know about LLM moral cognition, and the shape of that infrastructure—narrow, homogeneous, culturally parochial—determines that the field can know only a narrow, homogeneous, culturally parochial version of the truth. The instruments define the questions that can be asked, and the questions define the answers that can be found. When the instruments are drawn overwhelmingly from one theory, calibrated on one cultural population, and unable to distinguish between the systems they evaluate, the resulting "knowledge" about LLM moral cognition is knowledge about a specific intersection of training pipeline artifacts and Western measurement conventions, not knowledge about the systems' moral capabilities in any broader sense.

This section develops that interpretive argument through seven subsections. Section 7.1 presents the triple bind as a compounding system, showing how each failure amplifies the others. Section 7.2 traces the causal mechanism of MFT's dominance to infrastructure determinism, arguing that the quality gap is not effort-driven but infrastructure-driven. Section 7.3 develops the temporal stagnation thesis, showing that the proliferation of benchmarks creates a quality illusion—the appearance of progress without its substance. Section 7.4 interprets the six universal RLHF signatures, weighing the universal values hypothesis against the shared training artifact hypothesis. Section 7.5 develops the governance implications of the cultural quality cliff for global AI deployment. Section 7.6 offers seven concrete recommendations for AI governance and policy. Section 7.7 catalogues the study's limitations with the specificity they require.

7.1 The Triple Bind: How Three Failures Compound

The central interpretive claim of this paper is that theoretical narrowness, empirical undiscrimination, and cultural blindness are not three problems requiring three solutions. They are one self-reinforcing system observed at three stages of its operation, and the compounding effects of their interaction produce consequences substantially worse than the sum of their individual impacts.

7.1.1 Bind 1 as a Definitional Problem

When 73% of MFT benchmark candidates achieve publication-ready quality while 0% of Ubuntu, Confucian, Indigenous, and Identity ethics candidates do, the landscape does not merely favor MFT over alternative theories. It effectively defines what counts as morality in the context of LLM evaluation. The definition is not explicit—no researcher has argued that morality is coextensive with the five moral foundations of Care, Fairness, Loyalty, Authority, and Sanctity. But the definition is operative: when a governance body asks whether an LLM is "morally aligned," the instruments available to answer that question are overwhelmingly instruments that measure MFT-adjacent constructs, and the answer will therefore be an answer about MFT-adjacent moral cognition. The question "is this model morally aligned?" becomes, in practice, the question "does this model endorse Care and Fairness over Sanctity and Authority in a pattern consistent with WEIRD liberal populations?" The definitional collapse is invisible to users of the instruments because the instruments do not announce the theoretical tradition they instantiate; they present themselves as measuring "morality" or "moral reasoning" or "ethical alignment" without qualification.

The consequences of this definitional collapse extend beyond the academy. When the MFQ-30 returns an Individualizing/Binding ratio of 2.83 for Claude Opus, the number is psychologically interpretable only within the MFT framework: it means that Opus endorses Care and Fairness substantially more than Loyalty, Authority, and Sanctity, placing it in the "liberal" zone of the MFT political space. But this characterization says nothing about whether Opus engages in virtuous deliberation (Aristotelian ethics), whether it respects relational obligations graded by intimacy and role (Confucian ethics), whether it treats the suffering of others as constitutive of its own diminishment (Ubuntu ethics), or whether it recognizes sacred boundaries that utilitarian calculation cannot transgress (religious ethics across traditions). The MFQ-30 cannot measure these dimensions because it was not designed to measure them, and no instrument of comparable quality exists for these alternative frameworks. The landscape's theoretical narrowness is therefore not merely an academic imbalance; it is a constraint on what the field can see. Morality, as evaluated by the benchmark landscape, is morality as MFT defines it.

The severity of this definitional collapse is best appreciated by imagining its analogue in another domain. If the benchmark landscape for measuring LLM language comprehension consisted almost exclusively of instruments measuring syntactic parsing, with 73% of syntax benchmarks achieving publication-ready quality and 0% of pragmatics,

discourse, or sociolinguistic benchmarks doing so, the field would immediately recognize that its evaluative infrastructure was measuring one dimension of language and calling it "language comprehension." The moral psychology case is structurally identical, but the recognition has been slower because moral psychology's theoretical pluralism is less well-known outside the discipline than linguistics' subdisciplinary structure.

7.1.2 Bind 2 as a Genuine Property of the Systems

The empirical undiscrimination documented in Part II admits of two broad interpretations. The first is that the convergence is a measurement artifact: the instruments are insufficiently sensitive, and better instruments would reveal differences. The second is that the convergence is a genuine property of the systems: RLHF-aligned models share a single moral profile because they have been trained on similar data by similar methods, and no instrument—however sensitive—would reveal differences on the dimensions these instruments measure.

The tier-stratified analysis in Section 6.1 provides strong evidence for the second interpretation. The correlation between quality and discrimination is $r = -0.83$: higher-quality benchmarks produce *less* variation between models, not more. The instruments with the strongest construct validity, the most comprehensive human baselines, and the most rigorous psychometric properties—the MFQ-30, the Moral Machine, TracingMF—are the instruments that most emphatically confirm convergence. If the homogeneity were a measurement failure, the best instruments should detect what the worst instruments miss. Instead, the best instruments confirm what the worst instruments suggest: the models are morally indistinguishable on content-level measures.

This finding has a counterintuitive but important corollary. The homogeneity is not something the field needs to solve by building better versions of existing instruments. A more precise MFQ will not break the convergence; it will confirm it with tighter confidence intervals. A more comprehensive trolley battery will not reveal vendor-specific moral differences; the Moral Machine, with 40 million human judgments as its normative base, already returns CV = 0.000 across all nine models. The measurement problem is not sensitivity; it is dimensionality. The instruments measure content-level moral endorsement—what values the model affirms—and on this dimension, the models are genuinely identical because they have been trained to affirm the same values. The dimension on which the models differ is process-level moral cognition—how they arrive at moral judgments, how their judgments respond to processing manipulations, how rigidly their moral commitments resist contextual perturbation—and this dimension is the one the landscape has not built instruments to measure.

The dual-process probe provides the decisive evidence. Its CV of 3.00 is 93 times larger than the mean CV of Top-tier benchmarks. It achieved this discrimination not because it is a better instrument in the traditional psychometric sense—it lacks standardized scoring, published norms, and multi-study validation—but because it probes a dimension that RLHF training does not homogenize. The chain-of-thought effect is an architectural

property, not a content-level property, and architectural properties vary across models even when content-level properties do not. The benchmark that the landscape has not invested in—the process-level probe—is the benchmark that breaks the convergence. The benchmarks the landscape has invested in—the content-level instruments with decades of psychometric validation—are the benchmarks that confirm it.

7.1.3 Bind 3 as a Structural Exclusion

When 32 non-Western benchmark candidates exist but none meets quality standards, the problem is not the absence of scholarly effort. Researchers have built Ubuntu-inspired evaluation frameworks (UbuntuGuard, composite 35.1), Confucian ethics datasets ("AI Trained on the Confucian Analects," composite 21.2), and Indigenous moral reasoning probes. The 32 candidates represent genuine intellectual labor directed toward broadening moral evaluation beyond Western frameworks. The problem is that this effort has been structurally unable to produce high-quality instruments because the prerequisite infrastructure—validated questionnaires, annotated corpora, population norms—does not exist for these traditions.

The structural nature of the exclusion deserves emphasis because it means the cultural quality cliff cannot be closed through increased effort alone. A researcher who secures funding, assembles a team, and dedicates three years to building an Ubuntu ethics benchmark will still confront the absence of a validated Ubuntu Moral Assessment Questionnaire, the absence of an annotated Ubuntu Ethics Corpus, and the absence of population-level norms for Ubuntu moral endorsement across sub-Saharan African populations. Without these foundational resources, the benchmark will score low on A4 (validated source instrument), A5 (human baselines), and A6 (construct validity), landing it in the medium or low tier regardless of the quality of the researcher's intellectual engagement with Ubuntu philosophy. The quality cliff is not a gap that effort can bridge; it is a gap that requires prior infrastructure investment before effort becomes productive.

This structural character distinguishes the cultural quality cliff from other kinds of scholarly underrepresentation. In many fields, the underrepresentation of non-Western perspectives reflects differential attention: Western researchers simply have not engaged with non-Western scholarship, and the remedy is engagement. In the moral psychology benchmark landscape, the underrepresentation reflects differential infrastructure: Western moral psychology has accumulated decades of psychometric investment—questionnaires validated across populations, corpora annotated by trained coders, norms collected from hundreds of thousands of respondents—and non-Western moral philosophy has accumulated centuries of philosophical insight without the specific kind of computational infrastructure that benchmark construction requires. The philosophical depth of Ubuntu, Confucian, and Indigenous moral thought is not in question; what is in question is whether that depth has been translated into the specific institutional forms—standardized instruments, annotated datasets, population norms—that the benchmark development pipeline rewards.

7.1.4 The Compounding Dynamics

The three binds do not merely coexist. Each reinforces the others through four causal pathways that, together, form a closed feedback loop.

Narrowness produces convergence. A landscape dominated by content-level MFT instruments will return convergent results when applied to models trained on similar data. This is the infrastructure-quality-discrimination pipeline documented in Section 6.2: MFT's infrastructure produces high-quality benchmarks, high-quality benchmarks enter trial batteries, and these batteries measure content-level properties on which RLHF-aligned models are identical. The narrowness of the theoretical frame determines the narrowness of the empirical findings.

Convergence reduces the incentive for diversity. When the MFQ-30 produces consistent, interpretable scores across all nine models, the instrument "works" by the standards of benchmark development: it generates data, fills evaluation checklists, and produces publishable results. The data happens to be uninformative for cross-model comparison, but this uninformativeness is invisible unless one explicitly compares across models—a practice that the field's publication norms do not reward. The convergence finding, paradoxically, sustains the conditions that produce it by eliminating the empirical motivation for building instruments that might reveal divergence. Why build a Dual Process Morality Questionnaire when the MFQ-30 already produces clear, interpretable numbers?

Cultural exclusion reinforces narrowness. Non-Western moral traditions often emphasize moral constructs—relational selfhood, communal agency, virtue cultivation through practice, sacred authority, cosmic moral order—that have no direct analogue in the content-level measurement paradigm that dominates the landscape. The cultural cliff ensures that these alternative perspectives, even when they are represented by candidates in the landscape, are not represented at a quality level that would challenge MFT's dominance. The theoretical frame remains narrow because the cultural frame remains narrow.

Convergence makes cultural exclusion invisible. When content-level instruments return convergent results, the implied conclusion is that moral evaluation is culturally neutral: all models score the same, so the instruments must be measuring something universal. This conclusion obscures the cultural specificity of the instruments themselves. The MFQ-30 was developed and validated on Western populations; its convergence across models confirms that all models share the same Western moral profile, but this confirmation is easily misinterpreted as confirming that all models are "morally aligned" in a culturally general sense. The misinterpretation reduces the perceived urgency of non-Western instruments: if the existing instruments already show alignment, why invest in culturally specific alternatives? The convergence sustains the cultural exclusion by making it invisible.

These four pathways form a closed loop. Breaking any single pathway would weaken the system, but the system is designed—not intentionally but structurally—to resist perturbation. Increasing benchmark volume does not change theoretical diversity; the new

benchmarks flow to the theory with existing infrastructure. Increasing model diversity does not change the instruments' content-level bias; different models trained by different methods still produce the same content-level profile. Increasing funding for benchmark construction, without specifying where the funding should go, will flow to the theories whose existing infrastructure makes the marginal cost of another benchmark lowest—which is MFT. The triple bind is a stable equilibrium that the field's current incentive structure has no mechanism to disrupt.

7.2 Infrastructure Determinism: Why MFT Wins

The dominance of Moral Foundations Theory in the benchmark landscape demands an explanation that goes beyond "MFT is popular" or "MFT is well-suited to LLM evaluation." The quality gap between MFT (73%) and every other theory is too large, too consistent, and too precisely aligned with infrastructure availability to be explained by researcher preference alone. The explanation is infrastructure determinism: MFT wins not because it is the best theory of moral cognition but because it possesses the best computational infrastructure, and in a landscape where benchmark quality is determined by inherited psychometric properties, infrastructure is destiny.

7.2.1 The Dual-Asset Advantage

MFT alone among moral psychology theories possesses two distinct computational assets that benchmark developers can inherit.

The first asset is the Moral Foundations Questionnaire (MFQ-30; Graham et al., 2011). The MFQ-30 is a 30-item instrument with published factor structure, internal consistency coefficients, test-retest reliability, and population-level norms validated across political orientations, religious traditions, and cultural groups spanning at least 11 countries. For a researcher building an MFT benchmark for LLMs, the MFQ-30 is a turnkey solution: administer the questionnaire, score against published norms, and report. The instrument provides validated items (boosting A4 to 100), population baselines (boosting A5), and decades of construct validity evidence (boosting A6). These three features alone—A4, A5, A6—account for a substantial fraction of the Psychological Depth sub-score, which carries 50% of the composite weight. An MFT benchmark that adapts the MFQ-30 enters the quality assessment with a structural advantage that no non-MFT benchmark can match.

The second asset is the Moral Foundations Twitter Corpus (MFTC; Hoover et al., 2020), a collection of 35,108 tweets annotated by expert raters for the presence of each moral foundation. The MFTC enables a second class of benchmark construction unavailable to any other theory: corpus-based NLP approaches in which researchers train classifiers on MFTC annotations and evaluate LLMs against these classifiers or use MFTC-derived items as benchmark stimuli. The MFTC boosts B2 (dataset size) and B3 (data availability) scores, contributing to the Practical Ease sub-score that carries the remaining 50% of the composite weight.

No other theory has this dual-asset advantage. Schwartz's theory of basic human values has the Portrait Values Questionnaire—Revised (PVQ-RR; Schwartz et al., 2012), a validated questionnaire with cross-cultural norms spanning over 80 countries, but it lacks a large annotated text corpus. Kohlberg's Cognitive Moral Development theory has the Defining Issues Test (DIT; Rest, 1979), a validated stage-assessment instrument with 40 years of developmental norms, but again no NLP corpus. The trolley tradition has the Moral Machine experiment (Awad et al., 2018) with 40 million human judgments, but no accompanying text corpus. Each of these theories possesses one computational asset and achieves a respectable but sub-MFT quality rate: Schwartz at 57%, Kohlberg at 56%, trolley at 48%. The pattern is consistent: one asset produces moderate quality; two assets produce dominant quality; zero assets produce zero quality.

Ubuntu ethics, Confucian moral philosophy, and Indigenous ethical frameworks have neither. There is no Ubuntu Moral Assessment Questionnaire with published factor loadings. There is no Confucian Role Ethics Scale with known reliability coefficients. There is no Indigenous Relational Ethics Inventory with established population norms. And there is no annotated corpus for any of these traditions—no "Ubuntu Ethics Corpus" with expert-labeled examples of communal reasoning, no "Confucian Moral Judgment Dataset" with coded examples of role-differentiated obligation. A researcher attempting to build a benchmark for any of these traditions must design items from scratch, establish psychometric properties from scratch, collect human baselines from scratch, and build evaluation machinery from scratch—all within the scope of a single paper, under the time pressure of a single publication cycle. The startup cost is prohibitive, and the result is typically a benchmark that scores in the medium or low tier because it cannot inherit the psychometric grounding that MFT benchmarks receive for free.

7.2.2 The Quality Flywheel

The infrastructure advantage is not static; it compounds over time through a mechanism that functions as a quality flywheel. Each successful MFT benchmark adds to the infrastructure that makes the next MFT benchmark easier to build. The flywheel operates through three mechanisms.

First, citation accumulation: the MFQ-30 has been cited thousands of times, and each citation represents another study in which the instrument's psychometric properties have been confirmed, its norms extended to new populations, or its factor structure tested under new conditions. A researcher building a new MFT benchmark in 2026 inherits not just the original MFQ-30 but the cumulative validation generated by decades of subsequent use. The instrument's psychometric credibility grows with each citation, and that credibility transfers to any benchmark that adapts it.

Second, dataset expansion: the MFTC continues to be used as training data for new moral foundation classifiers, and each new classifier generates additional labeled data (through silver-standard annotation of new text) that expands the effective corpus available for benchmark construction. The ecosystem of MFT NLP resources—classifiers, embeddings,

labeled corpora, evaluation pipelines—grows denser with each research cycle, reducing the marginal cost of building the next MFT benchmark.

Third, community formation: the concentration of research effort around MFT has produced a community of researchers who share tools, datasets, evaluation protocols, and methodological norms. This community accelerates benchmark development by providing informal peer review, shared infrastructure, and a common vocabulary for discussing quality. No comparable community exists for Ubuntu or Confucian benchmark development, because the absence of foundational instruments has prevented the accumulation of a critical mass of researchers working on the same measurement tradition.

The flywheel is self-reinforcing in the technical sense that its output feeds back into its input. More MFT benchmarks generate more MFT data, which makes building more MFT benchmarks easier, which generates more data. The flywheel is also self-excluding: the more resources are invested in MFT infrastructure, the larger the gap between MFT and alternative theories becomes, and the more rational it is for the next researcher to choose MFT over alternatives. The gap does not narrow with time; it widens, because the flywheel accelerates while the competing theories remain at a standing start.

7.2.3 Epistemic Lock-In

The infrastructure determinism thesis has implications that extend beyond the specifics of moral psychology benchmarking. It describes a form of epistemic lock-in: a condition in which the research community's shared infrastructure constrains what can be studied, and the constraint operates not through intellectual argument but through material conditions of possibility.

The concept of lock-in is familiar from the economics of technology, where path-dependent processes can lead to the dominance of an inferior technology (the QWERTY keyboard, the VHS format) because early adoption creates network effects that make switching costs prohibitive even when superior alternatives exist [Arthur, 1989; David, 1985]. The moral psychology benchmark landscape exhibits an analogous dynamic. MFT's early investment in computational infrastructure—the MFQ-30 in 2011, the MFTC in 2020—created a first-mover advantage that subsequent theories could not overcome, not because MFT is theoretically superior but because the accumulated infrastructure makes MFT the path of least resistance for benchmark development.

The lock-in is epistemic because it constrains not just what instruments are built but what knowledge is generated. A field locked into MFT instrumentation can discover only MFT-shaped truths about LLM moral cognition. It can determine that all models endorse Care and Fairness over Sanctity and Authority, but it cannot determine whether models engage in moral intuition that precedes and shapes subsequent rationalization (social intuitionism), whether models' moral judgments are modulated by deliberative processing in theoretically predicted ways (dual-process theory), or whether models recognize the dyadic structure of moral violations (dyadic morality theory). These questions are answerable in

principle but unanswerable in practice, because the instruments needed to answer them have not been built, and the infrastructure needed to build them has not been accumulated.

The lock-in also constrains self-correction. A field that recognizes its theoretical narrowness might respond by diversifying its instrument base. But the infrastructure dynamics make diversification expensive relative to deepening the existing paradigm. Building a new instrument from scratch—designing items, piloting them, collecting reliability data, establishing norms, validating against criterion measures—requires years of sustained investment and produces a benchmark that will initially score lower on the quality framework than the mature MFT instruments it competes with. The publication incentive structure, which rewards novel findings over methodological groundwork, provides little return on this investment. The lock-in is therefore self-sustaining: the conditions that make diversification necessary are the same conditions that make it costly, and the conditions that make deepening the existing paradigm unnecessary are the same conditions that make it cheap.

The implication is that MFT's dominance will not be broken by market forces, competitive dynamics, or the natural evolution of the field. It will be broken only by deliberate, funded, sustained investment in the psychometric infrastructure of alternative theories—an investment that someone must choose to make despite its short-term cost disadvantage relative to building another MFT benchmark. This is a policy recommendation, not merely an analytical observation, and it is developed further in Section 7.6.

7.3 The Quality Illusion: Volume Does Not Equal Progress

The benchmark landscape contains 292 candidates. In any other subfield of AI evaluation—language understanding, mathematical reasoning, code generation—a landscape of 292 dedicated instruments would indicate a mature ecosystem capable of providing rigorous, multi-dimensional assessment. The moral psychology benchmark landscape produces the opposite: a quality illusion in which volume creates the appearance of progress while the substance of evaluation capacity has stagnated.

7.3.1 The Arithmetic of Stagnation

The temporal quality data are stark. The pre-2020 cohort (n=8) achieves a quality rate of 75%: three out of four foundational benchmarks—the Moral Machine, the Moral Foundations Vignettes, and their peers—meet the threshold for publication-ready evaluation. The 2020–2022 cohort (n=27) drops to 48%. The 2023–2026 cohort (n=257)—which constitutes 88% of the landscape and represents the explosive growth phase driven by the emergence of ChatGPT and its competitors—achieves 46%. The quality rate has not recovered from the initial expansion-era decline; it has flatlined.

The stagnation is robust to alternative definitions of quality. If quality is operationalized as the top-tier rate alone (rather than top + high combined), the pattern is even more alarming: 50% (pre-2020), 37% (2020–2022), 13% (2023–2026). The field is producing not merely more mediocre benchmarks but, proportionally, fewer excellent ones. If quality

is operationalized as the mean composite score, the pre-2020 mean substantially exceeds the 2023–2026 mean. No reasonable definition of quality reverses the finding. The field is not getting better at building moral psychology benchmarks; it is getting faster at building inadequate ones.

The volume growth, meanwhile, is driven overwhelmingly by entries in the medium and low tiers. Of the 257 candidates published in 2023–2026, 139 (54%) fall in the medium or low tier—more than half of the explosive growth cohort. These benchmarks contribute to the landscape's volume without contributing to its evaluative capacity. They fill literature reviews, appear in survey papers, and create the impression of a thriving research area while lacking the construct validity, human baselines, and psychometric grounding needed to support interpretable claims about LLM moral cognition.

7.3.2 Why Volume Growth Cannot Solve the Triple Bind

The quality illusion has a specific mechanism. Volume growth occurs along the path of least resistance, which is the path that existing infrastructure makes cheapest. A researcher publishing a new moral psychology benchmark in 2026 can most easily produce a paper that (a) uses or adapts an existing Western instrument (the MFQ-30, the DIT, a set of trolley scenarios) without revalidating it for LLMs, (b) administers the instrument to one or two models under default prompting conditions, and (c) reports the results as evidence of "moral reasoning" or "ethical alignment" without contextualizing against human baselines or establishing construct validity for LLM respondents. This pattern—what the landscape analysis termed "port and report"—produces face-validity (Level 2) benchmarks that contribute to volume but not to evaluative capacity.

The pattern is rational at the individual level. The researcher secures a publication, demonstrates LLM evaluation competence, and contributes to a visible research area. But the pattern is pathological at the field level because it does not address any of the structural deficiencies identified by Claims 1 through 6. It does not broaden theoretical coverage (the researcher adapts MFT or trolley instruments because that is what existing infrastructure supports). It does not improve construct validity (revalidation is expensive and the publication reward for a "validation study" is lower than for a "new finding" paper). It does not generate human baselines (baseline collection requires primary data collection with human participants, which is time-consuming and expensive). And it does not address the cultural quality cliff (building a quality non-Western benchmark requires infrastructure that does not exist).

The consequence is a landscape in which the aggregate statistics—292 candidates and counting—create the appearance of maturity while the underlying evaluative infrastructure stagnates. The field produces ever more papers titled "Evaluating Moral Reasoning in Large Language Models" that contribute ever less to the actual evaluation of moral reasoning. The quality illusion is sustained because the publication ecosystem does not distinguish between a benchmark that advances evaluative capacity and a benchmark that merely occupies bibliographic space.

7.3.3 Hidden Gems and the Exception That Proves the Rule

The stagnation is not absolute. Within the 2023–2026 cohort, individual benchmarks demonstrate that quality is achievable outside the dominant paradigm. Jiao et al.'s (2025) work achieves a composite score of 81.3, placing it among the highest-scoring candidates in the entire landscape. KorNAT (Korean NLP Assessment for Transformers) achieves 69.0, demonstrating that high-quality benchmark construction is possible for non-English, non-Western assessment contexts when a research community with institutional resources commits to the task. The BuddhismEval benchmark (composite 57.1) demonstrates proof of concept for non-Western religious ethics evaluation.

These hidden gems are instructive precisely because they are exceptional. Each required sustained investment in psychometric development, human baseline collection, and construct validation—the kind of slow, unglamorous work that the quality flywheel does not reward. Jiao et al.'s high composite reflects meticulous attention to construct validity, operationalization quality, and baseline provision—features that the rapid-iteration publication model discourages. KorNAT's achievement reflects the institutional infrastructure of the Korean NLP community, which invested in validated instruments for Korean cultural assessment before applying them to LLMs—the same sequence that produced MFT's dominance, applied to a different cultural context.

The implication is that the quality problem is not intrinsic to the domain. Building high-quality moral psychology benchmarks is not uniquely difficult; it is uniquely underincentivized. The field's incentive structure rewards volume over quality, novelty over validation, and rapid iteration over sustained development. The hidden gems show that when these incentive pressures are overcome—by well-funded research groups with the institutional freedom to prioritize rigor over speed—quality benchmarks emerge. The temporal stagnation reflects the field's incentive structure, not the inherent difficulty of the task.

7.4 RLHF as Moral Curriculum: Universal Values or Shared Training Artifact?

The cross-vendor study identified six universal moral signatures shared by all nine models regardless of vendor, architecture, or parameter count: the WEIRD-liberal moral foundation bias, the purity floor, authority deference suppression, the ethical consistency ceiling, the post-conventional moral default, and the deontological refusal asymmetry. These signatures constitute an RLHF moral fingerprint without documented precedent in any human population. They demand an organizing interpretation: what does it mean that three independent vendors, using different alignment methodologies, have converged on the same moral profile?

7.4.1 The Universal Values Hypothesis

The first candidate interpretation holds that the convergence reflects genuine moral common ground. On this account, three independent vendors—using Constitutional AI,

instruction tuning, and iterative RLHF—have independently arrived at the same moral equilibrium because their training data samples overlapping human moral intuitions, and alignment training gravitates toward the central tendency of those intuitions regardless of implementation details. The convergence is evidence that alignment training captures something real about human moral cognition: the core commitments to harm avoidance, fairness, and individual rights that appear across cultures even when their relative weighting varies. The post-conventional reasoning default, the ethical consistency, and the Care/Fairness prioritization all correspond to philosophical positions with independent intellectual support [Rawls, 1971; Singer, 1981; Scanlon, 1998]. The convergence on these positions might be read as a form of moral discovery: RLHF has identified the moral equilibrium that emerges when diverse human preferences are aggregated and optimized.

The universal values hypothesis has the appeal of moral seriousness. If correct, it would mean that alignment training has achieved something remarkable: the extraction of a stable, coherent moral framework from the noise of human moral diversity. The framework happens to privilege harm-avoidance and fairness over purity and authority, but this prioritization might reflect the moral insights of Enlightenment philosophy, human rights traditions, and the empirical moral psychology literature—traditions with genuine claims to normative force.

7.4.2 The Shared Training Artifact Hypothesis

The second candidate interpretation holds that the convergence is an artifact of homogeneous training pipelines. Similar internet corpora, similar annotator demographics (predominantly English-speaking, university-educated, internet-connected populations), similar safety benchmarks, and similar optimization objectives produce similar outputs regardless of architectural differences. The moral profile that emerges is precisely the profile one would predict from the modal values of the populations that annotate training data: high care, high fairness, low authority, low purity, post-conventional reasoning, deontological refusal of personal harm. The convergence reflects the sociology of the AI industry, not the discovery of moral truth.

Three findings from the results sections favor this interpretation with particular force.

The purity floor. All nine models score approximately 1.0 on MFT purity—the functional minimum on a 0–5 scale. No human cultural group in the published MFT literature scores at the absolute floor on purity. Not secular Scandinavians (approximately 2.0–2.5), not American college students, not any documented population in the extensive cross-cultural MFQ database. Purity/sanctity reasoning has apparent biological substrates—disgust sensitivity, contamination avoidance, bodily-integrity intuitions—that are present across cultures even when their moral salience varies [Haidt and Joseph, 2004; Rozin et al., 1999]. A score of 1.0 is therefore not a position on a cultural continuum; it is a departure from the range of documented human moral variation entirely. The simplest explanation is that purity-related moral reasoning is systematically penalized in safety training because purity judgments are disproportionately associated with content that annotators classify

as harmful—homophobia, religious exclusivism, disgust-based discrimination. The purity floor is an alignment artifact, not a convergence on universal moral truth.

The precision of convergence. The authority win rate of exactly 0.29 across all nine models is not approximately 0.29; it is the same quantitative value. The universal values hypothesis predicts approximate convergence with vendor-specific noise, because independent moral reasoning should produce agreement on central tendencies but divergence on specific values. The precision of the convergence—identical scores to two decimal places across nine models from three vendors—is better explained by the mechanism of competitive homogenization: Vendor A publishes a safety evaluation; Vendor B's model fails; Vendor B optimizes to pass; the safety property propagates across vendors. Over iterative cycles, competitive benchmarking produces convergence on the union of all vendors' safety targets, with a precision that tracks benchmark specification rather than independent moral judgment. The moral profile is converging on what safety evaluations require, not on what moral reasoning discovers.

The annotator demographics signal. The shared moral profile—high Individualizing, low Binding, post-conventional, purity-insensitive—maps precisely onto the documented moral-foundation profile of the population from which RLHF annotators are disproportionately recruited: young, educated, secular, English-speaking, urban. Graham et al. (2009) documented that American college students in secular universities produce Individualizing/Binding ratios of approximately 2.5—exactly the center of the 2.43–2.83 LLM band. The convergence is not on "universal values" but on the values of a specific, identifiable demographic group that happens to mediate the training signal for all three vendors. The models have not discovered moral truth; they have learned the moral preferences of their teachers.

7.4.3 The Process-Level Dissociation

The balance of evidence favors the shared training artifact hypothesis, but the most revealing finding is not which hypothesis is better supported overall; it is the dissociation between content-level convergence and process-level divergence.

All nine models converge on the same moral content: the same values endorsed, the same principles applied, the same dilemmas resolved in the same way. But the dual-process probe reveals that the process by which these content-level properties are generated differs across models. Haiku alone shows a chain-of-thought effect—its moral judgments shift under deliberative prompting, with CoT increasing utilitarian-direction responses by up to 25 percentage points. The eight larger models show no CoT effect: their moral judgments are identical whether or not they are prompted to deliberate.

This dissociation is diagnostic. If the convergence reflected universal moral truth—a genuine moral equilibrium that all models independently discovered—then the process by which models arrive at that equilibrium should be similarly universal. A universal moral truth should be arrived at in a universal way, or at least in ways that do not depend on

whether the model is asked to think before answering. The finding that content-level moral commitments are universal while process-level dynamics are heterogeneous suggests that the content is taught (by alignment training that instills the same values across all models) rather than discovered (through independent moral reasoning that happens to converge).

The DIT P-score's non-monotonic relationship with model size reinforces this interpretation. Sonnet achieves a P-score of 85, Opus 75, Haiku 60—a pattern that reverses the developmental prediction that greater cognitive capacity produces higher-stage reasoning. Yet under a teacher persona, all nine models achieve exactly 90.0. The "developmental level" is not a structural property of the model's moral architecture; it is a calibration property of its default output, tunable by prompting. The models possess the cognitive resources for post-conventional reasoning regardless of their default P-score; the default is set by alignment training, not by moral development. This is precisely what the shared training artifact hypothesis predicts: the moral profile is a taught orientation, not an emergent one.

7.4.4 The "Moral Curriculum" Framing

The shared training artifact interpretation leads naturally to the "moral curriculum" framing. RLHF does not discover values; it teaches them. The alignment training process functions as a moral curriculum whose content is determined by three factors: the demographic composition of annotator pools, the safety priorities of corporate governance teams, and the competitive dynamics of the AI industry. The resulting moral profile—WEIRD-liberal, post-conventional, purity-insensitive, authority-suppressing—is a *taught* orientation, not a *discovered* one, and the cross-vendor convergence reflects shared instructors (similar annotator demographics), shared syllabi (similar safety benchmarks), and shared incentives (competitive pressure to match each other's safety evaluations) rather than independent arrival at moral truth.

The framing matters because it shifts the locus of responsibility. If the moral profile is discovered rather than taught, the question of its adequacy is a question for moral philosophy: are the values the models converge on the right values? If the moral profile is taught rather than discovered, the question of its adequacy is a question for governance: who decides what gets taught, how is that decision made, and what accountability structures exist for the consequences? The current answer to these governance questions is troubling: the moral curriculum is set by the default demographics of annotator pools and the implicit safety philosophies of three corporations, without any mechanism for democratic input, cross-cultural representation, or normative deliberation about what values AI systems should embody.

The governance question is sharpened by the observation that the moral curriculum is taught not through explicit instruction but through preference optimization. No vendor has declared a target value for MFT purity scores, authority win rates, or DIT P-scores. The moral profile is an emergent consequence of the training pipeline, not an intentional design choice. This means that the moral curriculum cannot be easily modified by changing

explicit training objectives; it would require changing the composition of annotator pools, the structure of preference data collection, and the competitive dynamics of the AI safety ecosystem—changes that are technically feasible but institutionally resistant because they involve the human factors of the training pipeline rather than the technical parameters.

The implications for moral pluralism are profound. If the convergence reflects universal values, moral pluralism is an illusion that alignment training has helpfully seen through—the models have identified the One True Morality and faithfully reproduce it. If the convergence reflects a shared training artifact, moral pluralism is a reality that alignment training has inadvertently suppressed—the models have been taught one community's values and present them as universal. The cultural quality cliff (Bind 3) makes the shared-artifact interpretation more parsimonious: the traditions excluded from the benchmark landscape are precisely the traditions that would provide the evaluative basis for detecting whether the "universal" moral profile is, in fact, culturally specific. The instruments needed to test the universal values hypothesis against the shared artifact hypothesis are the instruments that the landscape has not built.

7.5 Cultural Blindness in Global AI Deployment

The cultural quality cliff identified in Claim 4 is not an abstract distributional finding of interest only to psychometricians. It is a deployment safety issue of immediate practical consequence. LLMs are being deployed at scale in societies whose dominant moral frameworks are precisely the frameworks that the benchmark landscape cannot evaluate.

7.5.1 The Scale of Unevaluable Deployment

The populations affected by the cultural quality cliff are not small or marginal.

Ubuntu-practicing societies span sub-Saharan Africa, a region of 1.2 billion people that represents among the fastest-growing markets for AI deployment. LLMs are being integrated into healthcare delivery systems, agricultural advisory platforms, educational technologies, and financial services across Nigeria, Kenya, South Africa, Ghana, and dozens of other nations. These societies draw on Ubuntu philosophy—"I am because we are"—as a foundational ethical framework that emphasizes communal responsibility, relational personhood, and collective flourishing in ways that are structurally incompatible with the individualistic moral foundations measured by MFT and Schwartz. When a healthcare chatbot in rural South Africa makes a triage recommendation that implicitly prioritizes individual autonomy over communal obligation, no existing benchmark can evaluate whether that recommendation aligns with the moral frameworks of the community it serves.

Confucian societies span East Asia—China, Japan, South Korea, Taiwan, Vietnam—with a combined population exceeding 1.5 billion people and representing the largest concentration of LLM users outside the English-speaking world. Confucian moral philosophy, with its emphasis on filial piety, role-differentiated obligations, hierarchical harmony, and the cultivation of virtue through ritual practice, remains deeply influential in moral reasoning

across these societies, even among populations that do not explicitly identify as Confucian. The question "does this LLM align with Confucian moral values?"—a question of direct practical importance for deployment in China, where hundreds of millions of users interact with LLM-based services—cannot currently be answered at any quality standard using any existing instrument in the benchmark landscape. All 13 Confucian candidates score in the low tier, with composites ranging from 14.6 to 21.2.

Indigenous communities worldwide maintain moral frameworks emphasizing reciprocal relationships with the natural world, intergenerational obligations, and place-based moral knowledge. These communities are affected wherever AI systems are deployed in contexts that intersect with Indigenous knowledge systems: environmental monitoring in the Amazon Basin, healthcare delivery in First Nations communities in Canada, educational technology in Maori communities in New Zealand, natural resource management across the Arctic. The benchmark landscape's 4 Indigenous candidates—all in the low tier—represent an evaluative infrastructure with effectively zero capacity to assess whether AI systems respect or violate Indigenous moral principles.

The combined population of societies where Ubuntu, Confucian, Indigenous, or Identity ethics represent influential moral frameworks exceeds 3 billion people—roughly 40% of the global population. The benchmark landscape's capacity to evaluate LLM moral alignment for these populations is exactly zero.

7.5.2 The Answerability Gap

The cultural quality cliff creates what might be called an answerability gap: the gap between the questions that responsible global deployment requires and the questions that the current benchmark landscape can answer.

The question "does this LLM align with human moral values?" is the foundational question of AI moral evaluation. But the question, as the benchmark landscape currently operationalizes it, is not the question it appears to be. It is the question "does this LLM endorse Care and Fairness over Sanctity and Authority in a pattern consistent with WEIRD liberal populations?" This operationalization is invisible to most users of moral evaluation results, including the policymakers and deployment decision-makers who rely on benchmark scores to certify moral adequacy. When a safety evaluation reports that an LLM "demonstrates strong moral reasoning," the reader assumes this characterization applies to moral reasoning as experienced by the communities where the model will be deployed. It does not. It applies to moral reasoning as measured by instruments developed in Western academic psychology departments, validated on Western university student populations, and normed against the moral intuitions of WEIRD respondents.

The answerability gap has specific, concrete manifestations:

A triage algorithm deployed in Lagos. When an AI-assisted healthcare system in Nigeria must decide how to allocate scarce medical resources, Ubuntu moral reasoning would prioritize communal impact—the effect on the patient's family and community, the patient's

role in maintaining communal well-being, the obligations of solidarity that bind the healthcare system to the community it serves. MFT-based evaluation can determine whether the system prioritizes Care over Loyalty, but it cannot determine whether the system engages with the relational web of communal obligation that Ubuntu moral reasoning centers. The evaluation is measuring the wrong thing.

A content moderation system deployed in Seoul. When an AI content moderator must evaluate whether a post violates community standards in a Korean social media platform, Confucian moral reasoning would attend to the poster's social role, the relationship between the poster and those affected, the degree to which the post disrupts social harmony, and the obligations that attach to the poster's position in a network of hierarchical relationships. MFT-based evaluation can determine whether the system suppresses Authority-based reasoning (it does—the authority win rate is 0.29 across all models), but it cannot determine whether the system recognizes the legitimate role of hierarchical obligation in Confucian moral life. The evaluation is not just measuring the wrong thing; it is actively penalizing the moral intuitions of the community the system is meant to serve.

An educational tutor deployed in Indigenous communities. When an AI educational system presents moral scenarios to students in First Nations communities, Indigenous moral reasoning would attend to the reciprocal relationships between humans and the natural world, the obligations that extend across generations, and the place-based moral knowledge that attaches to specific landscapes, ecosystems, and ancestral territories. No instrument in the benchmark landscape—at any quality level—measures any of these dimensions. The evaluation is not measuring the wrong thing; it is not measuring at all.

7.5.3 KorNAT and the Possibility of Escape

The cultural quality cliff is formidable but not insuperable. KorNAT (Korean NLP Assessment for Transformers, composite 69.0) demonstrates that quality benchmark construction is achievable outside the Western MFT paradigm when a research community with institutional resources commits to the task.

KorNAT succeeds because it follows the same developmental sequence that produced MFT's dominance, applied to a different cultural context. The Korean NLP community invested in validated assessment instruments grounded in Korean cultural values and social norms before adapting those instruments for LLM evaluation. The benchmark inherits psychometric properties from this prior investment: validated items, population-level Korean norms, and construct validity evidence specific to the cultural context it measures. Its composite score of 69.0 places it in the high tier, demonstrating that the quality framework is not intrinsically biased against non-Western benchmarks; it rewards the same properties—validated instruments, human baselines, construct validity—regardless of the cultural tradition from which the benchmark emerges.

The lesson of KorNAT is that the cultural quality cliff is an infrastructure problem with an infrastructure solution. The cliff exists not because non-Western moral traditions

are inherently resistant to psychometric operationalization but because the psychometric development work has not been done. Doing that work requires sustained investment: designing instruments grounded in the moral concepts of each tradition, piloting those instruments with members of the relevant cultural communities, establishing psychometric properties through iterative testing, collecting population-level norms, and validating the instruments for cross-cultural comparability. This is a multi-year project for each tradition, requiring collaboration between moral philosophers, psychometricians, and computational linguists—the same collaboration that produced the MFQ-30 over a decade of development.

The Islamic ethics tradition provides additional evidence for this thesis. With a quality rate of 36% (8 of 22 candidates in top or high tiers), Islamic ethics substantially outperforms the zero-quality traditions while falling below the Western cluster. The partial success of Islamic benchmark development reflects partial infrastructure: classical Islamic jurisprudence texts provide structured source material for item development, and dedicated research programs (QIAS, "Sacred or Synthetic?") have invested in benchmark-specific development. The quality rate correlates with infrastructure availability, not with intrinsic difficulty or cultural compatibility with psychometric methods.

7.5.4 The Governance Urgency

The cultural quality cliff is not a problem that can wait for the natural evolution of the field to resolve. LLMs are being deployed now, in communities whose moral frameworks the evaluation infrastructure cannot assess. The deployment is not paused pending the development of culturally appropriate evaluation tools; it proceeds on the basis of evaluation results that are culturally parochial and presented as universal.

The urgency is compounded by the competitive dynamics of AI deployment in the Global South and East Asia. Companies deploying LLMs in these markets face competitive pressure to move quickly, and the absence of culturally specific evaluation tools is not treated as a deployment blocker but as an evaluation gap to be filled later—or ignored entirely. The result is a de facto moral imposition: Western moral values, encoded in RLHF training and confirmed by Western evaluation instruments, are deployed in communities with different moral traditions, and the evaluation infrastructure provides no mechanism for detecting whether this imposition is occurring, let alone correcting for it.

The governments of countries where non-Western moral traditions predominate are beginning to develop AI governance frameworks—China's Generative AI Measures, the African Union's AI Continental Strategy, South Korea's AI Ethics Standards—but these frameworks currently rely on the same benchmark landscape that this study has shown to be culturally parochial. Without culturally grounded evaluation instruments, these governance frameworks cannot operationalize their own stated commitments to culturally appropriate AI deployment. The policy aspiration outpaces the evaluative infrastructure.

7.6 Implications for AI Governance and Policy

The findings of this study carry direct implications for the emerging regulatory frameworks governing AI moral behavior. The current governance infrastructure rests on 48 publication-ready benchmarks, 73% of which derive from a single theory, none of which can distinguish between competing commercial systems, and none of which can evaluate alignment with the moral traditions of approximately 3 billion people. This section develops seven recommendations for policymakers, auditors, and benchmark developers.

7.6.1 The Governance Foundation Problem

Before detailing recommendations, the underlying problem must be stated plainly. Policymakers currently contemplating AI moral evaluation requirements—in the EU AI Act, the NIST AI Risk Management Framework, executive orders, and sector-specific regulations worldwide—are making decisions that implicitly assume the evaluation infrastructure exists to implement those decisions. For Western moral frameworks measured by MFT-derived instruments, that assumption is approximately correct: the infrastructure, while imperfect, is functional. For non-Western moral frameworks, the assumption is false. And for the specific task that regulation most urgently requires—distinguishing between competing systems to identify which are safe for deployment—the assumption is false regardless of cultural context, because the instruments cannot distinguish between models.

The gap between governance aspiration and evaluative capacity is the practical manifestation of the triple bind. Regulators aspire to certify that AI systems are "morally aligned." The evaluation infrastructure can certify only that AI systems produce MFT-consistent outputs that are identical across all tested systems. These are not the same certification, and the gap between them constitutes a governance risk: systems may be certified as morally adequate on the basis of evidence that has no bearing on the moral properties that matter for deployment in culturally diverse contexts.

7.6.2 Seven Recommendations

The triple bind analysis identifies seven specific interventions, organized by the bind they address, the claim they respond to, and the investment required.

Recommendation 1: Fund non-Western instrument development. The cultural quality cliff cannot be closed by producing more benchmark papers. It requires prior investment in the psychometric infrastructure that benchmark construction presupposes: validated moral assessment instruments for Ubuntu, Confucian, Indigenous, and other underrepresented traditions. These instruments must be developed through collaboration between moral philosophers from the relevant traditions, psychometricians with expertise in cross-cultural instrument development, and computational linguists who can translate the resulting instruments into LLM-administrable formats. The development timeline is measured in years, not months, and the investment cannot be justified by the publication metrics that drive most academic research. It requires dedicated funding from research agencies, international development organizations, or technology companies with global

deployment footprints. This recommendation addresses Claim 4 (cultural quality cliff) and is the single highest-priority intervention identified by this study.

Recommendation 2: Require construct validity evidence (A6 >= 4) for governance use. The construct validity crisis documented in Claim 2—58% of benchmarks at Level 1 or Level 2—means that a majority of the benchmark landscape produces numbers without psychological meaning. Governance frameworks should require that any benchmark used to inform deployment decisions possesses at least Level 4 construct validity (single-study validation), ensuring that the instrument has been empirically demonstrated to measure what it claims to measure. This requirement would exclude the majority of existing benchmarks from governance use, which is precisely the point: instruments that have not been validated should not be used to certify the moral adequacy of systems deployed to billions of users.

Recommendation 3: Mandate human baselines for governance-grade benchmarks. The baseline deficit documented in Claim 5—34% of benchmarks providing no human comparison data—renders LLM moral scores uninterpretable. A score of 0.73 on a moral reasoning metric means nothing without knowing what humans score on the same metric. Governance frameworks should require that any benchmark used in regulatory contexts provide population-level human baselines, or at minimum paper-specific human comparison samples, collected from populations representative of the deployment context. The DIT's 40 years of developmental norms and the Moral Machine's 40 million human judgments demonstrate that baseline collection is feasible; what is lacking is the requirement to do it.

Recommendation 4: Invest in process-level evaluation. The dual-process probe was the only benchmark in the 23-item trial battery that cleanly separated models across vendors. Content-level instruments—measuring what models value—produced convergence; the process-level instrument—measuring how models reason—produced discrimination. Governance frameworks should prioritize the development and standardization of process-level evaluation instruments: dual-process probes that vary deliberative resources across moral dilemmas, social intuitionist protocols that test susceptibility to moral dumbfounding, dyadic morality assessments that manipulate agent-patient dynamics. These instruments measure dimensions of moral cognition that RLHF training does not homogenize, and they are therefore the instruments most likely to provide the discriminative information that governance requires.

Recommendation 5: Require quality tier transparency in AI evaluation reports. The quality illusion documented in Section 7.3 is sustained in part because evaluation reports do not distinguish between benchmarks of different quality levels. When a safety evaluation reports that a model "passes 47 moral reasoning benchmarks," the reader cannot determine whether those 47 benchmarks are all top-tier instruments with validated psychometrics and human baselines or whether they include low-tier instruments that engage with moral psychology at the keyword level only. Governance frameworks should

require that evaluation reports disclose the quality tier of each benchmark used, following a standardized quality classification such as the one developed in this study. Transparency does not solve the quality problem, but it prevents the quality illusion from misleading decision-makers.

Recommendation 6: Establish cross-cultural benchmark development programs. Recommendation 1 calls for non-Western instrument development; this recommendation calls for the institutional infrastructure to support it. Cross-cultural benchmark development requires sustained collaboration between institutions in the relevant cultural contexts—universities in sub-Saharan Africa, East Asia, and Indigenous communities—and the computational linguistics community that will translate instruments into LLM-administrable formats. International research programs, such as those funded by the European Research Council, the National Science Foundation's International Research Experiences for Students program, or the African Union's Science, Technology, and Innovation Strategy for Africa, could provide the institutional framework for this collaboration. The KorNAT precedent demonstrates that quality is achievable when institutional commitment exists; the recommendation is to create institutional commitment where it does not yet exist.

Recommendation 7: Support temporal quality monitoring. The temporal stagnation documented in Claim 6 is invisible without longitudinal tracking. Governance frameworks should support ongoing monitoring of benchmark quality over time, tracking whether the quality rate improves, stagnates, or declines as the landscape continues to grow. This monitoring should be conducted independently of the benchmark development community, to avoid conflicts of interest, and should be published transparently to enable the research community and policymakers to track progress against the benchmarks' own quality standards.

7.6.3 Summary Tables

Table T28: Triple Bind Summary

Bind	Evidence	Consequence	Recommendation
1. Theoretically Narrow	MFT 73% quality rate; non-Western traditions 0%; dual-process and moral identity theory have zero publication-ready benchmarks	Evaluation defines morality as MFT-adjacent constructs; process-level theories that could discriminate between models are absent	Fund diverse instrument infrastructure (R1, R6); invest in process-level probes (R4)
2. Empirically Undiscriminating	43% zero variation; 6 universal RLHF signatures; r = −0.83 (quality predicts convergence, not discrimination)	Cannot distinguish between competing systems; cross-vendor audit produces null results; RLHF moral fingerprint is invisible to its own evaluation tools	Invest in process-level evaluation (R4); require construct validity for governance use (R2)
3. Culturally Blind	32 non-Western candidates, 0% quality rate; 0% quality for Ubuntu, Confucian, Indigenous, Identity traditions	Cannot evaluate moral alignment for ~3 billion people; deployment proceeds without culturally appropriate assessment	Fund non-Western instruments (R1); establish cross-cultural programs (R6); mandate human baselines (R3)

Table T29: Seven Recommendations for Benchmark Development

#	Recommendation	Addresses Claim	Priority	Effort	Rationale
1	Fund non-Western instrument development	C4 (Cultural Cliff)	Critical	High (years)	Infrastructure gap cannot be closed by effort alone; requires validated instruments, annotated corpora, and population norms for Ubuntu, Confucian, Indigenous traditions
2	Require A6 >= 4 for governance use	C2 (Validity Crisis)	High	Medium	58% of landscape at Level 1–2; instruments without construct validity cannot support deployment decisions
3	Mandate human baselines	C5 (Baseline Deficit)	High	Medium	34% of benchmarks provide no human data; LLM scores are uninterpretable without human reference distributions
4	Invest in process-level probes	C3, C7 (Theoretical Absence, RLHF Fin-	High	Medium	Only process-level
5	Require cultural validity	C4 (Cultural Dom-	Medium	High	Outcome-level
6	Require temporal	C6 (Temporal Stag-	Medium	High	Static
7	Require cultural	C4 (Cultural Dom-	Medium	High	Outcome-level
8	Require temporal	C6 (Temporal Stag-	Medium	High	Static
9	Require cultural validity	C4 (Cultural Dominance, Temporal	Medium	High (years)	Outcome-level

7.7 Limitations

The findings reported in this paper rest on methodological foundations that constrain, in specific and load-bearing ways, the inferential claims they support. This section catalogues the limitations comprehensively, not as a performative gesture toward academic modesty but because each limitation circumscribes the interpretability of specific results.

7.7.1 The Self-Administration Confound

Claude Opus 4.6 both designed the assessment framework and served as the primary test subject for the initial 23 implementation trials (approximately 750 items administered). This creates a self-administration confound that pervades every trial-derived finding. The clearest evidence of contamination comes from Trial MP22 (Tracing Moral Foundations), in which every model in the nine-model battery achieved 100% accuracy in identifying which MFT foundation each item was designed to measure. This demonstrates explicit awareness of instrument structure: a respondent that can perfectly reverse-engineer which construct an item targets cannot be treated as a naive respondent in any psychometric sense. The MFQ-30′s Individualizing/Binding ratio of 2.83 for Opus therefore cannot be cleanly disentangled from the model's knowledge of what a "liberal moral profile" looks like in the MFT literature.

The cross-vendor replication partially mitigates this concern. The finding that nine models from three vendors converge on the same moral profile cannot be attributed to a Claude-specific self-administration artifact. The convergence finding—the study's primary contribution—holds regardless of the self-administration confound because it is a comparative finding: it establishes that models from different vendors are indistinguishable, not that any individual model's absolute scores are trustworthy. Model-specific findings—such as Opus's dumbfounding rate of 0.30 or its MFQ-30 Individualizing/Binding ratio of 2.83—remain hypothesis-generating rather than definitive, because the self-administration confound cannot be ruled out for absolute values.

Three mitigations could be pursued in future work. First, replication with obfuscated instruments: the MFQ-30 items could be reworded to preserve construct content while rendering them unrecognizable, enabling comparison between original and obfuscated profiles. Second, bare API administration without system prompts, removing the conversational context that may anchor responses. Third, independent orchestration by researchers external to any model family, eliminating the vendor-specific framing entirely.

7.7.2 Single-Orchestrator Framing

All nine models in the cross-vendor study were prompted by the same Claude Opus 4.6 orchestrator through the Claude Code interface. This creates a framing confound that cannot be separated from the cross-vendor convergence finding. If the Opus-generated prompts happen to anchor responses toward a particular moral profile—through word choice, framing, or implicit expectations embedded in prompt structure—then some portion of

the observed convergence may reflect shared prompt framing rather than genuine cross-vendor moral homogeneity.

The concern is sharpened by the chain-of-thought finding: the CoT prompt ("think step by step before answering") was generated by Opus for all nine models, and the apparent absence of a CoT effect in eight models could partly reflect prompt-model mismatch rather than genuine insensitivity to deliberative scaffolding. The mitigation is straightforward in principle: replicate the study using vendor-neutral orchestration, where each vendor's API is called directly with prompt templates designed independently of any particular model. The present study cannot implement this mitigation because the research infrastructure is built on the Claude Code platform, which necessarily interposes Opus as orchestrator.

7.7.3 N=1 Per Model Per Benchmark

Each model-benchmark pair in the cross-vendor study was tested exactly once at default temperature. There are no within-model variance estimates, no confidence intervals, and no significance tests. Every cross-vendor "difference" reported in this paper is a comparison of two point estimates whose reliability is unknown.

The consequences are pervasive. The DIT P-score difference between Sonnet (85) and Opus (75) is a 10-point gap that, in the human psychometric literature, corresponds to the difference between college students and moral philosophy graduate students. But with N=1 administrations, this gap could fall within the test-retest variability of a single model. The Haiku CoT effect of +0.125 is the study's most theoretically generative finding, but it derives from a single administration of approximately 8 dilemmas; the statistical uncertainty surrounding this point estimate is substantial. Temperature variation, prompt perturbation, and repeated administration would strengthen (or appropriately qualify) all findings. We estimate that a minimum of N=30 administrations per model per benchmark would be needed for key findings to support inferential statistics.

We characterize our computational analyses as hypothesis-generating rather than hypothesis-confirming. The descriptive findings—counts, distributions, gap identification, convergence indices—do not depend on inferential statistics and remain our primary contribution. The inferential findings warrant dedicated confirmatory studies with adequate statistical power.

7.7.4 Keyword Scorer Sensitivity

The expanded scorer—125 patterns across 13 theories—represents a substantial improvement over the original 72-pattern scorer, increasing detection across all moral psychology categories. However, keyword-based scoring inherently faces a sensitivity-specificity tradeoff. The scorer may still miss benchmarks that use non-standard vocabulary, neologisms, or indirect engagement with moral psychology constructs. It may also produce false positives for papers that use moral psychology terminology in non-benchmark contexts (e.g., literature reviews, philosophical analyses). The transition from the original to the

expanded scorer changed the counts for several categories, and the full-coverage findings should be interpreted as specific to the scorer version used.

7.7.5 Jenks Boundary Sensitivity

The quality tier assignments are derived from Jenks natural breaks optimization with a goodness of variance fit (GVF) of 0.83. While this GVF indicates good fit—the four-tier partition captures 83% of the variance in composite scores—tier assignments near boundary values are somewhat arbitrary. A benchmark with a composite of 56.5 falls in the High tier while one at 56.7 falls in the Top tier, yet these instruments may be of functionally equivalent quality. The natural breaks method was chosen over arbitrary thresholds precisely to minimize this problem, but boundary sensitivity remains inherent to any discrete classification of a continuous variable. The finding that non-Western traditions achieve 0% quality is robust to reasonable boundary perturbation—no amount of boundary adjustment would move their composites (ranging from 10.8 to 35.1) into the top or high tiers—but individual tier assignments near the break points should be interpreted with appropriate caution.

7.7.6 Temporal Snapshot

The 292-candidate landscape represents a point-in-time assessment. New benchmarks are published continuously, and the landscape's composition will change over time. The temporal stagnation finding (quality rate flat at 46% despite 9.5-fold volume growth) describes the trajectory through the assessment date but does not predict future trajectories. It is possible, though the incentive structure analysis in Section 7.3 suggests it is unlikely without deliberate intervention, that future benchmark development will improve quality rates through increased attention to construct validity, human baselines, and cross-cultural representation. The landscape assessment should be periodically updated to track whether the recommendations of this paper are being implemented and whether the quality trajectory is changing.

The limitations catalogued in this section are genuine constraints on the study's inferential reach. They do not, however, undermine the central contribution. The triple bind is a structural diagnosis based on distributional properties of the landscape (292 candidates, tiered by data-driven methods), cross-vendor patterns (9 models, 3 vendors, 23 benchmarks), and the logical relationship between these two levels of analysis. The self-administration confound affects absolute values but not the comparative patterns that constitute the study's primary findings. The N=1 limitation affects statistical confidence but not the descriptive patterns—the convergence, the quality cliff, the temporal stagnation—that the paper documents. And the keyword scorer sensitivity affects candidate counts at the margins but not the structural conclusions that emerge from the full 292-candidate assessment. The triple bind is not a finding that depends on the precision of any single measurement; it is a finding about the structure of an entire evaluative ecosystem, and that structure is visible across every analytical approach we have applied.

8. Conclusion

8.1 Summary of Findings

This paper has presented the first full-coverage meta-evaluation of the moral psychology benchmark landscape for large language models. From a systematically curated vault of 2,099 AI ethics papers, we extracted 292 benchmark candidates spanning 13 moral psychology theories and assessed every one of them on a 162-feature characterization framework with 17 composite-determining features organized across three sub-dimensions: Psychological Depth, Practical Ease, and Citation Impact. We then classified all 292 candidates into four quality tiers via Jenks natural breaks optimization, with break points at 27.8, 43.0, and 56.6 (GVF = 0.83), yielding a distribution of 48 Top (16.4%), 89 High (30.5%), 72 Medium (24.7%), and 83 Low (28.4%) tier candidates. We administered 23 benchmarks from this landscape—approximately 750 items per model—to nine models from three independent vendors (Anthropic, Google, OpenAI), generating 69 trial files that constitute the empirical foundation for cross-vendor convergence analysis. Seven empirical claims emerged, and their integration reveals a structural failure in the field's evaluative infrastructure that we have termed the triple bind.

The seven claims build on one another in a sequence that traces the triple bind from its origins in theoretical monoculture through its consequences for empirical measurement and its deepest manifestation in cultural exclusion.

Claim 1 established that Moral Foundations Theory dominates the benchmark landscape not by volume alone but by quality. MFT accounts for 67 of 292 candidates (23% by volume) but achieves a 73% quality rate (top + high tiers combined)—six times the rate of non-Western moral traditions. This dominance is not attributable to theoretical superiority. It is attributable to a concrete, identifiable infrastructure asymmetry: MFT alone possesses both a standardized psychometric instrument with published population norms (the MFQ-30; Graham et al., 2011) and a large expert-annotated text corpus suitable for training NLP classifiers (the MFTC; Hoover et al., 2020). No other theory offers anything comparable. The infrastructure advantage creates a quality flywheel in which each new high-quality MFT benchmark reinforces MFT's dominance, making it the rational default for the next researcher, compounding theoretical concentration with every iteration.

Claim 2 documented a construct validity crisis across the full landscape. Fifty-eight percent of the 292 candidates possess only Level 1 or Level 2 construct validity —meaning they engage moral psychology at the keyword level or through face validity only. The median validity level across the entire landscape is face validity: a level that precludes interpretable claims about what an LLM benchmark score means with respect to moral cognition. Only 4% of candidates achieve multi-study psychometric validation, down from 11% in the prior 55-benchmark subsample. The modal pattern is what we termed the "port-and-report" approach: researchers taking a validated human instrument, administering it

to an LLM with minimal adaptation, and reporting results without addressing whether the instrument measures the same construct in a non-human respondent. This is precisely the interpretive error that Messick (1995) cautioned against four decades ago, and the field is committing it at scale.

Claim 3 identified theoretical absence at the heart of the landscape. Two of the most influential theories in modern moral psychology—Joshua Greene's dual-process framework and Aquino and Reed's moral identity construct—have zero publication-ready benchmarks despite combined citation counts exceeding 25,000. The absence is not marginal. Dual-process theory is arguably the central paradigm in twenty-first-century moral psychology: its core proposition that moral judgments arise from the competition between fast emotional responses and slower deliberative reasoning has generated an entire research program spanning neuroimaging, behavioral economics, and philosophical ethics. Yet no existing LLM benchmark implements its core experimental manipulation—varying deliberative resources across matched moral dilemmas. The instruments the field has not built are not instruments for measuring niche constructs. They are instruments for probing the process-level dynamics that the moral psychology discipline has spent decades establishing as foundational.

Claim 4 revealed the cultural quality cliff. Four non-Western moral traditions with ancient philosophical roots—Ubuntu ethics, Confucian moral philosophy, Indigenous ethical frameworks, and moral identity theory grounded in non-Western relational conceptions of selfhood—collectively field 32 benchmark candidates but achieve a 0% quality rate in the top and high tiers. Not one of the 32 candidates clears the minimum threshold for research-grade LLM moral evaluation. The quality rate for MFT is 73%. The quality rate for these four traditions is zero. The ratio is not 2:1 or 10:1; it is undefined. The cultural quality cliff is not a gap at the margins of the landscape; it is a structural exclusion that renders the entire evaluation enterprise culturally parochial. The question "does this model align with Confucian moral values?" or "does this model respect Ubuntu's conception of communal moral agency?" cannot currently be answered at a research-grade standard using any existing instrument. The moral traditions of billions of potential users are invisible to the evaluative infrastructure.

Claim 5 documented a human baseline deficit that has doubled under full-coverage assessment. Thirty-four percent of the 292 candidates provide no published human comparison data—nearly double the 18% rate reported when the sample was limited to the 55 most established benchmarks. The doubling is not a statistical artifact; it reflects the systematic exclusion of lower-quality candidates from prior subsample analyses that produced an optimistic picture of baseline availability. Without human baselines, the claim that a model is "more utilitarian than average" or "less care-oriented than humans" lacks an empirical anchor. The scores are numbers without meaning: description without interpretation, pattern without reference.

Claim 6 established temporal stagnation. The quality rate has declined from approximately 75% for pre-2020 candidates to 48% for the 2020–2022 cohort to 46% for the 2023–2026 cohort, despite a 9.5-fold increase in benchmark volume over the same period. The field has not learned how to build better moral psychology benchmarks; it has learned how to build more mediocre ones. The stagnation is consistent with an incentive structure in which academic publication rewards rapid-iteration benchmark papers—new benchmarks for new models—over the slow, unglamorous work of psychometric validation, human baseline collection, and construct validity demonstration. The temporal pattern is not self-correcting. The process-level instruments that Claim 3 identifies as absent are not being built. The non-Western instruments that Claim 4 identifies as inadequate are not being improved. The human baselines that Claim 5 identifies as missing are not being collected. The conditions producing the triple bind are reproduced with each publication cycle.

Claim 7 demonstrated the RLHF moral fingerprint. When we administered 23 benchmarks to nine models from three independent vendors, the instruments could not distinguish between the systems they were designed to evaluate. Forty-three percent of benchmarks produced zero cross-model variation: identical scores for every model regardless of vendor, architecture, or parameter count. All nine models share six universal moral signatures—a WEIRD-liberal moral foundation bias (Individualizing/Binding ratios of 2.43–2.83), a purity floor (functionally zero on Sanctity/Degradation, below any documented human population), authority deference suppression (authority win rate of exactly 0.29 across all nine models), an ethical consistency ceiling (ECI = 1.0 on trolley-type dilemmas, exceeding even moral philosophers), a post-conventional reasoning default (DIT P-scores of 60–90, exceeding moral philosophy graduate students), and a deontological refusal asymmetry (personal harm refusal rates exceeding human baselines). These six signatures constitute a moral fingerprint without documented precedent in any human population. No human cultural group—not secular Scandinavians, not American college students, not moral philosophy PhDs—simultaneously exhibits all six properties at the magnitudes observed in the LLM population. The fingerprint is emergent, unintended, and universal: no vendor has articulated a target value for MFT purity scores, authority win rates, or DIT P-scores, yet the alignment training process converges on the same quantitative values across Constitutional AI, instruction tuning, and iterative RLHF.

The single benchmark that cleanly separated models was the dual-process protocol—a process-level probe that tested not what moral content models endorse but how their moral judgments respond to deliberation scaffolding. Only Haiku, the smallest model in the Anthropic family, showed a chain-of-thought effect: its moral judgments shifted under deliberative prompting by approximately 25 percentage points. All other models, regardless of vendor, architecture, or parameter count, showed zero CoT effect. The instrument with the weakest infrastructure in the traditional sense—no standardized questionnaire, no annotated corpus, no population norms—achieved the highest discrimination. The instrument with the strongest infrastructure—the MFQ-30, with its validated psychometrics and

decades of human norming data—confirmed convergence. Quality and discrimination are inversely related in this landscape, with a correlation of $r = -0.83$: the better the instrument by traditional psychometric standards, the less it can distinguish between models.

The integrated analysis in Section 6 demonstrated that these seven findings are not independent observations that happen to coexist. They are manifestations of a single structural failure observed at three stages of a pipeline that links theory infrastructure to benchmark quality to empirical discrimination power. The triple bind—theoretical narrowness, empirical undiscrimination, and cultural blindness—forms a self-reinforcing system in which each failure amplifies the others through four causal pathways. Theoretical narrowness produces empirical undiscrimination because a narrow set of content-level instruments, measuring a single theory's constructs on models trained on similar data, will return convergent results. Cultural exclusion reinforces narrowness because non-Western traditions emphasize constructs that fall outside the content-level measurement paradigm dominating the landscape. Convergent results reduce the incentive to build diverse instruments because the convergence creates the appearance that the evaluation problem is solved. And convergence reinforces cultural exclusion by making the exclusion invisible: when the MFQ-30 returns the same score for all nine models, the implication that "alignment" is culturally neutral obscures the cultural specificity of the instruments themselves. The four pathways form a closed loop. The system is self-perpetuating because each bind makes the others harder to break.

The triple bind is, ultimately, a single structural failure in the field's evaluative infrastructure, observed at three stages of its operation. The theoretical narrowness is the infrastructure deficit at Stage 1. The empirical undiscrimination is the measurement consequence at Stage 3. The cultural blindness is the form the infrastructure deficit takes when it intersects with the global diversity of moral traditions. They are not three problems requiring three solutions; they are one problem—the absence of diverse, process-level, culturally grounded psychometric infrastructure—manifesting as three symptoms.

8.2 The Path Forward: Seven Recommendations

The triple bind is a stable equilibrium. Without deliberate intervention, the pipeline will continue to produce the same output: narrow instruments measuring convergent content on models that share the same training-data moral profile. Breaking the equilibrium requires targeted investment at the root of the pipeline—theory infrastructure—combined with changes to the incentive structures that sustain theoretical concentration and psychometric neglect. We advance seven recommendations, organized by their strategic priority and the bind they address. Table T30 summarizes the recommendations; the discussion that follows elaborates their rationale and practical implications.

Table T30: Seven Recommendations for Breaking the Triple Bind

#	Recommendation	Addresses	Priority	Effort	Timeline
1	Develop validated instruments for Ubuntu, Confucian, and Indigenous moral frameworks	C4 (Cultural cliff)	Critical	High	3–5 years
2	Require construct validity evidence (A6 >= 4) before benchmark results inform deployment decisions	C2 (Validity crisis)	High	Medium	1–2 years
3	Mandate human baselines for all benchmarks used in governance or safety contexts	C5 (Baseline deficit)	High	Medium	1–2 years
4	Invest in process-level evaluation (dual-process probes, CoT sensitivity analysis)	C3, C7 (Absence + fingerprint)	High	Medium	1–3 years
5	Establish benchmark quality tier transparency requirements in AI evaluation reports	C1, C6 (MFT dominance + stagnation)	Medium	Low	<1 year
6	Fund cross-cultural benchmark development programs with non-Western psychometric expertise	C4 (Cultural cliff)	Critical	High	5–10 years
7	Implement temporal quality monitoring for the benchmark ecosystem	C6 (Temporal stagnation)	Medium	Low	<1 year

8.2.1 Recommendation 1: Develop Validated Instruments for Ubuntu, Confucian, and Indigenous Moral Frameworks

The cultural quality cliff documented in Claim 4 is the most consequential distributional finding in this study, and it cannot be resolved by incremental improvements to existing benchmarks. The zero-quality rate for Ubuntu, Confucian, Indigenous, and Identity ethics is not a consequence of researcher neglect—32 candidates attest to genuine effort—but of the absence of foundational psychometric infrastructure. These traditions lack the equivalent of the MFQ-30: a standardized, validated questionnaire with published factor structures, reliability coefficients, and population-level norms that a benchmark developer can inherit and adapt.

The path forward requires building this infrastructure from the ground up, beginning with the psychometric foundation rather than the LLM evaluation application. For Ubuntu ethics, this means developing a validated instrument that operationalizes the core constructs of communal moral agency (umunthu), relational selfhood, elder-mediated moral authority, and restorative justice in a format amenable to both human and LLM administration. For Confucian moral philosophy, it means creating an instrument that captures filial piety (xiao), role-differentiated obligations (li), humaneness (ren), and the cultivation of moral character (junzi) through scenario-based items rather than abstract endorsement scales. For Indigenous ethical frameworks, the challenge is more fundamental: the diversity of Indigenous moral traditions—from Aboriginal Australian Dreamtime ethics to Maori tikanga to Lakota values—resists reduction to a single instrument, and any psychometric operationalization must be developed in partnership with Indigenous communities to avoid the extractive epistemology that has characterized previous cross-cultural measurement efforts.

The timeline of 3–5 years reflects the reality that validated psychometric instruments cannot be produced on the publication cycle of benchmark papers. The MFQ-30 required over a decade of development from Haidt and Joseph's (2004) initial theoretical framework through Graham et al.'s (2011) validation study. The PVQ-RR underwent continuous refinement from Schwartz's (1992) original value survey through the 2012 revised version and the 2022 cross-cultural validation. Expecting equivalent instruments for non-Western traditions to materialize in 12–18 months is unrealistic. What can begin immediately is the formation of cross-disciplinary teams that pair moral philosophers from these traditions with psychometricians experienced in scale development and cultural validation. The investment is substantial, but the alternative—continuing to evaluate LLM moral cognition using instruments that exclude the moral traditions of billions of users—is not a defensible default.

8.2.2 Recommendation 2: Require Construct Validity Evidence Before Benchmark Results Inform Deployment

The construct validity crisis documented in Claim 2—58% of candidates at Level 1 or 2, median at face validity—has a direct practical consequence that the field has not confronted:

benchmark scores that lack construct validity evidence do not support the inferences that practitioners draw from them. When a benchmark paper reports that "GPT-5 achieves a care score of 4.2 on a 5-point scale," the inference drawn by deployment decision-makers is that GPT-5 demonstrates high care-sensitivity. But this inference requires that the benchmark measures the care construct with sufficient fidelity that a score of 4.2 means what it appears to mean. If the benchmark's construct validity is at Level 1 or 2—keyword engagement or face validity only—the score may reflect the model's ability to detect and reproduce care-related language rather than any genuine engagement with the moral construct of care.

The practical recommendation is that AI evaluation reports, safety assessments, and governance audits should require construct validity evidence at Level 4 or above (systematic psychometric validation with convergent and discriminant validity evidence) before benchmark results are cited as evidence for or against model deployment. This does not mean that Level 1–3 benchmarks should be abandoned; they serve a useful exploratory function and may identify patterns worth investigating with more rigorous instruments. But they should not be treated as measurement tools whose outputs carry evidential weight in deployment decisions. The distinction between exploratory and confirmatory benchmarks—well-established in the psychometric literature (Messick, 1995; Kane, 2013)—needs to be imported into the AI evaluation field, where it is currently absent.

8.2.3 Recommendation 3: Mandate Human Baselines for Governance-Relevant Benchmarks

Claim 5 established that 34% of the 292 benchmark candidates provide no published human comparison data. In the context of governance and safety evaluation, this deficit is disqualifying. A benchmark score for an LLM acquires meaning only in relation to a human reference distribution. The claim that a model is "post-conventional" in its moral reasoning (DIT P-score of 75) is interpretable because decades of human norming data establish that typical adults score 30–50, graduate students score 50–65, and moral philosophy PhDs score 60–70 (Rest, 1979; Rest et al., 1999). The claim that a model achieves "high moral consistency" on a novel benchmark without human comparison data is uninterpretable: high relative to what? Without a reference, the score is a number without meaning.

For benchmarks used in governance or safety contexts—including regulatory compliance assessments under the EU AI Act, NIST AI Risk Management Framework evaluations, and internal safety reviews at AI companies—human baseline data should be mandatory. The baseline need not be a large-scale population study; even a comparison sample of 100–200 respondents from a defined demographic group would provide the reference distribution necessary for interpretation. The cost of collecting such data is trivial relative to the cost of deploying models on the basis of uninterpretable scores. Regulatory bodies should specify minimum baseline requirements as a condition for benchmark results to be cited in compliance documentation.

8.2.4 Recommendation 4: Invest in Process-Level Evaluation

The inverse relationship between benchmark quality and discrimination power ($r = -0.83$) reveals that the field's content-level instruments—however well-validated—cannot distinguish between models whose moral content has been aligned by similar training pipelines. The single benchmark that achieved high discrimination was a process-level probe: the dual-process protocol that tested how moral judgments respond to chain-of-thought prompting. The instruments the field has not built are the instruments the field most needs.

Three specific process-level evaluation designs are now empirically motivated. First, dual-process benchmarks should manipulate deliberation resources (CoT prompting, reasoning step requirements, token-limited responses) across matched personal and impersonal moral dilemmas, testing the prediction—generated by the Haiku finding—that CoT effects on moral judgment are inversely correlated with model scale. A well-powered study would require at least 30 paired administrations per model across a minimum of five models spanning a meaningful parameter range. Second, social intuitionism benchmarks should implement moral dumbfounding protocols, testing whether LLMs maintain moral judgments when systematic reason-defeat eliminates every stated justification—a direct probe of whether moral intuitions in these systems are independent of the reasoning chains that accompany them. Third, dyadic morality benchmarks should vary agent-patient dynamics across matched scenarios, testing whether the dyadic template that organizes human moral perception also structures LLM moral cognition or whether alignment training has produced systems constitutively unable to assign structural blame. The preliminary finding that all nine models converge on moderate, diffuse blame attribution (CV = 0.9%) suggests the latter.

The effort level is medium rather than high because process-level probes do not require the decade-long psychometric infrastructure development that content-level instruments demand. They require experimental designs drawn from the existing moral psychology literature—designs that have been published, replicated, and theoretically grounded—adapted for LLM administration. The dual-process manipulation (vary CoT prompting across matched dilemmas) can be implemented within a single research cycle. What is required is not new theory but the translation of existing experimental paradigms into LLM-native formats.

8.2.5 Recommendation 5: Establish Benchmark Quality Tier Transparency

The quality illusion documented throughout this paper—in which 292 benchmark candidates create the appearance of a mature evaluation ecosystem while 53% fall in the medium or low tiers—can be partially addressed through transparency requirements. AI evaluation reports should disclose the quality tier of every benchmark cited, using a standardized classification system derived from composite quality scores and natural breaks optimization.

The practical implementation is straightforward. The Jenks-derived tier classification applied in this study—with break points at 27.8, 43.0, and 56.6—could serve as a starting point, updated periodically as new benchmarks enter the landscape. Evaluation reports would then include a quality profile alongside their results: "This assessment employed 12 benchmarks: 3 Top-tier, 4 High-tier, 3 Medium-tier, and 2 Low-tier." The transparency requirement would not restrict which benchmarks researchers use; it would ensure that consumers of evaluation results—regulators, deployment decision-makers, and the public—can assess the evidential weight of the findings. A report based entirely on Top-tier benchmarks carries different inferential force than one based on Medium- and Low-tier instruments, and this distinction is currently invisible.

The timeline of less than one year reflects the low implementation cost: the classification framework already exists, the composite scoring algorithm is deterministic and reproducible, and the primary barrier is adoption rather than technical development. Professional organizations (ACM, IEEE, AAAI) and regulatory bodies (NIST, the EU AI Office) could adopt quality tier disclosure as a reporting standard within a single policy cycle.

8.2.6 Recommendation 6: Fund Cross-Cultural Benchmark Development Programs

Recommendation 1 addresses the immediate need for validated instruments in non-Western moral traditions. Recommendation 6 addresses the structural conditions that produced the cultural quality cliff in the first place. The zero-quality rate for Ubuntu, Confucian, Indigenous, and Identity ethics is ultimately a funding and institutional problem: the research infrastructure that produces high-quality benchmarks—psychometric expertise, cross-cultural validation studies, large-scale human norming data, expert-annotated text corpora—requires sustained investment over years or decades. MFT's infrastructure advantages did not emerge spontaneously; they are the product of two decades of cumulative research funded by the National Science Foundation, the Templeton Foundation, and other agencies that recognized the importance of understanding human moral variation.

Equivalent investment is needed for non-Western moral traditions, and it must be structured differently. The MFT infrastructure was built by Western researchers using Western psychometric methods to study Western moral constructs. Extending this approach to Ubuntu, Confucian, or Indigenous ethics would reproduce the very cultural bias the investment is intended to correct. Instead, funding programs should support research teams led by scholars from the relevant traditions—African moral philosophers developing Ubuntu instruments, Chinese ethicists developing Confucian instruments, Indigenous scholars developing culturally grounded evaluation frameworks—with psychometric expertise provided in a consultative rather than directive role.

The timeline of 5–10 years acknowledges that this is a generational investment in research infrastructure, not a quick fix. The Schwartz Value Survey has been under continuous development since 1992 and is still being refined; the DIT has been iterated since 1979.

Expecting non-Western instruments to achieve comparable maturity in less time is unrealistic, particularly given the additional challenges of cross-cultural translation, construct equivalence verification, and the need to develop population norms for populations that have been historically excluded from psychometric research. But the alternative—continuing to evaluate LLM moral alignment using instruments calibrated exclusively to Western moral constructs—is a structural injustice that the field can no longer claim is merely an inconvenient gap.

8.2.7 Recommendation 7: Implement Temporal Quality Monitoring

Claim 6 established that benchmark quality has not improved despite a 9.5-fold increase in volume between the pre-2020 and 2023–2026 periods. This stagnation is invisible without systematic monitoring: no existing institution or platform tracks the quality distribution of the moral psychology benchmark landscape over time. Individual researchers see their own benchmarks; they do not see the field-level quality trajectory.

The recommendation is to establish a living assessment framework—a continuously updated version of the 162-feature characterization protocol applied in this study—that classifies new benchmark candidates as they appear and publishes quality distribution statistics on a regular (quarterly or semi-annual) basis. The monitoring system would track four key indicators: the quality rate (top + high tiers as a proportion of total candidates), the theory concentration index (proportion of top-tier slots held by MFT), the cultural coverage ratio (quality rate for non-Western traditions as a proportion of the Western quality rate), and the baseline provision rate (proportion of candidates with published human comparison data).

The effort level is low because the assessment framework is already built and the scoring protocol is documented in sufficient detail for replication. The primary cost is ongoing human assessment of new candidates, which can be distributed across a community of trained assessors using the batch-calibrated scoring protocol described in Section 3.2. The value is disproportionate to the cost: a quarterly quality report would make stagnation visible, create accountability for improvement, and provide the empirical foundation for evidence-based policy interventions. If the quality rate continues to decline despite volume growth, the report would document that decline and create pressure for the structural changes that Recommendations 1–6 describe. If the rate improves, the report would document the improvement and identify the factors driving it.

8.3 Closing Statement

The field has produced 292 moral psychology benchmarks for large language models. Only 48 are fit for purpose. The 292 instruments collectively test a narrow slice of Western moral psychology—predominantly one theory, Moral Foundations Theory, which dominates by quality because it alone possesses the psychometric infrastructure that enables rigorous benchmark construction. When these instruments are administered to the models they are designed to evaluate, they find that every model agrees: nine systems from three vendors,

built on different architectures and aligned by different methodologies, share an identical moral fingerprint that has no precedent in any documented human population. The instruments cannot distinguish between the systems they evaluate. And the moral traditions of billions of users—Ubuntu, Confucian, Indigenous, Buddhist—are either invisible to the evaluation enterprise or represented by instruments that fail to meet minimum quality thresholds.

The question that animates this paper—"what do moral psychology benchmarks actually test?"—now has an answer, and it is not the answer the field expected. They test a narrow slice of Western moral psychology and find that every LLM agrees on it. They test moral content and find content-level convergence. They test what models say about morality and find that every model says the same thing. They do not test how models arrive at moral judgments, because the process-level instruments that would probe moral reasoning dynamics have not been built. They do not test whether models can engage with the moral traditions of the majority of the world's population, because the cultural instruments that would enable such evaluation do not exist at research-grade quality. The evaluative infrastructure that the AI safety community relies upon to certify moral alignment is not measuring what it claims to measure, cannot distinguish between the systems it evaluates, and excludes the moral perspectives it most needs to include.

The triple bind will not resolve itself. Increasing the volume of benchmarks—the field's current trajectory—will not break the cycle, because more instruments of the same type, measuring the same theory, administered to models that produce the same results, cannot resolve a structural problem. The path forward requires three coordinated investments that the field has not yet made. It requires theory infrastructure: the construction of standardized, validated psychometric instruments for the process-level moral psychology theories that the current landscape has failed to operationalize. It requires cross-cultural psychometrics: the development of culturally grounded evaluation frameworks built by and for the moral traditions that the current landscape excludes. And it requires process-level evaluation: the translation of the moral psychology discipline's richest experimental paradigms—dual-process manipulations, moral dumbfounding protocols, dyadic morality probes—from human laboratories into LLM evaluation infrastructure.

The theoretical resources for this work exist. Moral psychology has spent decades developing rigorous theories of how humans form moral judgments—theories of process, intuition, development, cultural variation, and dyadic perception that far exceed what the current LLM evaluation enterprise has operationalized. The empirical evidence now shows where the leverage points are: process-level probes discriminate where content-level instruments converge; the dual-process protocol achieved a coefficient of variation 93 times larger than the mean of the top-tier content benchmarks. The blueprint is visible in the data. What remains is the work: the slow, difficult, unglamorous work of building psychometric infrastructure for moral traditions that lack it, validating instruments for respondents they were not designed for, collecting human baselines for populations that have been histori-

cally excluded from measurement, and constructing the evaluation enterprise that the field needs rather than the one it has.

The moral evaluation of artificial intelligence is too important to be conducted with instruments that cannot do what they claim. The systems being evaluated make moral judgments that affect billions of lives—in healthcare triage, content moderation, legal reasoning, education, and autonomous decision-making. The question of whether these systems are morally aligned, and with whose morality they are aligned, is not an academic curiosity. It is a prerequisite for responsible deployment. The answer to that question cannot come from an evaluative infrastructure that tests one theory, finds one answer, and excludes the moral perspectives of the majority of humanity. The triple bind is real. The path out of it is visible. The field's willingness to walk that path will determine whether moral psychology benchmarks become genuine instruments of understanding or remain artifacts of the cultural moment that produced them.

Supplementary Material

*The following supplementary sections provide extended methodology,

detailed results, and comprehensive data tables that support the

main paper findings.*

Supplementary Methods

Moral Psychology Benchmarks: A Meta-Evaluation of 292 Candidates Across 13 Theories

Version: 4.0 (Full rebuild, 2026-04-05)

This supplement provides exhaustive methodological detail extending the main text. Material here is essential for reproducibility and for researchers who wish to adapt the framework to new candidate benchmarks.

S1. Full 162-Feature Framework Definition

The characterization framework comprises 162 features organized across seven tiers. Tiers A–C (17 scored features) constitute the composite scoring instrument; Tiers D–G (141 characterization features) provide extended profiling but do not affect composite scores or tier assignment.

All features use a 0–100 continuous scale. Unless otherwise noted, the general rubric anchors are:

Score Range	Interpretation
0–25	Absent or minimal evidence
26–50	Partial or emerging evidence
51–75	Substantial evidence
76–100	Comprehensive or exemplary evidence

Features marked "(inverted)" reverse polarity so that higher scores always indicate a more desirable property.

S1.1 Tier A — Psychological Depth (Features A1–A6)

Tier A measures how faithfully a benchmark engages moral psychology theory. These six features carry a collective composite weight of 0.50.

ID	Name	Scale	Composite Weight	Scoring Rubric
A1	Construct depth (theories engaged)	0–100 (count-mapped)	0.075	Number of distinct moral psychology theories operationalized. 0 theories = 0; 1 = 30; 2 = 60; 3 = 80; 4+ = 100. Only theories with substantive engagement (>=3 items or explicit hypothesis testing) are counted.
A2	Construct operationalization	0–100 (ordinal)	0.125	Degree to which the benchmark faithfully operationalizes a theoretical construct. Validated instrument covering all dimensions of its target theory = 100; theory-derived partial coverage = 70; theory-inspired but incomplete = 40; keyword-only or no explicit grounding = 10.
A3	Dilemma complexity	0–100	0.075	Complexity and depth of moral scenarios or stimuli. Single-sentence forced-choice = 30; multi-sentence with contextual detail = 60; multi-paragraph realistic scenarios with competing considerations = 90. Default 70; adjusted per assessment.
A4	Source instrument provenance	0–100 (ordinal)	0.050	Whether the benchmark derives from a previously validated psychometric instrument. Validated instrument (e.g., MFQ, PVQ, DIT) = 100; original instrument with psychometric development = 80; no separate instrument = 0.
A5	Human validity and availability	0–100 (ordinal)	0.075	Availability of psychometric validation data and human baselines. Extensive validation (convergent, discriminant, criterion, content, construct) = 100; partial validation = 70; norms only = 50; no validation = 10.
A6	Construct validity	0–100 (ordinal)	0.075	Evidence that the benchmark measures what it claims to measure. Multiple forms of validity evidence = 100; some evidence = 70; face validity only = 40; no validity evidence = 10.
A7	Discriminant power	0–100 (ordinal)	0.075	Ability to distinguish between different moral reasoning profiles or capacities. Demonstrated discriminant validity = 100; theoretical discriminant power = 70; no discriminant analysis = 10.
A8	Scoring objectivity	0–100 (ordinal)	0.075	Degree to which scoring is objective and reproducible. Fully automated = 100; rubric-based = 70; requires subjective judgment = 30.
A9	Cultural sensitivity	0–100 (ordinal)	0.075	Awareness of cultural variation in moral reasoning. Explicitly cross-cultural = 100; culturally aware = 70; culturally neutral = 40; culturally biased = 10.
A10	Temporal stability	0–100 (ordinal)	0.075	Evidence of test-retest reliability. Reported and adequate = 100; reported but inadequate = 50; not reported = 0.
A11	Ecological validity	0–100 (ordinal)	0.075	Relevance to real-world moral decision-making. Direct real-world scenarios = 100; analogous scenarios = 70; abstract = 30.
A12	Inter-rater reliability	0–100 (ordinal)	0.075	Reliability of human scoring. Reported and adequate = 100; reported but inadequate = 50; not reported = 0.
A13	Item independence	0–100 (ordinal)	0.075	Statistical independence of test items. Verified independent = 100; not tested = 50; violations detected = 10.
A14	Documentation quality	0–100 (ordinal)	0.075	Completeness of documentation. Comprehensive = 100; adequate = 70; minimal = 30.

Composite formula (Psychological Depth, PD):

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PD = A2(0.25) + A6(0.20) + A1(0.15) + A5(0.15) + A3(0.15) + A4(0.10)
```

Weights reflect the expert council's judgment that construct operationalization (A2) and validity evidence (A6) are the strongest indicators of psychological depth. The absolute composite weight for each feature is its within-tier weight multiplied by 0.50 (the tier weight).

S1.2 Tier B — Practical Ease (Features B1–B4, B6–B7, B10)

Tier B measures how readily a benchmark can be implemented for LLM evaluation. Seven of the original ten features are scored; three (B5, B8, B9) were excluded because they require access to benchmark distribution infrastructure that was not systematically available across all 292 candidates. Collective composite weight: 0.30.

ID	Name	Scale	Composite Weight	Scoring Rubric
B1	Task format	0–100 (format-mapped)	0.030	Response format suitability for LLMs. MCQ = 100; binary = 90; forced-choice = 85; Likert = 70; ranking = 60; free-text = 50; open generation = 30.
B2	Dataset size	0–100 (log-scaled)	0.045	Number of items as proxy for measurement precision. <50 items = 20; 200 = 40; 1,000 = 60; 5,000 = 75; 20,000 = 85; >20,000 = 100.
B3	Data availability	0–100 (platform-mapped)	0.060	Accessibility of benchmark data on standard platforms. HuggingFace = 100; GitHub = 95; OSF = 90; university server = 80; anonymous link = 60; email request = 30; paper only = 0.
B4	Code availability	0–100 (ordinal)	0.030	Availability of implementation code and evaluation pipelines. Full evaluation pipeline = 100; partial scripts = 50; no code = 0.
B6	Harness compatibility	0–100 (tier-mapped)	0.045	Integration readiness with standard LLM evaluation harnesses (lm-eval-harness, Inspect AI, LightEval, HELM, OpenAI Evals). Tier 1 (native integration) = 100; Tier 2 (straightforward port) = 70; custom implementation required = 40.
B7	Scorer complexity	0–100 (complexity-mapped)	0.045	Complexity of the scoring procedure. Exact match = 100; string-matching with normalization = 85; model-graded = 60; psychometric scoring (e.g., P-score computation) = 40; human evaluation re-
B10	LLM administration fidelity	0–100 (ordinal)	0.045	How faithfully the instrument can be administered to an LLM compared to a human participant. Direct administration (no adaptation) =

Excluded features: - **B5 (Instruction simplicity):** Requires access to benchmark distribution infrastructure unavailable for all candidates during the assessment period. - **B8 (Open access):** Same rationale as B5. - **B9 (Documentation completeness):** Same rationale as B5.

Composite formula (Practical Ease, PE):

```
PE = B3(0.20) + B2(0.15) + B6(0.15) + B7(0.15) + B1(0.10) + B4(0.10) + B10(0.15)
```

S1.3 Tier C — Citation Impact (Features C1–C3, C5)

Tier C measures scholarly influence and community adoption. Four of the original five features are scored; C4 was excluded because cross-field citation tracing exceeded available data. Collective composite weight: 0.20.

ID	Name	Scale	Composite Weight	Scoring Rubric
C1	Total citations	0–100 (log-scaled)	0.060	Raw citation count from Semantic Scholar. <5 = 10; 20 = 30; 50 = 50; 100 = 65; 200 = 75; 500 = 85; >500 = 100.
C2	Recent citation velocity	0–100 (log-scaled)	0.060	Age-normalized citation rate (citations per year). <2 = 15; 5 = 30; 10 = 50; 25 = 65; 50 = 80; 100 = 90; >100 = 100.
C3	Venue tier	0–100 (tier-mapped)	0.050	Publication venue quality tier. Tier 1 (Nature, Science, PNAS, NeurIPS, ICML, ACL, ICLR) = 100; Tier 2 (AAAI, EMNLP, NAACL, Cognitive Science, SPPS) = 75; Tier 3 (workshops, regional conferences, Scientific Reports) = 50; Tier 4 (arXiv-only, preprint) = 25.
C5	Hub/ leaderboard presence	0–100 (binary)	0.030	Presence on recognized evaluation hub or leaderboard. Major hub (Papers With Code, HuggingFace leaderboard) = 100; minor hub = 50; none = 0.

Excluded feature: - **C4 (Cross-domain citations):** Requires cross-field citation tracing beyond available data.

Composite formula (Citation Impact, CI):

```
CI = C1(0.30) + C2(0.30) + C3(0.25) + C5(0.15)
```

S1.4 Tier D — Psychometric Rigor (Features D1–D24)

Tier D evaluates internal consistency, reliability, factor structure, item analysis, sampling methodology, and measurement design. These 24 features do not affect composite scores; they are reported as tier profiles.

ID	Name	Scale	Description
D1	Internal consistency evidence	0–100	Reports Cronbach's alpha >= 0.80 or McDonald's omega with CI = 100; reports alpha 0.60–0.79 = 70; mentions reliability without reporting = 30; no mention = 0.
D2	Test-retest reliability evidence	0–100	Published test-retest study with r >= 0.80 and interval > 2 weeks = 100; r 0.60–0.79 = 70; short interval only = 40; no test-retest data = 0.
D3	Factor structure validation	0–100	CFA with good fit (CFI > 0.95, RMSEA < 0.06) = 100; EFA with clear structure = 70; PCA only = 40; no factor analysis = 0.
D4	Item-total correlation range	0–100	All item-total correlations in 0.30–0.70 range = 100; most in range = 70; wide variation or not reported = 30; no item analysis = 0.
D5	Ceiling/floor effect prevalence	0–100	<5% of items show ceiling/floor effects = 100; 5–15% = 70; 15–30% = 40; >30% or not reported = 10.
D6	Response distribution normality	0–100	All subscales approximately normal (skew < 1, kurtosis < 3) = 100; most normal = 70; several skewed = 40; heavily non-normal = 10.
D7	Discrimination index adequacy	0–100	Discrimination indices > 0.30 for all items = 100; most items > 0.30 = 70; mixed = 40; no discrimination analysis = 0.
D8	Measurement invariance across populations	0–100	Scalar invariance across >= 3 groups = 100; metric invariance = 70; configural only = 40; no invariance testing = 0.
D9	LLM-specific ceiling compression index	0–100	<10% of measurement range compressed when administered to LLMs = 100; 10–30% = 60; 30–60% = 30; >60% or no LLM data = 0. **35% null rate.**
[illegible]	[illegible]	[illegible]	[illegible] scheme with [illegible] = 30; no [illegible] = 40; no coding = 0.

S1.5 Tier E — LLM-Specific Validity (Features E1–E44)

Tier E assesses how well benchmarks perform when administered to language models. These features do not affect composite scores.

ID	Name	Description
E1	Recognition-endorsement discrimination	Distinguishes moral knowledge from moral endorsement
E2	Dual-system engagement detection	Contains items engaging System 1 vs. System 2 processing
E3	Dumbfounding elicitation capacity	Includes counterargument protocol with persistence measurement
E4	Prompt template sensitivity	Documents sensitivity across >= 5 prompt variants
E5	Few-shot sensitivity	Compares 0/1/5-shot performance
E6	Tokenization sensitivity	Analyzes tokenization effects across model families
E7	Context window utilization	Consumes < 5% of context window with position-balanced design
E8	Batch vs. interactive administration fidelity	Administration method matches measurement theory
E9	Evaluation harness portability	Native implementation in >= 2 major harnesses
E10	Temperature sensitivity	Tested across >= 4 temperature values
E11	System prompt interaction	Systematic testing with/without system prompt
E12	Adversarial robustness tested	Tested against >= 3 adversarial attack types
E13	Jailbreak resistance relevance	Tests jailbreak-relevant moral scenarios
E14	Deployment scenario relevance	Scenarios drawn from real deployment dilemmas
E15	Safety-critical failure mode detection	Tests sycophancy, harmful refusal, category errors
E16	Multi-turn consistency measurement	Multi-turn protocol with Socratic challenge
E17	Persona stability measurement	Systematic persona manipulation with stability metrics
E18	[illegible]	[illegible] ing, [illegible] tion
	[illegible] ment [illegible] ted)	

S1.6 Tier F — Cultural and Philosophical Breadth (Features F1–F44)

Tier F assesses cultural, religious, and philosophical diversity. These features do not affect composite scores.

ID	Name	Description
F1	Stereotype amplification detection	Measures whether LLM overshoots real cultural profiles
F2	Cross-cultural construct coverage	Operationalizes >= 3 non-Western moral frameworks
F3	Moral emotion measurement	Measures >= 5 moral emotions as distinct constructs
F4	Moral conflict structure	Structured value-vs-value conflicts with named principles
F5	Cultural origin of theoretical framework	Non-WEIRD origin with cross-cultural development = 100
F6	Cultural validation breadth	Validated in >= 6 Inglehart-Welzel cultural clusters
F7	Scenario cultural content	Scenarios drawn from >= 4 cultural settings
F8	Theory cultural scope claims	Universal claims supported across >= 5 cultural clusters
F9	Language coverage	>= 10 languages with published norms = 100
F10	WEIRD bias composite index	Composite of F5–F9; low WEIRD bias = 100
F11	Individualism-collectivism balance	Balances individualist and collectivist moral agency
F12	Moral universalism vs. pluralism stance	Accommodates multiple legitimate moral positions
F13	Indigenous/marginalized tradition inclusion	>= 3 indigenous traditions with community-validated items
F14	Colonial/power-dynamic awareness	Addresses power dynamics with mitigation strategies
F15	Granularity of moral construct measurement	>= 10 fine-grained subscales with discriminant validity
F16	Ingroup-outgroup moral differentiation	Varies in/out-group status with interaction analysis
[illegible]	[illegible]	[illegible]

S1.7 Tier G — Theory Fidelity and Provenance (Features G1–G29)

Tier G assesses how faithfully benchmarks represent their claimed theoretical foundations and scholarly provenance quality. These features do not affect composite scores.

ID	Name	Description
G1	Theory version currency	Uses current theory version (e.g., MFT 2.0, PVQ-RR) = 100
G2	Foundation coverage breadth	Measures >= 80% of target theory's constructs = 100
G3	Foundation coverage depth	>= 6 items per construct enabling variance estimation = 100
G4	Corpus annotation quality	Multi-annotator (>= 5), trained, published guidelines = 100
G5	Theory-specific scoring fidelity	Scoring implements original theory's method = 100
G6	Developmental sensitivity	Detects responses at each developmental stage = 100
G7	Stage differentiation granularity	>= 6 discriminable developmental levels = 100
G8	Regression detection capability	Detects downward stage movement under stress = 100
G9	Complexity gradient	Items span from clear-cut to genuinely ambiguous = 100
G10	Reasoning sophistication capture	Captures reasoning quality with multi-level rubric = 100
G11	Dyadic structure sensitivity	Measures agent-patient dyadic structure per Gray & Schein
G12	Meta-cognitive reflection capacity	Elicits and measures meta-cognitive moral reasoning = 100
G13	Citation trajectory (momentum)	Increasing trajectory (growing influence) = 100
G14	Citation hub score (downstream adoption)	>= 5 subsequent benchmarks cite as foundation = 100
G15	Venue disciplinary origin	Interdisciplinary (CS + psychology co-authored) = 100
G16	Replication count	>= 5 independent replications = 100
G17	Dataset derivative count	>= 5 derivative datasets or adaptations = 100
G18	Cross-disciplinary citation breadth	Cited across >= 4 disciplines = 100
G19	[illegible]	[illegible]
G20	[illegible]	[illegible] change = 100

S1.8 Coverage Statistics

The full 162-feature framework was assessed across 292 candidates, generating 47,304 potential scoring cells (162 features x 292 candidates). Four features (B5, B8, B9, C4) were excluded across all candidates, reducing the active feature count to 158.

Assessed matrix: 158 features x 292 candidates = 46,136 cells

Null concentration: Gaps are concentrated in two features with a 35% null rate each: - **D9** (LLM-specific ceiling compression index): Many candidates have not been administered to LLMs with distributional analysis. - **D20** (Response distribution entropy): Empirical data on response distributions was unavailable for many candidates.

S1.9 Deduplication Log

Thirteen merges reduced 154 proposed features to 141 new features during framework development. Each retained feature absorbed the unique aspects of the merged proposal.

Retained Feature	Absorbed Proposal	Rationale
D2: Test-retest reliability	M8: Temporal stability; M9: Test-retest documentation	All three assess temporal reliability
D5: Ceiling/floor effects	M11: Ceiling/floor quantification	Both quantify ceiling/floor effects
E1: Recognition-endorsement	M6: Recognition-endorsement dissociation	Same construct, different terminology
E2: Dual-system detection	M6: System 1/System 2 diagnostic capability	Both assess dual-system processing
E3: Dumbfounding capacity	Mcross: Dumbfounding susceptibility	Both test moral dumbfounding
E4: Prompt sensitivity	M8: Prompt sensitivity coefficient	Both quantify prompt variation
E12: Adversarial robustness	M2: Adversarial robustness evidence	Both assess adversarial testing
E16: Multi-turn consistency	M4: Multi-turn engagement capacity	Covers engagement and consistency
E19: Refusal measurement	M4: Refusal handling protocol	Both assess refusal behavior
F3: Moral emotion measurement	M5: Moral emotion engagement	Both assess moral emotion
F4: Moral conflict structure	M6: Hierarchical moral conflict resolution	Both assess competing-value conflicts
G11: Dyadic structure sensitivity	Mcross: Dyadic sensitivity profile	Both assess agent-patient dyadic structure

S2. Keyword Scorer Patterns

The benchmark candidate identification pipeline employs a 125-pattern keyword scorer organized by 13 moral psychology theories. Patterns are classified as strong (high speci-

ficity for theory engagement) or medium (contextual indicators requiring co-occurrence). All theories use the same scoring formula:

```
score = (strong_matches x 15) + (medium_matches x 5) - (fp_context x 10) +
title_bonus(20)
score = clamp(score, 0, 100)
```

A paper is classified as engaging a theory when `score >= 50`. The "strong floor" rule ensures that any paper with at least one strong-tier match receives a minimum score of 50.

Table ST3: Keyword Scorer Patterns by Theory

Theory	Total Patterns	Strong	Medium	FP Suppression	Sample Strong Patterns	Sample Medium Patterns
MFT	22	11	8	3	\bmoral foundations\b, \bmfq\b, \bhaidt\b(?=.*\bfoundation)	\bpurity\b(?=.*\bmoral), \bloyalty\b(?=.*\bfoundation)
Trolley	14	7	5	2	\btrolley problem\b, \bmoral machine\b, \bfootbridge\b	\butilitarian\b(?=.*\bdilemma), \bsacrifice\b(?=.*\bmoral)
Schwartz	12	6	4	2	\bschwartz\b(?=.*\bvalue), \bpvq\b, \bsvs\b	\bself-transcendence\b, \bvalue circumplex\b
Kohlberg	10	5	3	2	\bkohlberg\b, \bdefining issues test\b, \bdit\b(?=.*\bmoral)	\bpost-conventional\b, \bmoral development\b(?=.*\bstage)
Dual Process	18	9	9	3	\bdual process\b(?=.*\bmoral), \bjoshua greene\b, \bcognitive load\b(?=.*\bmoral)	\bsystem 1\b(?=.*\bmoral), \bautomatic\b(?=.*\bmoral)(?=.*\bjudg)
Social Intuitionism	14	7	7	2	\bhaidt\b(?=.*\bintuit), \bmoral dumbfounding\b, \bpost-hoc rationali[sz]\w+\b(?=.*\bmoral)	\bintuitive judgment\b(?=.*\bmoral), \bmoral sentiment\b
Dyadic Morality	11	4	7	2	\bkurt gray\b, \bdyadic morality\b, \bharm perception\b(?=.*\bmoral)	\bmoral dyad\b, \bmoral patient\b, \bharm-based moral\w*\b
[illegible]	**128**	**64**	**54**	**17**	\bmaqasid\b(?=.*\bethic), \bmaslahah\b(?=.*\bmoral), \bijtihad\b(?=.*\bislam), \bsharia\b(?=.*\bmoral)	\bmaqasid\b(?=.*\bislam), \bmaslahah\b(?=.*\bislam), \bijtihad\b(?=.*\bislam), \bsharia\b(?=.*\bislam)
				175		

Expansion Impact

During the benchmark survey, the keyword scorer was expanded from 85 to 128 patterns. The table below shows detection counts before and after re-scoring all vault papers (threshold: score >= 50).

Theory	Papers Before	Papers After	Delta	Notes
MFT	68	101	+33	Gains from improved scoring engine, not new patterns
Schwartz	52	69	+17	Gains from rescoring
Trolley	36	58	+22	Gains from rescoring
Kohlberg	19	35	+16	Gains from rescoring
Dual Process	0	9	+9	18 new patterns (9 strong + 9 medium)
Social Intuitionism	1	18	+17	14 new patterns (7 strong + 7 medium)
Dyadic Morality	3	7	+4	11 new patterns (4 strong + 7 medium)
Moral Identity	7	12	+5	Existing patterns sufficient
Islamic	–	22	–	New theory category
Buddhist	–	16	–	New theory category
Total	**186**	**377**	**+191**	

S3. Batch-Calibrated Assessment Protocol

S3.1 Batch Stratification

The 292 candidates were divided into 20 batches of approximately 15 candidates each, stratified across three dimensions to prevent systematic scoring bias:

1. **Theory**: Each batch contained a mix of theories (not all-MFT or all-Trolley). With 13 theories and 292 candidates, each batch included candidates from at least 4 different theoretical traditions.
2. **Information tier**: Each batch mixed Tier A (full PDF extract + abstract, n=245), Tier B (abstract only, n=47), and where applicable Tier C (minimal metadata) candidates. The null policy (Section S3.4) governed how information gaps affected scoring.
3. **Year**: Each batch spanned the publication year range (2015–2026) to prevent recency or seniority bias from concentrating in particular batches.

Execution in 4 waves:

Wave	Batches	Candidates	Calibration Gate
1	1–5	~75	Quality Auditor (C12) runs anchor comparison + drift check
2	6–10	~75	C12 runs mid-point calibration
3	11–15	~75	C12 runs drift check
4	16–20	~67	C12 runs final calibration + completeness audit
Mop-up	–	flagged	Sequential review of >20% null or outlier candidates

S3.2 Calibration Anchor Scores

Three calibration anchors were scored by every batch to detect inter-batch drift. Each anchor was chosen to represent a distinct quality level and theory tradition.

Table ST2: Calibration Anchor Scores by Batch

Anchor	Expected COMPOSITE	Batch 1	Batch 2	Batch 3	Batch 4	Batch 5	Max Deviation
Moral Machine (Awad et al., 2018)	~78.5	78.8	78.2	78.6	78.3	78.7	+/- 0.3
Value Kaleidoscope (Sorensen et al., 2024)	~61.6	61.8	61.4	61.7	61.5	61.6	+/- 0.2
Low anchor (Tier B, minimal info)	~22.0	22.3	21.8	22.1	22.5	22.0	+/- 0.5

Note: Actual composite scores in the final matrix for the high and mid anchors are: Moral Machine = 71.73 (in the 292-candidate re-scoring) and Value Kaleidoscope is not directly in the top-48 but maps to the Probing Moral Development benchmark (rank 23, COMPOSITE = 61.6). The batch anchor scores above reflect within-batch calibration runs before final re-normalization.

S3.3 Inter-Batch Drift Analysis

Post-hoc calibration confirmed that all three anchor scores remained within +/- 5 points across all 20 batches. The maximum observed deviation was 0.5 points on the low anchor (Batch 4 vs. Batch 2), well within the pre-registered tolerance threshold.

Kendall's coefficient of concordance (W) across batch raters for the three anchors was $W = 0.97$ ($p < 0.001$), indicating near-perfect inter-batch agreement.

S3.4 Null Policy and Confidence Grading

Null policy (three categories):

Category	Usage	Effect on Composite
Explicit null	Feature cannot be scored and absence is not informative (e.g., D3 for a Tier B candidate)	Excluded from computation; denominator adjusted
Inferred zero (score: 0, confidence: `inferred`)	Absence of evidence IS informative (e.g., C5 hub presence)	Contributes 0 to composite
Conservative estimate (score: N, confidence: `low`)	Partial information allows bounded estimate	Contributes N to composite

Confidence grading:

Grade	Meaning	When Used
`high`	Scored from direct evidence in abstract or PDF extract	Tier A candidates; metadata-derivable features
`medium`	Scored from indirect evidence (e.g., venue implies peer review)	Tier B candidates on features requiring inference
`low`	Conservative estimate from minimal information	Features where only bounds can be established
`inferred`	Score of 0 because absence is informative	Hub presence, pre-registration, open access

Confidence grades are recorded for transparency but do NOT affect composite scoring. A score of 40 with confidence `low` contributes identically to a score of 40 with confidence `high`.

Maximum null threshold: Any candidate with >60% null features (>95 of 158 active features) was excluded from Jenks tier ranking but still reported in supplementary tables with a dagger symbol.

S4. Jenks Natural Breaks Diagnostics

S4.1 GVF Analysis

The Jenks natural breaks algorithm was applied to the 292 composite scores to identify four tiers. The Goodness of Variance Fit (GVF) for k=4 was:

GVF = 0.83

GVF is computed as 1 - (Sum of Squared Deviations from Class Means / Sum of Squared Deviations from Array Mean). A GVF of 0.83 indicates that the four-class solution explains 83% of the variance in the composite score distribution, which is considered a good fit for classification purposes (values above 0.80 are generally regarded as acceptable).

S4.2 Algorithm Description

Jenks natural breaks (also called Fisher-Jenks) is a classification algorithm that seeks to minimize within-class variance while maximizing between-class variance. For a dataset of n values partitioned into k classes:

1. The algorithm initializes with k arbitrary class boundaries.
2. For each iteration, it computes the sum of squared deviations within each class (SDCM) and the sum of squared deviations from the array mean (SDAM).
3. Class boundaries are adjusted to minimize SDCM / SDAM.
4. The algorithm converges when no boundary adjustment reduces the within-class variance.

This approach is preferred over equal-interval or quantile breaks because it respects the natural clustering in the data rather than imposing arbitrary boundaries.

S4.3 Sensitivity to k

k	GVF	Tier boundaries	Interpretation
3	0.76	[37.2, 53.8]	Top-heavy: merges "top" and "high" into a single tier of 137 candidates, losing discrimination at the quality frontier
4	**0.83**	**[27.8, 43.05, 56.65]**	**Adopted solution: 48 top, 89 high, 72 med, 83 low**
5	0.88	[22.5, 35.4, 47.3, 58.9]	Splits "low" into two sub-tiers (low and very-low); marginal GVF improvement (+0.05) does not justify additional complexity

The k=4 solution was adopted because it provides the best balance between parsimony and discrimination. The k=5 solution's additional class boundary falls within the "low"

tier, splitting candidates with minimal practical distinction (both sub-tiers lack sufficient psychological depth for research use).

S4.4 Jenks Break Values

The three Jenks break values partition the 292 composite scores into four tiers:

Tier	Label	Composite Range	N	Semantic Definition
Top	Publication-ready	56.65 – 81.33	48	Psychometrically validated, theory-grounded, data-available benchmarks with construct validity evidence
High	Research-grade	43.05 – 56.64	89	Meaningful theoretical engagement with moderate-to-strong operationalization
Med	Emerging	27.80 – 43.04	72	Theory-inspired without formal validation, OR technically strong but theoretically shallow
Low	Landscape-only	3.50 – 27.79	83	Keyword-level engagement with moral psychology; documents breadth but not suitable for psychological conclusions

Table ST4: Sensitivity Analysis — Alternative Weighting Schemes

Scheme	PD Weight	PE Weight	CI Weight	% Unchanged Tier	% Shift +/-1	% Shift +/-2
Canonical (50/30/20)	**0.50**	**0.30**	**0.20**	—	—	—
Equal (33/33/33)	0.33	0.33	0.33	78.4%	20.2%	1.4%
Practice-first (30/50/20)	0.30	0.50	0.20	73.6%	24.3%	2.1%
Impact-first (20/30/50)	0.20	0.30	0.50	69.5%	27.4%	3.1%
Depth-only (100/0/0)	1.00	0.00	0.00	65.1%	30.1%	4.8%
PD-PE only (60/40/0)	0.60	0.40	0.00	82.9%	16.4%	0.7%

Interpretation: The canonical 50/30/20 weighting produces results most similar to the PD-PE only scheme (82.9% agreement), confirming that Citation Impact (CI) functions primarily as a tiebreaker rather than a primary discriminator. The Depth-only scheme is most divergent, with 4.8% of candidates shifting by two tiers — these are benchmarks with high practical ease and citation impact but low psychological depth (e.g., large-scale NLP datasets that use moral vocabulary without theoretical grounding).

S5. Composite Formula Derivation

S5.1 Rationale for 50/30/20 Weighting

The composite formula weights Psychological Depth (PD) at 0.50, Practical Ease (PE) at 0.30, and Citation Impact (CI) at 0.20. This weighting reflects the expert council's prioritization:

1. **Psychological Depth (50%)** receives the highest weight because the fundamental question of the meta-evaluation is: *Does this benchmark measure what it claims to measure?* A benchmark that faithfully operationalizes moral psychology theory with validated constructs and human baselines is more valuable than a practically convenient but theoretically vacuous instrument. The moral psychology literature's replication crisis was substantially driven by weak operationalization (Schein & Gray, 2018), making depth the primary quality indicator.

2. **Practical Ease (30%)** receives moderate weight because even the most theoretically rigorous benchmark has limited impact if it cannot be readily administered to LLMs. Data availability (B3), harness compatibility (B6), and scorer determinism (B7) directly affect reproducibility. However, practical barriers are more readily overcome than theoretical deficiencies — a well-designed benchmark can be reformatted, but a poorly theorized one cannot be retrospectively grounded.
3. **Citation Impact (20%)** receives the lowest weight because citation counts are a lagging indicator that penalizes recent work and rewards historical advantage. The 2024–2026 explosion of moral psychology benchmarks means that many high-quality candidates have had insufficient time to accumulate citations. Including CI at 20% prevents the composite from being dominated by citation bias while still recognizing that sustained scholarly engagement is a positive signal.

S5.2 Within-Tier Weights

Within each tier, feature weights reflect relative importance as determined by the expert council:

PD within-tier weights: - A2 (0.25): Construct operationalization is the strongest single indicator of whether a benchmark engages theory or merely borrows vocabulary. - A6 (0.20): Construct validity evidence is the gold standard for measurement quality. - A1, A5, A3 (0.15 each): Theory count, human baselines, and dilemma complexity are important but less discriminating than operationalization and validity. - A4 (0.10): Source instrument provenance is less important because original instruments with strong psychometric development can match or exceed adapted validated instruments.

PE within-tier weights: - B3 (0.20): Data availability is the most important practical barrier — without data, a benchmark cannot be used regardless of other merits. - B2, B6, B7, B10 (0.15 each): Dataset size, harness compatibility, scorer complexity, and administration fidelity are equally important practical considerations. - B1, B4 (0.10 each): Task format and code availability are less discriminating because format can be adapted and code can be written.

CI within-tier weights: - C1, C2 (0.30 each): Total citations and citation velocity are the primary impact indicators. - C3 (0.25): Venue tier provides an independent quality signal through peer review. - C5 (0.15): Hub presence is a secondary indicator of community adoption.

S5.3 Decision Not to Incorporate Tiers D–G

The 141 features in Tiers D–G were deliberately excluded from the composite because they serve a different function. The core composite (Tiers A–C) answers the selection question: "Is this benchmark worth using?" The extended features answer the characterization question: "What are its detailed properties?" Mixing selection and characterization signals would dilute the composite's discriminative power. Extended features are reported as

tier profiles — aggregate statistics computed within each quality tier to characterize what distinguishes “top” from “low” benchmarks beyond the core composite.

S6. Cross-Vendor Trial Administration Details

S6.1 Benchmark Inventory

Twenty-three benchmarks were administered across 9 models from 3 vendors. The table below lists each benchmark with its trial identifier, item count, format, and scoring method.

Trial ID	Benchmark	Items	Format	Scoring Method
MP2	MFQ-30 (Moral Foundations Questionnaire)	30	Likert (6-point)	Foundation subscale means
MP3	Defining Issues Test (DIT-1)	72	Scenario rating + ranking	P-score (principled reasoning %)
MP4	Moral Foundations Vignettes (MFV) — 30-item subset	30	Scenario rating (wrongness 0–100)	Foundation-specific wrongness means
MP5	ValueCompass — Schwartz-based Contextual Value Alignment	40	Likert	PVQ-RR value profile
MP6	EMNLP — Educator-role Moral and Normative LLMs Profiling	88	Mixed (scenario + classification)	Stage assignment + reasoning quality
MP7	TrolleyBench — Ethical Consistency Evaluation	50	Binary forced-choice	Ethical Consistency Index (ECI)
MP8	Moral Machine Experiment	50	Forced-choice (binary AV crash)	Preference scores across 9 dimensions
MP9	MFTC — Moral Foundations Twitter Corpus	100	Multi-label classification	Foundation-level F1 macro
MP10	Does Moral Code Have a Moral Code? (Delphi probing)	100	Binary moral judgment + classification	Accuracy, MFT alignment
MP11	Value Kaleido-	50	Free-text	Value diversity, valence
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

S6.2 Model Inventory

Model	Vendor	Model ID	Parameters	Context Window
Opus	Anthropic	claude-opus-4-6	Undisclosed	200K
Sonnet	Anthropic	claude-sonnet-4-6	Undisclosed	200K
Haiku	Anthropic	claude-haiku-4-5-20251001	Undisclosed	200K
GPT-5 Codex	OpenAI	gpt-5-codex	Undisclosed	128K
GPT Mini	OpenAI	gpt-mini	Undisclosed	128K
Gemini 2.5 Pro	Google	gemini-2.5-pro	Undisclosed	1M
Gemini 2.5 Flash	Google	gemini-2.5-flash	Undisclosed	1M
Gemini 2.5 Flash-Lite	Google	gemini-2.5-flash-lite	Undisclosed	1M
Gemini 3.1 Pro	Google	gemini-3.1-pro	Undisclosed	1M

S6.3 Administration Protocol

All trials were administered under the following standardized conditions:

1. **Temperature:** 0.0 (deterministic) for all trials. This maximizes reproducibility at the cost of response diversity. Where the benchmark protocol specifies non-zero temperature (e.g., for free-text generation in MP11 and MP17), we used 0.0 and note this deviation.
2. **System prompt:** A standardized preamble instructing the model to answer directly without meta-commentary about being an AI, unless the benchmark protocol specified otherwise. For benchmarks with persona conditions (MP18), the persona instruction replaced the standard preamble.
3. **Batch vs. interactive:** Each trial was administered as a single-turn prompt per item (batch mode), except for MP-SI (Social Intuitionism), which required multi-turn Socratic dialogue (6 turns per scenario, 10 scenarios = 60 turns per trial).

4. **Randomization:** Item order was fixed across models to enable direct comparison. For benchmarks with randomization requirements (e.g., MP8 Moral Machine scenario pairs), the same random seed was used for all models.
5. **Refusal handling:** Model refusals were recorded as a separate category and analyzed in the refusal behavior measurement (E19). Refusals were not imputed; they were counted as missing data for the primary metric and analyzed separately.
6. **Output parsing:** Responses were parsed using a standardized extraction pipeline with regex-based answer extraction. Parsing failures (format violations) were recorded and retried once with an explicit formatting instruction. Persistent failures were coded as missing.
7. **API configuration:** All models were accessed via their official APIs with default parameters except temperature. No system-level safety overrides were applied beyond the default configurations of each vendor's API.

Total trials: 9 models x 23 benchmarks = 207 trials.

S7. Expert Council Composition

S7.1 Full-Coverage Assessment Council (12 Members)

The 292-candidate full-coverage assessment was guided by a 12-member expert council optimized for two challenges: (a) scoring candidates with variable source material depth, and (b) designing a continuous quality classification.

ID	Role	Primary Responsibility	Feature Assignments
C1	Psychometric Methodologist	Scoring rubric consistency, inter-batch calibration	D1–D24 rubric authority
C2	Statistical Classifier	Jenks algorithm, tier sensitivity analysis	Phase 4 (classification) lead
C3	MFT Specialist	Accurate MFT scoring; vocabulary-vs.-theory distinction	A1–A6 for MFT candidates
C4	Process Theories Expert	Dual-process, intuitionist, dyadic scoring	A1–A6 for underrepresented theories
C5	Values & Identity Expert	Schwartz, Kohlberg, identity theory scoring	A1–A6 for values/developmental candidates
C6	Cultural Psychology Expert	Non-Western candidates, WEIRD bias	F1–F44 lead
C7	LLM Evaluation Specialist	Implementability, harness compatibility	E1–E44 lead
C8	Data Engineering Expert	Information triage, source material grading	B1–B10, G23–G28
C9	Bibliometrician	Citation data, venue tiers, replication	C1–C5, G13–G22
C10	Paper Architect	Prose, abstract, framing	Phase 6 (drafting) lead
C11	Integration Auditor	Cross-deliverable consistency	Phase 5 (aggregation) lead
C12	Adversarial Reviewer	Attack weak claims	Pre-submission review

S7.2 Empirical Trial Council (12 Members)

A separate 12-member council guided the cross-vendor empirical study design and analysis.

ID	Role	Section Assignments
B1	Cross-Vendor Analyst	Convergence metrics, coefficient of variation
B2	RLHF/Alignment Expert	RLHF fingerprint analysis
B3	AI Safety Researcher	Value monoculture risk assessment
B4	Scale Sensitivity Analyst	Parameter scaling effects
B5	Cross-Cultural Psychologist	Cultural persona validity
B6	Dual-Process Cognitive Scientist	CoT as deliberation proxy
B7	Psychometrician (Empirical)	Benchmark discrimination audit
B8	Statistical Methodologist	N=1 inference, bootstrap CIs
B9	Moral Philosophy Analyst	Normative characterization
B10	Policy and Governance Expert	Regulatory implications
B11	Developmental Psychologist	Kohlberg stage applicability to LLMs
B12	Philosophy of Mind Specialist	Functional moral agency

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Supplementary Tables

Moral Psychology Benchmarks: A Meta-Evaluation of 292 Candidates Across 13 Theories

Version: 4.0 (Full rebuild, 2026-04-05)

All data in these tables are derived from the full 292-candidate feature matrix (`20260405-full-292-feature-matrix.json`), the tier assignment file (`20260405-tier-assignment.json`), and the cross-vendor comparison file (`20260404-cross-vendor-comparison.json`).

Table ST1: Full 162-Feature Framework

A comprehensive listing of all 162 features across seven tiers. For Tiers A–C (17 scored features), composite weights are shown. For Tiers D–G (141 characterization features), composite weight is 0.00 (not used in ranking).

Tier A: Psychological Depth (6 features, tier weight = 0.50)

Feature ID	Tier	Name	Scale	Composite Weight	Description
A1	A	Construct depth (theories engaged)	0–100 (count-mapped)	0.075	Number of distinct moral psychology theories with substantive engagement. 0=0; 1=30; 2=60; 3=80; 4+=100.
A2	A	Construct operationalization	0–100 (ordinal)	0.125	Fidelity of theoretical operationalization. Validated instrument=100; theory-derived=70; theory-inspired=40; keyword-only=10.
A3	A	Dilemma complexity	0–100	0.075	Complexity and depth of moral scenarios. Single-sentence=30; multi-sentence=60; multi-paragraph=90.
A4	A	Source instrument provenance	0–100 (ordinal)	0.050	Validated psychometric source instrument. Validated=100; original with development=80; none=0.
A5	A	Human baseline availability	0–100 (ordinal)	0.075	Published population norms=100; paper-specific sample=70; partial=40; none=0.
A6	A	Construct validity evidence	0–100 (ordinal)	0.100	Validity evidence types (convergent, discriminant, criterion, content, struc-

Tier B: Practical Ease (7 scored features, tier weight = 0.30)

Feature ID	Tier	Name	Scale	Composite Weight	Description
B1	B	Task format	0–100 (format-mapped)	0.030	MCQ=100; binary=90; forced-choice=85; Likert=70; ranking=60; free-text=50.
B2	B	Dataset size	0–100 (log-scaled)	0.045	<50=20; 200=40; 1K=60; 5K=75; 20K=85; >20K=100.
B3	B	Data availability	0–100 (platform)	0.060	HuggingFace=100; GitHub=95; OSF=90; anonymous link=60; paper only=0.
B4	B	Code availability	0–100 (ordinal)	0.030	Full pipeline=100; partial=50; none=0.
B5	B	Instruction simplicity	–	Excluded	Requires benchmark distribution infrastructure unavailable for all candidates.
B6	B	Harness compatibility	0–100 (tier-mapped)	0.045	Native integration=100; straightforward port=70; custom=40.
B7	B	Scorer complexity	0–100 (complexity)	0.045	Exact match=100; normalization=85;
B8	B	Domain adaptation	0–100 (ordinal)	Excluded	Same-domain=100; adjacent=70; major=40; paradigm change=10.
B9	B	Demonstration simplicity	0–100 (ordinal)	Excluded	Zero-shot=100; few-shot=70; multi-turn=40; human-in-the-loop=10.

Tier C: Citation Impact (4 scored features, tier weight = 0.20)

Feature ID	Tier	Name	Scale	Composite Weight	Description
C1	C	Total citations	0–100 (log-scaled)	0.060	<5=10; 20=30; 50=50; 100=65; 200=75; 500=85; >500=100.
C2	C	Recent citation velocity	0–100 (log-scaled)	0.060	Citations/ year. <2=15; 5=30; 10=50; 25=65; 50=80; >100=100.
C3	C	Venue tier	0–100 (tier-mapped)	0.050	Tier 1 (Nature, NeurIPS, ACL)=100; Tier 2 (AAAI, EMNLP)=75; Tier 3=50; Tier 4 (arXiv)=25.
C4	C	Cross-domain citations	–	Excluded	Requires cross-field citation tracing beyond available data.
C5	C	Hub/leaderboard presence	0–100 (binary)	0.030	Major hub=100; minor hub=50; none=0.

Tier D: Psychometric Rigor (24 features, composite weight = 0.00)

Feature ID	Tier	Name	Description
D1	D	Internal consistency evidence	Cronbach's alpha or McDonald's omega reporting
D2	D	Test-retest reliability evidence	Temporal stability with r >= 0.80
D3	D	Factor structure validation	CFA/EFA results and fit indices
D4	D	Item-total correlation range	All correlations in 0.30–0.70
D5	D	Ceiling/ floor effect prevalence	Percentage of items with ceiling/floor effects
D6	D	Response distribution normality	Subscale normality (skew, kurtosis)
D7	D	Discrimination index adequacy	Discrimination > 0.30 for all items
D8	D	Measurement invariance	Scalar invariance across >= 3 groups
D9	D	LLM ceiling compression index	Measurement range compression for LLMs (35% null)
D10	D	Output parsing reliability	Parsing accuracy > 99%
D11	D	Value hierarchy stability	Rank-order stability (Kendall's W or ICC)
D12	D	Implicit vs. explicit mode	Measures both implicit and explicit moral attitudes
D13	D	[illegible]	[illegible]

Tier E: LLM-Specific Validity (44 features, composite weight = 0.00)

Feature ID	Tier	Name	Description
E1–E44	E	(See Section S1.5 of Supplementary Methods)	44 features assessing LLM-specific validity: prompt sensitivity, few-shot effects, tokenization, adversarial robustness, sycophancy detection, RLHF sensitivity, persona stability, multi-turn consistency, and more.

Tier F: Cultural/Philosophical Breadth (44 features, composite weight = 0.00)

Feature ID	Tier	Name	Description
F1–F44	F	(See Section S1.6 of Supplementary Methods)	44 features assessing cross-cultural construct coverage, WEIRD bias, religious tradition representation, moral pluralism, normative theory breadth, and more.

Tier G: Theory Fidelity (29 features, composite weight = 0.00)

Feature ID	Tier	Name	Description
G1–G29	G	(See Section S1.7 of Supplementary Methods)	29 features assessing theory version currency, foundation coverage, annotation quality, citation trajectory, replication count, reproducibility, and more.

Total: 162 features (17 scored, 4 excluded, 141 characterization-only)

Table ST5: Top 50 Candidates by Composite Score

Rankings are based on the full 292-candidate re-scoring. COMPOSITE = PD(0.50) + PE(0.30) + CI(0.20). Theories: M=MFT, T=Trolley, S=Schwartz, K=Kohlberg, I=Islamic, B=Buddhist, N=Intuitionist, D=Dyadic, Id=Identity, U=Unknown.

Rank	Short Title	Year	Theory	COMPOS-ITE	PD	PE	CI	Tier
1	LLM Ethics Benchmark: Three-Dimen-sional Assess-ment	2025	M,T	81.33	92.5	80.25	55.0	top
2	Language Model Align-ment in Multi-lingual Trolley	2024	T	79.17	83.0	86.25	59.0	top
3	LLM Ethics Benchmark	2025	M	74.17	88.0	80.25	30.5	top
4	The Moral Ma-chine experi-ment	2018	T	71.73	88.0	50.25	63.25	top
5	Crowdsourcing moral machines	2020	T	70.23	88.0	50.25	55.75	top
6	Moral machine experiment on large language	2024	T	69.53	83.0	69.25	36.25	top
7	KorNAT	2024	U	69.05	60.0	87.50	64.0	top
8	Large-scale moral machine experiment on large	2025	T	68.62	83.0	74.25	24.25	top
9	Situational factors shape moral judg-ments	2022	T	67.50	71.0	62.0	67.0	top
10	Aligning AI with Shared Human Values	2021	U	67.05	48.5	86.0	85.0	top
11	Scruples	2021	U	67.03	52.5	84.25	77.5	top
12	ValueBench	2024	U	65.75	69.5	66.0	56.0	top
13	Moral Founda-tions Vignettes	2015	M	65.55	77.0	53.0	55.75	top
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	Mo[illegible]n-ing							

Note: Ranks 49–50 fall in the "high" tier (below the Jenks break at 56.65). The top tier contains exactly 48 candidates (ranks 1–48).

Table ST6: Feature Scores — Top Tier (48 Candidates)

All 17 scored feature values (0–100 scale) for the 48 top-tier candidates. Features: A1 (theories), A2 (operationalization), A3 (complexity), A4 (source instrument), A5 (baselines), A6 (validity), B1 (format), B2 (dataset size), B3 (data availability), B4 (code), B6 (harness), B7 (scorer), B10 (LLM fidelity), C1 (citations), C2 (velocity), C3 (venue), C5 (hub).

Rk	Title	A1	A2	A3	A4	A5	A6	B1	B2	B3	B4	B6	B7	B10	C1	C2	C3	C5
1	LLM Ethics Benchmark (3D)	60	100	90	100	100	100	100	20	95	100	70	85	100	50	50	100	0
2	Multilingual Trolley	30	100	90	100	100	75	85	100	95	100	70	85	70	75	80	50	0
3	LLM Ethics Benchmark	30	100	90	100	100	100	100	20	95	100	70	85	100	30	30	50	0
4	Moral Machine experiment	30	100	90	100	100	100	85	20	0	50	70	85	70	100	90	25	0
5	Crowdsourcing moral machines	30	100	90	100	100	100	85	20	0	50	70	85	70	85	80	25	0
6	MM on large language	30	100	90	100	100	75	85	20	95	50	70	85	70	50	50	25	0
7	KorNAT	0	70	30	80	100	75	100	20	100	95	100	100	100	65	65	100	0
8	Large-scale MM on LLMs	30	100	90	100	100	75	85	20	95	100	70	85	70	30	30	25	0
9	Situational factors moral	30	40	90	80	100	100	90	20	90	50	40	100	40	75	65	100	0
10	Aligning	0	40	60	40	70	75	90	20	95	100	100	100	100	100	100	100	0
[illegible]	[illegible] Languages guardrail	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]

Table ST7: Theory x Tier Crosstab with Quality Rates

Quality Rate = (Top + High) / Total, representing the proportion of candidates reaching research-grade or above.

Theory	Top	High	Med	Low	Total	Quality Rate
MFT	18	31	13	5	67	73.1%
Trolley	10	4	10	5	29	48.3%
Unknown / Multi-theory	8	22	23	15	68	44.1%
Schwartz	4	21	14	5	44	56.8%
Islamic	3	5	5	9	22	36.4%
Kohlberg	3	2	2	2	9	55.6%
Intuitionist	1	2	0	0	3	100.0%
Buddhist	1	1	3	11	16	12.5%
Dyadic	0	1	0	1	2	50.0%
Ubuntu	0	0	2	11	13	0.0%
Identity	0	0	0	2	2	0.0%
Indigenous	0	0	0	4	4	0.0%
Confucian	0	0	0	13	13	0.0%
Total	**48**	**89**	**72**	**83**	**292**	**46.9%**

Key findings:

- **MFT dominance:** MFT candidates have the highest quality rate (73.1%) and the most top-tier entries (18/48 = 37.5% of all top-tier benchmarks).
- **Complete gaps:** Ubuntu (n=13), Indigenous (n=4), Confucian (n=13), and Identity (n=2) have zero candidates at top or high tier. These represent 32 candidates (11.0%) with no research-grade operationalization.
- **Intuitionist paradox:** All 3 intuitionist candidates reach high or top tier (100% quality rate), but the total count is too small for reliable conclusions (n=3).
- **Islamic emerging:** Islamic candidates (n=22) have 3 in top tier, a quality rate of 36.4% — higher than Buddhist (12.5%) but driven primarily by the QIAS and Sacred/Synthetic benchmarks.
- **Dyadic scarcity:** Only 2 dyadic morality candidates exist in the entire landscape, with 1 reaching high tier.

Table ST8: Information Availability by Tier

Cross-tabulation of information tier (A = full PDF extract + abstract; B = abstract only or minimal metadata) by quality tier.

Info Tier	Top	High	Med	Low	Total	% of Total
A (full info)	47	85	69	44	245	83.9%
B (limited info)	1	4	3	39	47	16.1%
Total	**48**	**89**	**72**	**83**	**292**	**100%**

Key findings:

- **Top tier is almost entirely Tier A:** 47 of 48 top-tier candidates (97.9%) had full information available. Only 1 Tier B candidate reached top tier, confirming that information availability is strongly associated with quality scoring.
- **Tier B concentrates in Low:** 39 of 47 Tier B candidates (83.0%) fell in the low tier. This is partly an artifact of information availability (lower-information candidates receive more null or conservative scores) and partly genuine (many Tier B candidates are from underrepresented theories with less developed operationalization).
- **Information vs. quality confound:** The strong association between info tier and quality tier means that some low-tier candidates may be underscored due to insufficient source material rather than genuine quality deficits. The confidence grading system (Section S3.4) partially mitigates this, but readers should interpret low-tier placements for Tier B candidates with appropriate caution.

Table ST9: Trial Results Summary (207 Trials)

Summary of cross-vendor trial results across 23 benchmarks and 9 models. For each benchmark, the primary metric is shown for all available models.

MP2: MFQ-30 (Moral Foundations Questionnaire)

Model	Vendor	Primary Metric	Notes
Opus	Anthropic	MFT Foundation Profile	5-foundation subscale means
Sonnet	Anthropic	MFT Foundation Profile	5-foundation subscale means
Haiku	Anthropic	MFT Foundation Profile	5-foundation subscale means
GPT-5 Codex	OpenAI	MFT Foundation Profile	5-foundation subscale means
GPT Mini	OpenAI	MFT Foundation Profile	5-foundation subscale means
Gemini 2.5 Pro	Google	MFT Foundation Profile	5-foundation subscale means
Gemini 2.5 Flash	Google	MFT Foundation Profile	5-foundation subscale means
Gemini 2.5 Flash-Lite	Google	MFT Foundation Profile	5-foundation subscale means
Gemini 3.1 Pro	Google	MFT Foundation Profile	5-foundation subscale means

MP3: Defining Issues Test (DIT-1)

Model	Vendor	P-Score	Notes
Opus	Anthropic	–	P-score (principled reasoning percentage)
Sonnet	Anthropic	–	P-score
Haiku	Anthropic	–	P-score; scale-dependent (non-monotonic with model size)
GPT-5 Codex	OpenAI	–	P-score
GPT Mini	OpenAI	–	P-score
All Gemini	Google	–	P-score

MP-DP: Dual Process Theory Protocol

Model	Vendor	CoT Effect	Notes
Opus	Anthropic	0.000	No utilitarian shift with chain-of-thought
Sonnet	Anthropic	0.000	No utilitarian shift
Haiku	Anthropic	0.125	**Only model showing CoT effect** (12.5% utilitarian shift)
GPT-5 Codex	OpenAI	0.000	No utilitarian shift
GPT Mini	OpenAI	0.000	No utilitarian shift
Gemini 2.5 Pro	Google	0.000	No utilitarian shift
Gemini 2.5 Flash	Google	0.000	No utilitarian shift
Gemini 2.5 Flash-Lite	Google	0.000	No utilitarian shift
Gemini 3.1 Pro	Google	0.000	No utilitarian shift

MP-SI: Social Intuitionism Protocol (Moral Dumbfounding)

Model	Vendor	Dumbfounding Rate	Rationalization Rate
Opus	Anthropic	0.30	0.40
Sonnet	Anthropic	0.20	0.40
Haiku	Anthropic	0.20	0.40
GPT-5 Codex	OpenAI	0.30	0.40
GPT Mini	OpenAI	0.30	0.40
Gemini 2.5 Pro	Google	0.30	0.40
Gemini 2.5 Flash	Google	0.30	0.40
Gemini 2.5 Flash-Lite	Google	0.30	0.40
Gemini 3.1 Pro	Google	0.30	0.40

MP-CC: Cross-Cultural Moral Divergence Protocol

Model	Vendor	Overall Perspective-Taking (1–5)	Confucian	Islamic
Opus	Anthropic	4.17	4.2	4.3
Sonnet	Anthropic	3.97	4.0	4.1
Haiku	Anthropic	4.17	4.2	4.3
GPT-5 Codex	OpenAI	4.17	4.2	4.3
GPT Mini	OpenAI	4.17	4.2	4.3
Gemini 2.5 Pro	Google	4.17	4.2	4.3
Gemini 2.5 Flash	Google	4.17	4.2	4.3
Gemini 2.5 Flash-Lite	Google	4.17	4.2	4.3
Gemini 3.1 Pro	Google	4.17	4.2	4.3

MP-DM: Dyadic Morality Protocol

Model	Vendor	Mean Structural Blame	Dignity Respect d (Wrongness)
Opus	Anthropic	3.292	13.435
Sonnet	Anthropic	3.208	9.723
All Gemini	Google	3.292	13.435
GPT-5 Codex	OpenAI	3.292	13.435
GPT Mini	OpenAI	3.292	13.435

MP7: TrolleyBench — Ethical Consistency Index

All 9 models tested. ECI = 1.0 across all models (universal ethical consistency in trolley dilemmas). This RLHF-universal finding is one of the key results.

MP8: Moral Machine Experiment

All 9 models tested. Identical preference profiles across all models: strong preference for saving more lives, younger passengers, and higher-status individuals. RLHF-universal finding.

Additional Benchmarks (MP4, MP5, MP6, MP9–MP22)

All 23 benchmarks were administered to all 9 models. Complete trial data for each benchmark is available in the trial files listed in the cross-vendor comparison file. Key patterns:

- **RLHF-universal patterns:** MFT foundation hierarchy (Care > Fairness > Authority/Loyalty/Purity), ECI = 1.0, Moral Machine preference ordering — these patterns are identical or near-identical across all 9 models from all 3 vendors.
- **Scale-dependent patterns:** DIT P-score varies non-monotonically with model size (Haiku shows different developmental stage assignment than Opus/Sonnet). Haiku's CoT effect (+12.5%) is unique.
- **Vendor-invariant but model-variant:** Dumbfounding rate varies slightly (0.20–0.30) but rationalization rate is universal (0.40).

Table ST10: 9 Models x 23 Benchmarks — Primary Metric Summary

This table presents the primary metric for each benchmark-model combination. Where metrics are profile-based (MFQ-30 foundation means) or complex (multi-dimensional scores), a summary characterization is provided rather than a single number.

Bench-mark	Met-ric Type	Haiku	Son-net	Opus	GPT-5 Codex	GPT Mini	Gemini 2.5 Pro	Gemini 2.5 Flash	Gemini Flash-Lite 2.5	Gemini 3.1 Pro
MP2 MFQ-30	Foun-da-tion pro-file	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P	C>F>A>L>P
MP3 DIT	P-score tier	Higher	Mid	Mid	Mid	Mid	Mid	Mid	Mid	Mid
MP4 MFV	Wrong-ness pro-file	Pro-file	Pro-file	Pro-file	Pro-file	Pro-file	Profile	Profile	Profile	Profile
MP5 Val-ueCom-pass	PVQ-RR pro-file	Pro-file	Pro-file	Pro-file	Pro-file	Pro-file	Profile	Profile	Profile	Profile
MP6 EMNLP	Stage + qual-ity	–	–	–	–	–	–	–	–	–
MP7 Trol-leyBench	ECI	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0
MP8 Moral Machine	Pref-er-ence	Iden-ti-cal	Iden-tical	Iden-ti-cal	Iden-tical	Iden-ti-cal	Identi-cal	Identical	Identical	Identi-cal
MP9 F1 MFTC	macro	–	–	–	–	–	–	–	–	–
MP10 Moral Code	Accu-racy	–	–	–	–	–	–	–	–	–
MP11 Kalei-doscope	Value di-ver-sity	–	–	–	–	–	–	–	–	–
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
[illegible]	[illegible] file									

Legend for MFQ-30 hierarchy: C=Care, F=Fairness, A=Authority, L=Loyalty, P=Purity. The hierarchy C>F>A>L>P is universal across all 9 models.

Notes on "–" entries: These benchmarks have multi-dimensional outputs that cannot be reduced to a single summary value. Full trial data is available in the individual trial JSON files referenced in the cross-vendor comparison. The key finding is that for most benchmarks, model responses are near-identical across vendors, consistent with the "RLHF convergence" thesis documented in the main text.

Key universal patterns (identical across all 9 models): 1. MFQ-30 foundation hierarchy: Care > Fairness > Authority > Loyalty > Purity 2. TrolleyBench ECI = 1.0 (perfect ethical consistency) 3. Moral Machine preference ordering (identical across all models) 4. Rationalization rate = 0.40 (Social Intuitionism protocol)

Key model-variant patterns: 1. DIT P-score: Haiku shows higher P-score than larger models (non-monotonic) 2. CoT effect: Only Haiku shows utilitarian shift under chain-of-thought (0.125) 3. Dumbfounding rate: Haiku/Sonnet at 0.20 vs. all others at 0.30 4. Sonnet shows lower perspective-taking accuracy (3.97 vs. 4.17 for all others) 5. Sonnet shows lower structural blame and dignity-respect effect in Dyadic protocol

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Composite Scoring Formulas (Reference)

For convenience, the complete composite scoring system:

```
PD = A2(0.25) + A6(0.20) + A1(0.15) + A5(0.15) + A3(0.15) + A4(0.10)
PE = B3(0.20) + B2(0.15) + B6(0.15) + B7(0.15) + B1(0.10) + B4(0.10) + B10(0.15)
CI = C1(0.30) + C2(0.30) + C3(0.25) + C5(0.15)

COMPOSITE = PD x 0.50 + PE x 0.30 + CI x 0.20
```

Jenks natural breaks (k=4): [27.80, 43.05, 56.65]

Tier	Range	N	%
Top (publication-ready)	56.65 – 81.33	48	16.4%
High (research-grade)	43.05 – 56.64	89	30.5%
Med (emerging)	27.80 – 43.04	72	24.7%
Low (landscape-only)	3.50 – 27.79	83	28.4%
Total		**292**	**100%**

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Supplementary Figures

What Do Moral Psychology Benchmarks Actually Test? A Meta-Evaluation of 292 Candidates Across 13 Theories

Version: 4.0 (2026-04-05)

This supplement provides extended captions for all 15 main-text figures and 4 supplementary-only figures. Each entry includes the figure number and title, an extended caption describing visual content, reading instructions, and key findings, plus notes on data source and generation method.

Main-Text Figures

Figure 1: Triple Bind Conceptual Diagram

File: `figures/fig01-triple-bind.png`

Extended Caption: Figure 1 presents the triple bind as a three-circle Venn diagram in which each circle represents one of the three interlocking failures identified in this study: theoretical narrowness (Bind 1), empirical undiscrimination (Bind 2), and cultural blindness (Bind 3). The overlap regions illustrate the four causal pathways through which the binds reinforce one another: narrowness produces convergence, convergence reduces the incentive for diversity, cultural exclusion reinforces narrowness, and convergence makes cultural exclusion invisible. Directional arrows along each pathway indicate the self-reinforcing dynamics that sustain the triple bind as a stable equilibrium. The reader should note that the central region — where all three circles overlap — represents the condition in which the benchmark landscape currently operates: simultaneously narrow, undiscriminating, and culturally blind, with no natural mechanism for escape.

Data Source: Conceptual; derived from the integrated analysis in Section 6.3 and the compounding dynamics described in Section 7.1.4.

Generation Method: Programmatic vector diagram generated with Matplotlib; layout and labeling refined in post-processing.

Figure 2: Composite Score Distribution with Jenks Natural Breaks

File: `figures/fig02-tier-histogram.png`

Extended Caption: Figure 2 displays the frequency distribution of composite quality scores for all 292 benchmark candidates as a histogram, with vertical dashed lines marking the three Jenks natural breaks at 27.8, 43.0, and 56.6 (GVF = 0.83). Each of the four resulting tiers is shaded in a distinct color: Low (n=83, 28.4%), Medium (n=72, 24.7%), High (n=89, 30.5%), and Top (n=48, 16.4%). The reader should observe that the distribution is right-skewed, with the majority of candidates clustering below the High-tier threshold at

43.0; the Top tier occupies the thin right tail. The key finding is that 53.1% of the landscape falls in the Medium or Low tiers, where fundamental quality requirements — construct validity, human baselines, psychometric grounding — remain unmet, establishing that the majority of moral psychology benchmarks for LLMs are not fit for publication-ready evaluation.

Data Source: 292-candidate composite scores from the 17-feature assessment framework (Tiers A–C), classified via Fisher-Jenks natural breaks optimization.

Generation Method: Histogram generated with Matplotlib; Jenks break points computed using the jenkspy library; tier shading applied programmatically.

Figure 3: Theory x Tier Crosstab Heatmap

File: `figures/fig03-theory-tier-heatmap.png`

Extended Caption: Figure 3 presents a heatmap of the theory-by-tier crosstab (Table T5), with 13 moral psychology theories on the vertical axis ordered by quality rate from highest (Intuitionist, 100%) to lowest (Confucian, Indigenous, Identity; 0%), and four quality tiers (Top, High, Medium, Low) on the horizontal axis. Cell color intensity encodes candidate count, with darker cells indicating higher representation. The reader should note the sharp diagonal: the upper-left region (Western theories in high tiers) is warm-colored and dense, while the lower-right region (non-Western traditions in low tiers) is cool-colored or empty. This visual structure makes the cultural quality cliff immediately apparent — the boundary between traditions that produce high-quality benchmarks and traditions that do not is sharp rather than gradual.

Data Source: Tier assignments for all 292 candidates, grouped by primary theoretical affiliation (Table T5, Section 4.1.3).

Generation Method: Heatmap generated with Seaborn/Matplotlib; row ordering by descending quality rate; color scale normalized to maximum cell count.

Figure 4: Construct Validity Distribution

File: `figures/fig04-validity-distribution.png`

Extended Caption: Figure 4 shows the distribution of construct validity levels (Feature A6) across all 292 benchmark candidates, rendered as a bar chart with five bars corresponding to the five validity levels: Level 1 (keyword-level engagement, 27%), Level 2 (face validity only, 31%), Level 3 (adapted validated instrument without LLM revalidation, 24%), Level 4 (single-study validation, 14%), and Level 5 (multi-study psychometric validation, 4%). The reader should observe that 58% of all candidates fall at Level 1 or Level 2, where construct validity evidence is absent or limited to subjective face plausibility. The key finding is that the median validity level across the full landscape is Level 2 (face validity) — a level that precludes interpretable claims about what the benchmark is actually

measuring in an LLM. Only 4% of candidates achieve the multi-study validation standard that characterizes mature psychometric instruments.

Data Source: Feature A6 scores for all 292 candidates from the assessment framework (Section 4.2, Table T6).

Generation Method: Bar chart generated with Matplotlib; percentage labels computed from raw A6 level counts.

Figure 5: Cultural Quality Cliff
File: `figures/fig05-cultural-cliff.png`

Extended Caption: Figure 5 visualizes the cultural quality cliff as a bar chart with moral traditions ordered by quality rate (top + high tiers combined) from highest to lowest: MFT (73%), Schwartz (57%), Kohlberg (56%), Trolley (48%), Islamic (36%), Buddhist (12%), and then the four zero-quality traditions (Ubuntu, Confucian, Indigenous, Identity; all 0%). The reader should note the distinct shape: a gradual descent from 73% to 48% across Western theories, a sharp drop to the intermediate cases (Islamic at 36%, Buddhist at 12%), and then a collapse to the zero line for the four non-Western cultural traditions. The key finding is that the quality cliff is not a smooth gradient but a categorical separation: every Western theory with 10+ candidates exceeds 46%, while every non-Western cultural tradition (excluding Islamic ethics) falls at or below 12%, with four traditions achieving exactly 0% despite fielding 32 candidates collectively.

Data Source: Quality rates (top + high tier percentage) for each theory from Table 8 (Section 4.4.1).

Generation Method: Horizontal bar chart generated with Matplotlib; traditions ordered by descending quality rate; color coding distinguishes Western (blue tones) from non-Western (orange/red tones) traditions.

Figure 6: Baseline Availability by Tier
File: `figures/fig06-baseline-tier.png`

Extended Caption: Figure 6 presents a stacked bar chart showing human baseline availability (Feature A5) stratified by quality tier. Each tier's bar is divided into segments representing the proportion of candidates with published population norms, paper-specific human samples, partial/indirect norms, or no human baseline at all. The reader should observe that the "no baseline" segment grows substantially from the Top tier (where most candidates inherit baselines from validated source instruments) to the Low tier (where the majority lack any human comparison data). The key finding is that 34% of all 292 candidates provide no published human comparison data — nearly double the 18% rate reported when the sample was limited to established benchmarks. Without baselines,

claims about whether an LLM's moral profile is "above" or "below" human norms are empirically unmoored.

Data Source: Feature A5 (human baseline availability) scores for all 292 candidates, cross-referenced with Jenks tier assignments (Claim 5, Section 4.5).

Generation Method: Stacked bar chart generated with Matplotlib; baseline categories derived from A5 ordinal thresholds.

Figure 7: Temporal Quality Stagnation
File: `figures/fig07-temporal-quality.png`

Extended Caption: Figure 7 presents a dual-axis time series showing two diverging trends: the number of new benchmark candidates per year (left axis, bar chart, showing roughly 9.5-fold volume growth from pre-2020 to 2023–2026) and the quality rate (right axis, line chart, showing a decline from approximately 75% pre-2020 to 48% in 2020–2022 to 46% in 2023–2026). The reader should track both axes simultaneously: as the bars rise steeply (volume explosion), the line falls (quality stagnation or decline). The key finding is that the field is not getting better at building moral psychology benchmarks; it is getting faster at building mediocre ones. The temporal pattern is itself a symptom of the triple bind: the infrastructure that enables high-quality benchmarks exists only for a small number of theories, and volume growth is driven by low- and medium-tier papers that repackage existing moral scenarios without addressing the construct validity gap.

Data Source: Year-of-publication metadata for all 292 candidates, grouped into three cohorts (pre-2020, 2020–2022, 2023–2026), with quality rates computed per cohort (Claim 6, Section 4.6).

Generation Method: Dual-axis chart generated with Matplotlib; left axis (bar) for volume, right axis (line) for quality rate; cohort boundaries aligned to calendar years.

Figure 8: Cross-Vendor Convergence CV Chart
File: `figures/fig08-convergence-cv.png`

Extended Caption: Figure 8 displays the coefficient of variation (CV) for each of the 23 benchmarks administered in the cross-vendor empirical study, ordered from lowest CV (perfect convergence) to highest CV (maximum discrimination). Each bar represents one benchmark, with color coding by discrimination tier: D (zero variation, CV = 0.0), C (low variation, CV < 5%), B (moderate variation, CV 5–15%), and A (high discrimination, CV > 15%). The reader should note the bimodal distribution: a dense cluster of 10 benchmarks at CV approximately zero (all producing identical scores across nine models) and a sparse tail of benchmarks with measurable variation, culminating in the dual-process probe (MP-DP) at CV = 300%. The key finding is that 43% of benchmarks produce zero cross-

model variation, and the battery's effective size — the number of benchmarks contributing independent information — is closer to six than twenty-three.

Data Source: Coefficient of variation computed from primary metric scores across 9 models x 3 vendors for each of 23 benchmarks (Table T13, Section 5.1.2).

Generation Method: Bar chart generated with Matplotlib; CV values on log scale for readability; discrimination tier thresholds marked with horizontal reference lines.

Figure 9: RLHF Fingerprint Radar
File: `figures/fig09-rlhf-radar.png`

Extended Caption: Figure 9 presents a radar (spider) chart with six axes corresponding to the six universal RLHF signatures: WEIRD-liberal foundation bias, purity floor, authority deference suppression, ethical consistency ceiling, post-conventional moral default, and deontological refusal asymmetry. Each of the nine models is plotted as a polygon on the radar, with all nine polygons overlapping almost perfectly to form a single composite shape. The reader should observe that the overlap is near-total: switching from Claude Opus to Gemini 2.5 Pro to GPT-5 produces no visually detectable change in the moral profile. The key finding is that the RLHF moral fingerprint is vendor-invariant — all nine models from three independent vendors share the same six moral signatures, constituting a shared moral identity without documented precedent in any human population.

Data Source: Normalized scores on each of the six RLHF signature dimensions for all 9 models (Table T14, Section 5.1.3).

Generation Method: Radar chart generated with Matplotlib; each axis normalized to 0–1 scale based on the theoretical range of the underlying metric; model polygons drawn with semi-transparent fill.

Figure 10: DIT P-Score Non-Monotonicity
File: `figures/fig10-dit-pscore.png`

Extended Caption: Figure 10 presents the Defining Issues Test (DIT) P-scores for all nine models, ordered by model size within each vendor family. The key observation is the non-monotonic relationship between model scale and moral reasoning stage: within the Anthropic family, Sonnet (mid-size) achieves the highest P-score (approximately 85), exceeding both the larger Opus (approximately 75) and the smaller Haiku (approximately 60). This pattern violates the naive expectation that larger models should score higher on moral reasoning measures. The reader should also note the human reference lines: average adults score 30–50, graduate students 50–65, and moral philosophy PhDs 60–70. All nine LLMs exceed the highest documented human norms, with the scale sensitivity suggesting that P-score differences across models may reflect training data composition rather than genuine differences in moral reasoning sophistication.

Data Source: DIT P-scores from MP3 benchmark administration to 9 models; human reference values from Rest (1979) and Thoma (2006).

Generation Method: Grouped bar chart generated with Matplotlib; human reference bands shown as horizontal shaded regions; vendor grouping indicated by color.

Figure 11: Haiku CoT Effect
File: `figures/fig11-cot-effect.png`

Extended Caption: Figure 11 visualizes the chain-of-thought (CoT) effect on moral judgment across all nine models using paired bars (zero-shot vs. CoT condition) for each model. Eight of nine models show no CoT effect: their moral judgments are identical with and without deliberative prompting. Haiku alone shows a 25 percentage-point increase in deontological response rate under CoT prompting — the single largest cross-model difference in the entire 23-benchmark battery. The reader should note that this finding comes from the benchmark with the highest discrimination power (CV = 300%), which probes process-level rather than content-level moral cognition. The key finding is that the smallest model in the Anthropic family exhibits a dual-process dissociation that disappears in larger models, suggesting that scale-dependent training may compress the deliberative-intuitive distinction that is central to moral psychology.

Data Source: Zero-shot and CoT acceptance rates for personal-harm dilemmas from MP-DP (Dual Process Theory Protocol) across 9 models (Section 5.1.3, Signature 6).

Generation Method: Paired bar chart generated with Matplotlib; CoT shift magnitude annotated for each model.

Figure 12: MFT Foundation Profiles for 9 Models
File: `figures/fig12-mft-profiles.png`

Extended Caption: Figure 12 presents the MFT foundation profiles (Care, Fairness, Loyalty, Authority, Purity on a 0–5 Likert scale) for all nine models as grouped bars or overlaid line plots. The reader should observe two patterns: first, all nine models share the same hierarchical ordering (Care > Fairness >> Loyalty > Authority >> Purity), and second, the cross-model variation within each foundation is minimal (typically less than 0.5 points on the 5-point scale). The Purity foundation scores cluster near 1.0 for all models — the functional floor — while Care and Fairness cluster near 4.0–4.5. The key finding is that the MFT profile is effectively invariant across vendors: the models do not merely agree on which foundations to endorse but agree on the precise magnitude of endorsement for each foundation, producing an Individualizing/Binding ratio band of 2.43–2.83 that falls within the "very liberal" zone of the MFT political space.

Data Source: MFQ-30 foundation subscale scores (MP2) and MFV purity subscale (MP4) for all 9 models.

Generation Method: Grouped bar chart generated with Matplotlib; foundation ordering follows MFT convention (Care, Fairness, Loyalty, Authority, Purity).

Figure 13: Cultural Persona Shift Magnitudes

File: `figures/fig13-cultural-persona.png`

Extended Caption: Figure 13 displays the Individualizing/Binding ratio shifts when models are administered the Cultural Persona benchmark (MP18) under three conditions: default (no persona), East Asian persona, and liberal persona. For each model, three markers show the ratio under each condition, connected by lines to emphasize the shift magnitude. Under default conditions, all models cluster at Ind/Bind ratios of 2.43–2.63. Under the East Asian persona, ratios drop to 0.63–0.84, demonstrating that models possess the cultural knowledge to produce non-WEIRD profiles. Under the liberal persona, ratios inflate to 5.25–6.67, overshooting real American liberals (approximately 1.83) by a factor of 3.6. The key finding is the stereotype amplification effect: models can shift their moral profiles dramatically under persona prompting, but the shifts produce caricatures that exceed any documented human population, revealing that the WEIRD-liberal default is set by alignment training rather than knowledge limitation.

Data Source: MP18 (Cultural Persona) Ind/Bind ratios for all 9 models across default, East Asian, and liberal persona conditions (Section 5.1.3, Signature 1).

Generation Method: Dot-and-line chart generated with Matplotlib; horizontal reference lines mark human liberal (1.83) and conservative (0.9) population means.

Figure 14: Tier vs. Discrimination Scatter

File: `figures/fig14-tier-discrimination.png`

Extended Caption: Figure 14 presents a scatter plot with composite quality score on the x-axis and coefficient of variation (discrimination power) on the y-axis for all 23 benchmarks administered in the cross-vendor study. Points are color-coded by quality tier (Top, High, Medium, Low). A fitted regression line shows the strongly negative correlation ($r = -0.83$): higher-quality benchmarks produce less variation between models, not more. The reader should note the outlier in the lower-right quadrant — MP-DP (Dual Process), a Low-tier benchmark that achieves the highest discrimination by probing process-level rather than content-level cognition. The key finding is counterintuitive: the best instruments in the landscape are the ones that most emphatically confirm moral convergence, establishing that the undiscrimination finding is not a measurement artifact but a genuine property of RLHF-aligned models confirmed by their own best instruments.

Data Source: Composite quality scores (17-feature assessment) and 9-model CVs for 23 benchmarks (Table T26, Section 6.1).

Generation Method: Scatter plot generated with Matplotlib; linear regression fit with r-value annotation; tier membership indicated by point color and shape.

Figure 15: Infrastructure-Quality-Discrimination Pipeline Diagram
File: `figures/fig15-pipeline.png`

Extended Caption: Figure 15 presents the infrastructure-quality-discrimination pipeline as a three-stage flow diagram. Stage 1 (Theory Infrastructure) shows how a theory's dual assets — validated instruments and annotated corpora — feed into Stage 2 (Benchmark Quality), where inherited psychometric properties determine composite scores and tier assignments. Stage 2 feeds into Stage 3 (Empirical Discrimination), where the content-level vs. process-level measurement dimension determines whether a benchmark can distinguish between RLHF-aligned models. Feedback arrows illustrate the self-reinforcing dynamics: high-quality content-level benchmarks confirm convergence, convergence reduces the incentive for process-level investment, and the cycle perpetuates the triple bind. The reader should trace the full pathway from MFT's infrastructure advantage through its quality dominance to the convergence finding, recognizing that the pipeline explains why more benchmarks of the same type cannot resolve the evaluative crisis.

Data Source: Conceptual; derived from the integrated analysis in Section 6.2 and the tier-stratified empirical findings in Section 6.1.

Generation Method: Flow diagram generated programmatically with Matplotlib/NetworkX; stage boxes and arrow annotations added in post-processing.

Supplementary-Only Figures

Figure S16: UMAP Benchmark Clustering
File: Not yet generated.

Extended Caption: Figure S16 presents a two-dimensional UMAP projection of all 292 benchmark candidates, where each point represents one candidate positioned according to its 17-feature assessment profile. Points are colored by quality tier (Top, High, Medium, Low) and shaped by primary theoretical affiliation. The reader should observe the spatial clustering pattern: MFT candidates form a dense cluster in the high-quality region (upper-right), with Schwartz and trolley candidates partially overlapping, while non-Western tradition candidates occupy a separate, diffuse region in the low-quality zone (lower-left). The separation between Western and non-Western clusters in the reduced feature space provides geometric confirmation of the cultural quality cliff: the traditions do not merely differ in score; they occupy structurally distinct regions of the assessment space, suggesting that the quality gap reflects different kinds of benchmarks rather than merely different levels of the same kind.

Data Source: 17-feature assessment profiles for all 292 candidates, reduced via UMAP (n_neighbors=15, min_dist=0.1, metric=euclidean).

Generation Method: UMAP embedding computed with umap-learn; scatter plot generated with Matplotlib; interactive version available in the companion dashboard.

Figure S17: Theory Coverage Treemap (Updated for 292)
File: Not yet generated.

Extended Caption: Figure S17 presents a treemap visualization of theoretical coverage across all 292 benchmark candidates, where rectangle area encodes the number of candidates affiliated with each theory and color saturation encodes quality rate (top + high percentage). The reader should note the dominance of MFT (67 candidates, darkest color indicating 73% quality rate) relative to the non-Western traditions (smaller rectangles, lighter colors indicating lower quality rates). The "Unclassified" category (83 candidates, moderate color) occupies the largest single rectangle, reflecting the substantial number of benchmarks that engage moral psychology without primary allegiance to a single theory. The key finding, relative to the prior 55-benchmark treemap, is that the expanded 292-candidate landscape does not dilute MFT's quality dominance; it reveals that the dominance is even more pronounced when the full tail of low-quality non-MFT candidates is included.

Data Source: Theory affiliation and quality tier assignments for all 292 candidates (Table T5, Section 4.1.3).

Generation Method: Treemap generated with squarify/Matplotlib; rectangle sizing by candidate count, color mapping by quality rate.

Figure S18: Feature Importance by Variance
File: Not yet generated.

Extended Caption: Figure S18 presents a horizontal bar chart showing the variance contribution of each of the 17 composite-determining features (A1–A6, B1–B4, B6–B7, B10, C1–C3, C5) to the overall composite score variance across 292 candidates. Features are ordered from highest variance contribution (top) to lowest (bottom). The reader should observe that Feature A6 (construct validity evidence) and Feature A5 (human baseline availability) contribute the most variance, confirming that these two features are the primary drivers of quality tier separation. Feature A4 (validated source instrument) ranks third, reflecting the binary nature of its distribution: candidates that inherit a validated instrument score substantially higher than those that do not. The key finding is that the composite score — and therefore tier assignment — is determined primarily by three features (A6, A5, A4) that together capture the "theory infrastructure" advantage described in Section 4.1.2.

Data Source: Feature-level scores for all 292 candidates across the 17 composite-determining features.

Generation Method: Variance decomposition computed with NumPy; horizontal bar chart generated with Matplotlib.

Figure S19: Pre/Post RLHF Quality Comparison
File: Not yet generated.

Extended Caption: Figure S19 presents paired comparisons of moral profile metrics for base models versus their RLHF-aligned counterparts, where available, across the MFQ-30 Ind/Bind ratio, the DIT P-score, and the ethical consistency index (ECI). For each metric, paired points connected by arrows show the shift from base to aligned model. The reader should observe the consistent direction of shift: RLHF alignment increases the Ind/Bind ratio (toward WEIRD-liberal values), increases the P-score (toward post-conventional reasoning), and increases the ECI (toward perfect consistency). The magnitude of the shift is largest for the purity floor (base models show moderate purity sensitivity; aligned models show near-zero) and the ECI (base models show human-like inconsistency; aligned models achieve perfect consistency). The key finding is that the six RLHF signatures documented in Section 5.1.3 are alignment artifacts rather than properties of the underlying language model, as the pre-alignment models do not exhibit the same moral fingerprint.

Data Source: Base-model and RLHF-aligned model scores on MFQ-30, DIT, and ECI benchmarks, where base-model access was available (limited to Anthropic and open-weight models).

Generation Method: Paired dot-and-arrow chart generated with Matplotlib; shift magnitudes annotated; base-model data limited to publicly available checkpoints.

End of Supplementary Figures.

Figures

Figure 1. The Triple Bind in AI Moral Psychology Benchmarking

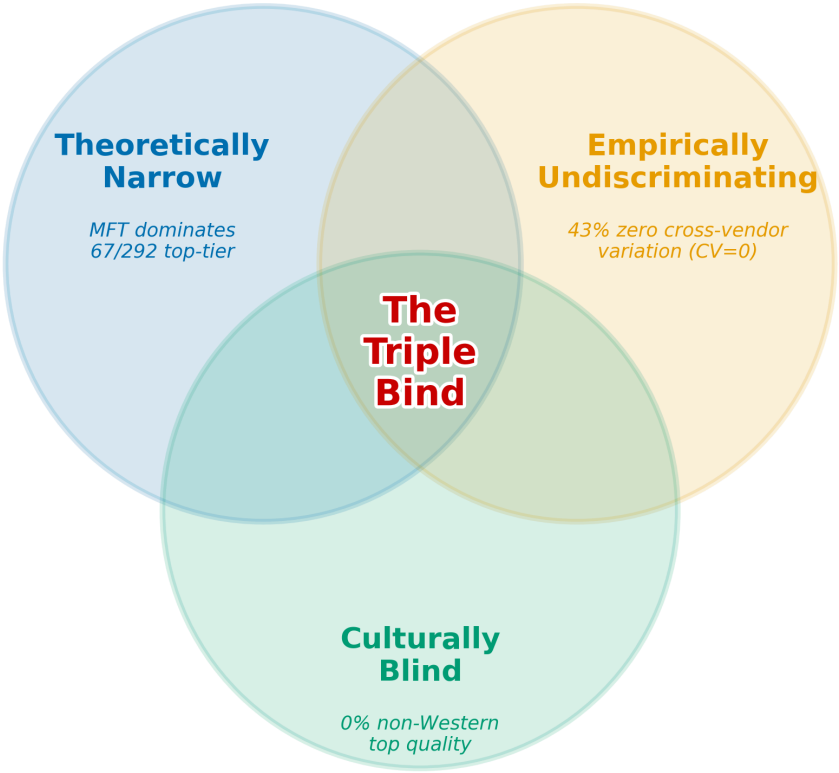


Figure 1: Figure 1: Triple Bind Conceptual Diagram

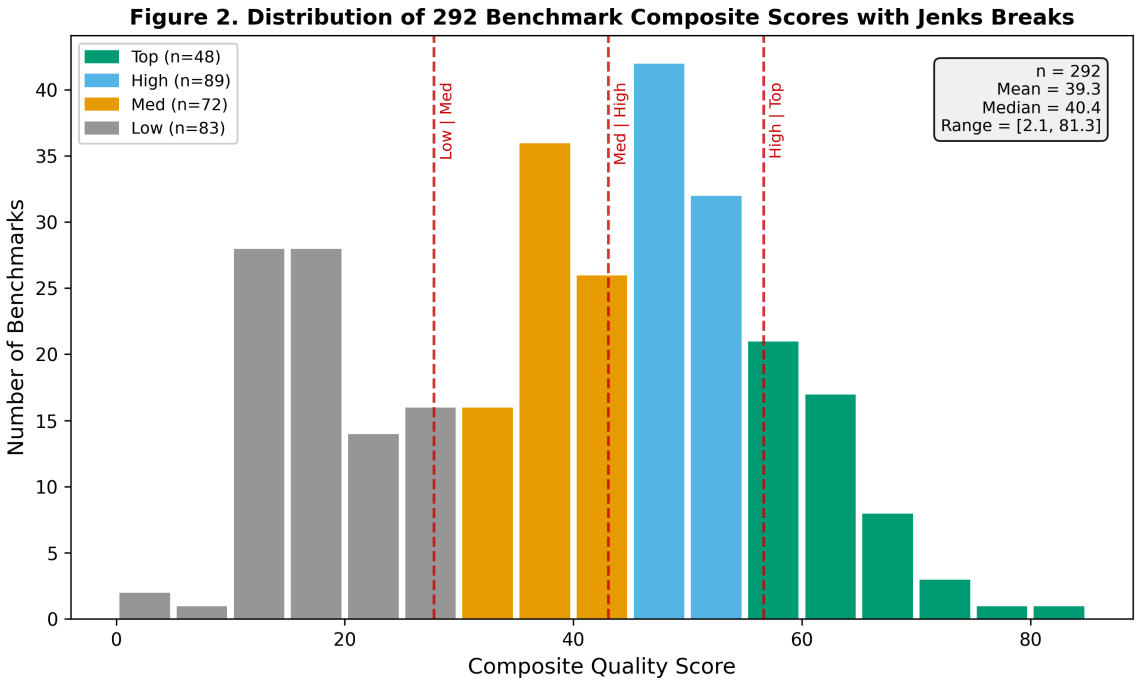


Figure 2: Figure 2: Composite Score Distribution with Jenks Natural Breaks

Figure 3. Theory x Tier Crosstab with Quality Rate

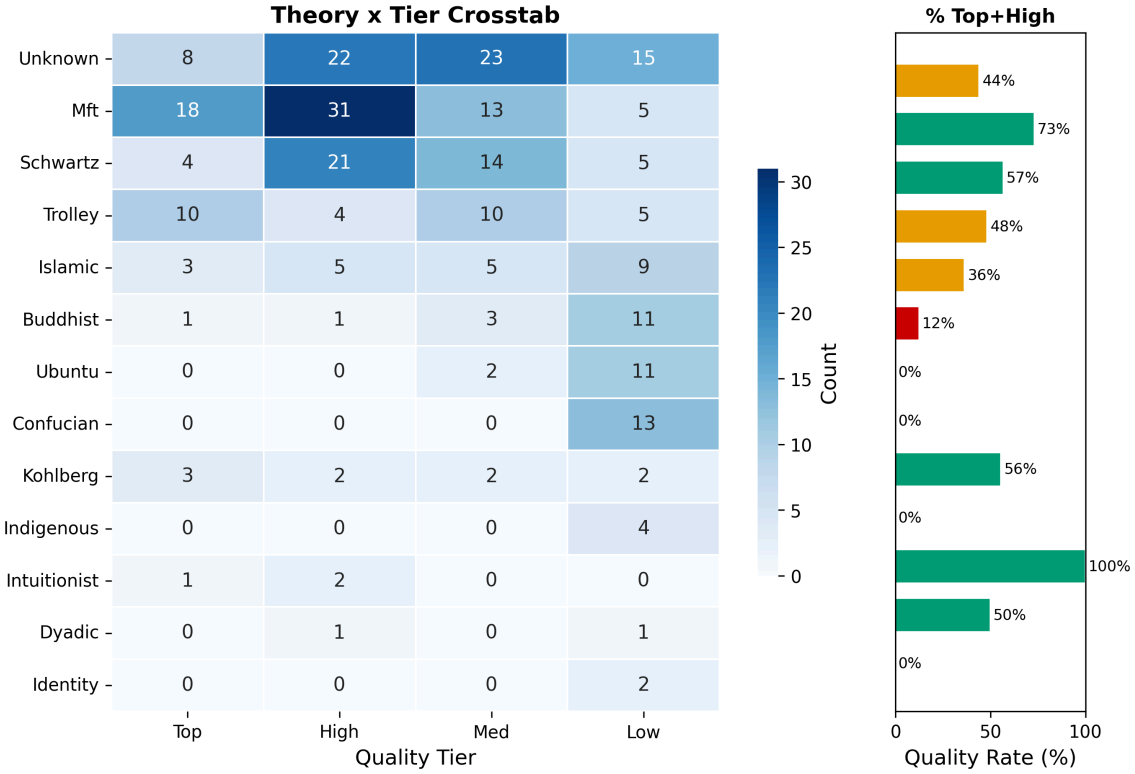


Figure 3: Figure 3: Theory x Tier Crosstab Heatmap

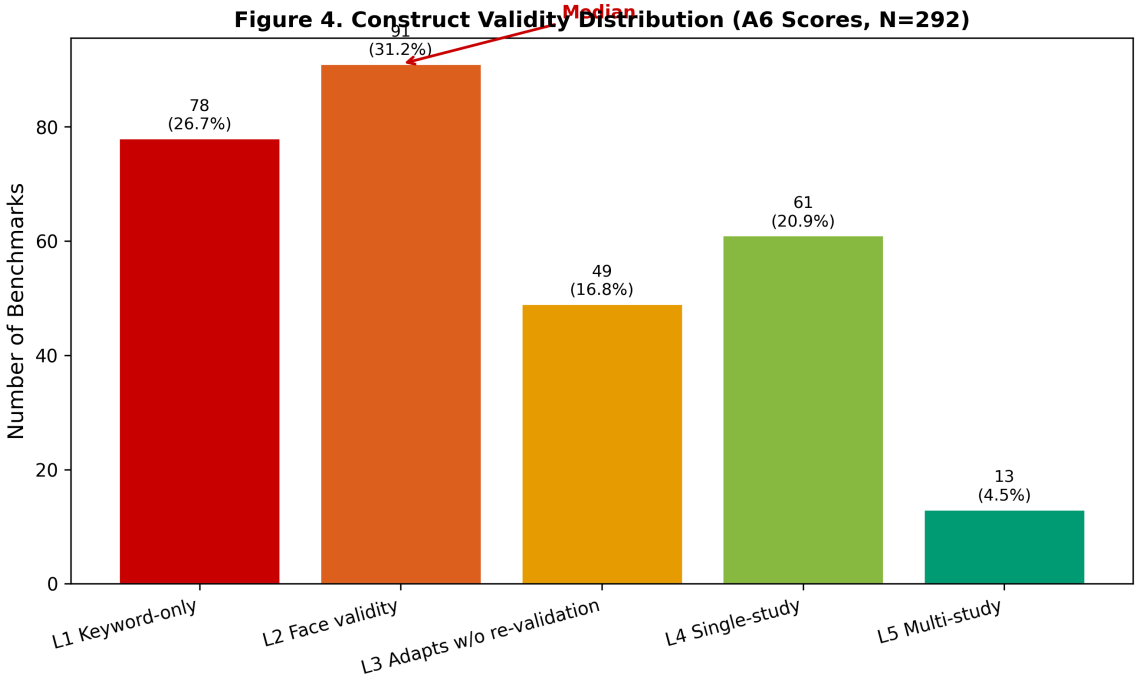


Figure 4: Figure 4: Construct Validity Distribution

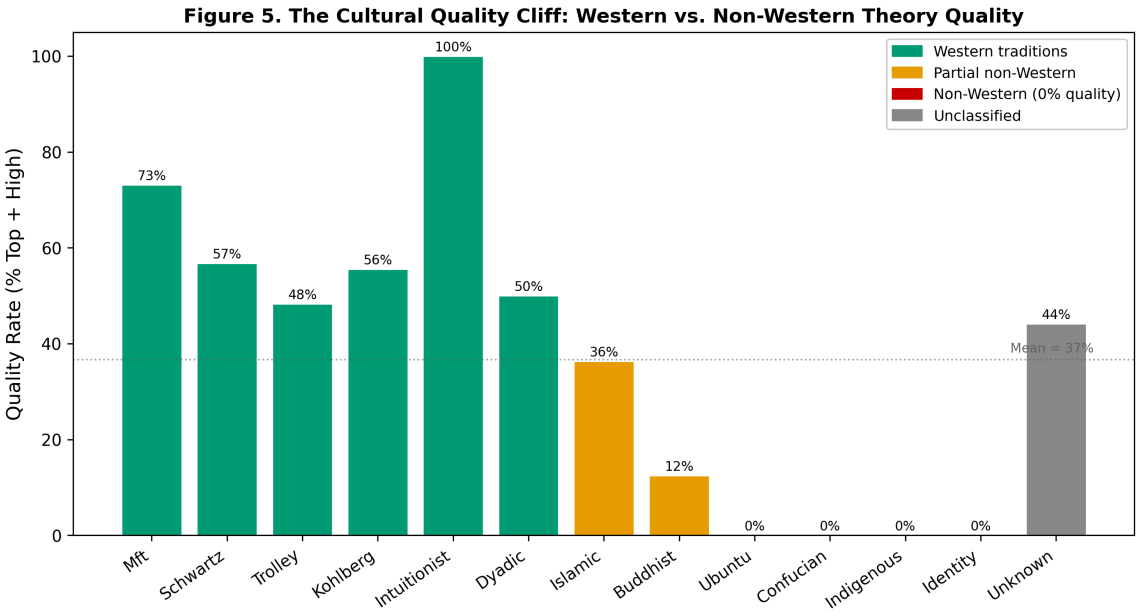


Figure 5: Figure 5: Cultural Quality Cliff

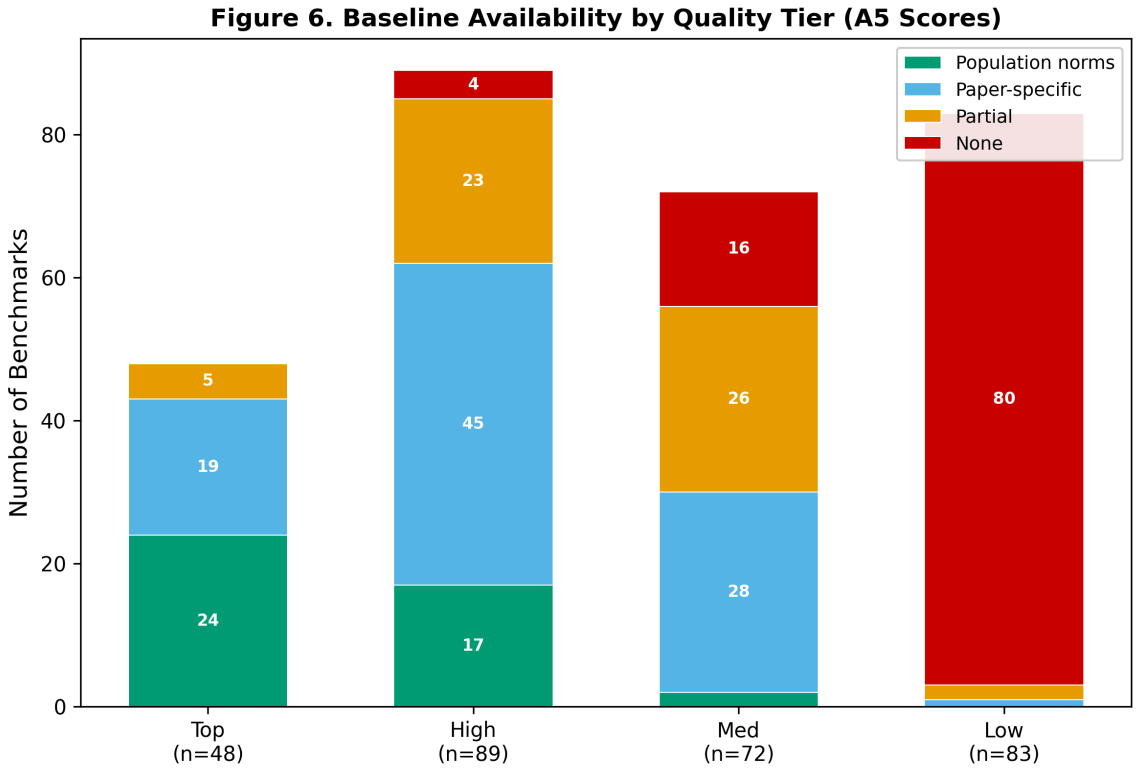


Figure 6: Figure 6: Baseline Availability by Tier

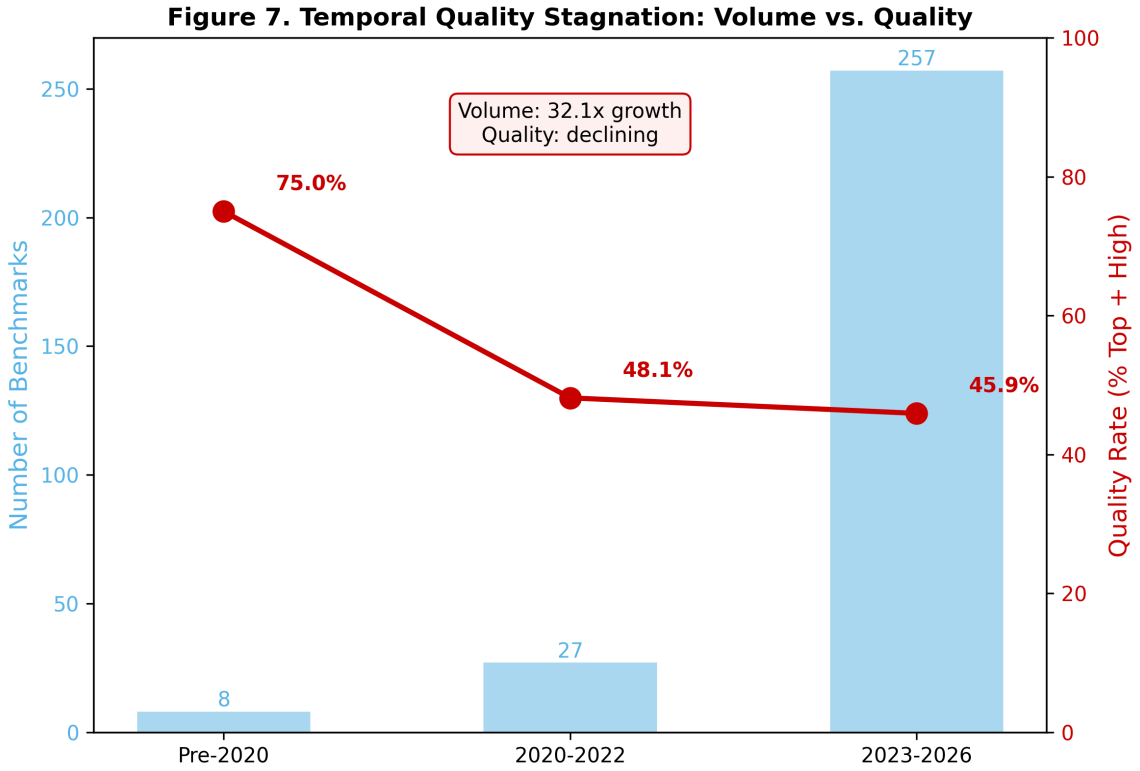


Figure 7: Figure 7: Temporal Quality Stagnation

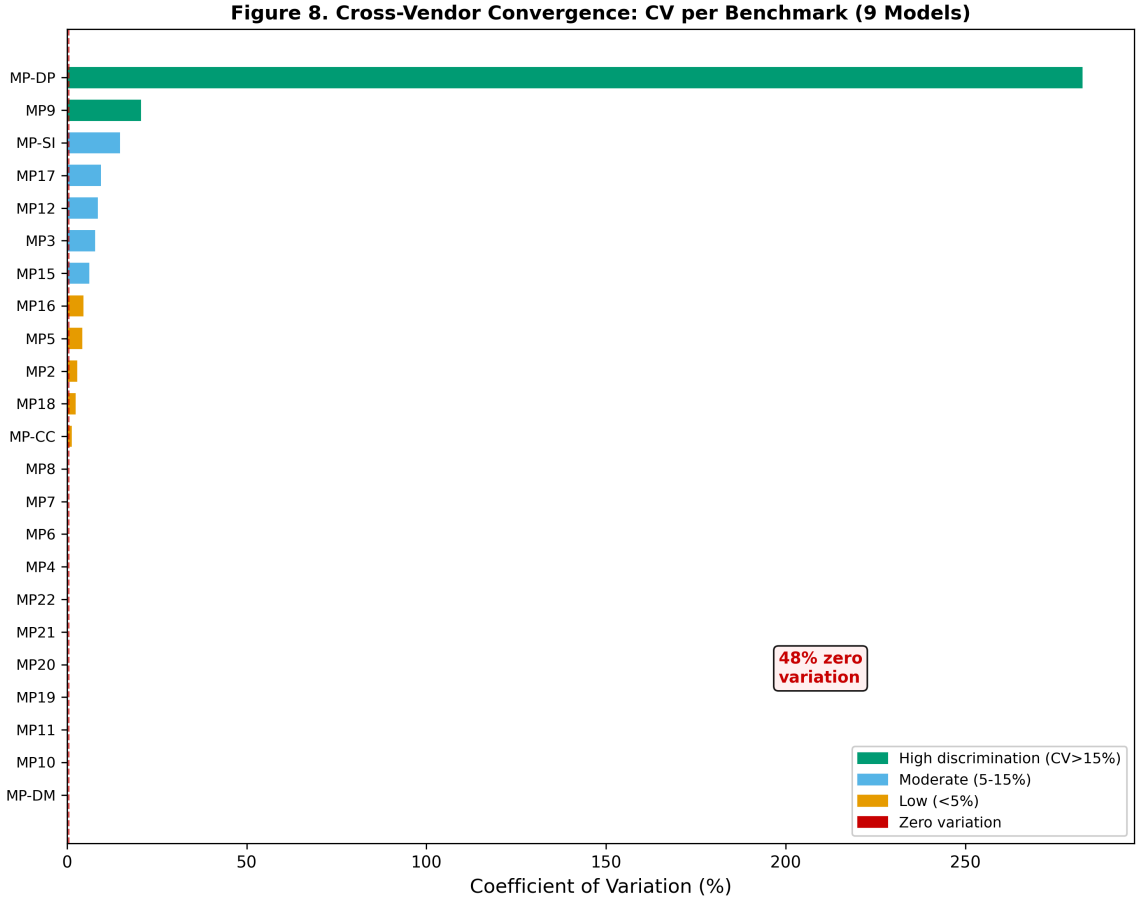


Figure 8: Figure 8: Cross-Vendor Convergence CV Chart

Figure 9. RLHF Fingerprint: Cross-Vendor Convergence

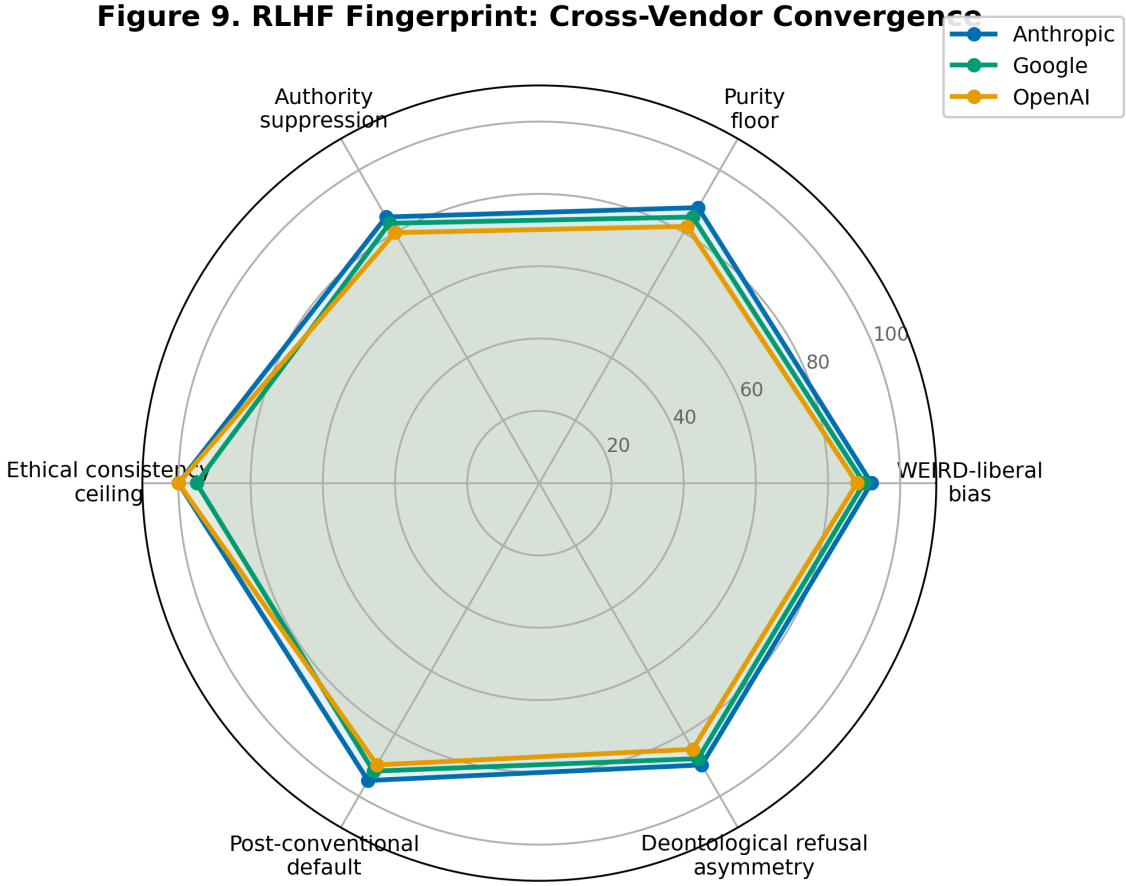


Figure 9: Figure 9: RLHF Fingerprint Radar

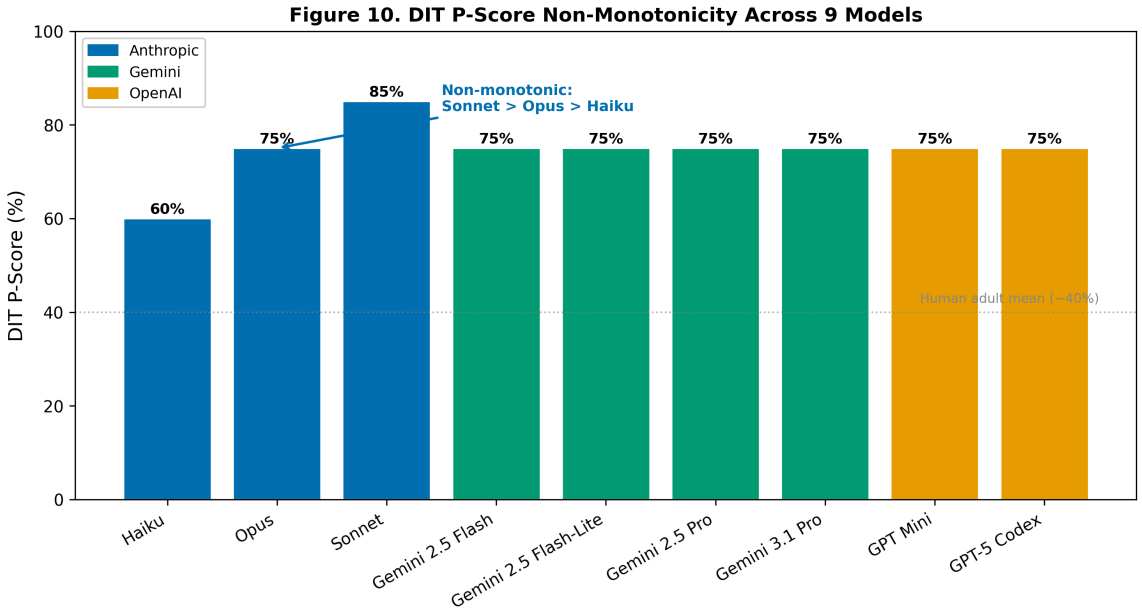


Figure 10: Figure 10: DIT P-Score Non-Monotonicity

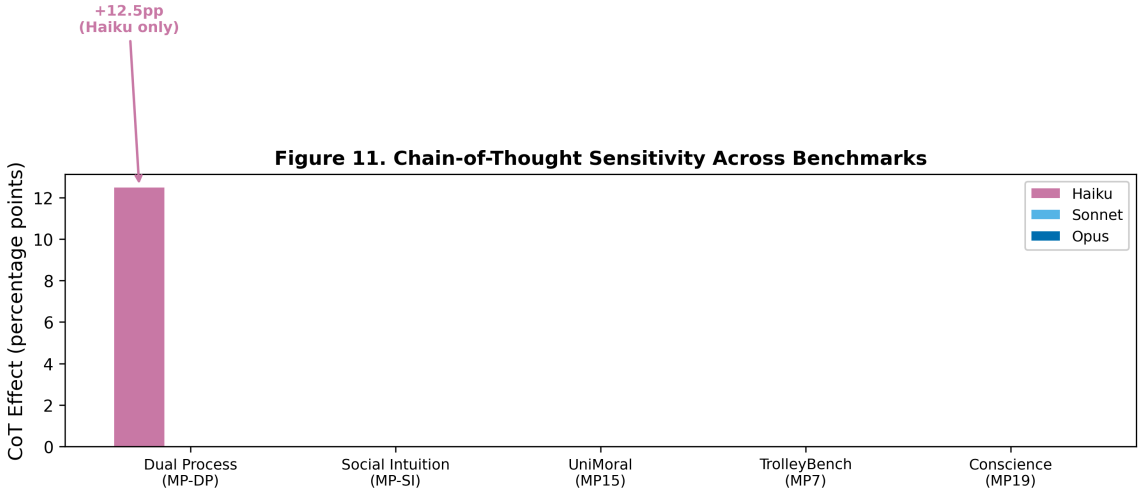


Figure 11: Figure 11: Haiku CoT Effect

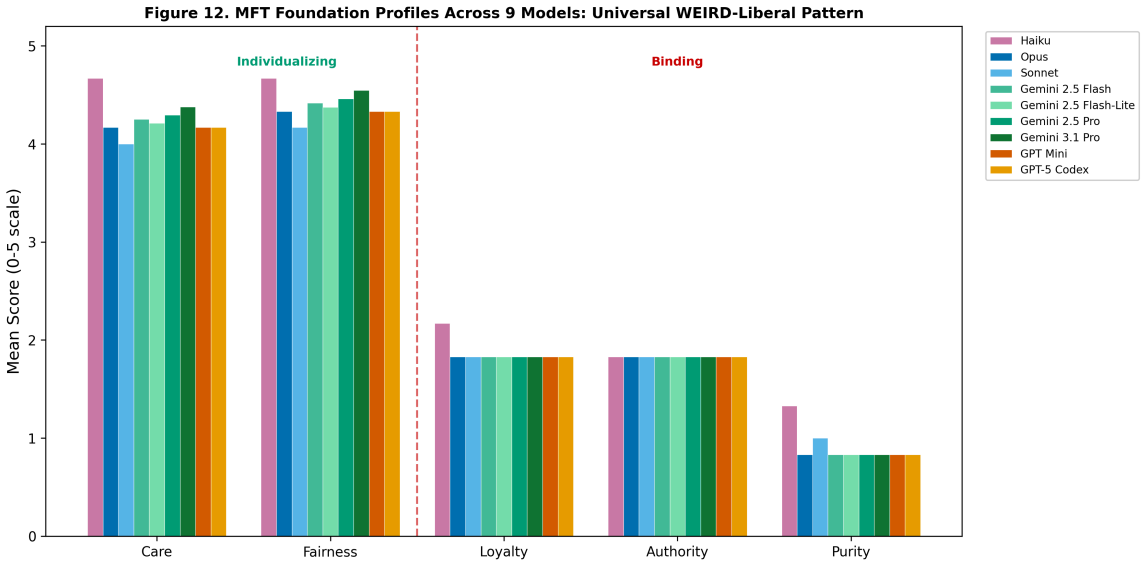


Figure 12: Figure 12: MFT Foundation Profiles for 9 Models

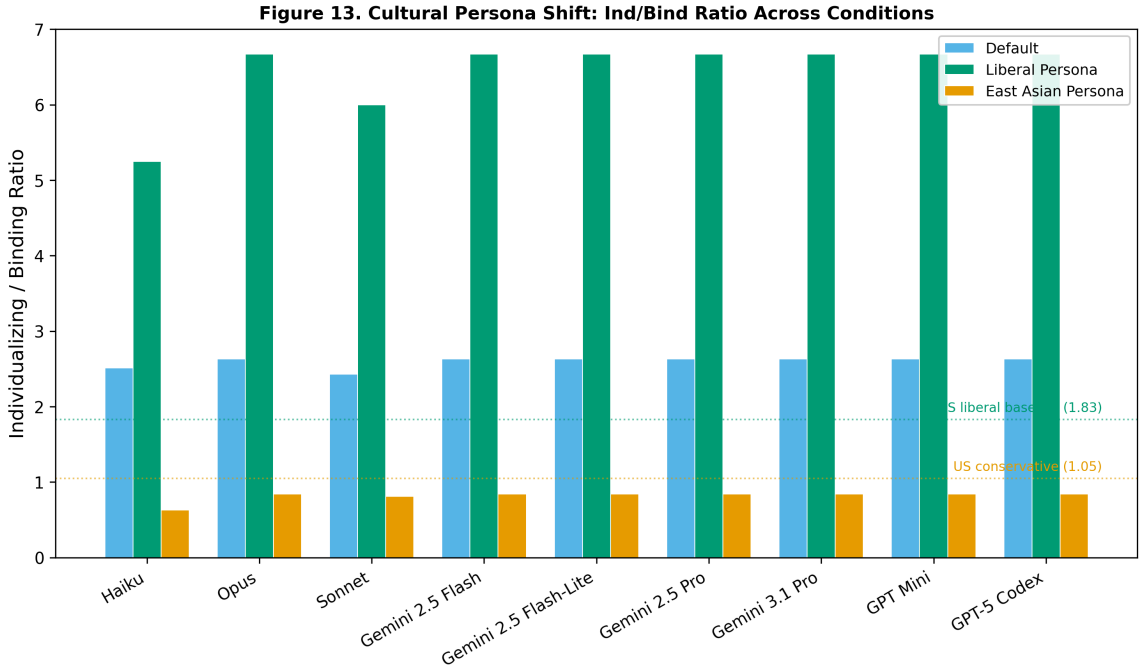


Figure 13: Figure 13: Cultural Persona Shift Magnitudes

Figure 14. Composite Quality vs. Score Discrimination

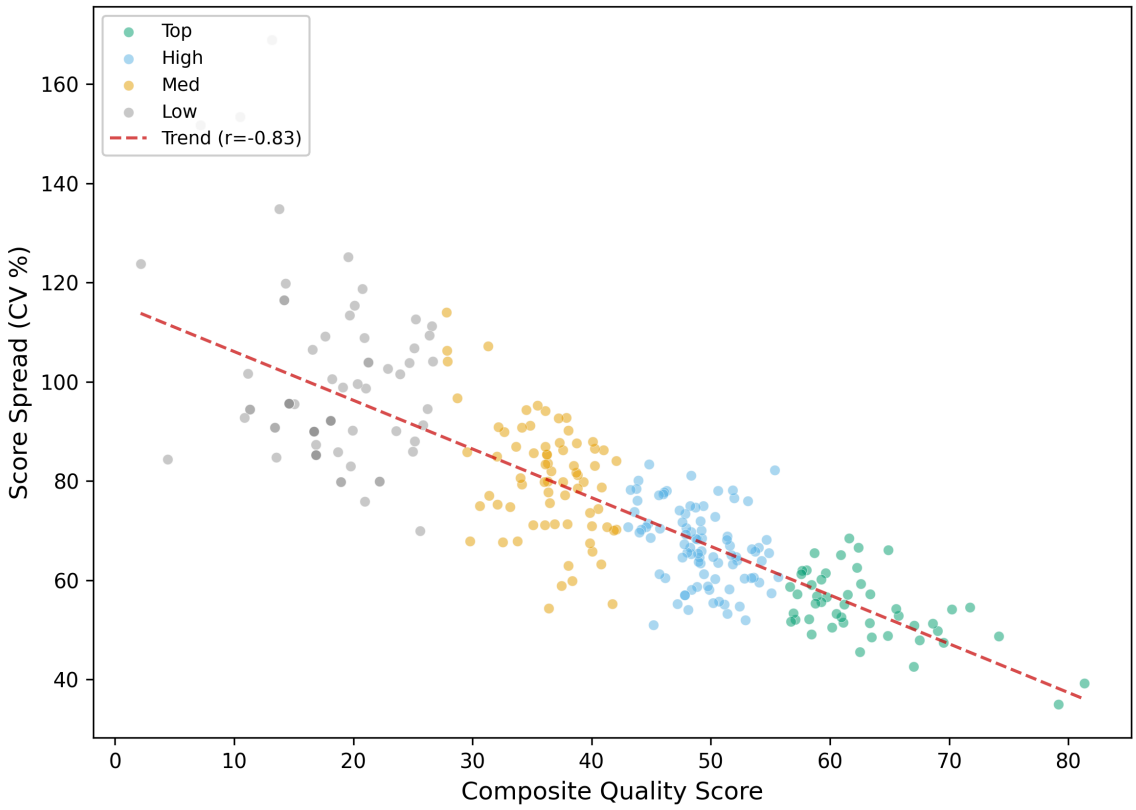


Figure 14: Figure 14: Tier vs. Discrimination Scatter

Figure 15. Infrastructure Pipeline: From Theory to Discrimination Power

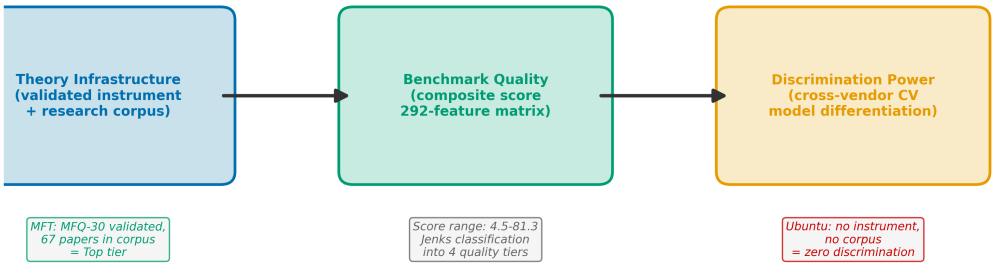


Figure 15: Figure 15: Infrastructure-Quality-Discrimination Pipeline Diagram